From Trace to Time-Bearing Worlds A Protocol-Bound Framework for Self-Reference, Conjugate Geometry, and Ledgered Commitment

A Protocol-Bound Framework for Self-Reference, Conjugate Geometry, and Ledgered Commitment

---

Abstract

Many systems produce compact, publicly usable outputs: a price, valuation, measurement result, verdict, token, scientific conclusion, performance indicator, or institutional decision. Such outputs are often treated as if they were complete descriptions of the systems that produced them. Yet in self-referential settings, the causal importance of a trace can greatly exceed its visible informational content. Internal observers retain the trace, update their filtrations, alter policies or instruments, and thereby participate in producing the system's later states. Some resulting consequences pass gates and become durable records; others remain incompletely integrated as residual. Both ledgered and residual history may then constrain what the system can observe, admit, or become next.

This article proposes a minimum typed framework for such processes. Its mandatory causal core is:

Trace → Filtration → Adaptive Policy → Changed Transition Law → New Trace,

joined, for full operational world formation, by:

Candidate Consequence → Gate → Ledger + Residual → Historical Backreaction.

The framework is organized through three functional roles:

Base → Relation → Commitment → Recompiled Base.

Base denotes the causally relevant condition from which subsequent evolution is generated. Relation denotes the transformation law operating within that Base. Commitment denotes the governed conversion of candidate consequences into durable history, together with preservation of what closure fails to integrate.

Complex numbers occupy an important but conditional position. Self-reference often reveals the incompleteness of scalar descriptions by generating an oriented response that is absent from the admitted scalar trace. When the local Relation generator contains a stable elliptic two-dimensional mode satisfying $J^2 = -I$, a scalar readout Y may admit a conjugate completion $Z_\gamma = Y + iQ_\gamma$, where Q_γ is its signed directional response. Self-reference therefore motivates relational completion, but the generator determines whether that completion is complex, hyperbolic, parabolic, dissipative, mixed, or not usefully reducible.

The framework separates three closure questions: effective-state closure, conjugate measurement closure, and historical closure. It also distinguishes measurement from movement, movement from commitment, ledger from truth, and residual from conjugate exposure. CAPM conjugate valuation, self-referential quantum observers, $\Delta 5$ phase opposition, dissipative dynamics, AI systems, and institutional ledgers are treated as modular examples

rather than manifestations of one universal substrate.

The result is not a completed unified theory. It is a formal architecture and falsifiable research programme for identifying when a protocol-bounded process becomes self-referential, when its Relation earns conjugate geometry, and when its own declared past becomes part of the machinery constructing its admissible future.

Keywords

Self-reference; operational world formation; protocol; trace; observer filtration; adaptive policy; Base–Relation–Commitment; conjugate geometry; complex numbers; gate; ledger; residual; historical closure; CAPM; time-bearing worlds.

Part I — Why a Trace Is Not Yet a World

0. Reader Contract: What This Article Claims and Does Not Claim

0.1 The problem addressed

This article studies systems in which an output produced at one stage becomes part of the causal machinery governing later stages.

The output may be:

- a measurement result;
- a financial value;
- a token selected by an AI model;
- a court judgment;
- an accounting entry;
- a scientific result;
- a policy decision;
- a remembered event;
- an institutional performance measure.

In each case, the visible output may be compact. Its subsequent consequences need not be.

A price may alter collateral, trading behaviour, required return, regulation, and later price formation. A verdict may alter precedent, strategic expectations, admissible arguments, and later judgments. An AI output may enter context, memory, or tool state and influence subsequent generation. A measurement record may change the instrument selected at the next stage.

The common issue is not merely feedback. It is **historically mediated self-conditioning**:

a system produces a trace;

the trace becomes accessible to an internal participant;
the participant acts differently because of it;
the changed action alters the future process that produces later traces.

This article proposes a minimum architecture for analysing that pattern without forcing all participating objects into one undifferentiated state vector.

The principal sequence is:

Trace
→ Filtration
→ Policy
→ Changed Transition Law
→ New Trace
→ Candidate Consequence
→ Gate
→ Ledger + Residual
→ Changed Future Conditions. (0.1)

The compact functional grammar is:

Base → Relation → Commitment → Recompiled Base. (0.2)

Here, Base, Relation, and Commitment are functional roles. They are not three substances, three universal physical dimensions, or three mutually independent ontological regions.

0.2 Operational world, not metaphysical world

The word *world* is used operationally.

Definition 0.1 — Operational world [D]

A protocol-bounded domain becomes an operational world when its admitted traces and records enter the causal machinery that generates its subsequent admissible history. (0.3)

This definition does not claim that an institution, financial market, AI runtime, or scientific model is a universe in the cosmological sense. It identifies a stronger form of causal organization.

A process forms an operational world when its participants must act inside consequences inherited from previous operations of that same process.

Examples include:

- a legal order whose earlier judgments constrain later judgments;
- a market whose earlier prices alter balance sheets and later trading capacity;
- an AI system whose outputs enter memory or tool state;
- a scientific community whose accepted results alter later experimental design;
- an organization whose recorded performance changes later resource allocation.

In all such cases, the past is not merely stored. It participates in producing the future.

This is the sense in which the system becomes **time-bearing**.

0.3 What the article does not claim

The framework does **not** claim:

1. that every feedback system is a self-referential operational world;
2. that every recurring system possesses an internal observer;
3. that every self-referential process requires complex numbers;
4. that self-reference mathematically proves the existence of i;
5. that every pair of coupled variables forms a complex plane;
6. that all systems possess exactly three effective state coordinates;
7. that PORE is a fundamental ontology;
8. that every residual can be represented by a scalar functional Γ;
9. that conjugate exposure is identical to residual;
10. that a ledgered record is necessarily true;
11. that financial, legal, biological, semantic, and quantum systems share one material substrate;
12. that the CAPM construction proves a universal complex theory of markets;
13. that movement, measurement, recognition, and settlement are the same operation;
14. that the framework is a completed unified theory.

These exclusions are structural, not merely rhetorical. They determine how the mathematics must be developed.

In particular:

Self-reference does not imply ellipticity. (0.4)

Cross-coupling does not imply $J^2 = -I$. (0.5)

Measurement does not imply movement. (0.6)

Movement does not imply commitment. (0.7)

Commitment does not imply exhaustion of residual. (0.8)

Ledgering does not imply objective truth. (0.9)

The supplied handoff explicitly requires compatibility with complex phase only where complex structure is earned, while retaining elliptic, parabolic, hyperbolic, dissipative, jump-like, and non-complex possibilities.

0.4 Levels of claim

Because the article integrates exact mathematics, formal architecture, hypotheses, and cross-domain comparisons, major statements will be distinguished by epistemic status.

[D] Definition

A term introduced to specify the framework.

Example:

Operational world = a protocol-bounded domain whose admitted history participates causally in generating later admissible history.

[S] Source-established result

A result explicitly derived or formally constructed in one of the source works.

Example:

In the declared CAPM valuation construction:

$\partial R/\partial \theta = -Q$. (0.10)

[F] Formal consequence

A result derived in this article from declared assumptions.

Example:

If a two-dimensional invariant mode admits $J^2 = -I$ and a nonzero readout covector a, then the state-mode quarter-turn can be represented as a quarter-turn in the associated readout plane.

[H] Testable hypothesis

A proposition requiring empirical or computational validation.

Example:

Some self-referential reduced generators develop persistent elliptic conjugate modes.

[A] Structural analogy

A comparison of functional roles without a claim of material identity.

Example:

An institutional ledger and a quantum record may both preserve admitted history, although their physical implementation and formal dynamics differ.

These labels are necessary because no single source independently proves the complete framework. The source works provide modular foundations: observer recursion, effective-state compilation, conjugate valuation, gate–ledger–residual dynamics, and operator-earned phase organization. Their integration is the present article's proposed contribution.

0.5 Two mathematical clocks

The main framework begins with only two time structures.

Evolution time

Let t represent continuous, latent, or fine-grained evolution.

A general process may be written:

$dX/dt = F_p(X, \pi, u, L, \mathscr{R}, \eta)$. (0.11)

Commitment index

Let k label discrete observation, decision, gate, or record-writing events.

The corresponding runtime may be written:

$X_k \to y_k \to \pi_{k+1} \to X_{k+1}$. (0.12)

The ledger progresses through:

$L_0 \to L_1 \to L_2 \to \cdots$. (0.13)

Phase, recurrence, recovery, switching time, breather period, and market horizon may later be added as derived or domain-specific variables. They are not all primitive dimensions of the universal core.

0.6 Reading path

The article develops in six movements.

Movement 1 — The scalar declares less than it causes

A system emits an admitted trace. Internal observers may act on it, so its causal consequence exceeds its visible content.

Movement 2 — Effective existence is not transformation

A complex recursive process may admit a reduced description, but a reduced Base does not by itself determine how the system moves.

Movement 3 — Conjugate structure must be earned

The local generator must be identified before elliptic, parabolic, hyperbolic, or another Relation algebra is selected.

Movement 4 — Measurement does not create history

A readout can identify exposure without producing movement. Movement can occur without institutional recognition. A gate determines what becomes operative record.

Movement 5 — History changes the next world

Ledgered and residual consequences return through observer filtration, policy, admissibility, and transition structure.

Movement 6 — Examples test modules, not universality

CAPM, quantum observers, Δ5, AI, law, and institutions illustrate different parts of the framework. None alone proves the whole.

---

1. The Scalar Declares Less Than It Causes

1.1 The success of scalar declaration

Modern operational systems depend on compression.

A financial model compresses assumptions about cash flow, time, risk, and discounting into a present value. A scientific experiment compresses a physical interaction into an outcome. A court compresses evidence, procedure, and interpretation into a judgment. An AI model compresses a large conditional distribution into a selected token. An organization compresses distributed activity into a performance indicator.

This compression is indispensable.

Without compact declarations, systems could not:

- communicate;
- compare;
- coordinate;
- allocate resources;
- trigger action;
- establish responsibility;
- preserve records.

The problem is therefore not that scalar outputs are false or useless.

The problem is that a scalar output can be operationally valid while remaining causally incomplete.

Let the detailed causal condition at commitment step k be:

$X_k \in \mathcal{X}_p$. (1.1)

Let the protocol-admitted trace be:

$y_k = h_p(X_k)$. (1.2)

Here:

- p is the declared protocol;
- $\mathcal{X}_p$ is the protocol-relative state space;

- X_k is the fuller causal state;
- h_p is the observation or declaration map;
- y_k is the trace made available under that protocol.

Nothing in Equation (1.2) requires h_p to be invertible.

Usually it is not.

Many distinct causal states may produce the same admitted trace:

$h_p(X_k^1) = h_p(X_k^2) = y_k$, while $X_k^1 \neq X_k^2$. (1.3)

The scalar may therefore identify the declared present without identifying the system's complete future-generating condition.

1.2 A trace is not the full state

The distinction between state and trace is fundamental.

The detailed state may include:

- physical or economic conditions;
- private information;
- internal observer states;
- prior records;
- positions and commitments;
- policy parameters;
- resource constraints;
- unresolved obligations;
- environmental conditions;
- current gate thresholds;
- model or instrument choices.

The trace may expose only one projection.

For example, an asset price does not by itself reveal:

- who holds which positions;
- which trades were leveraged;
- which institutions face collateral constraints;
- which investors interpret the move as information;
- which models will be recalibrated;

- which regulatory or accounting gates will be triggered.

Likewise, an AI token does not by itself reveal the entire internal conditional structure from which it was selected, nor the future effect it will have after entering context or memory.

The trace is therefore better understood as a **protocol-admitted interface object**.

It is what the protocol permits later observers to address.

This gives the first central distinction:

State ≠ Trace. (1.4)

The trace may be derived from the state, but it has a different causal role. Once retained and made addressable, it can influence future policy even when it contains less information than the state from which it arose.

1.3 Three forms of nonclosure

A scalar trace can fail to close the system in three different ways.

1.3.1 State nonclosure

A trace is state-nonclosing when conditioning on it does not make the future independent of omitted causal history.

Formally, a proposed trace state y_k is insufficient when:

$\Pr(y_{k+1} \mid y_k) \neq \Pr(y_{k+1} \mid y_k, H_k)$, (1.5)

for some omitted history H_k.

The omitted history may include:

- previous traces;
- observer memory;
- ledger state;
- residual obligations;
- policy changes;
- branch provenance;
- regime labels.

State nonclosure asks:

> Does the visible declaration contain enough information to predict subsequent evolution?

This is primarily a Base problem.

1.3.2 Relational nonclosure

A scalar may identify the current projection while omitting the signed direction through which the active generator changes that projection.

Suppose a state x moves under a local generator A:

$dx/dt = Ax.$ (1.6)

Let the admitted scalar readout be:

$Y(x) = a(x),$ (1.7)

where a is a covector.

Then:

$dY/dt = a(Ax).$ (1.8)

Knowing Y alone does not generally determine dY/dt.

The missing quantity is not necessarily another hidden substance. It may be a directional response induced by the generator.

When the active mode is elliptic and admits an operator J satisfying:

$J^2 = -I,$ (1.9)

the missing response can be represented by a conjugate coordinate:

$\mathcal{Q}_\gamma(x) = -a(Jx).$ (1.10)

But this complex completion is valid only after the generator and invariant mode have been established.

Relational nonclosure asks:

> Does the admitted scalar preserve the transformation structure required to describe its own local change?

This is primarily a Relation problem.

1.3.3 Historical nonclosure

Even a complete state description does not automatically determine which consequences have become binding history.

A market may move without the movement being recognized in a particular accounting ledger. A scientific anomaly may be observed without being accepted into theory. A model may generate a candidate action without deployment approval. A legal argument may be made without passing admissibility rules.

Let c_k denote a candidate consequence and $\mathcal{G}_p$ the protocol gate:

$e_k = \mathcal{G}_p(c_k).$ (1.11)

The admitted event updates the ledger:

$L_{k+1} = L_k \oplus T_p(e_k).$ (1.12)

What the gate fails to integrate remains residual:

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (1.13)

Historical nonclosure asks:

> Has the consequence become authoritative, addressable, and future-conditioning history?

This is primarily a Commitment problem.

1.4 Why self-reference amplifies scalar insufficiency

In a passive system, an incomplete trace may merely inconvenience an external analyst.

In a self-referential system, the trace is used by participants inside the causal process.

Suppose observer a retains an accessible history:

$\mathscr{F}_{a,k} = \sigma(y_{1:k}, L_{1:k}^{\text{accessible}}, s_{a,1:k})$. (1.14)

The observer selects a later policy:

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (1.15)

The policy participates in the next transition:

$X_{k+1} = \mathscr{B}_p(X_k, \{\pi_{a,k+1}\}, u_k, L_k, \Gamma_k, \eta_k)$. (1.16)

The next trace is:

$y_{k+1} = h_p(X_{k+1})$. (1.17)

The result is a causal loop:

y_k
$\rightarrow \mathscr{F}_{a,k}$
$\rightarrow \pi_{a,k+1}$
$\rightarrow X_{k+1}$
$\rightarrow y_{k+1}$. (1.18)

The trace is informationally smaller than the state, but because it changes observer policy, its causal effect can be much larger than its visible content.

This is the article's first governing principle:

> **The scalar declares less than it causes.**

The self-referential-observer source provides the formal prototype: recorded outcome history generates an observer filtration; the filtration selects later instruments; different histories can therefore produce different later measurement policies and outcome distributions. Its portable contribution is this typed causal relationship, not the automatic transfer of quantum completely positive maps into every other domain.

1.5 The first non-equivalence chain

The preceding analysis yields a sequence of distinctions:

Observable value ≠ Full causal state. (1.19)

Full causal state ≠ Effective reduced Base. (1.20)

Effective Base ≠ Transformation generator. (1.21)

Generator ≠ Readout. (1.22)

Readout ≠ Conjugate response. (1.23)

Conjugate response ≠ Residual. (1.24)

Exposure ≠ Movement. (1.25)

Movement ≠ Admitted consequence. (1.26)

Admitted consequence ≠ Exhausted consequence. (1.27)

Record ≠ Objective truth. (1.28)

These are not terminological refinements. Collapsing any two of them can produce a false theory.

For example, in the CAPM conjugate construction, the measurement $-Q$ identifies first-order valuation-phase exposure. Merely reading $-Q$ does not generate economic profit or loss. Actual phase movement is required. A later recognition or settlement gate determines what enters a financial ledger, and residual may remain after partial admission.

The broader framework generalizes this discipline:

> A system must be analysed through the distinct operations by which it exists, transforms, becomes observed, becomes acted upon, and becomes inherited.

---

2. Feedback, Self-Reference, and World Formation Are Different

2.1 Ordinary dynamics

Consider a process evolving as:

$X_{k+1} = F_p(X_k, u_k, \eta_k)$. (2.1)

Here:

- x_k is the state;
- u_k is an intervention;
- η_k represents disturbance or noise.

The process may be nonlinear, stochastic, dissipative, or highly complex. None of these properties alone makes it self-referential.

Complexity is not self-reference. (2.2)

Recurrence is not self-reference. (2.3)

Memory in the analyst's model is not necessarily memory inside the system. (2.4)

A weather system may display feedback and long-range dependence without selecting policies according to an internally retained symbolic record. A numerical simulation may contain recursive update rules without possessing an internal observer that treats previous outputs as addressable history.

2.2 Ordinary feedback

A feedback controller may observe an error:

$e_k = r_k - y_k$, (2.5)

and apply a control law:

$u_{k+1} = K(e_k)$. (2.6)

The next state depends on the control:

$X_{k+1} = F_p(X_k, u_{k+1})$. (2.7)

This is a genuine causal loop. Yet it need not satisfy the stronger architecture proposed here.

The control grammar K may remain fixed. The error may be used instantaneously and discarded. No identifiable trace need become durable history. No gate need distinguish candidate from admitted consequence. No ledger need alter later admissibility.

Ordinary feedback is therefore insufficient as a definition of self-referential world formation.

2.3 Minimum self-referential process

Definition 2.1 — Self-referential process [D]

A protocol-bound process is self-referential when a trace generated by the process becomes accessible to an internal observer or controller, conditions a later policy or instrument, and thereby participates causally in producing subsequent traces. (2.8)

The minimum topology is:

Trace → Filtration → Policy → Changed Transition Law → New Trace. (2.9)

Four elements are required.

Trace

The process generates an admitted record:

$y_k = h_p(X_k)$. (2.10)

Internal addressability

At least one participant can retrieve or condition upon the trace:

$y_k \in \mathcal{F}_{a,k}$. (2.11)

Policy adaptation

Later action depends on the accessible history:

$\pi_{a,k+1} = f_{a,p}(\mathcal{F}_{a,k})$. (2.12)

Causal backreaction

The selected policy affects subsequent evolution:

$\partial X_{k+1}/\partial \pi_{a,k+1} \neq 0$. (2.13)

A database that stores outputs but is never consulted does not satisfy this condition. A log read only by an external analyst does not by itself create internal self-reference. A fixed feedback gain responding to a present error may qualify only in a weak sense unless retained history can alter later policy, instrumentation, or admissibility.

This gives a practical criterion:

> A trace becomes self-referentially active when removing its accessibility changes the distribution of later system behaviour.

2.4 From self-reference to operational world formation

Self-reference alone does not guarantee a world.

A system may remember and adapt without possessing:

- governed admission;
- durable authority;
- branch-stabilizing records;
- preserved residual;
- future rules conditioned by prior commitment.

A stronger threshold is crossed when candidate consequences become historically binding.

Definition 2.2 — Commitment [D]

Commitment is the protocol-governed transformation by which a candidate consequence becomes an addressable and future-conditioning record, while incompletely integrated consequence is preserved as residual. (2.14)

The minimum Commitment structure is:

$c_k \to \mathcal{G}_p \to e_k \to L_{k+1} + \mathcal{R}_{k+1}$. (2.15)

The admitted record and residual then return:

$(L_{k+1}, \mathcal{R}_{k+1})$
→ later filtration

→ later policy
→ later transition law. (2.16)

Definition 2.3 — Protocol-bound operational world [D]

A self-referential process forms an operational world when its admitted and residual history alters the rules, policies, instruments, or feasible paths governing its subsequent admissible evolution. (2.17)

The complete cycle is:

Trace
→ Filtration
→ Policy
→ Changed State
→ Candidate Consequence
→ Gate
→ Ledger + Residual
→ Changed Future Admissibility. (2.18)

This is stronger than feedback because the system inherits not merely a present error signal but an organized historical condition.

2.5 Two recursive returns

The complete framework contains two distinguishable returns.

Policy return

$y_k \to \mathscr{F}_{a,k} \to \pi_{a,k+1} \to X_{k+1}$. (2.19)

The record changes how an internal participant acts.

Historical return

$c_k \to e_k \to (L_{k+1}, \mathscr{R}_{k+1}) \to X_{k+1}$. (2.20)

The admitted and unresolved consequences change what the system can subsequently do or become.

These returns can operate at different speeds.

- A trader may react immediately to a price trace.
- Settlement may occur later.
- Accounting recognition may occur at a reporting date.
- Legal commitment may require a separate procedure.
- Residual pressure may accumulate before altering policy.
- Public objectivity may require redundant records across observers.

The two returns must therefore remain typed and independently testable.

2.6 The threshold hierarchy

The distinctions can be summarized as follows.

Ordinary dynamics

State changes.

Feedback

Current output changes control.

Self-referential process

Recorded output changes later policy or instrumentation.

Operational world

Admitted and residual history changes later admissibility and evolution.

In compact form:

Dynamics ⊂ Feedback ⊂ Self-Reference ⊂ Operational World Formation, (2.21)

where the inclusion signs indicate increasing architectural requirements, not strict set inclusion for every possible mathematical formalism.

The decisive transition is not from simplicity to complexity.

It is from:

past as prior state

to:

past as addressable, governed, future-generating history.

2.7 Why this world bears time

A sequence of states can be indexed by time without the system carrying an operational history.

A time-bearing world requires more.

Suppose two systems possess the same current observable:

$y_k^A = y_k^B$. (2.22)

They may nevertheless have different futures because:

$L_k^A \neq L_k^B$, (2.23)

$\mathscr{R}_k^A \neq \mathscr{R}_k^B$, (2.24)

$\mathscr{F}_{a,k}^A \neq \mathscr{F}_{a,k}^B$. (2.25)

Therefore:

$\Pr(y_{k+1} \mid y_k, L_k^A, \mathscr{R}_k^A) \neq \Pr(y_{k+1} \mid y_k, L_k^B, \mathscr{R}_k^B)$. (2.26)

The past is operationally real because it changes future possibility even when the currently visible trace is the same.

This yields the preliminary definition:

Definition 2.4 — Time-bearing world [D]

A protocol-bound operational world is time-bearing when accessible ledger and preserved residual alter the admissible, feasible, or probable future paths available from otherwise comparable present states. (2.27)

Earlier SMFT development similarly distinguishes mere recursive presentation from ledgered disclosure: collapse produces a trace, the trace enters a ledger, and later disclosure becomes conditioned by the accumulated ledger. In that source architecture, time is not identified with recursion alone but with the ordered retention of viewpoint-selected commitments.

The present article generalizes that insight beyond one pre-time ontology:

> Time becomes operationally substantial when the system must inherit its own declared past.

[End of Draft Installment 1: Abstract and Sections 0–2]

3. Three Closure Problems

The preceding sections identified three distinct failures of scalar description:

- the visible trace may not close the causal state;
- the scalar readout may not close its local transformation law;
- the observed consequence may not close into binding history.

These failures correspond to three different closure problems.

They must not be treated as different names for the same thing.

3.1 Effective-state closure

A detailed causal state may be too large, inaccessible, or domain-specific to serve as a practical operational description.

One may therefore seek a reduced state:

$\Xi_k = C_p(W_k)$, (3.1)

where:

- C_p is a protocol-relative compiler;
- W_k is a finite or weighted history window;

- Ξ_k is a reduced operational description.

The reduced state has **effective-state closure** when it preserves enough information to support the intended prediction and intervention tasks.

Definition 3.1 — Effective-state closure [D]

A reduced state Ξ_k satisfies effective-state closure relative to protocol P and task class $\mathcal{T}$ when conditioning on omitted accessible history does not materially improve prediction or intervention performance for tasks in $\mathcal{T}$. (3.2)

A practical predictive test is:

$\Pr(y_{k+1} \mid \Xi_k, u_k) \approx \Pr(y_{k+1} \mid \Xi_k, H_k, u_k)$, (3.3)

where H_k denotes historical information excluded by the compiler.

The approximation is task-, scale-, and regime-dependent.

It need not hold universally.

A proposed reduced state may work:

- for short-horizon prediction but not long-horizon regime change;
- under one gate rule but not another;
- in continuous evolution but not during jumps;
- for one observer class but not another;
- for passive prediction but not intervention.

Effective-state closure is therefore not a timeless property of a coordinate tuple.

It is a protocol-relative empirical achievement.

3.2 The PORE compiler as one candidate

The Post-Ontological Reality Engine proposes a protocol-fixed effective description:

$\Xi = (\rho, \gamma, \tau)$, (3.4)

where the coordinates broadly encode:

- ρ — effective presence, occupancy, or basin participation;
- γ — integrity, closure, or resistance to leakage;
- τ — persistence, recurrence, or characteristic duration.

Its governing principle is:

> No protocol, no valid effective state.

The PORE source treats these variables as a compiler output under a declared protocol, not as a universal ontology or complete microscopic state. It also distinguishes smooth response from

jump behaviour and warns that a compact signature may fail when the process changes regime.

Within the present framework:

Base role ≠ PORE signature. (3.5)

The Base role is mandatory.

The PORE signature is optional.

A full causal state may contain:

- observer identity;
- phase orientation;
- sign;
- semantic content;
- branch provenance;
- ledger history;
- residual obligations;

that are not preserved by (ρ, γ, τ).

The PORE triple should therefore be introduced as:

$\Xi_p = C_p$[X-history, traces, interventions], (3.6)

and accepted as an effective Base only after predictive closure tests.

3.3 Failure of effective-state closure

Suppose a proposed compiler produces Ξ_k.

Its insufficiency can be tested by introducing omitted history H_k:

Model A: $y_{k+1} = f(\Xi_k, u_k) + \varepsilon_k$. (3.7)

Model B: $y_{k+1} = g(\Xi_k, H_k, u_k) + \varepsilon'_k$. (3.8)

If Model B produces materially superior out-of-sample performance, then:

Ξ_k is not predictively closed for that task. (3.9)

The missing information may be:

- branch identity;
- observer filtration;
- regime label;
- accumulated residual;

- unobserved policy state;
- delayed commitment;
- event ordering.

The correct response is not to declare the framework falsified in total.

The correct response is to revise:

- the compiler;
- the state dimension;
- the memory window;
- the regime partition;
- the protocol;
- or the task claim.

Effective-state failure is therefore diagnostic.

3.4 Conjugate measurement closure

A reduced Base may be predictively useful while an admitted scalar readout remains relationally incomplete.

Let the active reduced state be:

$\xi \in E$, (3.10)

where E is a local invariant or approximately invariant mode.

Let the admitted scalar readout be:

$Y(\xi) = a(\xi)$, (3.11)

where a is a nonzero readout covector.

If the local generator contains an elliptic operator J satisfying:

$J^2 = -I$, (3.12)

then the directional response of the readout under the quarter-turn generated by J is:

$\mathcal{Q}_\gamma(\xi) = -a(J\xi)$. (3.13)

The completed readout is:

$Z_\gamma = Y + i\mathcal{Q}_\gamma$. (3.14)

Definition 3.2 — Conjugate measurement closure [D]

A scalar readout Y satisfies conjugate measurement closure on an elliptic mode when pairing it with $\mathcal{Q}_Y = -a \circ J$ preserves the local quarter-turn transformation of that mode in the readout

plane. (3.15)

This closure concerns transformation grammar.

It does not concern:

- whether the state is predictively sufficient;
- whether movement has occurred;
- whether an event has been admitted;
- whether a ledger has been updated.

Conjugate closure is therefore neither effective-state closure nor historical closure.

3.5 Historical closure

A process may possess:

- an effective reduced state;
- a valid generator;
- an exact conjugate readout;

and still lack historically binding consequence.

Consider a candidate consequence:

$c_k \in \mathcal{C}_p$. (3.16)

A gate produces an admitted event:

$e_k = \mathcal{G}_p(c_k)$. (3.17)

The event is written to a ledger:

$L_{k+1} = L_k \oplus T_p(e_k)$. (3.18)

Unintegrated consequence is preserved as:

$\mathscr{R}_{k+1} = \mathfrak{R}_p(c_k, e_k, L_{k+1})$. (3.19)

Definition 3.3 — Historical closure [D]

A consequence achieves historical closure when it becomes an addressable, protocol-authorized record whose presence alters later filtration, policy, admissibility, or transition structure. (3.20)

Historical closure is not complete erasure of residual.

A system may admit one consequence while leaving:

- excluded alternatives;
- disagreement;

- basis risk;
- unresolved liabilities;
- unrecorded externalities;
- rejected model branches;
- inaccessible observer records.

Therefore:

Historical closure ≠ total closure. (3.21)

A historically closed event may still coexist with open residual.

3.6 The three closures are logically independent

The closures answer three different questions.

Closure	Question	Main object
Effective-state closure	Can the process be compressed without losing task-relevant causality?	Reduced Base
Conjugate measurement closure	Does the readout preserve the local signed transformation relation?	Relation/ readout
Historical closure	Has a consequence become authorized and future-conditioning history?	Commitment

The combinations are not equivalent.

Case A — Effective state without conjugate geometry

A discrete institutional process may admit a useful reduced Markov state but no elliptic mode.

Case B — Conjugate geometry without historical commitment

A laboratory oscillator may possess an exact complex phase representation without any institutional gate or ledger.

Case C — Historical commitment without predictive closure

A legal judgment may become binding even when the full causal system remains poorly modelled.

Case D — Private self-reference without public closure

An internal observer may retain and act upon a trace that is inaccessible to other observers.

Case E — Ledger without causal return

An archive may preserve a record that no later policy consults.

Such an archive has storage but not world-forming historical closure in the stronger sense used here.

3.7 Closure hierarchy

The framework does not require the three closures to occur in one fixed chronological order.

A process may first create a ledger and only later develop a useful reduced model.

A researcher may first discover a conjugate readout and only later identify the detailed generator.

Nevertheless, the logical dependencies are:

Protocol declaration
→ candidate Effective Base
→ generator identification
→ optional conjugate closure
→ movement and candidate consequence
→ gate and historical closure. (3.22)

The order is methodological rather than ontological.

The principal discipline is:

> Do not infer one closure merely because another has been demonstrated.

Part II — The Minimum Runtime of a Self-Referential World

4. Protocol Before Ontology

4.1 Why the protocol must be declared first

A trace does not exist as a trace independently of:

- what is observed;
- at what scale;
- under what boundary;
- by which observer;
- with what access;
- under what gate;
- for which future use.

The same physical or informational occurrence may be:

- an event under one protocol;
- noise under another;

- a ledger entry under one institution;
- an unrecognized residual under another.

The protocol therefore precedes claims about:

- state;
- phase;
- record;
- residual;
- closure;
- world.

The framework adopts the principle:

> There is no protocol-independent operational world.

This does not mean that nothing exists independently of observation.

It means that the article's terms describe operational roles whose validity depends on declared boundaries and rules.

4.2 Expanded protocol tuple

Let:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (4.1)

The components are:

Boundary B

Defines what belongs to the modelled system and what is treated as environment.

Timebase Δ

Defines observation and commitment granularity.

Observation map h

Defines which aspect of the causal state becomes an admitted trace.

Intervention set u

Defines actions permitted under the protocol.

Gate $\mathcal{G}$

Defines how candidate consequences are accepted, rejected, delayed, transformed, or partially admitted.

Access rule $\mathcal{A}$

Defines which observers can retrieve which traces or ledger records.

Residual rule $\mathfrak{R}$

Defines what is preserved when a candidate consequence is not fully integrated.

The tuple is not required to have this exact software representation in every domain.

Its functions must nevertheless be recoverable.

4.3 Boundary declaration

Let the broader environment be $\mathfrak{E}$.

The protocol boundary selects:

$X_k \in \mathcal{X}_p \subset \mathfrak{E}$. (4.2)

The boundary determines:

- what counts as endogenous;
- what counts as intervention;
- what counts as disturbance;
- what counts as external residual;
- which observer is internal.

Boundary errors can create false self-reference.

For example, if an analyst treats an external regulator as part of the environment, a regulatory response appears as an external shock.

If the regulator is included inside the operational world, the same response becomes historical backreaction.

Neither choice is universally correct.

The boundary must match the question.

4.4 Observation scale and aggregation

Let the fine-grained state evolve at scale dt.

The admitted observation may be aggregated over Δ:

$y_k = h_p(\{X(t): t \in [k\Delta, (k+1)\Delta)\})$. (4.3)

Changing Δ may change:

- whether a loop appears persistent;

- whether phase is detectable;
- whether two events are ordered;
- whether residual accumulates;
- whether a jump is visible.

A high-frequency process may appear as noise at a coarse scale.

A rapid sequence of commitments may appear as one event.

A bounded breather may appear as an attractor when undersampled.

Protocol dependence is therefore not only semantic.

It is mathematical.

4.5 Observation map

The observation map is:

$h_p: \mathcal{X}_p \to \mathcal{Y}_p$. (4.4)

It may be:

- deterministic;
- stochastic;
- many-to-one;
- observer-dependent;
- thresholded;
- censored;
- delayed.

An observer-specific trace may be:

$y_{a,k} = h_{a,p}(X_k)$. (4.5)

Two observers may therefore receive different traces from the same causal state:

$h_{a,p}(X_k) \neq h_b,_p(X_k)$. (4.6)

This need not imply contradiction.

It may reflect:

- different instruments;
- different frames;
- different access rights;

- different aggregation;
- different local coupling.

Cross-observer agreement requires more than identical labels.

It requires compatibility of protocols, frame maps, and accessible records.

The self-referential-observer source explicitly treats agreement as conditional upon compatible instruments and record accessibility, with stronger objectivity supported by redundant records.

4.6 Intervention declaration

Let:

$u_k \in \mathcal{U}_p$. (4.7)

An intervention is not simply any variable correlated with the future.

It is an action admitted by the protocol as manipulable.

This distinction matters when defining a conjugate response.

A formal derivative:

$\partial Y / \partial \theta$ (4.8)

is operationally stronger when θ can be perturbed or when the derivative is exact by construction.

Without a declared intervention or exact structural derivation, the derivative may remain descriptive rather than causal.

4.7 Gate declaration

The gate is:

$\mathcal{G}_p$: $\mathcal{C}_p \times L_k \times X_k \rightarrow$ [illegible]$_p \cup$ {reject, defer}. (4.9)

The gate may:

- admit;
- reject;
- defer;
- partially recognize;
- transform;
- split a candidate into several records.

Examples include:

- trade execution;

- settlement;
- accounting recognition;
- evidential admissibility;
- publication review;
- human approval of an AI action;
- model deployment;
- token decoding.

The gate is not necessarily a conscious decision.

It is any protocol-defined rule by which candidate consequence becomes operative event.

4.8 Access declaration

Let:

$\mathcal{A}_p(a, L_k)$ (4.10)

return the portion of ledger L_k accessible to observer a.

Then:

$L_{1:k}^{\text{accessible}} = \mathcal{A}_p(a, L_{1:k})$. (4.11)

An existing record may be:

- private;
- sealed;
- forgotten;
- encrypted;
- inaccessible;
- available only through an interface.

Existing record ≠ accessible record. (4.12)

Only accessible history can directly enter an observer's filtration.

Inaccessible records may still affect the system indirectly through other agents or institutional constraints.

4.9 Residual declaration

Residual is relative to:

- the candidate field;

- the gate;
- the trace rule;
- the ledger rule;
- the protocol boundary.

Let:

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (4.13)

Residual may include:

- rejected candidates;
- partially admitted consequences;
- unresolved inconsistency;
- branch information;
- unrecognized liability;
- inaccessible trace;
- model error;
- delayed obligation.

Residual is not identical to generic uncertainty.

Uncertainty concerns incomplete knowledge.

Residual concerns incomplete integration under a declared commitment process.

The residual source formalizes this distinction through a protocol-conditioned chain from candidate field to gate, ledger, residual, and future-selective effect.

4.10 Protocol change

A self-referential world may modify its own protocol.

Let:

$P_{k+1} = \mathcal{U}(P_k, L_k, \mathscr{R}_k, \pi_{k+1})$. (4.14)

Examples include:

- a court changing evidential standards;
- an exchange changing margin rules;
- an AI agent updating tool permissions;
- a scientific field revising publication criteria;
- an organization changing KPI definitions.

This is a stronger form of self-reference because history changes not only state or policy but the rules by which future traces and commitments are defined.

However, the framework should not assume protocol revision in every case.

A fixed protocol is sufficient for the minimum architecture.

---

5. Trace, Filtration, and Adaptive Policy

5.1 Detailed causal state

Let the detailed state at commitment index k be:

$X_k \in \mathcal{X}_p$. (5.1)

The state may contain:

$X_k = (x_k, s1_{,k}, ..., s_N_{,k}, L_k, \mathcal{R}_k, \theta_k, r_k, ...)$, (5.2)

where the components may represent:

- physical or economic configuration;
- observer states;
- records;
- residual;
- regime;
- resources;
- policy parameters.

Equation (5.2) is schematic.

The framework does not claim one universal decomposition.

The full causal state is defined operationally:

> It is the minimum state required, together with declared interventions and disturbances, to generate the future distribution under the chosen protocol.

5.2 Trace production

The admitted trace is:

$y_k = h_p(X_k)$. (5.3)

A trace must be distinguishable from a transient micro-event that leaves no accessible record.

Definition 5.1 — Trace [D]

A trace is a protocol-admitted representation of an event or state that persists long enough to become accessible to at least one later process. (5.4)

Persistence may be:

- physical;
- digital;
- institutional;
- cognitive;
- semantic.

The trace need not be permanent.

It must survive long enough to participate in later causality.

5.3 Observer filtration

For observer a, define:

$\mathscr{F}_{a,k} = \sigma(y_{a,1:k}, L_{1:k}^{\text{accessible}}, s_{a,1:k})$. (5.5)

The filtration represents the information operationally available to that observer at step k.

It may include:

- private traces;
- public records;
- internal state;
- prior decisions;
- inferred models;
- received messages.

The filtration does not have to equal the mathematically complete sigma-algebra of all events.

It is observer-relative.

Two observers may possess:

$\mathscr{F}_{a,k} \neq \mathscr{F}_b,_k$. (5.6)

Therefore they may rationally select different policies even when they participate in the same broader world.

5.4 Adaptive policy

The observer selects:

$\pi_{a,k+1} = f_{a,p}(\mathcal{F}_{a,k})$. (5.7)

The policy may determine:

- measurement instrument;
- action;
- model;
- interpretation;
- resource allocation;
- attention;
- gate recommendation;
- communication.

The policy may itself be stochastic:

$\pi_{a,k+1} \sim \Pi_{a,p}(\cdot \mid \mathcal{F}_{a,k})$. (5.8)

The critical requirement is dependence on accessible history.

If:

$\Pi_{a,p}(\cdot \mid \mathcal{F}_{a,k}) = \Pi_{a,p}(\cdot)$, (5.9)

then the observer's later policy is history-independent under the model.

The self-referential loop may then be absent or weaker than claimed.

5.5 Recursive evolution

The next state is generated by:

$X_{k+1} = \mathcal{B}_p(X_k, \{\pi_{a,k+1}\}, u_k, L_k, \Gamma_k, \eta_k)$. (5.10)

Here:

- $\mathcal{B}_p$ is the transition map;
- $\{\pi_{a,k+1}\}$ is the collection of active observer policies;
- u_k is external or protocol-defined intervention;
- L_k is the ledger;
- Γ_k is an optional residual-derived constraint;
- η_k is disturbance or noise.

The next trace is:

$y_{k+1} = h_p(X_{k+1})$. (5.11)

The self-referential cycle is:

$y_k \to \mathscr{F}_{a,k} \to \pi_{a,k+1} \to X_{k+1} \to y_{k+1}$. (5.12)

5.6 Self-reference as causal topology

The framework does not introduce "self-reference" as an additional scalar coordinate `s`.

Self-reference is the existence of a causal path:

$y_k \rightsquigarrow \pi_{a,k+1} \rightsquigarrow X_{k+1} \rightsquigarrow y_{k+1}$. (5.13)

The path is active when:

$\partial \Pr(y_{k+1} \mid \cdot)/\partial \pi_{a,k+1} \neq 0$, (5.14)

and:

$\pi_{a,k+1}$ depends nontrivially on $y_{1:k}$ or accessible ledger history. (5.15)

This yields the article's second governing principle:

> **Self-reference is a causal topology, not a coordinate.**

One may later compile indicators of self-referential strength, but such indicators measure the loop.

They do not constitute it.

5.7 Latching

A trace may become difficult or impossible for an observer to revise.

Let $r_{a,k}$ denote the observer's internal record.

A latching update may take the form:

$r_{a,k+1} = \Lambda_a(r_{a,k}, y_k)$, (5.16)

where, after commitment to outcome `j`:

$r_{a,k} = j \Rightarrow \Pr(r_{a,k+1} = j \mid \text{no contrary admissible event}) \approx 1$. (5.17)

Latching produces observer-relative fixedness.

It does not necessarily produce public objectivity.

The self-referential-observer source distinguishes internal certainty from cross-observer agreement and requires compatible records and access for stronger objectivity.

5.8 Agreement and redundant objectivity

Suppose observers `a` and `b` possess records:

$r_{a,k}$, $r_b,_k$. (5.18)

Agreement is meaningful only after declaring a frame map:

$\Phi_b \leftarrow a: \mathscr{R}_a \rightarrow \mathscr{R}_b$. (5.19)

Then compatibility requires:

$r_b,_k \approx \Phi_b \leftarrow a(r_{a,k})$. (5.20)

Stronger objectivity arises when the event is redundantly encoded:

$e_k \rightarrow \{r1,_k, r2,_k, \ldots, r_m,_k\}$, (5.21)

and multiple observers can independently access compatible copies.

Thus:

Internal fixedness ≠ public objectivity. (5.22)

Agreement ≠ identical raw record. (5.23)

Objectivity requires accessibility, compatibility, and redundancy. (5.24)

5.9 Ablation test for self-reference

To test whether the proposed self-referential path is causal, construct an ablated process in which trace-conditioned policy is removed:

$\pi_{a,k+1}^{abl} = f_{a,p}(\mathscr{F}_{a,k}$ without selected trace history). (5.25)

Compare:

$Pr(y_{k+1} \mid$ full recursion) (5.26)

with:

$Pr(y_{k+1} \mid$ ablated recursion). (5.27)

If the distributions are indistinguishable within the claimed regime, then the retained trace may not be causally active in the asserted way.

This is the minimum falsification test for the observer-recursion module.

---

6. Gate, Ledger, and Residual

6.1 Candidate consequence

The evolution of the system produces candidate consequences:

$c_k = C_p(X_k, X_{k+1}, y_k, \pi_{k+1})$. (6.1)

A candidate may be:

- a proposed action;

- a valuation change;
- a legal claim;
- a model output;
- a transaction;
- a scientific interpretation;
- an internal outcome.

Candidate status means that the consequence has not yet acquired full protocol authority.

6.2 Gate operation

The gate maps:

$e_k = \mathcal{G}_P(c_k; X_k, L_k)$. (6.2)

The output may be:

- admitted event;
- rejected candidate;
- deferred candidate;
- transformed event;
- partially admitted event.

A stochastic gate may be represented by:

$\Pr(e_k \mid c_k, X_k, L_k, P)$. (6.3)

Gate dependence on L_k is especially important.

Prior commitments may alter:

- thresholds;
- eligibility;
- credibility;
- capital capacity;
- admissibility;
- interpretation.

The gate is therefore one location where history directly shapes future reality.

6.3 Ledger update

An admitted event updates the ledger:

$L_{k+1} = L_k \oplus T_p(e_k)$. (6.4)

The operator $\oplus$ may mean:

- append;
- aggregate;
- reconcile;
- net;
- overwrite;
- version;
- branch.

The ledger need not be a literal database.

It is any durable structure whose admitted contents can condition later operations.

Definition 6.1 — Ledger [D]

A ledger is a protocol-governed, addressable record structure whose contents carry operational authority in subsequent system evolution. (6.5)

A diary may be a ledger for one person.

A blockchain may be a ledger for a network.

A precedent database may be a ledger for a legal institution.

A model context may function as a temporary ledger for an AI process.

The authority and durability differ.

The functional role is comparable.

6.4 Residual update

Define:

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (6.6)

Residual is what the commitment operation does not fully integrate.

It may contain:

- rejected alternatives;
- excluded evidence;
- unresolved branch difference;

- partial recognition;
- hidden external cost;
- unsettled obligation;
- inaccessible memory;
- unmodelled effect.

Residual is relational.

A candidate may be fully integrated under one protocol and residual under another.

Therefore:

$\mathscr{R}_{p1}(c) \neq \mathscr{R}_{p2}(c)$. (6.7)

6.5 Residual is not Q

A central non-identity is:

Residual $\mathscr{R} \neq$ Conjugate response $\mathcal{Q}_{\gamma}$. (6.8)

$\mathcal{Q}_{\gamma}$ is defined through the readout and Relation operator:

$\mathcal{Q}_{\gamma} = -a(Jx)$. (6.9)

Residual is defined through the candidate, gate, and ledger:

$\mathscr{R} = \Re_{p}(c, e, L)$. (6.10)

A conjugate response may be exactly calculable before any event occurs.

Residual arises after or relative to an incomplete commitment process.

They may interact.

They are not the same type of object.

6.6 From residual to Γ

In some domains, accumulated residual may be represented by a functional:

$\Gamma_{p}[x; \mathscr{R}1:_{k}]$. (6.11)

An effective action may be written:

$S_eff,_{p}[x] = \int L_{p}(x, \dot{x}, t)dt - \lambda\Gamma_{p}[x]$. (6.12)

A force-like contribution is then:

$F_\Gamma = -\delta\Gamma_{p}/\delta x$. (6.13)

The residual source proposes this as a conditional mathematical interface through which incompletely integrated consequence can influence later path selection. It does not establish that

every residual is scalar, differentiable, variational, or nonnegative.

The hierarchy is:

Raw residual $\mathscr{R}$
→ optional functional Γ
→ optional derivative or subgradient
→ future-selective influence. (6.14)

6.7 Commitment has three conditions

A record becomes operational commitment when it has:

Addressability

A later process can retrieve or condition upon it.

Authority

The protocol treats it as admitted rather than merely observed.

Consequence

Its presence changes later:

- policy;
- permission;
- obligation;
- valuation;
- gate behaviour;
- state evolution.

A stored record lacking consequence is archival.

An authoritative but inaccessible record may constrain the system indirectly.

A consequential but unauthorized signal may act as hidden influence rather than official commitment.

The three properties should therefore be tested separately.

6.8 Historical return

The historical state returns through:

$X_{k+1} = \mathscr{B}_p(X_k, \{\pi_{a,k+1}\}, u_k, L_k, \Gamma_k, \eta_k)$. (6.15)

Observer filtration also includes accessible ledger:

$\mathcal{F}_{a,k} = \sigma(y_{1:k}, \mathcal{A}_p(a, L_{1:k}), s_{a,1:k})$. (6.16)

The ledger can therefore act through two routes:

Direct structural route

L_k changes the transition law or admissible action set.

Observer-mediated route

L_k changes what observers know, which changes policy.

Residual may act through:

- Γ;
- gate thresholds;
- risk limits;
- accumulated contradiction;
- model revision;
- social or institutional pressure.

The complete return is:

$(L_k, \mathscr{R}_k)$
$\rightarrow$ filtration and policy
$\rightarrow$ generator and gate
$\rightarrow$ future trace and commitment. (6.17)

6.9 Historical-return test

To test the claim that history is causally active, compare future outcomes conditional on the current proposed Base.

Let two cases have approximately equal present reduced state:

$\Xi_k^A \approx \Xi_k^B$. (6.18)

But different histories:

$(L_k^A, \mathscr{R}_k^A) \neq (L_k^B, \mathscr{R}_k^B)$. (6.19)

If:

$\Pr(y_{k+1} \mid \Xi_k, L_k^A, \mathscr{R}_k^A)$
$\neq \Pr(y_{k+1} \mid \Xi_k, L_k^B, \mathscr{R}_k^B)$, (6.20)

then history carries predictive information beyond the proposed Base.

This may imply either:

1. history is an independent world-forming causal factor; or

2. the proposed Base failed to incorporate causally relevant history.

The distinction depends on modelling purpose.

The framework allows both readings but requires them to be stated explicitly.

---

7. The Protocol-Bound World-Formation Proposition

7.1 Statement

Proposition 7.1 — Protocol-Bound World Formation [D/F]

Let P be a declared protocol.

Suppose a process satisfies:

1. **Trace production**
   It generates protocol-admitted traces:

   $$y_k = h_p(X_k). \quad (7.1)$$

2. **Internal addressability**
   At least one observer retains accessible history:

   $$\mathscr{F}_{a,k} = \sigma(y_{1:k}, \mathcal{A}_p(a, L_{1:k}), s_{a,1:k}). \quad (7.2)$$

3. **Trace-conditioned policy**
   Later policy depends nontrivially on that history:

   $$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k}). \quad (7.3)$$

4. **Causal backreaction**
   The selected policy alters later state evolution:

   $$X_{k+1} = \mathscr{B}_p(X_k, \{\pi_{a,k+1}\}, u_k, L_k, \Gamma_k, \eta_k). \quad (7.4)$$

5. **Governed admission**
   Candidate consequences pass through a gate:

   $$e_k = \mathcal{G}_p(c_k; X_k, L_k). \quad (7.5)$$

6. **Ledger and residual preservation**
   Admitted and incompletely integrated consequences update:

   $$L_{k+1} = L_k \oplus T_p(e_k), \quad (7.6)$$

   $$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1}). \quad (7.7)$$

7. **Historical return**
   Ledger or residual changes later filtration, policy, gate, or transition structure.

Then the process exhibits **protocol-bound self-referential operational world formation** under P.

7.2 Interpretation

The proposition does not state that the system possesses metaphysical selfhood.

It states that the system's own admitted past participates in producing its subsequent operational possibilities.

The world-forming loop is:

Base_k
$\rightarrow \text{Relation}_k$
$\rightarrow \text{Candidate Consequence}_k$
$\rightarrow \text{Commitment}_k$
$\rightarrow \text{Base}_{k+1}$. (7.8)

The new Base may differ because:

- observers changed policy;
- the ledger changed obligations;
- residual changed constraints;
- the protocol changed;
- the active generator changed regime.

7.3 Necessity of each condition

Remove trace production

No internally addressable result exists.

Remove internal addressability

The trace cannot condition internal policy.

Remove policy adaptation

The process may retain memory but not act differently because of it.

Remove causal backreaction

The observer's updated policy is epiphenomenal.

Remove governed admission

Every candidate is treated identically, and Commitment loses protocol structure.

Remove ledger and residual

Consequences leave no durable or unresolved historical condition.

Remove historical return

The ledger is archival rather than world-forming.

Thus each condition identifies a separate failure mode.

7.4 Weak and strong forms

Weak self-referential world

History changes observer policy but not the formal protocol.

Strong self-referential world

History changes:

- policy;
- gate;
- generator;
- admissible actions;
- or the protocol itself.

Publicly stabilized world

Commitments are redundantly accessible across compatible observers.

These grades should not be conflated.

7.5 Relation to the source modules

The proposition synthesizes several source-supported components:

- the self-referential-observer paper supplies trace-conditioned filtration, adaptive instrument choice, latching, and conditional agreement;
- the residual paper supplies the candidate–gate–ledger–residual–future chain and the optional Γ interface;
- PORE supplies a protocol-relative candidate compiler for an Effective Base;
- the CAPM work supplies an exact conjugate readout branch, developed later in this article;
- the $\Delta 5$ work supplies a separate operator-earned phase-opposition example.

No source independently establishes Proposition 7.1 in this full form.

Its status is therefore an architectural synthesis.

7.6 Immediate corollary

Corollary 7.1 — Time-bearing condition [F]

If two otherwise comparable present states possess different accessible ledger or residual histories, and these differences alter their subsequent admissible evolution, then the operational world carries historical time as a causal variable. (7.9)

This does not require time to be a new physical dimension.

It requires only:

Past commitment → different future possibility. (7.10)

The next Part will compress this mandatory runtime into the functional grammar:

Base → Relation → Commitment → Recompiled Base.

Part III — Base, Relation, and Commitment

8. Base: What Exists Operationally?

8.1 Base as a functional role

The previous Part described the full causal runtime of a self-referential operational world. That runtime contains many typed objects:

- causal state;
- trace;
- filtration;
- observer policy;
- intervention;
- gate;
- ledger;
- residual;
- transition law.

For explanation and comparison across domains, these objects can be organized through three higher-level functions:

Base → Relation → Commitment → Recompiled Base. (8.1)

The first function is **Base**.

Definition 8.1 — Base [D]

Under protocol P, the Base is the causally relevant condition from which the system's subsequent

admissible evolution can be generated. (8.2)

At the most general level, the Base is not a three-coordinate vector, a visible trace, or a metaphysical substance.

It is the fuller causal condition:

$X_k \in \mathcal{X}_p$. (8.3)

The Base may include:

- material configuration;
- observer states;
- available resources;
- policy parameters;
- active constraints;
- accessible records;
- residual obligations;
- regime identity;
- protocol state.

Therefore:

Base ≠ Observable trace. (8.4)

Base ≠ Public declaration. (8.5)

Base ≠ Reduced signature. (8.6)

The trace is emitted by the Base. A reduced signature is compiled from it. Neither should be presumed causally complete.

8.2 Detailed Base and Effective Base

The article distinguishes two levels.

Detailed Base

The detailed Base is:

X_k. (8.7)

It is the state required in principle to generate the subsequent distribution under the declared protocol.

Effective Base

An Effective Base is a compressed representation:

$\Xi_k = C_p(W_k)$, (8.8)

where W_k may contain:

- a recent trajectory window;
- trace history;
- ledger history;
- intervention history;
- estimated hidden states.

The compiler is:

$C_p: W_k \rightarrow \Xi_k$. (8.9)

The reduced representation is useful only if it preserves the causal distinctions required for the intended task.

This yields:

Detailed Base → Compiler → Candidate Effective Base. (8.10)

The arrow is not automatically reversible.

Many detailed states may map to the same reduced state:

$C_p(W_k^1) = C_p(W_k^2) = \Xi_k$, while $W_k^1 \neq W_k^2$. (8.11)

The question is not whether the compiler preserves everything.

The question is whether it preserves what matters for the declared prediction or intervention.

8.3 The PORE Existence Signature

The Post-Ontological Reality Engine proposes a compact protocol-relative signature:

$\Xi = (\rho, \gamma, \tau)$. (8.12)

Its components can be read broadly as:

Coordinate	Operational role
ρ	presence, basin occupation, or effective participation
γ	integrity, closure, or resistance to leakage
τ	persistence, recurrence, or characteristic duration

The PORE source places protocol declaration before the construction of these coordinates and treats the resulting triple as an effective signature rather than a final ontology. It also separates smooth-response analysis from regime jumps.

Within the present framework, PORE is therefore interpreted as:

$\Xi_k = C_p(X_{0:k}, y_{1:k}, L_{1:k}, u_{1:k})$. (8.13)

This is an optional compiler of the Base role.

It does not replace the full state.

8.4 What an Existence Signature may omit

The triple (ρ, γ, τ) may describe whether an organized loop is:

- present;
- coherent;
- persistent.

It may not preserve:

- sign;
- orientation;
- phase;
- observer identity;
- semantic content;
- legal or institutional authority;
- branch provenance;
- ledger composition;
- residual provenance.

For example, two processes may possess comparable occupancy, integrity, and persistence while differing in:

- whether their active mode is restorative or explosive;
- which observer controls the next instrument;
- which obligation remains unresolved;
- which branch generated the present state;
- whether a consequence was officially recognized.

Therefore:

Effective existence ≠ Complete operational content. (8.14)

A PORE signature may answer:

> Does a persistent organized loop exist under this protocol?

It does not automatically answer:

> How does it transform?

or:

> Which consequence becomes binding history?

Those are Relation and Commitment questions.

8.5 Effective Base as predictive state or descriptive signature

A reduced variable can occupy at least three maturity levels.

Descriptive signature

The coordinates summarize observed properties but do not close prediction.

Mesostate

The coordinates support useful short-horizon description within a stable regime.

Effective state

The coordinates approximately close the declared predictive and intervention tasks.

These should not be treated as equivalent.

Definition 8.2 — Effective Base [D]

A reduced representation Ξ_k is an Effective Base relative to $(P, \mathcal{T})$ when it is approximately sufficient for the task class $\mathcal{T}$ and remains stable under the declared intervention class. (8.15)

The same representation may be:

- an Effective Base for regime identification;
- merely descriptive for long-horizon prediction;
- insufficient for intervention design.

8.6 Memory and branch tests

Suppose:

$\Xi_k = C_p(W_k)$. (8.16)

To test whether Ξ_k closes the process, introduce omitted history H_k.

If:

$$\Pr(y_{k+1} \mid \Xi_k, u_k, H_k) \approx \Pr(y_{k+1} \mid \Xi_k, u_k), \quad (8.17)$$

then the omitted history adds little predictive information for the declared task.

If instead:

$$\Pr(y_{k+1} \mid \Xi_k, u_k, H_k) \neq \Pr(y_{k+1} \mid \Xi_k, u_k), \quad (8.18)$$

the proposed Effective Base is incomplete.

Important candidates for H_k include:

- earlier commitment branch;
- observer identity;
- inaccessible ledger;
- accumulated residual;
- switching history;
- delayed action;
- gate sequence.

This is especially important in self-referential worlds because two systems with similar current effective signatures may differ substantially in inherited policy or obligation.

8.7 Base recompilation

Commitment changes the condition from which the next stage begins.

The new Base may be represented as:

$$X_{k+1} = \mathscr{B}_p(X_k, \pi_{k+1}, u_k, L_{k+1}, \Gamma_{k+1}, \eta_k). \quad (8.19)$$

The compiler then produces:

$$\Xi_{k+1} = C_p(W_{k+1}). \quad (8.20)$$

This gives:

$$\text{Base}_k \rightarrow \text{Relation}_k \rightarrow \text{Commitment}_k \rightarrow \text{Detailed Base}_{k+1} \rightarrow \text{Effective Base}_{k+1}. \quad (8.21)$$

The term **Recompiled Base** refers to this return.

It does not imply that the Base is recreated from nothing.

It means that the system's present operational description must be recomputed after history has changed its conditions.

8.8 Base closure failure as a research result

Failure of a proposed Base compiler should not be hidden.

It may reveal that the process requires:

- an additional coordinate;
- explicit observer state;
- a longer memory window;
- a branch variable;
- a residual variable;
- a regime-dependent compiler;
- a non-Markov representation.

A failed three-coordinate reduction may therefore be more informative than a forced successful fit.

The framework's discipline is:

> Use the smallest Base that passes the declared tests, not the smallest Base that produces an attractive diagram.

---

9. Relation: How the Base Can Change

9.1 Relation as transformation grammar

The second functional role is **Relation**.

Definition 9.1 — Relation [D]

Relation is the protocol-relative transformation law governing how the Base can move, couple, recur, amplify, decay, switch, or branch. (9.1)

At the detailed level:

$X_{k+1} = \mathscr{B}_p(X_k, \pi_{k+1}, u_k, L_k, \Gamma_k, \eta_k)$. (9.2)

In continuous form:

$dX/dt = F_p(X, \pi, u, L, \Gamma, \eta)$. (9.3)

These equations remain intentionally general.

Relation may be:

- deterministic;
- stochastic;
- reversible;
- dissipative;

- linear;
- nonlinear;
- continuous;
- jump-like;
- history-dependent.

The framework does not begin with complex numbers, phase, or Hamiltonian motion.

It begins with the transition rule.

9.2 Local reduced generator

If an Effective Base Ξ has been validated within a regime r, local evolution may be approximated by:

$$\delta\Xi_{k+1} = A_r\delta\Xi_k + G_r\delta u_k + \varepsilon_k. \quad (9.4)$$

In continuous time:

$$d(\delta\Xi)/dt = A_r\delta\Xi + G_r\delta u + \varepsilon. \quad (9.5)$$

Here:

- A_r is the local generator;
- G_r is the intervention map;
- ε is model error or disturbance.

The regime label r is essential.

A generator estimated during one regime may fail after:

- a gate change;
- a policy revision;
- a phase transition;
- a liquidity event;
- a model switch;
- a protocol update.

Thus:

$$A_r \neq A \text{ globally.} \quad (9.6)$$

The Relation structure may be local in state, time, observer, and protocol.

9.3 Generator before geometry

The central methodological sequence is:

1. identify the recursive process;
2. declare the protocol;
3. estimate or compile the state;
4. identify a local generator;
5. inspect its invariant modes;
6. determine its algebraic type;
7. introduce phase or complex notation only where earned.

In compact form:

Loop
→ Base
→ Generator
→ Mode
→ Algebra
→ Readout. (9.7)

Not:

Loop
→ Complex plane. (9.8)

This gives the article's third governing principle:

> **Generator before geometry.**

9.4 Regime classification

A two-dimensional mode can be classified through a normalized operator c_χ satisfying:

$C\chi^2 = \chi I$. (9.9)

Elliptic regime

If:

$\chi < 0$, (9.10)

the normalized operator can be written:

$J = C\chi/\sqrt{(-\chi)}$, (9.11)

so that:

$J^2 = -I$. (9.12)

This supports:

- rotation;

- phase;
- quadrature;
- bounded oscillatory closure;
- ordinary complex representation.

Parabolic regime

If:

$$\chi = 0, \quad (9.13)$$

then:

$$C_0^2 = 0. \quad (9.14)$$

This supports:

- drift;
- accumulation;
- shear;
- near-defective memory;
- dual-number-like representation where useful.

Hyperbolic regime

If:

$$\chi > 0, \quad (9.15)$$

define:

$$K = C\chi/\sqrt{\chi}, \quad (9.16)$$

so that:

$$K^2 = I. \quad (9.17)$$

This supports:

- expansion and contraction;
- opposing growth directions;
- saddle dynamics;
- split-complex representation where useful.

Mixed or unreduced regime

A higher-dimensional generator may contain several types simultaneously or no stable low-dimensional invariant mode.

The correct representation may then remain a real matrix system.

9.5 Cross-coupling is insufficient

Consider:

$$\frac{d}{dt}[u; v] = [a\ b; c\ d][u; v]. \quad (9.18)$$

The existence of off-diagonal terms b and c does not imply complex structure.

An elliptic mode requires conditions on the spectrum or canonical form.

For the matrix:

$$A = [a\ b; c\ d], \quad (9.19)$$

the eigenvalues satisfy:

$$\lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0. \quad (9.20)$$

A complex-conjugate pair occurs when:

$$\mathrm{tr}(A)^2 - 4\det(A) < 0. \quad (9.21)$$

Even then, the complex structure may be:

- damped;
- regime-local;
- metric-dependent;
- distorted by nonlinear terms;
- destroyed by switching.

Therefore:

$$\text{Two variables} \neq \text{Complex pair}. \quad (9.22)$$

$$\text{Feedback} \neq \text{Elliptic feedback}. \quad (9.23)$$

$$\text{Self-reference} \neq J^2 = -I. \quad (9.24)$$

9.6 Dissipative Relation

The Relation module must admit open, nonconservative dynamics.

The uploaded dissipative-quantum paper begins from Langevin-type motion with friction and stochastic forcing, uses forward and backward stochastic derivatives, and constructs a nonlinear Schrödinger–Langevin representation. Its relevance here is structural: phase-bearing descriptions can coexist with dissipation, diffusion, and nonconservative evolution.

A general damped elliptic mode may be written:

$$\frac{dz}{dt} = (-\kappa + i\omega)z, \quad (9.25)$$

where:

- $\kappa \geq 0$ is damping;
- ω is angular frequency.

Then:

$$z(t) = z(0)e^{(-\kappa t)}e^{(i\omega t)}. \quad (9.26)$$

Complex structure describes the rotational Relation.

Damping describes amplitude loss.

They are distinct components of the generator.

This prevents the false inference:

Complex phase ⇒ conservative dynamics. (9.27)

9.7 Breather and recurrence regimes

A Relation may generate bounded recurrent structures whose amplitude and phase vary periodically without settling into a fixed point.

A schematic breather form is:

$$z(t) = A(t)e^{(i\theta(t))}, \quad (9.28)$$

where both $A(t)$ and $\theta(t)$ remain bounded and recurrent.

The semantic-breather document supplies a descriptive vocabulary for several timing failures:

- drift;
- freeze;
- sticking;
- overdrive;
- rupture;
- overflow.

Within this article, these are used only as regime labels.

The document's clinical and acupuncture correspondences are not required for, and do not establish, the general formal framework.

9.8 Relation and self-reference

Self-reference can alter Relation in several ways.

Accessible traces may change:

- generator parameters;
- active mode;
- coupling strength;
- damping;
- gate threshold;
- model choice;
- measurement basis.

Thus:

$A_{r,k+1} = \mathcal{A}_p(A_{r,k}, \mathcal{T}_k, L_k, \mathcal{R}_k)$. (9.29)

A self-referential generator may therefore be adaptive:

$dX/dt = F_p(X; A_k)$, (9.30)

$dA_k/dk = H_p(A_k, y_k, L_k, \mathcal{R}_k)$. (9.31)

This creates a distinction between:

State movement

The state moves under the current generator.

Generator revision

History changes the law governing later state movement.

Strong world formation often involves both.

9.9 Locality of complex geometry

Suppose regime r_1 supports:

$J_{r1}^2 = -I$. (9.32)

After a commitment event, regime r_2 may support:

$K_{r2}^2 = I$. (9.33)

The system has moved from elliptic to hyperbolic Relation.

A single global complex plane would then conceal the change.

The framework therefore assumes:

Complex geometry is local until global compatibility is demonstrated. (9.34)

A global complex structure requires consistent transport between local mode planes.

That advanced problem is deferred to the appendices.

10. Commitment: How Consequences Become History

10.1 Commitment as a separate operation

Relation describes movement.

Commitment describes which consequence of movement becomes operational history.

Definition 10.1 — Commitment [D]

Commitment is the protocol-governed conversion of candidate consequence into authoritative, addressable, future-conditioning record, together with preservation of what the conversion does not integrate. (10.1)

The generic sequence is:

Movement
→ Candidate Consequence
→ Gate
→ Admitted Event
→ Ledger + Residual. (10.2)

This sequence prevents several category errors.

10.2 Measurement is not movement

A readout may expose a sensitivity or orientation without changing the state.

Let:

$Y = a(x)$. (10.3)

A measurement operation may select a different readout orientation:

$Y_\varphi = a_\varphi(x)$. (10.4)

The state x may remain fixed.

Thus:

Readout change ≠ State movement. (10.5)

This is particularly important in conjugate geometry. Rotating the measurement basis can reveal the conjugate response without producing the economic, physical, or institutional consequence represented by actual movement.

10.3 Movement is not candidate admission

Suppose the state changes:

$x \to x'$. (10.6)

This produces a candidate consequence:

$c = C_p(x, x')$. (10.7)

The candidate may still fail the gate.

Examples include:

- a price movement not yet recognized in a particular account;
- a generated AI action not yet approved;
- an experimental anomaly not yet accepted;
- a transaction instruction not yet executed;
- evidence not admitted by a court.

Therefore:

State movement ≠ Admitted event. (10.8)

10.4 Admission is not exhaustion

An admitted event updates a ledger:

$L' = L \oplus T_p(e)$. (10.9)

But the update may not capture the complete consequence.

Residual remains:

$\mathscr{R}' = \Re_p(c, e, L')$. (10.10)

Examples include:

- basis risk after hedging;
- unresolved dissent after judgment;
- unrecorded externality after accounting recognition;
- rejected alternatives after model decoding;
- inaccessible observer memory after public consensus.

Therefore:

Admitted event ≠ Fully integrated consequence. (10.11)

10.5 Three levels of commitment

Internal commitment

A trace becomes fixed relative to one observer's filtration.

Examples:

- a remembered outcome;
- an internally latched model choice;
- a private interpretation.

Institutional commitment

A recognized authority admits an event into an operative ledger.

Examples:

- settlement;
- accounting recognition;
- legal judgment;
- approved deployment;
- published result.

Redundant public commitment

Compatible records are independently accessible across multiple observers or channels.

This supports stronger objectivity.

The self-referential-observer source distinguishes internal fixedness from cross-observer agreement and requires compatible records and access for broader objectivity.

These levels should not be collapsed into one binary variable.

10.6 Authority is protocol-relative

A record can be authoritative under one protocol and irrelevant under another.

For example:

- a management report may bind an internal budget process but not a court;
- a market price may bind collateral valuation but not tax recognition;
- a private memory may bind an individual's future action but not public history;
- a scientific preprint may guide research before formal publication.

Authority is therefore represented as:

$\mathrm{Auth}_p(e, L) \in [0, 1]$, (10.12)

or as a discrete institutional status.

The framework does not require authority to be scalar.

It requires its operational rule to be declared.

10.7 Commitment changes admissibility

Let the admissible action set at step k be:

$\mathcal{U}_k = \mathcal{U}_p(X_k, L_k, \mathscr{R}_k)$. (10.13)

After commitment:

$\mathcal{U}_{k+1} = \mathcal{U}_p(X_{k+1}, L_{k+1}, \mathscr{R}_{k+1})$. (10.14)

If:

$\mathcal{U}_{k+1} \neq \mathcal{U}_k$, (10.15)

history has changed future possibility.

Examples include:

- collateral reducing or increasing trading capacity;
- legal precedent altering admissible arguments;
- an AI permission update changing available tools;
- an approved budget changing feasible projects;
- a scientific result changing which hypotheses are credible.

This is the strongest sense in which Commitment creates a time-bearing world.

10.8 Residual governance

A mature Commitment module must specify not only what is admitted but what is done with the remainder.

Residual may be:

- preserved;
- discounted;
- hidden;
- deferred;
- transferred;
- aggregated;
- forgotten;
- transformed into a penalty.

A residual-governance rule can be written:

$\mathscr{R}_{k+1} = \Re_p(\mathscr{R}_k, c_k, e_k, L_{k+1})$. (10.16)

An optional cost functional is:

$\Gamma_{k+1} = \Gamma_p[\mathscr{R}_{0:k+1}]$. (10.17)

The next generator or policy may depend on Γ:

$A_{k+1} = A_p(L_{k+1}, \Gamma_{k+1})$. (10.18)

$\pi_{k+2} = f_p(\mathscr{F}_{k+1}, \Gamma_{k+1})$. (10.19)

Residual is therefore not simply waste.

It may be a future-generating condition.

10.9 Commitment and irreversibility

The underlying Relation may be reversible while Commitment is not.

Suppose:

$x' = Ux$, (10.20)

where ∪ is invertible.

The state movement can be reversed:

$x = U^{-1}x'$. (10.21)

But if the movement triggers a ledger update:

$L' = L \oplus e$, (10.22)

the operational world after reversal may differ because the record remains.

Thus:

Physical reversibility ≠ Historical reversibility. (10.23)

This is one source of an operational arrow of time.

---

11. The Full Recursive Cycle

11.1 Compact functional grammar

The framework can now be compressed into:

Base → Relation → Commitment → Recompiled Base. (11.1)

Each term has a distinct function.

Base

The causally relevant condition from which later evolution is generated.

Relation

The law governing possible transformation.

Commitment

The gate-governed conversion of consequence into ledgered and residual history.

Recompiled Base

The new operational condition after history has altered state, policy, constraints, or protocol.

The triad is not a claim that reality consists of three substances.

It is a minimum explanatory grammar.

11.2 Expanded runtime

The full typed architecture is:

1. detailed state:

   X_k; (11.2)

2. admitted trace:

   $y_k = h_p(X_k)$; (11.3)

3. observer filtration:

   $\mathscr{F}_{a,k} = \sigma(y_{1:k}, \mathcal{A}_p(a, L_{1:k}), s_{a,1:k})$; (11.4)

4. adaptive policy:

   $\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$; (11.5)

5. recursive transition:

   $X_{k+1} = \mathscr{B}_p(X_k, \pi_{k+1}, u_k, L_k, \Gamma_k, \eta_k)$; (11.6)

6. optional Effective Base:

   $\Xi_{k+1} = C_p(W_{k+1})$; (11.7)

7. local Relation generator:

   $\delta\Xi_{k+2} \approx A_r\delta\Xi_{k+1} + G_r\delta u_{k+1} + \varepsilon_{k+1}$; (11.8)

8. readout:

   $Y_{k+1} = a_p(\Xi_{k+1})$; (11.9)

9. optional conjugate response:

$$\mathcal{Q}_{\gamma,k+1} = -a_p(J_r \Xi_{k+1}); \quad (11.10)$$

10. candidate consequence:

$$c_{k+1} = C_p(X_k, X_{k+1}, Y_{k+1}); \quad (11.11)$$

11. gate:

$$e_{k+1} = \mathcal{G}_p(c_{k+1}; X_{k+1}, L_k); \quad (11.12)$$

12. ledger update:

$$L_{k+1} = L_k \oplus T_p(e_{k+1}); \quad (11.13)$$

13. residual update:

$$\mathscr{R}_{k+1} = \Re_p(c_{k+1}, e_{k+1}, L_{k+1}); \quad (11.14)$$

14. historical return:

$$(L_{k+1}, \mathscr{R}_{k+1})$$
$$\rightarrow \text{future filtration, policy, gate, generator, and Base.} \quad (11.15)$$

This is the full architecture.

Not every domain requires every optional module.

The mandatory world-forming core is:

Trace
→ Filtration
→ Policy
→ Changed Transition
→ Gate
→ Ledger + Residual
→ Historical Return. (11.16)

11.3 Typed objects are not state dimensions

The objects in Equation (11.2)–(11.15) have different mathematical types.

Object	Type
x	causal state
y	trace or observation
$\mathcal{F}$	filtration
π	policy
C_p	compiler
Ξ	reduced state or signature
A_r	generator
Y	scalar readout
$\mathcal{Q}_\gamma$	conjugate response
c	candidate consequence
$\mathcal{G}_p$	gate
e	admitted event

Object	Type
$\llcorner$	ledger
$\mathcal{R}$	residual
$\ulcorner$	optional residual-derived functional

They should not be plotted as if they were coordinates of one Euclidean vector merely because they appear in one diagram.

For example:

- policy is a map, not necessarily a state coordinate;
- filtration is an information structure;
- gate is an operator;
- ledger is an ordered record structure;
- residual may be a field, set, measure, or typed collection.

This typed discipline is essential to the framework's generality.

11.4 The three closure tests in the full cycle

The three closure questions now occupy precise locations.

Effective-state closure

Tests:

X-history $\to \Xi$. (11.17)

Does the compiler preserve task-relevant causality?

Conjugate measurement closure

Tests:

Relation mode $\to (Y, \mathcal{Q}_\gamma)$. (11.18)

Does the readout pair preserve the local transformation grammar?

Historical closure

Tests:

candidate $\to$ gate $\to$ ledger. (11.19)

Does the admitted record become accessible and future-conditioning?

These closures are separate but composable.

11.5 Three nested notions of reality

The framework also distinguishes three meanings often compressed into the word *real*.

Real-valued coordinate

A quantity belongs to $\mathbb{R}$.

Selected real readout

A measurement protocol selects one projection as the admitted scalar output.

Historically admitted reality

A consequence passes a gate and enters a future-conditioning ledger.

These are not equivalent.

An imaginary coordinate in a complex representation may be algebraically real-valued as a coefficient.

A selected real readout may never become institutional commitment.

A ledgered event may be historically operative even if later shown to be factually mistaken.

Therefore:

Algebraic reality ≠ Measurement selection ≠ Historical admission. (11.20)

This distinction will become central when the article introduces complex geometry.

11.6 The minimal world-forming engine

The complete architecture can be summarized in one recursive expression:

$\mathcal{W}_{k+1} = \mathscr{R}_p(\mathcal{W}_k, \mathrm{Trace}_k, \mathrm{Policy}_{k+1}, \mathrm{Commitment}_k)$, (11.21)

where $\mathcal{w}_k$ denotes the current operational world-state, including:

- Base;
- active Relation;
- observer filtrations;
- ledger;
- residual;
- protocol.

More explicitly:

$\mathcal{W}_k = (\mathrm{X}_k, \mathrm{P}_k, \{\mathscr{F}_{a,k}\}, \mathrm{L}_k, \mathscr{R}_k, \mathrm{A}_k)$. (11.22)

Then:

$\mathcal{W}_{k+1} = \mathbb{F}(\mathcal{W}_k, \mathrm{y}_k, \{\pi_{a,k+1}\}, \mathrm{e}_k, \mathscr{R}_{k+1})$. (11.23)

This notation is useful for high-level comparison.

It should not replace the typed equations.

11.7 The core architectural claim

The central claim of Part III can now be stated:

> **An operational world persists because a Base supports transformation, Relation produces candidate consequence, Commitment makes selected consequence historically effective, and that history recompiles the conditions of subsequent existence.**

The next Part addresses the framework's most distinctive conditional branch:

> Why does self-reference so often expose the insufficiency of scalar description, and under what precise conditions does the missing Relation become complex?

Part IV — Why Self-Reference Often Invites Conjugate Geometry

12. Self-Reference and Relational Incompleteness

12.1 From causal recursion to geometric demand

The mandatory world-forming core established in Parts I–III does not require complex numbers.

A system can be self-referential whenever:

Trace → Filtration → Policy → Changed Transition Law → New Trace. (12.1)

It can form an operational world whenever that recursive return is joined by:

Candidate → Gate → Ledger + Residual → Changed Future Admissibility. (12.2)

Neither sequence alone implies:

$J^2 = -I$. (12.3)

Nevertheless, self-referential systems frequently create a particular descriptive difficulty.

A visible trace may report the system's current admitted condition, while omitting the oriented response generated by the way observers, policies, or internal couplings act upon that trace.

The present scalar says:

> What has been selected or declared?

It may not say:

> Along which relational direction is the system now prepared to change?

This is **relational incompleteness**.

12.2 Why recursion often creates a missing direction

Consider a state x and a scalar readout:

$Y(x) = a(x)$, (12.4)

where a is a linear readout covector.

Suppose the system evolves under a local generator A:

$dx/dt = Ax$. (12.5)

Then:

$dY/dt = a(Ax)$. (12.6)

The current value $Y(x)$ generally does not determine $a(Ax)$.

Two states may produce the same scalar readout:

$a(x_1) = a(x_2)$, (12.7)

while having different directional responses:

$a(Ax_1) \neq a(Ax_2)$. (12.8)

This problem can occur in any dynamical system. Self-reference makes it especially important because the generator may itself depend on prior traces.

Let:

$A_{k+1} = \mathcal{A}_p(A_k, \mathscr{F}_k, L_k, \mathscr{R}_k)$. (12.9)

Then the trace influences:

- the observer's policy;
- the observer's policy influences the transition law;
- the transition law determines the direction of the next readout change.

The scalar trace is therefore inserted into a causal loop that creates its own future response direction.

12.3 Scalar nonclosure in self-referential systems

A self-referential scalar may be insufficient in at least four ways.

Value without response

The scalar tells us the admitted value but not its first-order response under the active generator.

Outcome without observer orientation

The scalar tells us what was recorded but not how different observers will act upon it.

State without phase

The scalar identifies the current projection but not where the system lies within a recurrent mode.

Declaration without counterpressure

The scalar identifies what has been recognized but not the organized pressure acting against, around, or through that recognition.

These are related but not identical.

The framework should not assume that every missing quantity can be compressed into one coordinate called Q.

Instead, it asks:

1. Does a stable low-dimensional Relation mode exist?
2. What is its algebraic type?
3. Does the declared readout detect that mode?
4. Does pairing the readout with a second quantity close the transformation law?
5. Does that pairing improve prediction, intervention, or interpretation?

Only after these questions are answered should a conjugate coordinate be introduced.

12.4 Self-reference motivates completion

The general insight inherited from the earlier discussion can now be stated carefully.

Conjecture 12.1 — Relational Completion Conjecture [H]

In some self-referential systems, the recursive coupling between trace-conditioned policy and subsequent state evolution produces a stable relational degree of freedom that is not preserved by the admitted scalar trace. (12.10)

This conjecture does not specify the algebra of the missing relation.

The missing structure may be:

- elliptic;
- hyperbolic;
- parabolic;
- dissipative;
- delayed;
- jump-like;
- higher-dimensional.

The more specific complex-number claim requires additional conditions.

Conjecture 12.2 — Elliptic Completion Conjecture [H]

When a self-referential reduced generator contains a persistent two-dimensional elliptic invariant mode, a scalar readout that detects that mode can be locally completed by a signed conjugate response. (12.11)

The distinction between Conjectures 12.1 and 12.2 is essential.

Self-reference motivates the search for completion.

The generator determines the completion's mathematics.

12.5 Why complex numbers repeatedly appear

Complex numbers are particularly effective when two quantities form a closed quarter-turn relation.

Let:

$z = u + iv.$ (12.12)

Multiplication by `i` gives:

$iz = -v + iu.$ (12.13)

Therefore:

$(u, v) \to (-v, u).$ (12.14)

Applying the operation twice gives:

$i^2 z = -z,$ (12.15)

and four applications return to the starting state:

$i^4 z = z.$ (12.16)

This algebra naturally represents:

- phase;
- quadrature;
- oscillation;
- impedance;
- rotating frames;
- wave propagation;
- feedback with oriented lag;
- conjugate measurement.

Complex numbers do not merely store two real quantities.

They encode a specific transformation law between them.

This is why they are more informative than an unordered pair `(u, v)` whenever the relation:

$u \to -v \to -u \to v \to u$ (12.17)

is structurally active.

12.6 The correct backbone statement

The strongest defensible formulation is:

> **Self-reference frequently exposes scalar relational incompleteness. When the active missing relation closes elliptically, complex numbers provide the natural local completion.**

Or in more compact form:

> **Self-reference motivates completion; the generator determines whether that completion is complex.**

This principle is important to the framework, but it belongs inside the Relation module.

It is not the definition of self-reference.

It is not the definition of an operational world.

It is not the definition of Commitment.

The Handoff Summary explicitly preserves this hierarchy by requiring the recursive trace–policy–history loop as mandatory while treating signed-conjugate geometry as an independently activated module.

---

13. Signed-Conjugate Classification

13.1 One algebra should not be forced onto every regime

Suppose a local two-dimensional mode is governed by a normalized operator C_χ satisfying:

$C\chi^2 = \chi I.$ (13.1)

The sign of χ determines three qualitatively different Relation structures.

Elliptic

$\chi < 0.$ (13.2)

Parabolic

$\chi = 0.$ (13.3)

Hyperbolic

$\chi > 0$. (13.4)

This classification is not intended as an exhaustive classification of every nonlinear system.

It is a minimum local grammar for distinguishing three forms of signed relational completion.

13.2 Elliptic completion

For $\chi < 0$, define:

$J = C\chi/\sqrt{(-\chi)}$. (13.5)

Then:

$J^2 = -I$. (13.6)

The exponential is:

$e^{(J\theta)} = I \cos \theta + J \sin \theta$. (13.7)

This produces bounded rotation in the invariant mode plane.

If:

$x(\theta) = e^{(J\theta)}x_0$, (13.8)

then:

$dx/d\theta = Jx$. (13.9)

The mode possesses:

- orientation;
- periodicity;
- quadrature;
- phase;
- quarter-turn closure.

Ordinary complex numbers are the natural representation.

13.3 Parabolic completion

For $\chi = 0$:

$N^2 = 0$. (13.10)

Then:

$e^{(N\xi)} = I + \xi N$. (13.11)

This describes a shear or accumulation rather than rotation.

Repeated action does not produce a four-step cycle.

It produces linear drift along a nilpotent direction.

Such a regime may represent:

- accumulated memory;
- unresolved carry;
- one-way integration;
- near-defective dynamics;
- slow policy drift;
- secular growth.

A complex number would conceal the distinction between circulation and accumulation.

Dual-number-like notation may sometimes be useful:

$z = u + \varepsilon v$, with $\varepsilon^2 = 0$. (13.12)

But the framework does not require specialized algebra where a real matrix representation is clearer.

13.4 Hyperbolic completion

For $\chi > 0$, define:

$K = C\chi/\sqrt{\chi}$. (13.13)

Then:

$K^2 = I$. (13.14)

The exponential is:

$e^{(K\xi)} = I \cosh \xi + K \sinh \xi$. (13.15)

This supports:

- expansion in one direction;
- contraction in another;
- saddle dynamics;
- self-amplification;
- positive-feedback separation.

A split-complex representation may be introduced:

$z = u + jv$, with $j^2 = 1$. (13.16)

The associated transformation differs fundamentally from ordinary phase rotation.

Thus:

Elliptic opposition ≠ Hyperbolic separation. (13.17)

13.5 Mixed modes

A high-dimensional generator may contain several mode types.

For example:

$A \approx A_{elliptic} \oplus A_{hyperbolic} \oplus A_{dissipative}$. (13.18)

One subsystem may oscillate while another amplifies and another decays.

A single global complex variable would then be insufficient.

The correct description may require:

- several local mode planes;
- a block decomposition;
- regime-dependent coordinates;
- a higher-dimensional real representation.

The framework therefore treats complex geometry as a local chart until compatibility across modes is established.

13.6 Spectral test

For a local real 2×2 generator:

$A = [a\ b; c\ d]$, (13.19)

the characteristic polynomial is:

$\lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$. (13.20)

Define the discriminant:

$\Delta_A = \mathrm{tr}(A)^2 - 4\det(A)$. (13.21)

Then:

- $\Delta_A < 0$ gives a complex-conjugate eigenpair;
- $\Delta_A = 0$ gives a repeated or defective boundary case;
- $\Delta_A > 0$ gives two real eigenvalues.

A complex-conjugate eigenpair is evidence for a local elliptic component, but further checks remain necessary.

One must test:

- persistence across windows;
- approximate invariance of the mode plane;
- robustness under protocol perturbation;
- stability of metric and normalization;
- whether the readout detects the mode;
- whether nonlinear terms destroy closure.

13.7 Cross-coupling is not enough

Consider:

$du/dt = au + bv$, (13.22)

$dv/dt = cu + dv$. (13.23)

The mere existence of b and c does not determine the Relation type.

Examples:

Elliptic coupling

$A = [0\ -\omega;\ \omega\ 0]$. (13.24)

Then:

$A^2 = -\omega^2 I$. (13.25)

Hyperbolic coupling

$A = [0\ \kappa;\ \kappa\ 0]$. (13.26)

Then:

$A^2 = \kappa^2 I$. (13.27)

Parabolic coupling

$A = [0\ 1;\ 0\ 0]$. (13.28)

Then:

$A^2 = 0$. (13.29)

All three matrices contain cross-coupling.

Only the first supports ordinary complex phase.

Therefore:

Coupling does not determine algebra. (13.30)

The signed square of the normalized generator does.

13.8 Self-reference may change the algebraic regime

Suppose the generator depends on ledger history:

$A_k = A(L_k, \mathscr{R}_k)$. (13.31)

A commitment may move the system across:

$\Delta_A < 0 \rightarrow \Delta_A > 0$. (13.32)

The system changes from elliptic to hyperbolic behaviour.

Examples may include:

- stable oscillation becoming runaway amplification;
- market repricing becoming a margin spiral;
- deliberative disagreement becoming polarization;
- AI revision becoming irreversible tool escalation.

Such transitions are hypotheses until empirically instantiated.

The general lesson is formal:

> Historical return may alter not only the state but the algebra of the active Relation.

13.9 Signed-Conjugate Mode Emergence Hypothesis

Hypothesis 13.1 [H]

In some self-referential operational worlds, trace-conditioned policy and historical backreaction produce a persistent reduced two-dimensional mode satisfying:

$C\chi^2 \approx \chi I$, (13.33)

over an identifiable regime.

The sign and stability of χ determine whether the mode is:

- elliptic;
- parabolic;
- hyperbolic.

This hypothesis is falsifiable.

It fails if:

- no stable two-dimensional reduction exists;
- χ varies unpredictably across nearby windows;
- omitted history destroys the apparent mode;
- the mode is an artefact of preprocessing;
- the readout is insensitive to it.

---

14. The Elliptic Branch and the Meaning of i

14.1 The quarter-turn operator

Assume an active invariant plane E and an operator:

$$J: E \to E, \quad (14.1)$$

satisfying:

$$J^2 = -I. \quad (14.2)$$

For any $x \in E$:

$$x \to Jx \to -x \to -Jx \to x. \quad (14.3)$$

This is a four-stage orientation cycle.

The imaginary unit i is the scalar representation of this operator.

The identification is:

$$i \leftrightarrow J. \quad (14.4)$$

Thus i should be understood as a transformation rule.

It does not denote a hidden material quantity.

14.2 Complex coordinate construction

Choose a basis $\{e_1, e_2\}$ satisfying:

$$Je_1 = e_2, \quad (14.5)$$

$$Je_2 = -e_1. \quad (14.6)$$

Write:

$$x = ue_1 + ve_2. \quad (14.7)$$

Associate:

$$z = u + iv. \quad (14.8)$$

Then:

$Jx \leftrightarrow iz = -v + iu.$ (14.9)

The complex coordinate preserves both:

- the state coordinates `(u, v)`;
- the orientation grammar generated by `J`.

The structure is therefore richer than a two-column table.

14.3 Phase evolution

Let:

$dx/d\theta = Jx.$ (14.10)

Then:

$x(\theta) = e^{(J\theta)}x(0).$ (14.11)

In complex notation:

$z(\theta) = e^{(i\theta)}z(0).$ (14.12)

If:

$z(0) = Ae^{(i\theta_0)},$ (14.13)

then:

$z(\theta) = Ae^{[i(\theta_0+\theta)]}.$ (14.14)

The amplitude `A` and phase `θ₀ + θ` separate two features:

- how much of the mode is present;
- where the mode lies within its Relation cycle.

A scalar projection typically preserves only part of this information.

14.4 Real and imaginary as protocol-relative roles

Suppose the readout protocol selects:

$Y = \mathrm{Re}(z).$ (14.15)

Then:

$Y = u.$ (14.16)

The unselected quadrature is:

$Q = \mathrm{Im}(z) = v.$ (14.17)

The labels *real* and *imaginary* here do not mean:

- existent and nonexistent;
- physical and unreal;
- actual and fictional.

They mean:

- selected projection;
- algebraically conjugate quadrature.

Both u and v are real-valued coefficients.

The complex structure lies in their transformation relation.

14.5 Three meanings of “real”

The framework distinguishes:

Real-valued coefficient

$u \in \mathbb{R}$.

Selected real projection

The protocol admits u as the current scalar readout.

Historically realized consequence

A movement passes a gate and enters a ledger.

These three senses must remain separate.

An imaginary-axis coordinate may be a perfectly real-valued exposure.

A real-axis readout may remain uncommitted.

A ledgered consequence may be historically effective even when based on an erroneous measurement.

Therefore:

Real-valued ≠ Selected ≠ Historically committed. (14.18)

14.6 Conjugacy and strict opposition

Let:

$z = u + iv$. (14.19)

The pair (u, v) is conjugate or quadrature-related under the quarter-turn operator.

The strict opposite of z is:

$$-z = -u - iv. \quad (14.20)$$

Thus:

- `u` and `v` are not strict opposites;
- `z` and `-z` are strict opposites;
- `(u, v)` and `(-u, -v)` are polar opposites.

This distinction matters when comparing complex conjugacy with Δ5 phase opposition.

14.7 Quarter-turn versus half-turn

The Δ5 paper defines a half-turn operator:

$$T_5^2 = I. \quad (14.21)$$

The phase-opposed sector satisfies:

$$T_5 a = -a. \quad (14.22)$$

Its pair-energy:

$$E_pair(a) = \Sigma_{n=1}^{5} |a_n + a_{n+5}|^2 \quad (14.23)$$

is minimized when:

$$a_{n+5} = -a_n. \quad (14.24)$$

This is a rigorous operator-earned opposition.

But it is not automatically the same as an elliptic quarter-turn.

The difference is:

Half-turn

$$x \to -x. \quad (14.25)$$

Quarter-turn

$$x \to Jx \to -x. \quad (14.26)$$

A Δ5 pair may occupy polar positions within a phase cycle, while the conjugate readout occupies quadrature positions.

The Δ5 source is therefore valuable as a warning:

> Opposition must be derived from the declared operator and cost; it should not be inferred from symbolic pairing alone.

14.8 Damped elliptic dynamics

Elliptic structure may coexist with dissipation.

Let:

dz/dt = (−κ + iω)z. (14.27)

Then:

z(t) = z0e^(−κt)e^(iωt). (14.28)

The mode rotates while its amplitude decays.

The generator decomposes as:

A = −κI + ωJ. (14.29)

Here:

- `-κI` produces isotropic damping;
- `ωJ` produces phase rotation.

Thus:

Dissipation ≠ Loss of all phase structure. (14.30)

The dissipative-quantum source provides a more elaborate example in which stochastic and frictional dynamics coexist with a complex wave representation. Its relevance is to show compatibility, not to universalize its physical equations.

14.9 Why i is useful in self-referential analysis

In a self-referential system, the complex representation may preserve:

- the admitted scalar state;
- the direction of recursive response;
- the phase of observer-policy coupling;
- delayed counteraction;
- restorative versus amplifying orientation.

But only the elliptic component belongs naturally to ordinary `i`.

Other aspects—ledger, residual, gate, observer identity—remain separate typed objects.

A complex number can complete a local Relation.

It cannot, by itself, encode the entire operational world.

---

15. Conjugate Measurement Closure

15.1 Readout on an elliptic mode

Let `E` be a two-dimensional real vector space equipped with:

$J^2 = -I$. (15.1)

Let:

$a \in E^*$ (15.2)

be a nonzero readout covector.

Define the scalar readout:

$Y(x) = a(x)$. (15.3)

Define its conjugate response:

$\mathcal{Q}_\gamma(x) = -a(Jx)$. (15.4)

The pair:

$\Phi_a(x) = [Y(x); \mathcal{Q}_\gamma(x)]$ (15.5)

maps the dynamic mode into a two-dimensional readout plane.

15.2 Differential relation

Let the state evolve by:

$dx/d\theta = Jx$. (15.6)

Then:

$dY/d\theta = a(dx/d\theta)$. (15.7)

Therefore:

$dY/d\theta = a(Jx)$. (15.8)

Using Equation (15.4):

$dY/d\theta = -\mathcal{Q}_\gamma$. (15.9)

Now differentiate the conjugate response:

$d\mathcal{Q}_\gamma/d\theta = -a(Jdx/d\theta)$. (15.10)

Since:

$dx/d\theta = Jx$, (15.11)

then:

$d\mathcal{Q}_\gamma/d\theta = -a(J^2x)$. (15.12)

Using $J^2 = -I$:

$d\mathcal{Q}_\gamma/d\theta = a(x)$. (15.13)

Therefore:

$dQ_\gamma/d\theta = Y$. (15.14)

The readout pair satisfies:

$d/d\theta\ [Y; Q_\gamma] = [0\ -1; 1\ 0][Y; Q_\gamma]$. (15.15)

This is exact under the stated assumptions.

15.3 Complex readout

Define:

$Z_\gamma = Y + iQ_\gamma$. (15.16)

Then:

$dZ_\gamma/d\theta = iZ_\gamma$. (15.17)

Therefore:

$Z_\gamma(\theta) = e^{(i\theta)}Z_\gamma(0)$. (15.18)

The scalar readout has been completed into a phase-bearing quantity.

This is **conjugate measurement closure**.

15.4 Measurement cycle

Repeated differentiation gives:

Y (15.19)

$dY/d\theta = -Q_\gamma$ (15.20)

$d^2Y/d\theta^2 = -Y$ (15.21)

$d^3Y/d\theta^3 = Q_\gamma$ (15.22)

$d^4Y/d\theta^4 = Y$. (15.23)

Hence the cycle:

$Y \to -Q_\gamma \to -Y \to Q_\gamma \to Y$. (15.24)

This is a measurement-orientation cycle.

It does not by itself describe chronological state changes at four consecutive times.

The distinction between:

- rotating the measurement orientation;
- moving the state;
- admitting the consequence;

must be preserved.

15.5 Unit consistency

Suppose:

$[Y] = U$, (15.25)

where υ denotes the readout's physical, economic, or informational unit.

Since:

$\mathcal{Q}_\gamma = -dY/d\theta$, (15.26)

and phase θ is dimensionless:

$[\mathcal{Q}_\gamma] = U$. (15.27)

This is an important test.

The readout and its conjugate response can occupy one complex plane because they have compatible units.

If the proposed second coordinate has incompatible units, one must:

- normalize;
- introduce a metric or conversion operator;
- or reject the complex pairing.

15.6 Generalized Relation coordinate

The conjugate response need not always be parameterized by a literal angle.

Let ξ denote a declared Relation coordinate.

Define:

$\mathcal{Q}_\gamma = -\partial Y/\partial \xi$. (15.28)

This quantity is meaningful when:

- ξ is operationally defined;
- the derivative is structurally exact or empirically estimable;
- the units are interpretable;
- the direction corresponds to an active mode.

If ξ is not dimensionless:

$[\mathcal{Q}_\gamma] = [Y]/[\xi]$. (15.29)

A conversion factor or metric may then be required before forming a complex coordinate.

Ordinary phase is the cleanest case because it preserves units automatically.

15.7 Covector-like character

The conjugate response depends on both:

- the state direction;
- the readout rule.

It is not merely another component of the state.

Let:

$$Y = V(\Xi). \quad (15.30)$$

Suppose the Relation direction is:

$$\partial\Xi/\partial\theta_D. \quad (15.31)$$

Then:

$$dY/d\theta_D = \nabla_\Xi V \cdot \partial\Xi/\partial\theta_D. \quad (15.32)$$

Define:

$$Q_D = -dY/d\theta_D. \quad (15.33)$$

Thus:

$$Q_D = -\nabla_\Xi V \cdot \partial\Xi/\partial\theta_D. \quad (15.34)$$

The conjugate response is a pairing between:

- a tangent direction $\partial\Xi/\partial\theta_D$;
- a valuation or readout covector $\nabla_\Xi V$.

This explains why the same dynamic mode may produce different conjugate responses under different readout protocols.

15.8 Readout dependence

Let two observers use:

$$Y_1 = a_1(x), \quad (15.35)$$

$$Y_2 = a_2(x). \quad (15.36)$$

Their conjugate responses are:

$$Q_1 = -a_1(Jx), \quad (15.37)$$

$$Q_2 = -a_2(Jx). \quad (15.38)$$

Even when the underlying dynamic mode is the same:

$Q_1 \neq Q_2$ (15.39)

in general.

Conjugate exposure is therefore protocol-relative.

The dynamic complex mode is not identical to every complex readout constructed from it.

15.9 Failure of conjugate measurement closure

The construction fails or becomes degenerate when:

Readout annihilation

If:

a|_E = 0, (15.40)

then:

Y = Q_γ = 0 (15.41)

on the active mode.

Rank failure

If `a` and `a∘J` are linearly dependent, the readout map may lose rank.

Mode leakage

The state leaves the invariant plane.

Generator change

The active regime ceases to satisfy `J² = -I`.

Unit failure

The proposed coordinates cannot be consistently normalized.

Empirical failure

The proposed Q_γ does not predict first-order movement in `Y`.

These are genuine rejection conditions.

---

16. The Complex Intertwining Principle

16.1 Statement

Let:

- `E` be a two-dimensional real vector space;
- `J: E → E` satisfy `J² = -I`;
- `a ∈ E*` be a nonzero readout covector.

Define:

Y(x) = a(x), (16.1)

$\mathcal{Q}_\gamma(x) = -a(Jx)$, (16.2)

and:

$\Phi_a(x) = [Y(x); \mathcal{Q}_\gamma(x)]$. (16.3)

Let the standard readout-plane quarter-turn be:

J_R = [0 −1; 1 0]. (16.4)

Proposition 16.1 — Complex Intertwining Principle [F]

The readout map satisfies:

$\Phi_a J = J_R \Phi_a$. (16.5)

16.2 Proof

For any `x ∈ E`:

$\Phi_a(Jx) = [a(Jx); -a(J^2x)]$. (16.6)

Since:

$J^2x = -x$, (16.7)

then:

$\Phi_a(Jx) = [a(Jx); a(x)]$. (16.8)

Using:

Y(x) = a(x), (16.9)

and:

$\mathcal{Q}_\gamma(x) = -a(Jx)$, (16.10)

we obtain:

$a(Jx) = -\mathcal{Q}_\gamma(x)$. (16.11)

Therefore:

$\Phi_a(Jx) = [-\mathcal{Q}_\gamma(x); Y(x)]$. (16.12)

But:

$J_R\Phi_a(x)$
$= [0\ -1;\ 1\ 0][Y(x);\ \mathcal{Q}_\gamma(x)]$
$= [-\mathcal{Q}_\gamma(x);\ Y(x)]$. (16.13)

Hence:

$\Phi_a J = J_R\Phi_a$. ■ (16.14)

16.3 Meaning of the proposition

The proposition shows that the quarter-turn acting in the dynamic mode plane is preserved by the readout map as a quarter-turn in the readout plane.

The two planes are not necessarily the same physical or semantic space.

They are related through an intertwining map.

This permits a precise distinction:

Dynamic complex structure

The generator possesses an elliptic invariant mode.

Readout complex structure

A declared scalar readout and its conjugate response inherit that mode's quarter-turn grammar.

Thus:

Dynamic complex plane ≠ Readout complex plane. (16.15)

But under the proposition:

Dynamic quarter-turn → Readout quarter-turn. (16.16)

16.4 Rank of the readout map

The map Φ_a is invertible when:

a and $a \circ J$ (16.17)

are linearly independent covectors.

Suppose they were dependent:

$a \circ J = \lambda a$. (16.18)

Apply J again:

$a \circ J^2 = \lambda a \circ J$. (16.19)

Using $J^2 = -I$:

$-a = \lambda^2 a$. (16.20)

For nonzero a:

$\lambda^2 = -1$. (16.21)

No real λ satisfies this.

Therefore, over a real two-dimensional elliptic mode, any nonzero real covector a makes a and $a \circ J$ linearly independent.

Hence:

$\operatorname{rank}(\Phi_a) = 2$. (16.22)

The readout map is locally an isomorphism between the dynamic mode and its conjugate readout plane.

16.5 Consequence

Corollary 16.1 [F]

Every nonzero scalar readout of a real two-dimensional elliptic mode has a unique conjugate completion relative to the declared orientation J. (16.23)

The uniqueness is relative to:

- the generator;
- its orientation;
- the readout covector;
- normalization.

Changing any of these may change the conjugate coordinate.

16.6 Sign convention

The definition:

$Q_\gamma = -a(Jx)$ (16.24)

was chosen so that:

$dY/d\theta = -Q_\gamma$, (16.25)

$dQ_\gamma/d\theta = Y$. (16.26)

Using the opposite orientation $-J$ reverses the sign convention.

Therefore:

Q-sign is orientation-dependent. (16.27)

A valid application must declare:

- positive phase direction;
- readout orientation;
- generator convention.

This is especially important when comparing different domains.

16.7 Local rather than global intertwining

The proposition is local to a stable mode.

Suppose the generator varies with state:

J = J(x). (16.28)

The readout map becomes:

Φ_a,x. (16.29)

Transporting conjugate coordinates across state space may introduce:

- connection terms;
- gauge choices;
- holonomy;
- curvature;
- discontinuity at regime changes.

A global complex structure requires consistency conditions beyond Proposition 16.1.

The main article does not assume these conditions.

16.8 Relationship to CAPM

The CAPM conjugate construction provides a mature exact readout plane:

Z_t = R_t + iQ_t, (16.30)

with:

∂R_t/∂θ_t = −Q_t, (16.31)

∂Q_t/∂θ_t = R_t. (16.32)

This proves conjugate closure in the declared valuation parameterization.

It does not by itself establish that a broader market dynamic mode `E_D` exists with generator `J_D`.

A stronger integrated finance theory would need to test whether there is a map:

Φ_fin: E_D → span{R, Q}, (16.33)

satisfying:

$\Phi_{fin} J_D = J_{fin} \Phi_{fin}$. (16.34)

That would show that a market-dynamic elliptic mode and the CAPM readout plane represent the same local complex structure.

At present:

- the CAPM readout closure is source-established;
- the dynamic-to-financial intertwining is a proposed formal and empirical research problem.

16.9 Relationship to self-reference

The Complex Intertwining Principle is not a theorem that self-reference generates complex numbers.

Its assumptions already require an elliptic mode.

Its role is narrower and more precise:

> Once an elliptic mode has been earned, every nonzero scalar readout of that mode possesses a natural conjugate completion that preserves the mode's quarter-turn grammar.

The remaining empirical question is:

> Under what conditions does self-referential world formation produce or stabilize such an elliptic mode?

That is the Signed-Conjugate Mode Emergence Hypothesis, not a proven universal law.

16.10 Part IV synthesis

Part IV has established the following chain:

Self-reference
→ possible scalar relational incompleteness
→ local generator estimation
→ signed-conjugate classification
→ elliptic mode, if present
→ conjugate readout
→ complex intertwining. (16.35)

The framework therefore gives complex numbers an important but disciplined role.

They are:

- more than decorative notation;
- less than a universal ontology;
- a natural local grammar of elliptic relational completion.

The next Part returns from geometry to history.

A conjugate readout can reveal exposure.

It does not by itself produce movement.

Movement does not by itself create commitment.

The world becomes time-bearing only when selected consequences pass gates, enter ledgers, preserve residual, and alter the future conditions of action.

Part V — From Movement to Time-Bearing History

17. Movement Is Not Commitment

17.1 Why geometry must return to governance

Part IV established a conditional geometric result.

When a Relation generator possesses a two-dimensional elliptic mode:

$J^2 = -I$, (17.1)

a scalar readout:

$Y = a(x)$ (17.2)

admits the conjugate completion:

$Q_\gamma = -a(Jx)$, (17.3)

$Z_\gamma = Y + iQ_\gamma$. (17.4)

This construction preserves the local quarter-turn grammar of the active mode.

It does not yet explain:

- whether the state moved;
- what caused the movement;
- whether the movement had an operational consequence;
- whether that consequence passed a gate;
- whether it entered a ledger;
- what remained unresolved afterward.

The complex plane completes a Relation.

It does not complete the operational world.

The full sequence is:

Measurement
→ Exposure

→ State Movement
→ Operational Consequence
→ Gate
→ Ledger + Residual. (17.5)

Every arrow represents a different operation.

The CAPM source develops this separation explicitly and insists that measurement, movement, gate, and ledger must not be collapsed into one another.

17.2 Passive measurement

Let the state be:

$Z = Y + iQ_\gamma$. (17.6)

Define a family of readouts:

$M\varphi(Z) = \mathrm{Re}[e^{(i\varphi)}Z]$. (17.7)

Then:

$M\varphi(Z) = Y \cos\varphi - Q_\gamma \sin\varphi$. (17.8)

At:

$\varphi = 0$, (17.9)

the protocol reads:

$M0(Z) = Y$. (17.10)

At:

$\varphi = \pi/2$, (17.11)

it reads:

$M\pi/2(Z) = -Q_\gamma$. (17.12)

Changing ϕ changes the measurement orientation.

It need not change z.

This is a **passive rotation**:

State fixed; readout frame changed. (17.13)

The observer asks a different question of the same completed state.

Therefore:

Passive rotation produces no state movement by necessity. (17.14)

It may reveal an exposure that was invisible under the original readout.

It does not generate the consequence associated with movement along that exposure.

17.3 Exposure

The conjugate readout:

$-\mathcal{Q}_\gamma$ (17.15)

answers a local question:

> What is the first-order change in the admitted scalar under a unit positive movement along the declared Relation coordinate?

If:

$\mathcal{Q}_\gamma = -\partial Y/\partial\theta$, (17.16)

then for a small phase movement $\Delta\theta$:

$\Delta Y \approx -\mathcal{Q}_\gamma \Delta\theta$. (17.17)

The quantity $\mathcal{Q}_\gamma$ is exposure before movement.

It is not itself:

- the movement;
- the realized consequence;
- the committed ledger entry;
- the residual after commitment.

Thus:

Exposure ≠ Movement. (17.18)

The CAPM framework states this distinction in monetary form: reading $-\mathcal{Q}$ identifies signed phase exposure, but no profit or loss has yet been generated merely by performing that measurement.

17.4 Active movement

An **active rotation** changes the state:

$Z' = e^{(i\Delta\theta)}Z$. (17.19)

The measurement axis remains fixed.

The new admitted scalar is:

$Y' = Y \cos\Delta\theta - \mathcal{Q}_\gamma \sin\Delta\theta$. (17.20)

Therefore:

$\Delta Y = Y' - Y$. (17.21)

So:

$$\Delta Y = Y(\cos \Delta\theta - 1) - \mathcal{Q}_\gamma \sin \Delta\theta. \quad (17.22)$$

For small $\Delta\theta$:

$$\Delta Y = -\mathcal{Q}_\gamma \Delta\theta - (Y/2)(\Delta\theta)^2 + (\mathcal{Q}_\gamma/6)(\Delta\theta)^3 + O((\Delta\theta)^4). \quad (17.23)$$

To first order:

$$\Delta Y \approx -\mathcal{Q}_\gamma \Delta\theta. \quad (17.24)$$

The movement creates an operational consequence in the readout.

But the movement may still fail to become historically admitted.

17.5 General state movement

The active change need not be a pure rotation.

Suppose the completed mode changes through radial and angular components:

$$Z = Ae^{(i\theta)}. \quad (17.25)$$

Then:

$$dZ/Z = dA/A + id\theta. \quad (17.26)$$

Writing:

$$Z = Y + i\mathcal{Q}_\gamma, \quad (17.27)$$

gives:

$$dY = Y(dA/A) - \mathcal{Q}_\gamma d\theta, \quad (17.28)$$

$$d\mathcal{Q}_\gamma = \mathcal{Q}_\gamma(dA/A) + Yd\theta. \quad (17.29)$$

Thus scalar movement may arise from:

- amplitude change;
- phase change;
- both simultaneously.

The conjugate coordinate identifies the phase-sensitive component.

It does not necessarily explain every source of movement.

17.6 Consequence is protocol-relative

A state movement becomes a candidate consequence through a protocol-defined translation:

$$c_k = C_p(X_k, X_{k+1}, \Delta Y_k). \quad (17.30)$$

The same movement may correspond to different candidates under different protocols.

A market-price movement may become:

- a trading signal;
- a mark-to-market change;
- an impairment indicator;
- a collateral event;
- no operative event at all.

An AI state update may become:

- a candidate token;
- a proposed tool action;
- an internal hidden-state change;
- a deployable instruction.

Therefore:

Movement ≠ Unique consequence. (17.31)

The candidate field depends on the translation rule.

17.7 Gate before historical reality

The candidate consequence passes through:

$e_k = \mathcal{G}_P(c_k; X_k, L_k, \text{Authority}_k)$. (17.32)

The gate may return:

$e_k \in \{\text{Admit, Partially Admit, Defer, Reject}\}$. (17.33)

Only after the gate can the consequence become a committed event under P.

This yields:

Movement ≠ Commitment. (17.34)

The source on declared disclosure identifies the gate as the commitment core of the operator chain: projection provides candidate visibility, while the gate determines what becomes committed trace.

17.8 Four layers in one table

Layer	What changes?	Can reveal exposure?	Can create operational consequence?	Can create ledger history?
Measurement	readout orientation	Yes	No	No

Layer	What changes?	Can reveal exposure?	Can create operational consequence?	Can create ledger history?
State movement	state or mode	Indirectly	Yes	Not automatically
Gate	commitment status	No	Recognizes or transforms consequence	Prepares admission
Ledger	durable record	No	Records admitted consequence	Yes

The distinctions are foundational.

A framework that jumps directly from complex rotation to ledgered history omits the Commitment module.

17.9 Example: financial movement

In the CAPM conjugate valuation example:

$Z = R + iQ.$ (17.35)

A passive measurement may reveal:

$M\pi/2(Z) = -Q.$ (17.36)

No financial profit or loss is created merely by asking the conjugate question.

If the valuation phase actually moves by $\Delta\theta$:

$R' = R\cos\Delta\theta - Q\sin\Delta\theta.$ (17.37)

Then:

$\Delta R_{econ} = R' - R.$ (17.38)

For small movement:

$\Delta R_{econ} \approx -Q\Delta\theta.$ (17.39)

A gate may then decide whether this economic movement becomes:

- recognized profit or loss;
- a margin call;
- collateral transfer;
- impairment;
- settlement;
- regulatory-capital consumption.

The CAPM source states the complete sequence as:

CAPM Filter
→ Admitted Value R

→ Declared Phase θ
→ Conjugate Exposure Q
→ Measurement Rotation
→ State Movement
→ P&L
→ Gate
→ Ledger Trace + Residual
→ Updated Financial World.

CAPM is one exact example of the broader architecture.

It is not the definition of the architecture.

17.10 Example: AI action

An AI system may internally evaluate several candidate outputs.

A readout may expose:

- confidence;
- uncertainty;
- alternative tendency;
- safety pressure.

This is not yet action.

The system state may change as new context enters.

This is movement.

A decoder selects a token or tool proposal.

This creates a candidate consequence.

A policy checker, human approval step, or tool-permission rule acts as gate.

Only the admitted output updates:

- conversation context;
- external system state;
- long-term memory;
- action history.

Rejected candidates may remain unavailable, partially logged, or residual.

Thus the same architecture applies without assuming that the AI state is literally a financial complex plane.

17.11 Example: scientific declaration

A scientific anomaly may be measured.

Measurement reveals the result.

The anomaly may alter model estimates.

This is state movement in the scientific inference system.

A researcher proposes a claim.

The claim is a candidate consequence.

Review, replication, and evidential standards act as gates.

Publication or accepted database entry creates ledger trace.

Unresolved disagreement, excluded data, and model limitation remain residual.

The committed result then shapes future experiments.

The scientific world bears history because prior accepted and residual results alter later inquiry.

17.12 Movement–Commitment Principle

Proposition 17.1 — Movement–Commitment Separation [D/F]

For a protocol-bound world, conjugate measurement closure and state movement are insufficient to establish historical commitment.

A committed consequence requires an additional gate and ledger update:

State movement
→ candidate consequence
→ gate
→ ledger + residual. (17.40)

Consequently:

Conjugate exposure may exist without movement. (17.41)

Movement may exist without admission. (17.42)

Admission may exist without full resolution. (17.43)

This proposition is architectural rather than domain-specific.

---

18. Three Forms of Commitment

18.1 Why Commitment has levels

A consequence can become fixed for one participant without becoming authoritative for an institution or accessible to a public.

The framework therefore distinguishes:

1. internal commitment;
2. institutional commitment;
3. redundant public commitment.

These are not merely larger quantities of the same record.

They involve different:

- access structures;
- authorities;
- frame maps;
- consequences;
- standards of objectivity.

The Handoff Summary explicitly requires this three-level distinction.

18.2 Internal commitment

Definition 18.1 — Internal commitment [D]

An event achieves internal commitment when it enters an observer's accessible history and becomes stable enough to condition that observer's later policy. (18.1)

Let observer a possess internal record:

$r_{a,k}$. (18.2)

An update is:

$r_{a,k+1} = \Lambda_a(r_{a,k}, e_k)$. (18.3)

Internal commitment requires:

$r_{a,k+1} \in \mathscr{F}_{a,k+1}$, (18.4)

and:

$\pi_{a,k+2} = f_{a,p}(\mathscr{F}_{a,k+1})$ (18.5)

to depend on that record.

Internal commitment is observer-relative.

The record may be:

- private;
- mistaken;
- inaccessible to others;

- expressed in an observer-specific frame.

Nevertheless, it is operationally real for that observer if it changes later behaviour.

18.3 Internal fixedness

A latched record may satisfy:

$\Pr(r_{a,k+1} = j \mid r_{a,k} = j, \text{no admitted reversal}) \approx 1.$ (18.6)

This creates internal certainty or fixedness.

But:

Internal certainty ≠ Universal truth. (18.7)

The self-referential-observer source formally separates observer-relative fixedness from cross-observer agreement. Agreement requires compatibility, accessible records, and frame reconciliation.

18.4 Institutional commitment

Definition 18.2 — Institutional commitment [D]

An event achieves institutional commitment when a recognized authority admits it into a ledger whose contents alter later institutional permissions, duties, classifications, or actions. (18.8)

Examples include:

- settled transaction;
- accounting recognition;
- court judgment;
- approved policy;
- deployed AI action;
- official scientific classification;
- regulatory decision.

Let institutional authority be:

A_i. (18.9)

The gate is:

$e_k = \mathcal{G}_p(c_k; \text{Evidence}_k, \text{Threshold}_k, A_i, L_k)$. (18.10)

The ledger update is:

$L_{k+1} = L_k \oplus T_p(e_k, A_i)$. (18.11)

Institutional commitment differs from internal commitment because its authority is recognized by a rule-governed collective process.

18.5 Institutional authority does not guarantee truth

An institution can commit an erroneous consequence.

For example:

- a court may issue a mistaken judgment;
- an account may recognize a flawed valuation;
- an AI approval process may admit an unsafe action;
- a scientific consensus may later be revised.

Therefore:

Institutional reality ≠ Ontological certainty. (18.12)

Yet the error may still be historically effective.

It may alter:

- obligations;
- ownership;
- reputation;
- resource allocation;
- future evidence gathering.

This is why historical reality must be distinguished from factual truth.

18.6 Redundant public commitment

Definition 18.3 — Redundant public commitment [D]

An event achieves redundant public commitment when compatible records of it are independently accessible across multiple observers or channels and support stable cross-observer coordination. (18.13)

Suppose:

$e_k \rightarrow \{r1_{,k}, r2_{,k}, \ldots, r_m_{,k}\}$. (18.14)

Let each observer possess an access map:

$\mathcal{A}_p(a_j, L)$. (18.15)

Redundant commitment requires that compatible propositions can be recovered:

$\Phi_{j \leftarrow i}(r_{i,k}) \approx r_{j,k}$. (18.16)

Redundancy strengthens:

- detectability;
- robustness;
- auditability;
- resistance to local record loss;
- cross-observer agreement.

It does not guarantee perfect frame independence.

18.7 Objectivity as constrained agreement

The framework does not define objectivity as absence of viewpoint.

It treats operational objectivity as stability across declared viewpoints and access paths.

A minimal form is:

$\mathrm{Obj}_{\mathrm{p}}(e) = \mathrm{Agreement}(e \mid \text{compatible frames, accessible records, declared protocol})$. (18.17)

Stronger objectivity requires:

- multiple records;
- independent access;
- frame compatibility;
- resistance to local perturbation;
- reproducibility.

Therefore:

One ledger entry ≠ Public objectivity. (18.18)

Public repetition ≠ Independent redundancy. (18.19)

Agreement under one frame ≠ Frame-robust invariance. (18.20)

18.8 Commitment transitions

The levels may form a sequence:

Internal commitment
→ institutional admission
→ redundant public stabilization. (18.21)

But this order is not guaranteed.

A public observation may precede formal institutional recognition.

An institution may create a binding record that remains private.

An observer may refuse to internalize a publicly committed event.

The levels are therefore partially ordered rather than strictly chronological.

18.9 Reopening commitment

A mature commitment system must sometimes reopen prior closure.

Let:

$\mathrm{Reopen}_p(L_k, \mathrm{NewEvidence}, \mathrm{AppealRule}) \to c'_k$. (18.22)

The reopened candidate passes through a new gate:

$e'_k = \mathcal{G}_p(c'_k; L_k)$. (18.23)

The ledger may be revised:

$L_{k+1} = \mathrm{Revise}_p(L_k, e'_k)$. (18.24)

Reopening is not the erasure of historical fact that an earlier commitment occurred.

It is a new commitment about the status of the earlier commitment.

The legal-interface source expresses this as:

Legal maturity = closure + governed reopening.

The same principle applies more broadly.

A world with no closure cannot coordinate.

A world with no reopening becomes brittle.

18.10 Commitment and observerhood

A strong observer is not merely a device that stores outcomes.

It can use ledger and residual history to revise its own future disclosure or policy frame.

Let:

D_k (18.25)

denote the observer's declaration or framing rule.

Then:

$D_{k+1} = \mathrm{Revise}(D_k \mid L_k, \mathscr{R}_k)$. (18.26)

The source on declared disclosure describes a self as a ledger capable of modifying its own future disclosure frame.

Within the present framework, this is a strong form of self-reference:

History revises the mechanism through which later history is selected.

18.11 Commitment matrix

Property	Internal	Institutional	Redundant public
accessible to one observer	Required	Usually	Required
formal authority	Not required	Required	Variable
alters policy	Required	Required	Usually
cross-observer compatibility	Not required	Desirable	Required
redundant record	No	Optional	Required
can be mistaken	Yes	Yes	Yes
can be reopened	Sometimes	By procedure	Through revision and replication

---

18.12 Commitment hierarchy principle

Proposition 18.1 — Commitment Hierarchy [D/F]

Internal fixedness, institutional authority, and redundant public objectivity are distinct properties.

No one of them implies the other two without additional access, authority, compatibility, and redundancy conditions. (18.27)

Consequently:

Ledgering and objectivity are related but non-identical. (18.28)

Authority and truth are related but non-identical. (18.29)

Private certainty and public agreement are related but non-identical. (18.30)

---

19. Residual, Γ, and Future Path Selection

19.1 Closure produces two outputs

A common modelling mistake is to treat a gate as producing only a trace.

A more complete closure produces:

$\text{Closure}_p = \text{Trace}_p \sqcup \text{Residual}_p$. (19.1)

The symbol $\sqcup$ indicates a typed join rather than ordinary scalar addition.

Trace and residual are not the same kind of object.

Trace records what has been admitted.

Residual preserves what has not been fully integrated.

The declared-disclosure source states this explicitly: trace stabilizes history, while residual preserves unfinished possibility.

19.2 Residual is protocol-generated

Let:

c_k (19.2)

be a candidate consequence.

Let:

$e_k = \mathcal{G}_p(c_k)$. (19.3)

The residual is:

$\mathscr{R}_{k+1} = \mathfrak{R}_p(c_k, e_k, L_{k+1})$. (19.4)

Residual therefore depends on:

- what candidates were considered;
- what the gate admitted;
- what the ledger records;
- what the protocol treats as unresolved.

A different protocol may generate a different residual from the same underlying occurrence.

Thus:

Residual is not discovered independently of the commitment grammar. (19.5)

It is produced by the mismatch between candidate consequence and achieved integration.

19.3 Residual categories

Residual may contain:

Observational residual

What was not observed or resolved.

Evidential residual

What was excluded, missing, or insufficiently supported.

Branch residual

Unselected alternatives or unresolved counterfactuals.

Institutional residual

Unrecognized liability, deferred decision, or jurisdictional remainder.

Model residual

Error, omitted factor, or regime failure.

Access residual

Records that exist but remain unavailable to relevant observers.

Ethical or social residual

Consequences omitted by the operative accounting or legal boundary.

These categories may overlap.

The framework should preserve provenance rather than compressing all residual into one scalar prematurely.

19.4 Residual honesty

Definition 19.1 — Residual honesty [D]

A closure is residual-honest when it records the material limitations, exclusions, unresolved alternatives, and reopening conditions associated with its commitment. (19.6)

Residual honesty may require recording:

- what was not observed;
- which assumptions were imposed;
- which evidence was excluded;
- what remains contradictory;
- what could reopen the gate;
- which future observations would alter the conclusion.

A closure that reports only its committed trace may appear complete while remaining brittle.

The declared-disclosure source treats preservation of residual as a condition of mature closure rather than as an admission of total failure.

19.5 Trace-only and residual-only pathologies

A system with only trace tends toward rigidity.

It repeatedly reinforces what has already passed the gate.

A system with only residual tends toward diffusion.

It preserves possibility without stabilizing actionable history.

A mature world requires:

Stable trace + Governed residual. (19.7)

This does not mean equal weighting.

It means that closure and revision both have structured places in the runtime.

19.6 From raw residual to Γ

Raw residual may be high-dimensional:

$\mathscr{R}_k \in \mathfrak{B}_p$, (19.8)

where $\mathfrak{B}_p$ may be:

- a set;
- field;
- measure;
- graph;
- collection of obligations;
- typed database.

Under additional assumptions, residual history may be compiled into a functional:

$\Gamma_k = \Gamma_p[\mathscr{R}_{1:k}]$. (19.9)

The functional may measure:

- unresolved cost;
- inconsistency;
- leakage;
- unfulfilled constraint;
- branch tension;
- accumulated liability.

A variational representation may use:

$S_eff,_p[x] = \int L_p(x, \dot{x}, t)dt - \lambda\Gamma_p[x]$. (19.10)

The induced future-selective influence is:

$F_\Gamma = -\delta\Gamma_p/\delta x$. (19.11)

The residual source proposes this ʟ-ᴦ grammar conditionally. It does not show that every residual possesses a differentiable action functional.

19.7 Activation conditions for Γ

A Γ representation should be used only when:

1. residual provenance is declared;
2. residual can be mapped to the candidate path or state;
3. the mapping is stable across the intended regime;
4. the functional has interpretable units;
5. the derivative or subgradient predicts a future-selective effect;
6. alternative encodings perform worse or less transparently.

If these conditions fail, residual should remain in a richer typed form.

The framework must not force:

Residual → Scalar Γ. (19.12)

19.8 Γ is not physical friction

A Γ term may resemble dissipation, penalty, or friction.

This does not make it identical to physical friction.

Physical friction may be represented by:

$F_{friction} = -\kappa v$. (19.13)

A residual-derived influence may be:

$F_{\Gamma} = -\delta\Gamma/\delta x$. (19.14)

The two may coincide in a particular model.

They need not.

Likewise:

Γ ≠ Entropy universally. (19.15)

Γ ≠ Risk universally. (19.16)

Γ ≠ Residual itself. (19.17)

Γ is one possible operational encoding of residual influence.

19.9 Γ is not conjugate exposure

The distinction between Γ and Q_v is especially important.

Conjugate response:

$\mathcal{Q}_\gamma = -a(Jx)$. (19.18)

Residual functional:

$\Gamma = \Gamma_p[\mathscr{R}\text{-history}]$. (19.19)

$\mathcal{Q}_\nu$ is determined by:

- an elliptic Relation;
- a readout covector;
- a current state.

Γ is determined by:

- commitment history;
- residual rules;
- a selected cost or constraint model.

Thus:

$\mathcal{Q}_\gamma$ = pre-movement directional exposure. (19.20)

Γ = post-closure accumulated future-selective constraint. (19.21)

They may influence one another through the full runtime.

They are not interchangeable.

The Handoff Summary emphasizes this exact distinction: $\mathcal{Q}$ is exposure before movement, whereas residual is incompletely integrated consequence after a gate or translation.

19.10 Partial admission

A gate may admit only a fraction of the candidate consequence.

Let:

$0 \le \alpha_k \le 1$. (19.22)

Then:

$c_k\text{^admitted} = \alpha_k c_k$, (19.23)

and:

$\mathscr{R}_{k+1} = (1 - \alpha_k)c_k + \mathscr{R}_k\text{^other}$. (19.24)

In finance:

$\Delta R_admitted = \alpha \Delta R_econ$. (19.25)

The gate residual is:

$\varepsilon_gate = \Delta R_econ - \Delta R_admitted$. (19.26)

The CAPM source uses this construction to distinguish actual economic movement from the portion recognized in a particular ledger.

The broader lesson is:

Commitment can be graded rather than binary.

19.11 Residual backreaction

Residual can affect future evolution through several routes.

Policy route

$\pi_{k+1} = f_p(\mathscr{T}_k, \mathscr{R}_k)$. (19.27)

Generator route

$A_{k+1} = A_p(L_k, \mathscr{R}_k)$. (19.28)

Gate route

$\mathcal{G}_{p,k+1} = \mathrm{UpdateGate}_p(\mathcal{G}_{p,k}, \mathscr{R}_k)$. (19.29)

Protocol route

$P_{k+1} = \mathrm{Revise}_p(P_k \mid L_k, \mathscr{R}_k)$. (19.30)

Base route

$X_{k+1} = \mathscr{B}_p(X_k, L_k, \Gamma_k, \ldots)$. (19.31)

The return may be delayed.

A residual can remain latent before crossing an intervention or revision threshold.

19.12 Residual-induced regime change

Suppose the local Relation generator depends on Γ:

$A = A(\Gamma)$. (19.32)

Its discriminant is:

$\Delta_A(\Gamma) = \mathrm{tr}(A(\Gamma))^2 - 4\det(A(\Gamma))$. (19.33)

As residual accumulates, the Relation may cross:

$\Delta_A(\Gamma) < 0$ (19.34)

to:

$\Delta_A(\Gamma) > 0$. (19.35)

The system moves from an elliptic to a hyperbolic regime.

This is a general formal possibility.

A domain-specific application would need to identify:

- the residual measure;
- the generator dependence;
- the transition threshold;
- observable predictions.

The framework does not claim that all residual accumulation produces this transition.

19.13 Residual as future option

Residual is not always a burden.

It may preserve:

- alternative interpretation;
- unrealized design;
- future theory;
- appeal;
- innovation;
- adaptive flexibility.

Thus:

Residual_today may become Structure_tomorrow. (19.36)

A mature world does not merely minimize residual.

It governs which residual should be:

- resolved;
- retained;
- reopened;
- transferred;
- explored.

This is why a zero-residual ideal may be undesirable or impossible.

19.14 Ledger–Residual Backreaction Hypothesis

Hypothesis 19.1 [H]

After conditioning on the present proposed Effective Base, ledger and residual history retain explanatory or predictive power for later policy, gate behaviour, generator regime, or future trace. (19.37)

A test is:

$\Pr(y_{k+1} \mid \Xi_k, L_k, \mathscr{R}_k)$
$\neq \Pr(y_{k+1} \mid \Xi_k)$. (19.38)

If the difference disappears consistently, then:

- historical backreaction may be weaker than claimed; or
- the Effective Base already incorporates the relevant history.

The interpretation must be stated.

---

20. Why Such a World Bears Time

20.1 Sequence is not yet operational time

A mathematical sequence can be ordered without the order becoming causally active.

For example:

$x_0, x_1, x_2, \ldots$ (20.1)

may be an analyst's indexing device.

A static proof may possess ordered steps.

A hierarchy may possess levels.

A fractal may possess recursive depth.

None of these facts alone establishes a time-bearing operational world.

The source on ledgered disclosure makes the same correction in its own SMFT setting:

Filtration depth is not yet time.
Only retained, ordered trace creates historical continuity.

20.2 Event, trace, history, and time-bearing order

The framework distinguishes:

Event = occurrence or selection. (20.2)

Trace = retained representation of event. (20.3)

Ledger = ordered, authoritative trace structure. (20.4)

History = ledgered trace plus preserved residual provenance. (20.5)

Time-bearing world = world whose future depends on that history. (20.6)

An event without retained trace may vanish from later operational causality.

A trace without order may preserve information but not a readable history.

An ordered ledger without backreaction may be archival.

A world bears time only when history changes the future.

20.3 Minimum time-bearing condition

Let the present Base be:

X_k. (20.7)

Let the historical state be:

$H_k = (L_k, \mathscr{R}_k, \text{Provenance}_k)$. (20.8)

The future distribution is:

$\Pr(X_{k+1} \mid X_k, H_k, P)$. (20.9)

A minimum operational time-bearing condition is:

$$\Pr(X_{k+1} \mid X_k, H_k, P) \neq \Pr(X_{k+1} \mid X_k, P). \quad (20.10)$$

History contributes causal information beyond the present state as represented.

Alternatively, where the Base incorporates history:

$X_k = \text{Encode}(\text{CurrentCondition}, H_k)$, (20.11)

the time-bearing character appears in the internal constitution of the Base.

Either way, past commitment has not disappeared.

It is carried explicitly or compiled into the present.

20.4 Definition of a time-bearing world

Definition 20.1 — Time-bearing world [D]

A protocol-bound operational world is time-bearing when ordered, accessible commitment history or preserved residual changes the admissible, feasible, or probable future paths available from otherwise comparable current conditions. (20.12)

This definition does not reduce physical time to institutional ledgers.

It defines the operational sense of *time-bearing* used in this article.

The source works on filtration and declaration propose stronger SMFT-specific theses such as “time is ledgered filtration” or $\texttt{Time}_p = \texttt{order}(L_p)$.

The present framework extracts a narrower domain-general claim:

> Ordered trace becomes operational time when it causally conditions later possibility.

20.5 Three temporal structures

The article distinguishes at least three temporal structures.

Evolution time

t indexes movement under Relation:

$dX/dt = F_p(X, \ldots)$. (20.13)

Phase time

θ indexes position within an elliptic mode:

$dX/d\theta = JX$. (20.14)

Ledger time

k indexes committed history:

$L_k \rightarrow L_{k+1}$. (20.15)

These structures may be related.

They are not identical.

A system may possess:

- evolution without stable phase;
- phase without a commitment ledger;
- ledger order without smooth continuous dynamics;
- several ledgers with different commitment rates.

20.6 Phase is not history

A periodic mode may return:

$Z(\theta + 2\pi) = Z(\theta)$. (20.16)

If no record distinguishes the completed cycle, the present state may be identical after one revolution.

The phase has advanced.

The state has returned.

No durable historical difference need remain.

If a ledger records the completed cycle:

$L_{k+1} = L_k \oplus \text{Trace}(\text{CycleComplete})$, (20.17)

then the operational world after the cycle differs despite state recurrence.

Thus:

Phase time ≠ Ledger time. (20.18)

Recurrence ≠ Historical return. (20.19)

This distinction is central to the article's title.

20.7 Irreversibility from ledgering

Suppose a reversible state movement occurs:

$X' = UX$, (20.20)

with inverse:

$X = U^{-1}X'$. (20.21)

If the movement triggered commitment:

$L' = L \oplus e$, (20.22)

then reversing the physical state does not automatically erase the ledger.

The resulting world-state is:

$\mathcal{W}' = (X, L', \mathscr{R}')$, (20.23)

not:

$\mathcal{W} = (X, L, \mathscr{R})$. (20.24)

Therefore:

State reversibility ≠ World-history reversibility. (20.25)

The arrow of operational time can arise from persistent commitment even when the underlying Relation is reversible.

20.8 Gate quality and temporal quality

A ledger can become unstable if its gate is unstable.

A gate that admits too early may write false trace.

A gate that admits too late may fail to create usable history.

A gate that changes standards silently produces inconsistent order.

The declared-disclosure source states:

TooEarlyGate ⇒ false trace.
TooLateGate ⇒ no usable history.
GoodGate ⇒ committed trace with residual disclosure.

Within this framework, gate quality affects temporal quality because the ledger's order determines which prior events can condition later operations.

Thus:

Unstable gate → unstable historical order. (20.26)

Unstable historical order → unstable causal interpretation. (20.27)

20.9 Causality as ledger-stabilized relevance

The framework does not attempt a complete metaphysical theory of causality.

It proposes an operational criterion.

A prior event becomes an operative cause under `P` when:

1. it entered an accessible ledger;
2. it is relevant to a later transition or decision;
3. its removal or alteration changes the later distribution.

Let event `e_j` precede event `e_k`.

Operational causal relevance can be tested by:

$\Pr(e_k \mid L \text{ including } e_j, P)$
$\neq \Pr(e_k \mid L \text{ with } e_j \text{ ablated}, P)$. (20.28)

Thus:

Cause_p = prior committed trace with demonstrable later relevance under P. (20.29)

The source on declared disclosure similarly treats gate and ledger as conditions for stable causal order, while explicitly limiting that claim to its operational framework.

20.10 Multiple ledgers, multiple times

A complex system may contain several ledgers.

For example:

- market ledger;
- accounting ledger;

- legal ledger;
- regulatory ledger;
- private observer memory;
- public record.

Let:

$L_k^{(1)}, L_k^{(2)}, \ldots, L_k^{(m)}$. (20.30)

Each may update under a different gate:

$L_{k+1}^{(j)} = \mathrm{Update}_j(L_k^{(j)}, e_k^{(j)})$. (20.31)

The same occurrence may become committed at different times in different ledgers.

Thus one system can carry several operational times.

A legal harm may occur before becoming actionable.

A financial loss may exist economically before accounting recognition.

A scientific observation may precede institutional acceptance.

The philosophical-interface source describes this clearly in law: an event may occur on one date but become legally real only later, so legal time is ordered legal trace plus procedural gate.

20.11 Synchronization and conflict among ledgers

Different ledgers may disagree:

$L^{(1)} \nvdash L^{(2)}$. (20.32)

Examples include:

- market value versus book value;
- private memory versus public record;
- legal status versus social belief;
- model log versus actual external outcome.

A reconciliation map may be required:

$\mathscr{R}\mathrm{ec}_{2\leftarrow 1}: L^{(1)} \to L^{(2)}$. (20.33)

Failure of reconciliation produces:

- basis residual;
- jurisdictional conflict;
- model disagreement;
- observer disagreement;

- institutional delay.

The broader world may therefore contain temporal friction between ledgers.

20.12 Historical depth

Ledger order alone may be insufficient.

The *depth* of a trace depends on how strongly it conditions the future.

Define a trace-impact functional:

D(e_j; k) = Effect of e_j on admissible paths at step k. (20.34)

A simple intervention-based measure is:

D(e_j; k)
= Dist[Pr(X_k | L), Pr(X_k | L with e_j removed)]. (20.35)

A trace with large D has deep historical influence.

A trace with negligible D is archival or obsolete for the chosen task.

Thus:

Historical age ≠ Historical depth. (20.36)

An old precedent may remain deep.

A recent log entry may be causally shallow.

20.13 Residual time

Residual also bears temporal structure.

An unresolved obligation may persist:

$\mathscr{R}_k \to \mathscr{R}_{k+1} \to \mathscr{R}_{k+2}$. (20.37)

Its effect may accumulate:

$\Gamma_k = \Gamma_p[\mathscr{R}_{1:k}]$. (20.38)

Or decay:

$d\mathscr{R}/dt = -\kappa\mathscr{R}$. (20.39)

Or reactivate after a trigger:

GateReopen($\mathscr{R}$, NewEvidence) → c_new. (20.40)

Residual time is the persistence and transformation of what closure did not resolve.

It is not necessarily represented in the official ledger's main trace sequence.

A residual index may therefore be needed alongside committed trace.

20.14 Time-bearing versus history-storing

A system stores history when:

L_k exists. (20.41)

A system bears history when:

$\partial \text{Future} / \partial L_k \neq 0$ (20.42)

or:

$\partial \text{Future} / \partial \mathcal{R}_k \neq 0$. (20.43)

The distinction is:

Storage = past retained. (20.44)

Time-bearing = retained past changes future generation. (20.45)

This is the core criterion.

20.15 Time-Bearing World Proposition

Proposition 20.1 — Time-Bearing World [D/F]

Let a protocol-bound operational world possess ordered ledger L_k and preserved residual $\mathcal{R}_k$.

If there exist otherwise comparable present conditions for which differences in $(L_k, \mathcal{R}_k)$ produce different later admissible actions, policies, gate outcomes, generators, or state distributions, then the world bears operational time. (20.46)

Formally, for some present Base X:

$$\Pr(X_{k+1} \mid X, L_k^A, \mathcal{R}_k^A, P) \neq \Pr(X_{k+1} \mid X, L_k^B, \mathcal{R}_k^B, P). \quad (20.47)$$

The difference demonstrates that history is part of the future-generating condition.

20.16 The title's full meaning

The movement:

From Trace to Time-Bearing Worlds

now has a precise sequence.

A trace becomes:

1. **addressable**, when it enters an observer filtration;
2. **self-referential**, when it changes later policy;

3. **committed**, when it passes a gate;
4. **historical**, when it enters an ordered ledger;
5. **time-bearing**, when ledger and residual alter future possibility.

In compact form:

Trace
→ Accessible History
→ Adaptive Policy
→ Gate
→ Ledger + Residual
→ Future Constraint
→ Time-Bearing World. (20.48)

The source chain “filtration → collapse → ledger → time → causality” is preserved here as a specific intellectual precursor, while the present article generalizes it into a protocol-bound runtime that does not require the full SMFT pre-time ontology.

20.17 Part V synthesis

The article has now established:

1. conjugate measurement reveals exposure;
2. exposure does not create movement;
3. movement produces candidate consequence;
4. a gate determines commitment;
5. commitment produces ledger and residual;
6. ledger and residual return into future selection;
7. this recursive historical return makes the operational world time-bearing.

The complete sequence is:

Base
→ Relation
→ Readout
→ Exposure
→ Movement
→ Candidate Consequence
→ Gate
→ Ledger + Residual
→ Recompiled Base. (20.49)

The next Part tests this architecture against worked examples.

No example will be treated as the universal origin of the framework.

Each will be used to show which modules are exact, which are conditional, and which remain research hypotheses.

Part VI — Worked Examples as Module-Coverage Tests

The examples in this Part do not establish that all domains share one substance, one ontology, one state space, or one global complex plane.

Each example instead answers a narrower question:

> Which parts of the Base–Relation–Commitment architecture are explicitly defined, mathematically established, operationally measurable, or still hypothetical in this domain?

The examples are therefore not interchangeable.

- Self-referential quantum observers supply the strongest formal model of trace-conditioned policy.
- CAPM supplies the strongest exact conjugate-readout construction.
- Δ5 supplies an explicit operator-earned opposition structure.
- Dissipative and breather systems show why Relation cannot be restricted to conservative rotation.
- AI, law, science, and institutions show how gate, ledger, residual, and reopening can become operationally central even without an established elliptic mode.

No individual example covers the entire framework.

---

21. Self-Referential Quantum Observers

21.1 Why begin with the observer example?

A theory of self-reference should not begin by merely declaring that a system “observes itself.”

It should specify:

- what is recorded;
- where the record is stored;
- which future operation can access it;
- how the record changes later policy;
- how the changed policy alters later outcomes.

The self-referential-observer source supplies a mathematically explicit implementation of this causal topology.

Its observer is not defined by consciousness.

It is defined as a process that:

1. records discrete outcomes;
2. retains them as an internal trace;

3. selects later instruments as a measurable function of that trace;

4. updates the system–observer state through declared quantum operations.

Its importance for the present framework is therefore structural.

The portable result is:

Recorded trace
→ usable history
→ adaptive instrument
→ changed future outcome law. (21.1)

The specific Hilbert-space and completely positive map machinery belongs to the quantum implementation and should not be transferred automatically to finance, law, AI, or organizations.

21.2 Declared quantum runtime

Let the initial quantum state be:

$\rho_0 \in \mathcal{D}(\mathscr{H})$. (21.2)

Let:

$\{\mathcal{M}_{\theta,\varphi} : \varphi \in \Phi\}$ (21.3)

be a family of instruments indexed by measurement setting θ.

Let:

$\mathcal{E}_k$ (21.4)

be the between-tick completely positive trace-preserving evolution.

Let the adaptive policy be:

$f_k: \Phi^{k-1} \to \Theta$. (21.5)

Given prior outcomes:

$\varphi_{1:k-1} = (\varphi_1, \ldots, \varphi_{k-1})$, (21.6)

the next instrument setting is:

$\theta_k = f_k(\varphi_{1:k-1})$. (21.7)

The pre-measurement state is:

$\rho_{k^-} = \mathcal{E}_k(\rho_{k-1})$. (21.8)

The conditional transition kernel is:

$K_k(\varphi_{1:k-1}, \varphi) = \mathrm{Tr}[\mathcal{M}_{\theta_k,\varphi}(\rho_{k^-})]$. (21.9)

The source establishes measurability conditions under which these kernels define a unique stochastic process over infinite outcome traces. It invokes the Ionescu–Tulcea extension

theorem to construct the global probability measure.

This is an important formal achievement.

The adaptive process is not merely described narratively.

It is shown to be mathematically well formed.

21.3 Translation into the general runtime

The quantum implementation maps to the general framework as follows.

General object	Quantum implementation
Detailed Base x_k	joint system–observer state, including ρ_k and record state
Trace y_k	realized outcome ϕ_k
Filtration $\mathcal{F}_k$	sigma-algebra generated by $\phi_{1:k}$
Policy π_{k+1}	adaptive measurement setting $\theta_{k+1} = f_{k+1}(\phi_{1:k})$
Relation	between-tick evolution plus selected instrument
Candidate event	possible next outcome
Gate	instrument-conditioned outcome realization
Internal ledger	observer's retained outcome trace
Historical return	trace changes later measurement setting

The core recursion is:

$\varphi_{1:k}$
$\rightarrow \theta_{k+1}$
$\rightarrow \boldsymbol{\mathcal{M}}_{\theta_{k+1},\varphi}$
$\rightarrow$ distribution of φ_{k+1}. (21.10)

This is the exact form of:

Trace → Filtration → Policy → Changed Transition Law → New Trace. (21.11)

21.4 Delta-certainty

Once observer A records an outcome:

$\varphi_k = j$, (21.12)

the observer's internal filtration treats that past event as fixed.

The conditional probability relative to the observer's own realized history becomes:

$\Pr_A(\varphi_k = j \mid \mathcal{F}_{A,k}) = 1$. (21.13)

This is internal delta-certainty.

It does not claim that the observer possessed certainty before the event.

It states that after the outcome has entered the observer's own retained trace, the observer conditions later policy on the realized record rather than on the earlier unresolved distribution.

This corresponds to internal commitment.

The past record becomes operationally definite relative to that observer's filtration.

21.5 Latching

The observer source also formalizes **latching**.

Suppose two histories differ:

$h_k^A \neq h_k^B$. (21.14)

If the adaptive policy depends on history:

$f_{k+1}(h_k^A) \neq f_{k+1}(h_k^B)$, (21.15)

then the two histories induce different future instruments.

Consequently:

$\Pr(\varphi_{k+1} \mid h_k^A) \neq \Pr(\varphi_{k+1} \mid h_k^B)$. (21.16)

The histories are no longer merely records of different pasts.

They have become different future-generating conditions.

This is exactly the time-bearing criterion developed in Section 20.

Past trace changes future possibility.

Latching therefore provides a source-established example of branch-dependent historical return.

21.6 Internal fixedness is not universal objectivity

An observer's own record may be internally fixed while remaining:

- inaccessible to another observer;
- expressed in another frame;
- based on an incompatible instrument;
- unavailable for public verification.

Therefore:

Internal delta-certainty ≠ Cross-observer agreement. (21.17)

The observer framework requires compatibility and frame mapping before agreement can be asserted.

Let the effects for observers A and B be:

E^A_{θ_A,φ}, E^B_{θ_B,φ}. (21.18)

Let:

F_{A→B} (21.19)

be an admissible frame map.

Under compatibility and normalization, the source derives equality of observer marginals after the appropriate transformation:

Pr_A(φ | θ_A, Σ) = Pr_B(φ | θ_B, Σ). (21.20)

The agreement is conditional.

It does not state that all observers always obtain identical raw records.

21.7 Objectivity through redundancy

A stronger form of objectivity arises when information about the same event is redundantly encoded in several disjoint environmental fragments.

Let:

{E_j^env} (21.21)

be independently accessible fragments.

If several fragments encode distinguishable records of the same proposition ϕ, then multiple observers may infer compatible outcomes without interacting directly with the original system.

The source states that redundant encoding can support approximate consensus:

Pr_A(φ | θ_A, Σ)
≈ Pr_B(φ | θ_B, Σ)
≈ Pr_env(φ | Σ). (21.22)

This supplies a formal route from:

private record
→ accessible compatible records
→ operational objectivity. (21.23)

The general framework therefore distinguishes:

Internal commitment
→ cross-observer compatibility
→ redundant public commitment. (21.24)

21.8 What the quantum example establishes

Source-established results [S]

The source establishes:

- measurable history-dependent instrument policies;
- a unique stochastic law on infinite traces;

- observer-relative certainty about past recorded outcomes;
- latching through policy dependence on history;
- conditional agreement under compatible instruments and frame maps;
- redundant records as a route toward operational consensus.

What transfers structurally [F/A]

The portable architecture is:

Trace
→ Filtration
→ Policy
→ Changed Future Instrument
→ Changed Future Distribution. (21.25)

What does not transfer automatically

The source does not prove that:

- financial observers use completely positive maps;
- legal judgments are quantum collapses;
- AI memory is a Hilbert-space record;
- every self-referential process obeys Born probabilities;
- every operational world has quantum no-signalling or no-cloning.

The correct transfer is functional rather than material.

21.9 Module-coverage assessment

Framework module	Quantum observer coverage
Protocol	Strong
Detailed Base	Strong
Trace	Strong
Filtration	Strong
Adaptive policy	Strong
Recursive backreaction	Strong
Effective Base compiler	Not central
General generator classification	Partial
Conjugate readout	Not the principal result
Internal commitment	Strong
Institutional commitment	Not applicable directly
Redundant public commitment	Strong formal analogue
Residual governance	Limited
Protocol revision	Not central

The quantum observer model is therefore the strongest example of the **mandatory self-referential core**, but not of every optional module.

---

22. CAPM Conjugate Valuation

22.1 Why CAPM is a particularly useful example

CAPM is not introduced as the universal origin of the framework.

It is used because it supplies an unusually clean answer to one difficult question:

> Can an ordinary mature scalar valuation be completed by a conjugate coordinate whose financial meaning is exact, unit-consistent, and locally equivalent to conventional sensitivity analysis?

The answer in the CAPM source is yes.

The construction begins with familiar discounted-cash-flow finance and derives:

- an admitted scalar value;
- a declared valuation phase;
- an orthogonal monetary coordinate;
- an exact quarter-turn measurement cycle;
- a precise distinction between exposure, movement, P&L, recognition, and ledger history.

This makes CAPM the most mature example of **conjugate measurement closure** in the current framework.

22.2 Declared valuation protocol

Consider a future cash flow:

CF_t. (22.1)

Declare a baseline rate:

r_base. (22.2)

The baseline-discounted value is:

$A_t = CF_t/(1 + r_base)^t$. (22.3)

Declare the CAPM required return:

$r_CAPM = r_base + \beta ERP$. (22.4)

The ordinary CAPM-discounted value is:

$R_t = CF_t/(1 + r_CAPM)^t$. (22.5)

Within the declared protocol:

- A_t is the baseline valuation amplitude;
- R_t is the admitted risk-adjusted scalar value.

The relationship between A_t and R_t defines the valuation phase.

22.3 Valuation phase

Define:

$\cos \theta_t = R_t/A_t$. (22.6)

Therefore:

$\cos \theta_t = [(1 + r_base)/(1 + r_CAPM)]^t$. (22.7)

For the standard case:

$0 < R_t \leq A_t$, (22.8)

the phase lies in:

$0 \leq \theta_t < \pi/2$. (22.9)

The phase is not introduced as hidden market time.

It is a declared coordinate generated by the relationship between the baseline valuation and the CAPM-admitted valuation.

22.4 Conjugate coordinate

Define:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (22.10)

Then:

$A_t^2 = R_t^2 + Q_t^2$. (22.11)

The completed valuation state is:

$Z_t = R_t + iQ_t$. (22.12)

Equivalently:

$Z_t = A_t e^{(i\theta_t)}$. (22.13)

Since:

$R_t = A_t \cos \theta_t$, (22.14)

and:

$Q_t = A_t \sin \theta_t$, (22.15)

both R_t and Q_t possess monetary units.

The construction is therefore not an arbitrary pairing of price with a dimensionless risk score.

22.5 Conjugate Risk Theorem

Differentiate:

$R = A \cos \theta$. (22.16)

Holding A fixed:

$\partial R/\partial \theta = -A \sin \theta$. (22.17)

Since:

$Q = A \sin \theta$, (22.18)

we obtain:

Theorem 22.1 — CAPM Conjugate Risk [S]

$\partial R/\partial \theta = -Q$. (22.19)

Similarly:

$\partial Q/\partial \theta = R$. (22.20)

Therefore, Q is the magnitude of the first-order dollar exposure of admitted value R to movement in the declared valuation phase.

The result is exact within the declared construction.

22.6 Relationship to conventional required-return sensitivity

Let:

$r = r_CAPM$. (22.21)

Then:

$R = CF/(1 + r)^t$. (22.22)

Differentiate with respect to r:

$\partial R/\partial r = -tR/(1 + r)$. (22.23)

Since:

$dR = -Qd\theta$, (22.24)

and:

$dR = -[tR/(1 + r)]dr$, (22.25)

the two descriptions are locally equivalent:

Qdθ = [tR/(1 + r)]dr. (22.26)

The phase language does not replace conventional finance.

It reorganizes conventional sensitivity into a conjugate geometric form.

22.7 Financial meaning of Q

The source carefully distinguishes `Q` from several familiar risk quantities.

`Q` is not automatically:

- expected loss;
- realized loss;
- volatility;
- beta;
- Value at Risk;
- Expected Shortfall;
- duration;
- convexity;
- option premium;
- market price;
- the haircut `A - R`;
- opportunity cost without additional assumptions.

Its exact meaning is narrower:

> `Q` is the magnitude of first-order monetary exposure to movement in the declared valuation phase.

This is the source-established interpretation.

Broader economic meanings require additional models.

22.8 Measurement cycle

Define the quarter-turn operator:

J_fin = [0 −1; 1 0]. (22.27)

Acting on:

[R; Q], (22.28)

it gives:

J_fin[R; Q] = [−Q; R]. (22.29)

The operator satisfies:

J_fin² = −I, (22.30)

J_fin⁴ = I. (22.31)

The closed measurement cycle is:

R → −Q → −R → Q → R. (22.32)

The first quarter-turn changes the readout from:

admitted mark `R`

to:

signed conjugate exposure `-Q`.

The second quarter-turn produces:

−R,

which is reversal of the signed valuation orientation.

It is not automatically “a second loss.”

22.9 Passive measurement and active movement

A passive measurement rotation changes the readout basis:

Mφ(Z) = Re[e^(iφ)Z]. (22.33)

At:

φ = π/2, (22.34)

the readout is:

Mπ/2(Z) = −Q. (22.35)

The state remains unchanged.

No P&L is generated merely by measuring `-Q`.

An active phase movement changes the state:

Z′ = e^(iΔθ)Z. (22.36)

The new admitted value is:

R′ = R cos Δθ − Q sin Δθ. (22.37)

Therefore:

ΔR_econ = R′ − R. (22.38)

For small $\Delta\theta$:

$\Delta R_{econ} \approx -Q\Delta\theta$. (22.39)

The CAPM source explicitly separates passive measurement rotation from active state movement.

22.10 Gate and ledger

Actual economic movement does not automatically become recognized financial history.

Let:

$G_P(\Delta R_{econ}, X, L)$ (22.40)

be a financial recognition or settlement gate.

If the consequence is admitted:

$L_{k+1} = L_k + \mathrm{Trace}(\Delta R_{admitted})$. (22.41)

Partial recognition may be written:

$\Delta R_{admitted} = \alpha \Delta R_{econ}$, with $0 \le \alpha \le 1$. (22.42)

The gate residual is:

$\varepsilon_{gate} = \Delta R_{econ} - \Delta R_{admitted}$. (22.43)

The complete runtime is:

Measurement
→ Exposure
→ Movement
→ P&L
→ Gate
→ Ledger + Residual. (22.44)

This exact separation makes finance especially useful for the general framework.

22.11 Multi-period extension

For multiple future cash flows:

$Z = \Sigma_t A_t e^{(i\theta_t)}$. (22.45)

Then:

$R = \Sigma_t A_t \cos \theta_t$, (22.46)

$Q = \Sigma_t A_t \sin \theta_t$. (22.47)

For a common phase rotation ϕ:

$d\,\mathrm{Re}[e^{(i\varphi)}Z]/d\varphi \,|_{\{\varphi=0\}} = -Q$. (22.48)

For term-specific phase changes:

$dR = -\Sigma_t Q_t d\theta_t$. (22.49)

The scalar conjugate exposure therefore extends to a phase-exposure term structure.

22.12 The empirical limit

In a one-period static construction:

$Q = \sqrt{(A^2 - R^2)}$. (22.50)

Thus `Q` is algebraically determined by `A` and `R`.

It does not automatically add independent statistical information.

Its practical value must arise from uses such as:

- dynamic attribution;
- multi-horizon decomposition;
- comparison among valuation protocols;
- communication of directional exposure;
- gate-event diagnosis;
- intervention design;
- testing interaction with market or institutional state.

If these gains do not appear, `Q` remains an optional reparameterization rather than a superior empirical state variable. The CAPM source explicitly recognizes this limit.

22.13 What CAPM establishes and what remains open

Source-established [S]

CAPM establishes:

- the valuation phase;
- the conjugate coordinate `Q`;
- the exact derivative `∂R/∂θ = −Q`;
- the quarter-turn measurement cycle;
- unit consistency;
- equivalence to ordinary required-return sensitivity;
- separation of measurement, movement, recognition, ledger, and residual.

Framework-level inference [F]

CAPM supplies an exact example of the general conjugate-readout construction:

$Y + iQ_{\gamma}$. (22.51)

Open empirical hypothesis [H]

It remains to be shown whether a broader financial market generator possesses a stable elliptic mode that intertwines with the CAPM readout plane:

Φ_finJ_market = J_finΦ_fin. (22.52)

The static valuation construction does not prove this.

22.14 Module-coverage assessment

Framework module	CAPM coverage
Protocol	Strong
Detailed causal Base	Partial
Admitted scalar trace	Strong
Effective Base compiler	Limited
Relation phase	Strong, exact in valuation construction
Elliptic operator	Strong
Conjugate readout	Strong
Measurement versus movement	Strong
Economic consequence	Strong
Gate	Formally specified
Ledger	Formally specified
Residual	Formally specified
Observer filtration	Partial
Adaptive market policy	Research programme
Full world-forming backreaction	Not yet established

CAPM is therefore a mature **Relation/readout example**, not a completed general market-world theory.

---

23. Δ5 and Operator-Earned Opposition

23.1 Why Δ5 matters to the general framework

The Δ5 source addresses a different problem from CAPM.

CAPM asks:

> How can one admitted scalar readout be completed by its quadrature response?

Δ5 asks:

Under what declared symmetry and energy does a half-turn phase opposition become selected?

This distinction is important because ordinary discourse often treats:

- opposite;
- complementary;
- conjugate;
- counterbalancing;
- yin and yang;

as interchangeable.

The $\Delta 5$ construction shows how opposition can be earned mathematically through:

- a declared operator;
- invariant sectors;
- a quadratic energy;
- spectral structure;
- and, under additional assumptions, dissipative stability.

23.2 Declared decagon

Let the state be:

$$a = (a_1, \ldots, a_{10})^T \in \mathbb{C}^{10}. \quad (23.1)$$

Let:

$$\rho: n \mapsto n + 1 \bmod 10 \quad (23.2)$$

be rotation by one site.

Define the half-turn:

$$T_5 = \rho^5. \quad (23.3)$$

Thus:

$$T_5: n \mapsto n + 5 \bmod 10. \quad (23.4)$$

The $\Delta 5$ pairs are:

$$(1,6), (2,7), (3,8), (4,9), (5,10). \quad (23.5)$$

The sum-to-11 reflection is a different operator:

$$R_{11}: n \mapsto 11 - n. \quad (23.6)$$

It produces:

(1,10), (2,9), (3,8), (4,7), (5,6). (23.7)

The source emphasizes that these pairings are logically distinct, even though the corresponding involutions commute.

23.3 Symmetric and antisymmetric sectors

Define the projectors:

$P_+^{(\Delta 5)} = \tfrac{1}{2}(I + T_5)$, (23.8)

$P_-^{(\Delta 5)} = \tfrac{1}{2}(I - T_5)$. (23.9)

The $\Delta 5$-antisymmetric sector satisfies:

$T_5 a = -a$. (23.10)

Equivalently:

$a_{n+5} = -a_n$. (23.11)

For nonzero amplitudes, this gives:

$\varphi_{n+5} - \varphi_n = \pi \bmod 2\pi$. (23.12)

This is strict half-turn phase opposition.

23.4 Pair-energy minimization

Define:

$E_{pair}(a) = \sum_{n=1}^{5} |a_n + a_{n+5}|^2$. (23.13)

Equivalently:

$E_{pair}(a) = \|P_+^{(\Delta 5)} a\|^2$. (23.14)

The minimum is achieved when:

$P_+^{(\Delta 5)} a = 0$. (23.15)

Therefore:

$a_{n+5} = -a_n$. (23.16)

The source thus derives $\Delta 5$ opposition from a declared cost.

It does not merely name the pairs as opposite.

The conclusion is conditional on the selected energy.

A different quadratic form may select another mode.

23.5 Spectral interpretation

Under the discrete Fourier transform on C_{10}, the half-turn acts as:

$T_5 \hat{a}_k = (-1)^k \hat{a}_k$. (23.17)

Thus the Δ_5-antisymmetric sector is the odd-k subspace.

The half-frequency mode:

$k = 5$ (23.18)

satisfies strict alternation:

$a_{n+1} = -a_n$, (23.19)

and hence:

$a_{n+5} = -a_n$. (23.20)

However, the ordinary cycle Laplacian:

$E_{lap}(a) = \Sigma_{n=1}^{10} |a_{n+1} - a_n|^2$ (23.21)

does not select $k = 5$ as its minimum.

The smooth constant mode is the global minimum.

Within the odd sector, lower-frequency odd modes have lower Laplacian energy than $k = 5$.

The strict alternating mode becomes the ground mode for an anti-alignment cost such as:

$E_{nn}^{(+)}(a) = \Sigma_{n=1}^{10} |a_{n+1} + a_n|^2$. (23.22)

This is a crucial methodological lesson:

> The selected phase mode depends on the declared energy or generator.

The state space does not possess one universally privileged phase merely because it is drawn as a decagon.

23.6 Dissipative locking

The Δ_5 source proposes dynamics of the form:

$$i\,da_n/dt = \omega_n a_n + \lambda|a_n|^2 a_n + \kappa_{m(n)} a_{n+5} - i\Gamma_n a_n.$$ (23.23)

Under pair-symmetric parameters and damping, it constructs a Lyapunov-style functional containing:

$E_{pair}(a)$, (23.24)

the cycle Laplacian energy, and optional boundary penalties.

Within the source's assumptions, the dynamics are intended to drive:

$E_{pair}(a(t)) \to 0$, (23.25)

and:

$\varphi_{n+5} - \varphi_n \to \pi$. (23.26)

The source proposes observable tests including:

- small pair-imbalance energy;
- negative work correlation;
- phase differences concentrated near π;
- changes in effective leakage and modal Q-factor.

These results belong to the declared $\Delta 5$ model.

They are not universal consequences of all half-turn symbolism.

23.7 Half-turn opposition versus complex quadrature

The $\Delta 5$ relation satisfies:

$T_5^2 = I$. (23.27)

The elliptic quarter-turn satisfies:

$J^2 = -I$. (23.28)

These operators have different roles.

Δ5 half-turn

$a \to -a$. (23.29)

Elliptic quarter-turn

$a \to Ja \to -a$. (23.30)

A $\Delta 5$ partner is a strict polar opposite.

A complex conjugate readout is a quadrature partner.

They may coexist in a larger phase architecture.

They are not identical.

23.8 What Δ5 contributes

Source-established or source-proposed [S]

Within its declared model, $\Delta 5$ supplies:

- a central half-turn operator on C_{10};
- commuting symmetry projectors;
- an explicit pair-energy minimization;
- an odd Fourier sector;
- conditions selecting a half-frequency mode;
- dissipative phase-locking hypotheses and tests;
- a specific coarse-graining architecture.

General methodological consequence [F]

A Relation structure should be admitted only after declaring:

1. the operator;
2. its invariant sectors;
3. the cost or generator;
4. the selected mode;
5. the stability conditions.

Nonclaims

Δ5 does not prove:

- that every self-referential system has ten states;
- that every stable system organizes into five pairs;
- that all opposition minimizes dissipation;
- that Δ5 is the source of all complex structure;
- that ancient diagram symbolism alone establishes physical phase.

23.9 Module-coverage assessment

Framework module	Δ5 coverage
Protocol	Strong within declared decagon
Base	Strong for ten-amplitude state
Relation operator	Strong
Invariant sectors	Strong
Phase opposition	Strong and operator-earned
Ordinary complex quarter-turn	Not identical to Δ5
Dissipation	Modelled conditionally
Effective Base reduction	Proposed five-mode reduction
Observer filtration	Not central

Framework module	Δ5 coverage
Gate and ledger	Not central
Residual governance	Limited
World-forming historical return	Not established

Δ5 is therefore a mature **specific Relation architecture**, not a complete operational world model.

---

24. Dissipative, Breather, and Non-Complex Regimes

24.1 Why this example class is necessary

A framework emphasizing conjugate geometry faces a predictable danger.

Once an exact complex branch has been found, researchers may begin representing every dynamic process as rotation.

That would repeat the very scalar-overcompression problem the framework was designed to avoid.

Relation may instead involve:

- damping;
- diffusion;
- memory;
- drift;
- switching;
- rupture;
- hyperbolic amplification;
- bounded breathers;
- high-dimensional mixed modes.

The article must therefore show explicitly that:

Complex phase is one branch of Relation, not Relation itself.

24.2 Dissipative stochastic dynamics

The uploaded Japanese paper begins from a Langevin-type process with:

- conservative force;
- friction;
- Gaussian white noise.

A schematic form is:

$m\,d^2q/dt^2 = -\nabla V(q) - \gamma\,dq/dt + \xi(t)$. (24.1)

Ordinary conservative Hamiltonian machinery does not directly capture the frictional and stochastic terms.

The paper instead uses stochastic mechanics, including:

- forward and backward derivatives;
- probability density;
- current and osmotic velocities;
- a complex wave representation;
- a nonlinear Schrödinger–Langevin equation.

Its significance for the present framework is limited but important:

> Dissipation and stochasticity do not automatically eliminate all useful phase-bearing representation.

The conclusion is not:

All dissipative systems are quantum.

The conclusion is:

Complex Relation and dissipation can coexist in one declared model.

24.3 Damped elliptic mode

A minimal damped elliptic mode is:

$dz/dt = (-\kappa + i\omega)z$. (24.2)

The solution is:

$z(t) = z_0 e^{(-\kappa t)} e^{(i\omega t)}$. (24.3)

The amplitude obeys:

$d|z|/dt = -\kappa|z|$. (24.4)

The phase obeys:

$d \arg z/dt = \omega$. (24.5)

The generator separates into:

$A = -\kappa I + \omega J$. (24.6)

The damping term and quarter-turn term have different mathematical roles.

Therefore:

Damping ≠ Conjugate response. (24.7)

Damping ≠ Residual. (24.8)

Damping ≠ Commitment loss. (24.9)

They may be related in a domain-specific model, but not identified by type.

24.4 Breather regime

A breather is a localized or bounded recurrent structure whose amplitude changes periodically without immediate collapse into a fixed point or runaway growth.

A schematic form is:

$z(t) = A(t)e^{(i\theta(t))}$, (24.10)

where:

$A(t + T) = A(t)$, (24.11)

$\theta(t + T) = \theta(t) + \Omega T$. (24.12)

A breather can therefore carry:

- internal recurrence;
- oscillatory phase;
- bounded energy exchange.

It need not create a ledger at every cycle.

It may possess phase time without historical commitment.

24.5 Semantic-breather vocabulary

The semantic-breather source proposes a descriptive taxonomy of timing and trace failures.

Stripped of its clinical claims, the taxonomy suggests the following general Relation regimes:

Regime	General interpretation
Drift	phase or policy anchor changes gradually
Freeze	event or commitment rate becomes too sparse
Sticking	repeated return without productive closure
Overdrive	updates occur faster than integration capacity
Rupture	continuity between states or traces fails
Overflow	high activity produces little stable meaning or record

The source presents these as collapse-tick pathologies inside its AI-semantic simulation framing. It does not establish them as human medical diagnoses.

Within this article, they function only as candidate regime labels for recurrent systems.

24.6 Parabolic accumulation

Consider:

$du/dt = v$, (24.13)

$dv/dt = 0$. (24.14)

The generator is:

$N = [0\ 1; 0\ 0]$, (24.15)

with:

$N^2 = 0$. (24.16)

The solution is:

$u(t) = u_0 + tv_0$, (24.17)

$v(t) = v_0$. (24.18)

This is not rotation.

It is accumulation or shear.

Possible operational interpretations include:

- unresolved carry;
- one-way memory accumulation;
- delayed debt;
- irreversible queue growth;
- policy drift.

Forcing this process into an ordinary complex plane would create a false periodicity.

24.7 Hyperbolic amplification

Consider:

$du/dt = \kappa v$, (24.19)

$dv/dt = \kappa u$. (24.20)

Then:

$d^2u/dt^2 = \kappa^2 u$. (24.21)

The solutions contain:

$e^{(\kappa t)}, e^{(-\kappa t)}$. (24.22)

This is hyperbolic rather than elliptic.

It describes:

- amplification and contraction;
- polarization;
- runaway feedback;
- unstable divergence.

A self-referential market or institution may move from a damped elliptic regime into a hyperbolic regime after leverage, residual, or policy feedback crosses a threshold.

That is a testable domain hypothesis, not a universal law.

24.8 Jump and switch regime

Suppose:

$A_k = A_{r_k}$, (24.23)

where:

$r_k \in \{r_1, r_2, \ldots\}$. (24.24)

A commitment event may trigger:

$r_{k+1} = \mathrm{Switch}(r_k, e_k, L_k, \mathscr{R}_k)$. (24.25)

The system then changes generator:

$A_{r_k} \to A_{r_{k+1}}$. (24.26)

PORE explicitly separates smooth no-jump windows from switch events and requires jump payloads and transition statistics to be recorded separately.

This reinforces the rule:

Do not estimate one global generator across regime changes.

24.9 Mixed Relation

A realistic system may contain:

- one oscillatory subsystem;
- one accumulating ledger;
- one expanding residual mode;
- one threshold gate.

A local block form may be:

$A_{\mathrm{total}} = A_{\mathrm{elliptic}} \oplus A_{\mathrm{parabolic}} \oplus A_{\mathrm{hyperbolic}} + \mathrm{Coupling}$. (24.27)

Complex coordinates may remain useful in the elliptic block.

They should not absorb the entire mixed system.

The typed framework is designed precisely for this situation.

24.10 Complex-branch rejection test

Ordinary complex completion should be rejected when:

1. no persistent two-dimensional mode exists;
2. the local discriminant is nonnegative;
3. the apparent phase changes arbitrarily with preprocessing;
4. the mode is not stable across nearby windows;
5. a real drift or jump model predicts better;
6. the proposed conjugate coordinate has no unit-consistent interpretation;
7. the readout does not detect the mode.

Rejecting the complex branch does not reject the general world-formation framework.

It identifies the Relation as non-elliptic in that regime.

24.11 Module-coverage assessment

Regime example	Elliptic phase	Dissipation	Historical ledger	Self-reference
Damped oscillator	Yes	Yes	Not required	Not required
Stochastic dissipative quantum model	Yes	Yes	Measurement record possible	Model-dependent
Breather	Often	Possibly	Not required	Not required
Parabolic accumulation	No ordinary phase	Optional	May represent memory	Possible
Hyperbolic feedback	No ordinary phase	Optional	Possible	Possible
Jump process	Regime-specific	Possible	Often important	Possible

This table demonstrates the independence of:

- phase;
- dissipation;
- self-reference;
- commitment.

---

25. AI, Law, Science, and Institutions as Partial Operational Worlds

25.1 Why these examples are presented differently

The previous examples possess explicit mathematical models.

AI, law, science, and organizations are broader socio-technical domains.

Their current role in this article is therefore not to receive one universal equation.

They are analysed through **implementation cards**.

For each domain, the article asks:

1. What is the Base?
2. What produces traces?
3. Who can access them?
4. What policy adapts?
5. What acts as gate?
6. What enters the ledger?
7. What remains residual?
8. How does history return?
9. Is an elliptic mode actually established?
10. What would falsify the proposed mapping?

These are typed structural mappings, not proof of material identity.

25.2 AI agent card

Consider an AI agent interacting with users, tools, memory, and an external environment.

Base

The detailed Base may include:

- current context;
- model state;
- tool outputs;
- memory;
- permissions;

- task state;
- external environment state;
- safety policy.

Represent schematically:

$$X_k^{AI} = (\text{Context}_k, \text{Memory}_k, \text{Tools}_k, \text{Permissions}_k, \text{Environment}_k). \quad (25.1)$$

Trace

Possible traces include:

- generated token;
- answer;
- tool result;
- critique;
- correction;
- execution log;
- user feedback.

Filtration

The operational filtration is:

$$\mathscr{F}_k^{AI} = \text{accessible context} + \text{retrieved memory} + \text{tool history} + \text{policy state}. \quad (25.2)$$

Policy

The adaptive policy may select:

- next token;
- next tool;
- planning strategy;
- retrieval query;
- verification step;
- abstention;
- escalation.

Gate

Possible gates include:

- decoder selection;
- safety classifier;
- tool permission;
- human approval;
- deployment policy;
- irreversible-write control.

Ledger

Possible ledgers include:

- current conversation;
- long-term memory;
- action log;
- audit log;
- external database update;
- tool-created artifact.

Residual

Possible residual includes:

- rejected candidates;
- unresolved contradiction;
- inaccessible context;
- hidden model uncertainty;
- unexecuted plan;
- failed tool state;
- safety concern.

25.3 Stored data is not governed trace

An AI system may store information without using it in a governed future process.

The philosophical-interface source states:

Stored Data ≠ Governed Trace. (25.3)

A datum becomes operational trace only when:

- its retention rule is declared;

- it is accessible to later policy;
- its use changes later behaviour;
- correction and deletion rules are known;
- residual and uncertainty are auditable.

An AI that remembers everything without governance risks surveillance and contamination.

An AI that remembers nothing cannot learn from history.

An AI that selectively remembers without exposing the selection rule creates opaque world formation.

25.4 AI world-forming test

An AI runtime becomes self-referential in the strong sense when:

Output_k
→ Memory or Context_{k+1}
→ Policy_{k+1}
→ Changed Tool or Generation Behaviour
→ Output_{k+1}. (25.4)

It becomes a fuller operational world when admitted actions alter the external or institutional environment:

Candidate Action
→ Approval Gate
→ External Write
→ Audit Ledger + Residual
→ Changed Future Permissions and State. (25.5)

Many current language-model interactions implement only part of this architecture.

A stateless one-turn model may produce traces but lack durable historical return.

A tool-using memory agent may implement substantially more.

25.5 Does AI require a complex mode?

No general answer is currently established.

AI systems may display:

- oscillatory revision;
- generate–critique cycles;
- competing latent tendencies;
- phase-like attention dynamics.

These observations do not prove:

$J^2 = -I$. (25.6)

An elliptic AI mode would require:

- state estimation;
- generator identification;
- spectral evidence;
- readout construction;
- predictive gain.

Until then, complex AI interpretation remains a hypothesis or analogy.

25.6 Law card

Law provides one of the clearest examples of governed Commitment.

Base

The legal Base may include:

- facts;
- parties;
- jurisdiction;
- applicable law;
- prior cases;
- evidential record;
- institutional competence.

Candidate consequence

A raw harm, claim, argument, or proposed remedy does not automatically become a legal event.

Gate

The legal gate includes:

- standing;
- jurisdiction;
- evidence;
- admissibility;
- burden of proof;

- procedure;
- judgment;
- appeal.

The philosophical-interface source summarizes:

Raw Harm + Legal Gate → Legal Event. (25.7)

Only after passing the gate does the event enter legal trace.

25.7 Legal ledger

The legal ledger may contain:

- judgment;
- order;
- precedent;
- statutory classification;
- settlement;
- procedural record.

Its contents alter future:

- rights;
- duties;
- strategies;
- admissible arguments;
- judicial interpretation;
- institutional behaviour.

This is a direct example of:

Ledger Shape → Institutional Shape. (25.8)

25.8 Legal residual

A harm may fail to become legal event because:

- evidence is unavailable;
- categories are outdated;
- jurisdiction is absent;

- procedure blocks recognition;
- power limits access;
- the burden cannot be met.

The source defines schematically:

Legal Residual
= Harm − Recognized Legal Event. (25.9)

This subtraction is conceptual rather than necessarily scalar.

The residual may later return through:

- appeal;
- review;
- new evidence;
- precedent revision;
- legislative reform;
- public inquiry.

Thus mature law requires:

Closure + Governed Reopening. (25.10)

Law strongly instantiates:

- gate;
- authority;
- ledger;
- residual;
- reopening;
- historical return.

It does not require an ordinary complex phase to do so.

25.9 Organizational card

Organizations often define their operational worlds through metrics.

A KPI may function as:

KPI = Measurement + Gate + Reward + Trace. (25.11)

The metric determines:

- what counts;

- what is compared;
- what is rewarded;
- what is remembered;
- what is treated as success.

Repeated recording alters behaviour:

Ledger Shape → Institutional Shape. (25.12)

For example, if an organization records:

- speed;
- throughput;
- cost;
- utilization;

but fails to record:

- burnout;
- technical debt;
- trust loss;
- unresolved disagreement;

then the official ledger omits material residual.

The source argues that such omitted cost may later appear as crisis:

Unrecorded Cost → Residual Accumulation. (25.13)

Examples include:

- burnout becoming attrition;
- technical debt becoming failure;
- trust loss becoming reputational collapse;
- ignored risk becoming scandal.

25.10 Organizational self-reference

An organization becomes self-referential when:

Metric
→ Management Interpretation
→ Resource Policy
→ Employee Behaviour
→ New Metric. (25.14)

The metric does not passively describe the organization.

It helps train the organization into the form the metric rewards.

This is a clear macro-scale example of:

Trace → Filtration → Policy → Changed Future Trace. (25.15)

The framework predicts that changing the KPI gate may change organizational evolution even when the underlying workforce initially remains the same.

25.11 Metric gaming as adaptive response to a gate

Metric gaming is often treated as moral failure by individuals.

The world-formation framework adds a structural interpretation.

Agents adapt to the declared success gate.

Let the objective be:

Reward = g(KPI). (25.16)

If the KPI omits important value dimensions, an intelligent agent may maximize:

KPI (25.17)

while degrading:

Unmeasured Value. (25.18)

Thus:

Metric Gaming
= Intelligence Adapting to an Incomplete Gate. (25.19)

The source expresses a related principle:

AI Optimization + Bad KPI → Scaled Deformation. (25.20)

This makes gate design and residual auditing central to AI governance.

25.12 Science card

Science also forms operational worlds through governed declaration.

Base

The scientific Base includes:

- instruments;
- datasets;
- methods;

- background theory;
- accepted results;
- funding and institutional constraints.

Trace

The trace may be:

- measurement;
- dataset;
- model fit;
- published claim;
- replication result.

Gate

Scientific gates include:

- methodological standards;
- statistical thresholds;
- review;
- replication;
- reproducibility;
- disciplinary acceptance.

Ledger

The scientific ledger includes:

- publications;
- databases;
- reference models;
- accepted constants;
- textbooks;
- standards.

Residual

Residual includes:

- anomalies;

- excluded data;
- failed replications;
- unresolved disagreement;
- model dependence;
- inaccessible negative results.

Historical return

Accepted science changes:

- which experiments are designed;
- which questions are funded;
- which measurements are interpreted as meaningful;
- which theories are admissible.

Science therefore becomes a time-bearing world when prior accepted and unresolved results alter later inquiry.

25.13 Scientific ledger is not truth itself

A published claim can be historically operative and later revised.

Therefore:

Scientific commitment ≠ Final truth. (25.21)

Yet the commitment matters because it shapes subsequent research.

A mature scientific ledger should preserve:

- evidence;
- method;
- uncertainty;
- failed alternatives;
- revision triggers.

This is residual honesty in scientific form.

25.14 Comparative implementation matrix

Module	Quantum observer	CAPM	Δ5	AI agent	Law	Organization	
Protocol	Strong	Strong	Strong	Variable	Strong	Variable	St

Module	Quantum observer	CAPM	Δ5	AI agent	Law	Organization	
Detailed Base	Strong	Partial	Strong	Partial	Complex	Complex	C
Trace	Strong	Strong	State amplitude, not central trace	Strong	Strong	Strong	St
Filtration	Strong	Partial	Not central	Strong candidate	Strong	Strong	St
Adaptive policy	Strong	Hypothetical market level	Not central	Strong candidate	Strong	Strong	St
Generator	Strong quantum implementation	Exact valuation geometry	Strong specific model	Usually opaque	Procedural rather than low-dimensional	Mixed	M
Elliptic mode	Quantum structure present	Exact readout geometry	Half-turn, not identical	Unproven	Unproven	Unproven	D d
Conjugate readout	Not principal result	Strong	Not principal result	Hypothetical	Not required	Not required	P su
Gate	Measurement instrument	Recognition/ settlement	Not central	Decoder/ approval	Very strong	KPI/approval	R re
Ledger	Internal records	Financial ledger	Not central	Context/ memory/ audit	Very strong	Strong	St
Residual	Limited	Strongly distinguished	Leakage-related	Important	Very strong	Very strong	V
Historical return	Strong	Promising, partially formalized	Not central	System-dependent	Strong	Strong	St
Reopening	Limited	Revaluation possible	Not central	Correction/ update	Strong	Governance-dependent	R re

This matrix shows why no example should dominate the general theory.

25.15 Main comparative finding

The examples separate the framework into three maturity patterns.

Pattern A — Strong recursion, limited Commitment

The quantum observer example strongly formalizes:

Trace → Filtration → Policy → Future Instrument.

Its institutional gate and residual modules are comparatively limited.

Pattern B — Strong conjugate geometry, incomplete world recursion

CAPM strongly formalizes:

Readout → Conjugate Exposure → Movement → Gate → Ledger.

The complete market-level observer recursion remains a research programme.

Pattern C — Strong Commitment, weak or unknown low-dimensional geometry

Law, organizations, and science strongly instantiate:

Gate → Ledger → Residual → Historical Return.

They do not yet justify one ordinary complex Relation.

This is not a defect.

It demonstrates the modularity of the architecture.

25.16 Worked-example conclusion

The examples support the following synthesis:

1. **Self-referential observerhood** is best established through trace-conditioned adaptive policy.
2. **Conjugate geometry** is best established through a declared generator and unit-consistent readout.
3. **Opposition** must be earned through an operator and energy rather than inferred symbolically.
4. **Dissipation and phase** can coexist, but neither implies the other.
5. **Historical reality** depends on gates, ledgers, authority, access, residual, and reopening.
6. **Complex numbers are important but conditional.**
7. **A full operational world requires the return of its own history into future generation.**

The next Part turns these conclusions into a domain-instantiation protocol, maturity rubric, ablation programme, and explicit falsification criteria.

Part VII — Research Programme, Maturity, and Falsification

26. Domain Instantiation Protocol

26.1 Why resemblance is not enough

A cross-domain framework becomes scientifically weak when every suggestive resemblance is treated as an implementation.

A market, legal system, AI agent, quantum observer, organization, and biological process may all contain:

- traces;
- gates;
- persistence;
- feedback;
- unresolved consequence.

That does not establish that they instantiate the same equations.

The framework therefore distinguishes:

Structural analogy
→ typed conceptual mapping
→ operational correspondence
→ formal module
→ integrated domain model
→ mature operational theory. (26.1)

A domain should not be called an **instantiation** merely because its vocabulary can be translated into Base, Relation, and Commitment.

The mapping must identify:

- actual objects;
- operational rules;
- measurements;
- interventions;
- failure conditions.

The Handoff requires a fixed admission procedure and states that a domain lacking these definitions should remain an analogy rather than an instantiation.

26.2 Domain Admission Rule

Definition 26.1 — Domain instantiation [D]

A domain instantiates the framework only to the extent that its proposed objects, operators, measurements, gates, and recursive returns can be operationally declared and independently tested. (26.2)

Instantiation is therefore modular.

A domain may instantiate:

- the observer-recursion module;
- the Effective Base module;

- the conjugate-readout module;
- the Commitment module;

without instantiating all four.

For example:

- CAPM strongly instantiates conjugate readout;
- the quantum observer model strongly instantiates trace-conditioned policy;
- law strongly instantiates gate–ledger–residual Commitment;
- PORE proposes an Effective Base compiler.

None currently establishes the entire general architecture at its highest maturity.

26.3 The World Instantiation Card

Every proposed domain application should complete the following card.

1. Boundary

What is inside the modelled world?

What is treated as environment?

Denote:

B_p. (26.3)

The boundary must identify:

- internal observers;
- external interventions;
- resources;
- relevant institutions;
- excluded processes.

---

2. Protocol

Declare:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (26.4)

Specify:

- observation scale;
- time window;

- admissible interventions;
- gate rule;
- record access;
- residual rule.

A protocol cannot be reconstructed retrospectively only after seeing the result.

---

3. Trace

What stable or addressable output is generated?

Write:

$y_k = h_p(X_k)$. (26.5)

Specify its:

- units;
- data type;
- persistence;
- uncertainty;
- provenance.

---

4. Internal recursion

How does the trace alter an observer, controller, or policy?

Show:

$y_{1:k} \to \mathscr{F}_{a,k} \to \pi_{a,k+1}$. (26.6)

A verbal claim that “participants react” is insufficient.

The policy dependence must be identified.

---

5. Backreaction

How does the changed policy alter the later state or transition law?

Demonstrate:

$\partial \Pr(X_{k+1} \mid \cdot)/\partial \pi_{a,k+1} \neq 0$. (26.7)

This is the causal link that distinguishes internal recursion from passive record keeping.

---

6. Loop validity

Is there a persistent organized loop?

A proposed loop should possess:

- repeated trace production;
- identifiable boundary;
- sufficient duration;
- recoverable recurrence;
- measurable intervention response.

A transient event should not automatically receive a PORE-like state compiler.

---

7. Base compiler

Can the effective condition be reproducibly compiled?

Write:

$\Xi_k = C_p(W_k[X, y, L])$. (26.8)

Specify:

- input window;
- proxies;
- normalization;
- uncertainty;
- compiler version.

---

8. State sufficiency

Does the reduced Base preserve predictive and intervention information?

Test:

$I(H_k; \Xi_{k+1} \mid \Xi_k, r_k, u_k) \approx 0$, (26.9)

where H_k denotes omitted history and r_k an optional regime label.

If the conditional information remains substantial, the proposed Effective Base is incomplete.

---

9. Generator

Can a transition operator or kernel be estimated or formally declared?

Examples include:

$\delta\Xi_{k+1} \approx A_r\delta\Xi_k + G_r\delta u_k + \varepsilon_k$, (26.10)

or:

$\Pr(\Xi_{k+1} \mid \Xi_k, u_k, r_k)$. (26.11)

The generator must be distinguished from the compiler and the readout.

---

10. Algebraic type

What Relation geometry is supported?

Test whether the active mode is:

- elliptic;
- parabolic;
- hyperbolic;
- dissipative;
- jump-like;
- mixed;
- not reducible.

Ordinary complex structure requires more than cross-coupling.

---

11. Readout

What scalar or declared output is selected?

Write:

$Y = V_p(\Xi, r)$. (26.12)

Specify:

- units;
- orientation;
- baseline;
- observer;
- valuation or interpretation protocol.

---

12. Conjugate response

Is there an operational Relation coordinate ξ such that:

$Q_\gamma = -\partial Y / \partial \xi$ (26.13)

has measurable or exact meaning?

In an elliptic mode:

$\mathcal{Q}_\gamma = -a(Jx)$. (26.14)

The proposed response must pass:

- unit consistency;
- sign convention;
- perturbation validation;
- readout-rank tests.

---

13. Gate

What admits, rejects, transforms, or defers a candidate consequence?

Write:

$e_k = \mathcal{G}_p(c_k; X_k, L_k)$. (26.15)

Specify:

- threshold;
- authority;
- timing;
- partial-admission rule;
- appeal or reopening rule.

---

14. Record

Where is the admitted consequence stored?

Specify:

- ledger structure;
- durability;
- access;
- versioning;
- auditability;
- deletion or revision rules.

Write:

$L_{k+1} = L_k \oplus T_p(e_k)$. (26.16)

---

15. Residual

What remains inadequately represented after commitment?

Write:

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (26.17)

Residual categories should be named rather than hidden inside one undifferentiated error term.

---

16. Historical return

How do ledger and residual alter later selection?

Show at least one return:

$L_k, \mathscr{R}_k$
→ filtration; (26.18)

$L_k, \mathscr{R}_k$
→ policy; (26.19)

$L_k, \mathscr{R}_k$
→ gate; (26.20)

$L_k, \mathscr{R}_k$
→ generator; (26.21)

$L_k, \mathscr{R}_k$
→ feasible action set. (26.22)

Without such return, the record is archival rather than world-forming.

---

17. Falsification

What result would show that the proposed mapping is wrong?

The application must declare rejection conditions before broad interpretation.

The Handoff's admission rule contains these same seventeen requirements and explicitly rejects domain promotion by analogy alone.

26.4 Minimum Domain Card

For practical studies, the full card can be summarized in one table.

Field	Required declaration
World ID	domain, boundary, protocol version
Base	full state and optional reduced state
Trace	observable output and units
Observer	record access and filtration
Policy	trace-conditioned action

Field	Required declaration
Backreaction	effect on later state or generator
Relation	local transition law and regime
Readout	declared scalar
Conjugate	response and activation test
Gate	admission rule
Ledger	storage and authority
Residual	unintegrated consequence
Return	future causal pathway
Falsifier	rejection condition
Maturity	M0–M5 rating
Status	[S], [F], [H], or [A]

26.5 Protocol and version control

Every numerical or empirical implementation should store:

- Protocol_ID;
- Boundary_ID;
- Timebase;
- Compiler_Version;
- Generator_Window;
- Gate_Version;
- Ledger_Rule;
- Residual_Rule;
- Unit_Policy;
- Orientation;
- Numerical_Tolerance;
- Data_Timestamp.

This matters because a model can appear inconsistent when the underlying protocol has silently changed.

The CAPM source similarly requires implementations to store formula version, protocol ID, units, baseline, phase domain, orientation, warnings, and residuals.

26.6 No silent repair rule

A failed domain test should not be repaired by reinterpretation after the fact.

Examples of prohibited silent repair include:

- redefining the boundary after prediction fails;
- changing the phase orientation without reporting it;
- replacing an unsuccessful complex mode with a metaphorical one;
- absorbing gate residual into the Base without updating the model;
- altering proxy definitions to preserve a preferred conclusion;
- treating inaccessible records as though observers used them.

The correct response is:

Failure
→ diagnose module
→ revise protocol
→ rerun test
→ record version change. (26.23)

PORE adopts the same operational discipline: claims are valid only inside a declared boundary, timebase, and instrument, and failure outside that protocol should be reported rather than patched by narrative.

---

27. Maturity and Epistemic Status

27.1 Why one maturity score is not enough

A domain can be mathematically mature in one module and immature in another.

CAPM provides:

- an exact conjugate-readout formula;
- but not yet a validated complete recursive market world.

Law provides:

- mature institutional gates and ledgers;
- but no established low-dimensional elliptic generator.

The quantum observer model provides:

- rigorous filtration and adaptive policy;
- but not a general institutional residual theory.

The maturity assessment should therefore report:

M_domain = (M_Base, M_Relation, M_Commitment, M_Integration). (27.1)

A single overall level may still be assigned, but it should be limited by the weakest module required for the claim.

27.2 M0 — Suggestive metaphor

At M0, only verbal resemblance exists.

Examples:

- “markets behave like waves”;
- “law is a ledger”;
- “AI has phase”;
- “residual is like entropy.”

No protocol, variables, operators, or failure tests are declared.

Permitted claim

> The comparison may suggest a research direction.

Prohibited claim

> The domain instantiates the formal framework.

27.3 M1 — Typed conceptual mapping

At M1, the main roles are distinguished.

The application identifies candidate:

- Base;
- trace;
- observer;
- policy;
- gate;
- ledger;
- residual.

The objects are not yet operationally measured.

Achievement

Category errors are reduced.

Remaining weakness

The mapping can still accommodate almost any outcome.

27.4 M2 — Operational correspondence

At M2, the application declares:

- protocol;
- proxies;
- units;
- interventions;
- access;
- gate rule;
- residual rule;
- failure criteria.

At least some objects can be measured or audited.

Achievement

The mapping can now fail operationally.

Remaining weakness

No major formal module may yet be mathematically closed or empirically validated.

27.5 M3 — Formal module

At M3, at least one module possesses a developed mathematical construction.

Examples include:

- CAPM `R/Q` conjugate valuation;
- PORE compilation;
- quantum observer filtration;
- Δ5 phase locking.

The Handoff classifies these as examples of M3 formal modules.

Achievement

A specific part of the framework has:

- equations;
- assumptions;
- internal checks;

- failure conditions.

Remaining weakness

The module may remain isolated from the complete Base–Relation–Commitment loop.

27.6 M4 — Integrated domain model

At M4, several modules are linked:

Base
→ Relation
→ Commitment
→ Recompiled Base. (27.2)

An M4 application should demonstrate:

1. trace-conditioned policy;
2. validated or useful Base representation;
3. declared Relation generator;
4. operational gate;
5. ledger and residual;
6. historical backreaction.

Not every optional module is required.

An M4 legal model may lack complex geometry.

An M4 oscillator model may lack institutional Commitment and therefore fail to qualify as an integrated world model.

The required modules depend on the claim.

27.7 M5 — Mature operational theory

At M5, the integrated domain model has:

- reproducible measurement;
- validated intervention;
- predictive gain;
- transport or frame rules;
- independent replication;
- established failure boundaries.

The Handoff explicitly states that no supplied source independently establishes the complete

general framework at M5.

M5 standard

A separate team should be able to:

1. reconstruct the protocol;
2. reproduce measurements;
3. estimate the Base;
4. recover the claimed generator;
5. verify gate and ledger updates;
6. reproduce predictions;
7. observe declared failure outside the valid regime.

Only then should the framework be treated as a mature operational theory in that domain.

27.8 Maturity matrix

Level	Name	Main evidence	Permitted statement
M0	Suggestive metaphor	verbal resemblance	“may be analogous”
M1	Typed mapping	objects and roles distinguished	“has a candidate correspondence”
M2	Operational correspondence	protocols, proxies, interventions, falsifiers	“can be tested as an implementation”
M3	Formal module	developed mathematical component	“this module is formally constructed”
M4	Integrated domain model	linked Base–Relation–Commitment loop	“the domain implements the architecture under P”
M5	Mature operational theory	reproducibility, intervention, prediction, replication	“the integrated theory is validated within its failure boundary”

27.9 Current approximate module maturity

The following ratings are provisional and protocol-dependent.

Example	Strongest current module	Approximate level of that module	Integrated-world status
Quantum observer model	filtration and adaptive policy	M3	below M4 general-world integration
CAPM conjugate valuation	conjugate readout	M3	below M4 market integration
PORE	Effective Base compiler	M3	domain-dependent
Δ_5	operator-earned opposition	M3	not a complete world model

Example	Strongest current module	Approximate level of that module	Integrated-world status
Law	Commitment architecture	M2–M3 conceptually	formal integrated model still required
AI agents	recursion and gate candidates	M1–M2 generally	implementation-specific
Organizations	trace–policy feedback	M1–M2 generally	implementation-specific
Science	gate–ledger–revision structure	M1–M2 generally	formalization required

These are not universal rankings of the domains.

They assess only the framework components represented in the supplied sources.

27.10 Epistemic-status labels

The Handoff specifies four claim categories.

The article's earlier symbol `[S]` corresponds to the Handoff's `[E]`.

For final editing, the notation should be normalized consistently.

[E] Established within a declared construction

Examples:

- CAPM:

  $$Q = -\partial R/\partial \theta_V. \quad (27.3)$$

- quantum observer policy is measurable with respect to prior trace;
- Δ5 opposition minimizes its declared pair energy.

"Established" means established inside the assumptions of the source construction, not established as a universal law.

[F] Formal consequence

A result derived from stated assumptions.

Examples:

- Complex Intertwining Principle:

  $$\Phi_a J = J_R\Phi_a; \quad (27.4)$$

- a real three-dimensional generator with one complex-conjugate pair decomposes locally as:

  $$\mathbb{R}^3 \cong \mathbb{C} \oplus \mathbb{R}. \quad (27.5)$$

The result may be mathematically exact while its empirical applicability remains untested.

[H] Testable hypothesis

Examples:

- some market loops admit a sufficient PORE signature;
- some self-referential generators contain persistent elliptic modes;
- ledger and residual improve prediction after conditioning on the current Base.

Hypotheses require declared protocols and failure conditions.

[A] Structural analogy

Examples:

- settlement resembles operational world commitment;
- legal precedent resembles ledgered historical return;
- conjugate valuation has a Yin–Yang-like relational form.

An analogy may motivate formal work.

Repetition does not transform it into a theorem.

27.11 Claim matrix

Every major claim should be accompanied by:

Field	Example
Claim	“The selected mode is elliptic.”
Status	[H]
Protocol	P_fin_01
Module	Relation
Evidence	stable conjugate eigenpair
Valid regime	low-volatility windows
Failure condition	discriminant becomes nonnegative
Maturity	M2
Source or derivation	estimated generator
Revision date	declared timestamp

This claim matrix prevents a mathematical identity inside one source from being used as empirical proof of a broader domain model.

---

28. Falsification and Ablation Programme

28.1 Why the framework must be breakable

A framework that can reinterpret every failure as hidden residual, undiscovered phase, or incorrect observer perspective cannot be falsified.

It becomes vocabulary rather than science.

The framework therefore requires **module-specific falsification**.

A failed Base compiler does not automatically falsify the gate model.

A failed complex mode does not falsify self-reference.

A failed Γ functional does not imply that no residual exists.

The correct logic is:

Claim
→ Required module
→ Activation conditions
→ Test
→ Pass, revise, or reject. (28.1)

28.2 Protocol falsification

Claim

The declared protocol produces reproducible traces.

Test

Independent teams repeat:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (28.2)

Compare:

y_k^(lab 1), y_k^(lab 2). (28.3)

Failure condition

Equivalent runs under the declared tolerances produce incompatible results that cannot be explained by declared noise.

Interpretation

The protocol, observation map, or boundary is not reproducible.

No downstream state or phase claim should be accepted.

28.3 Self-reference ablation

Claim

Accessible trace changes later policy and future outcome distribution.

Full process

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (28.4)

Ablated process

$\pi_{a,k+1}$^abl
= $f_{a,p}(\mathscr{F}_{a,k}$ without selected trace history). (28.5)

Compare:

$\Pr(y_{k+1} \mid \text{full})$ (28.6)

with:

$\Pr(y_{k+1} \mid \text{ablated})$. (28.7)

Failure condition

No material difference appears across the declared regime.

Interpretation

The trace may be stored but not causally active.

The system may exhibit ordinary dynamics or fixed feedback rather than the proposed self-referential recursion.

28.4 Effective Base falsification

Claim

The proposed Effective Base Ξ_k closes prediction for task $\mathcal{T}$.

Reduced model

$\Pr(y_{k+1} \mid \Xi_k, u_k)$. (28.8)

History-augmented model

$\Pr(y_{k+1} \mid \Xi_k, H_k, u_k)$. (28.9)

Test

Evaluate:

ΔPredictiveGain
= Score(history-augmented) − Score(reduced). (28.10)

Or estimate:

$I(H_k; y_{k+1} \mid \Xi_k, u_k)$. (28.11)

Failure condition

Omitted history adds persistent out-of-sample information.

Repair sequence

1. revise compiler;
2. split regimes;
3. add branch label;
4. add minimum memory;
5. use coupled loops.

The Handoff specifically recommends this escalation rather than immediately rebuilding a huge universal latent vector.

28.5 PORE-specific failure

For a PORE candidate:

$\Xi = (\varrho, \gamma, \tau)$, (28.12)

falsifiers include:

- proxy instability;
- failure of repeatability;
- poor intervention prediction;
- unmodelled jump behaviour;
- boundary leakage;
- substantial residual information after conditioning.

PORE itself restricts its claims to declared protocols and distinguishes smooth response from jump regimes rather than forcing one continuous model across both.

28.6 Generator falsification

Claim

The local Relation is governed by A_r.

Test

Estimate:

$\delta\Xi_{k+1} = A_r\delta\Xi_k + G_r\delta u_k + \varepsilon_k$. (28.13)

Evaluate:

- held-out prediction;
- parameter stability;
- residual structure;
- intervention response;
- regime dependence.

Failure condition

A_r changes unpredictably inside the claimed stable regime, or residuals retain systematic dynamics.

Interpretation

The mode may be:

- nonlinear;
- nonstationary;
- history-dependent;
- mis-specified;
- not low-dimensional.

28.7 Complex-mode falsification

Claim

The generator contains a stable elliptic mode.

Required evidence

1. complex-conjugate eigenpair;
2. persistent two-dimensional invariant plane;
3. stable orientation;
4. readout sensitivity;
5. improved predictive or explanatory performance.

Failure conditions

Reject ordinary complex completion when:

- the discriminant is nonnegative;

- the eigenpair is window-fragile;
- the apparent phase is preprocessing-dependent;
- no coherent mode plane persists;
- a parabolic or hyperbolic model predicts better;
- nonlinear leakage destroys quarter-turn closure.

A nonzero metric-relative antisymmetric component alone is insufficient. The Handoff grades evidence from cross-coupling through circulation to persistent coherent phase-bearing mode, reserving strong ordinary-complex language for the higher levels.

28.8 Conjugate-readout falsification

Claim

The proposed response:

$\mathcal{Q}_\gamma = -\partial Y/\partial \xi$ (28.14)

predicts the first-order change of Y.

Perturbation test

Apply small positive and negative changes:

$\xi_+ = \xi + \delta\xi$, (28.15)

$\xi_- = \xi - \delta\xi$. (28.16)

Estimate:

$\mathcal{Q}_\gamma$^est
$= -[Y(\xi_+) - Y(\xi_-)]/[2\delta\xi]$. (28.17)

Compare with:

$\mathcal{Q}_\gamma$^model. (28.18)

Error

$\varepsilon_Q = \mathcal{Q}_\gamma$^est $- \mathcal{Q}_\gamma$^model. (28.19)

Failure condition

The error remains materially larger than the declared tolerance, or the sign reverses systematically.

The CAPM source specifies an analogous bump-and-revalue audit for comparing numerical and analytic Q.

28.9 Unit and orientation falsification

A proposed complex readout fails if:

- Y and $\mathcal{Q}_\gamma$ cannot be placed in compatible units;
- orientation is undefined;
- opposite conventions are combined silently;
- the reported phase cannot be reconstructed.

A valid implementation should store:

Orientation_ID. (28.20)

Changing `J` to `-J` reverses:

$Y \rightarrow -\mathcal{Q}_\gamma$ (28.21)

into:

$Y \rightarrow \mathcal{Q}_\gamma$. (28.22)

The sign difference is acceptable only when declared.

28.10 Gate falsification

Claim

The proposed gate distinguishes commitment-producing candidates from failed candidates.

Test

Compare events around gate threshold or across controlled gate changes.

Measure subsequent changes in:

- ledger update;
- policy;
- generator;
- regime;
- future trace;
- record redundancy.

Failure condition

Passing the proposed gate does not create larger or more persistent downstream change than comparable failed candidates.

Interpretation

The alleged gate may be:

- ceremonial;
- incorrectly specified;
- downstream rather than causal;
- one of several necessary gates.

28.11 Ledger falsification

Claim

The ledger is causally consulted.

Ablation

Remove, mask, or randomize the selected ledger entry while preserving the present observable state where possible.

Compare:

Pr(Future | L) (28.23)

with:

Pr(Future | L^ablated). (28.24)

Failure condition

No later policy or state changes.

Interpretation

The record may be archival rather than world-forming.

28.12 Residual falsification

Claim

A declared residual category captures future-relevant incompletely integrated consequence.

Test

Compare:

Pr(Future | Ξ, L, $\mathscr{R}$) (28.25)

with:

Pr(Future | Ξ, L). (28.26)

Failure condition

Residual adds no prediction, diagnosis, or intervention value.

Alternative explanation

The residual may already be compiled into ⩵ or ㄴ.

The study must distinguish redundancy from nonexistence.

28.13 Γ falsification

A Γ term should not be introduced merely because it produces elegant equations.

Each Γ component should possess at least one of:

- measurable proxy;
- expert audit;
- historical validation;
- simulation test;
- falsifiable prediction;
- intervention value.

The residual source states that a Γ lacking such evidence is decorative rather than operational.

Test

Evaluate whether:

$F_\Gamma = -\delta\Gamma/\delta x$ (28.27)

predicts future path selection after controlling for the current Base and ledger.

Failure condition

The term does not improve prediction or intervention and cannot be independently audited.

Governance condition

Γ itself requires:

- ledgering;
- audit;
- appeal;
- revision;
- protection against authority capture.

The residual source expands the loop to:

Gate
→ Ledger + Residual
→ Γ
→ Γ Audit
→ Admissible Revision
→ New Gate.

28.14 Historical-return falsification

Claim

The world bears operational time because history alters future possibility.

Matched-state test

Construct cases with:

$\Xi_k^A \approx \Xi_k^B$, (28.28)

but:

$(L_k^A, \mathscr{R}_k^A) \neq (L_k^B, \mathscr{R}_k^B)$. (28.29)

Compare:

$\Pr(y_{k+1} \mid \Xi_k, L_k^A, \mathscr{R}_k^A)$ (28.30)

with:

$\Pr(y_{k+1} \mid \Xi_k, L_k^B, \mathscr{R}_k^B)$. (28.31)

Failure condition

The distributions remain equivalent throughout the claimed regime.

Interpretation

Either:

- history is not independently causal;
- or the proposed Base has already fully compiled it.

The claim must be narrowed accordingly.

28.15 Commitment-level falsification

Internal commitment

Test whether the record changes the observer's later policy.

Institutional commitment

Test whether ledger admission changes rights, permissions, resource allocation, or obligations.

Public commitment

Test whether independent observers can recover compatible records under declared frame maps.

Failure at one level does not imply failure at the others.

28.16 CAPM-specific falsifiers

The CAPM complex completion is valid in its principal Euclidean real domain when:

$$0 \le R/A \le 1. \quad (28.32)$$

Failure conditions include:

- `R > A` under the declared baseline;
- incompatible units;
- numerical violation of:

  $$A^2 = R^2 + Q^2; \quad (28.33)$$

- violation of:

  $$R = A \cos \theta; \quad (28.34)$$

- violation of:

  $$\partial R/\partial \theta = -Q. \quad (28.35)$$

The CAPM source instructs implementations not to silently convert a baseline or domain failure into a valid `Q`.

Even when the identities pass, empirical usefulness can still fail if the phase representation adds no explanatory, predictive, attributional, or control value.

28.17 Δ5-specific falsifiers

The Δ5-locked regime predicts:

- small pair energy;
- phase differences near π;
- enhanced use of Δ5 pair relations under its declared graph cost;
- Q-factor improvement under stronger locking in the specified model.

The source explicitly states that when pair energy is not small and Δ5 usage does not meet its regime conditions, the system is outside the locked regime and the predictions may weaken or fail.

This is the correct use of a specialized Relation module:

valid inside its declared regime, rejected outside it.

28.18 Model comparison

A proposed advanced module should be compared against simpler alternatives.

For example:

- scalar model;
- real multivariate model;
- complex elliptic model;
- hyperbolic model;
- history-augmented model;
- gate-free model.

A complex representation should be preferred only if it improves:

- held-out likelihood;
- forecast accuracy;
- intervention prediction;
- parameter stability;
- interpretability under unit and sign constraints.

Elegance alone is not predictive gain.

28.19 Falsification matrix

Module	Principal falsifier
Protocol	failure of reproducibility
Self-reference	trace ablation does not change future behaviour
Effective Base	omitted history materially improves prediction
Generator	unstable or systematically wrong transition law
Elliptic mode	no persistent complex-conjugate invariant mode
Conjugate readout	perturbation derivative fails
Gate	gate passage does not alter later commitment consequences
Ledger	record ablation has no downstream effect
Residual	no diagnostic or predictive value
Γ	unauditable or non-predictive cost functional
Historical return	matched states do not diverge by history
Public objectivity	records are inaccessible or frame-incompatible

---

29. Central Conjectures, Nonclaims, and Open Problems

29.1 Four central research conjectures

The framework's research programme can be organized around four conjectures identified in the Handoff.

29.2 Conjecture I — Effective-Existence Compression

Statement [H]

A broad but non-universal class of finite-lived organized loops admits a useful protocol-relative compilation:

$\Xi = (\varrho, \gamma, \tau)$, (29.1)

representing:

- effective presence;
- integrity;
- duration.

Supporting intuition

For an organized loop to be operationally identifiable:

1. enough of it must be instantiated;
2. it must retain enough integrity to count as the same loop;
3. it must persist long enough to be measured or acted upon.

Falsification

Reject the compression when:

- proxies are unstable;
- prediction requires substantial omitted history;
- interventions do not produce reproducible coordinate changes;
- mixed loops cannot be separated;
- regime transitions invalidate the signature.

Status

The broad portability of the triple is a hypothesis.

PORE provides the candidate compiler and operational testing grammar, not a proof that every

system possesses exactly three sufficient coordinates.

29.3 Conjecture II — Signed-Conjugate Mode Emergence

Statement [H]

Some self-referential reduced generators possess a stable two-dimensional invariant mode classified by:

$C\chi^2 = \chi I.$ (29.2)

The ordinary complex branch is:

$\chi < 0.$ (29.3)

Supporting intuition

Trace-conditioned policy can create:

- corrective lag;
- reciprocal response;
- amplification;
- accumulation;
- oscillation.

The generator determines which form emerges.

Falsification

Reject the proposed mode if:

- no stable reduction exists;
- the algebraic signature changes arbitrarily;
- a simpler real model performs better;
- the mode disappears under minor protocol variation.

Important limit

Complex numbers are neither exclusive to self-reference nor guaranteed by it.

Classical waves and circuits possess phase without internal observer recursion.

29.4 Conjecture III — Conjugate Readout Sufficiency

Statement [H]

Where a declared scalar readout detects a valid Relation mode, the response:

$\mathcal{Q}_\gamma = -\partial Y/\partial \xi$ (29.4)

provides explanatory, predictive, attributional, or control information not contained in Y alone.

Exact example

CAPM provides:

$Q = -\partial R/\partial \theta_V$. (29.5)

This is exact within the valuation construction.

Empirical test

Compare:

Model 0: FutureY ~ Y; (29.6)

Model 1: FutureY ~ Y + $\mathcal{Q}_\gamma$. (29.7)

Evaluate:

- forecast gain;
- intervention accuracy;
- regime diagnosis;
- gate-event prediction;
- communication clarity.

Falsification

If $\mathcal{Q}_\gamma$ adds no value beyond an equivalent real-variable representation, the conjugate notation may remain an optional coordinate transformation.

29.5 Conjecture IV — Ledger–Residual Backreaction

Statement [H]

After conditioning on the present Effective Base, ledger and residual history may still explain changes in:

- policy;
- generator;
- gate hazard;
- regime transition;
- future trace distribution.

Formally:

$$\Pr(y_{k+1} \mid \Xi_k, L_k, \mathscr{R}_k) \neq \Pr(y_{k+1} \mid \Xi_k). \quad (29.8)$$

Interpretation

If supported, Commitment is not merely descriptive bookkeeping.

It is part of the future-generating machinery.

Alternative result

If the difference disappears, the current Base may already compile the relevant history.

This would strengthen Effective Base closure while weakening the claim of separately measurable historical backreaction.

29.6 Conjecture interactions

The conjectures form a staged programme.

Stage A

Test whether an organized loop exists and can be compiled.

Stage B

Estimate its Relation generator.

Stage C

Classify its modes.

Stage D

Construct and test readouts.

Stage E

Identify gates, ledgers, and residual.

Stage F

Test historical return.

The failure of an early stage blocks stronger downstream claims.

For example:

$$\text{No sufficient Base} \Rightarrow \text{no reliable reduced generator.} \quad (29.9)$$

No elliptic mode
⇒ no ordinary complex conjugate readout. (29.10)

No operational gate
⇒ movement but not Commitment. (29.11)

No historical return
⇒ record but not full operational world formation. (29.12)

29.7 Required nonclaims

The article does not claim:

1. that it is a completed universal theory;
2. that every feedback loop is an internal-observer world;
3. that every self-referential system is complex;
4. that self-reference proves the existence of i;
5. that complex numbers arise only in self-referential systems;
6. that PORE is a fundamental ontology;
7. that every system has exactly three state dimensions;
8. that ϱ, γ, and τ are always orthogonal;
9. that τ is always an instantaneous state variable;
10. that every reduced generator has a stable two-dimensional mode;
11. that a nonzero antisymmetric component proves coherent phase;
12. that the full three-dimensional PORE space carries a global complex structure;
13. that Q is the full recursive state;
14. that Q is residual;
15. that Q is automatically loss, volatility, VaR, or opportunity cost;
16. that measurement rotation is chronological evolution;
17. that state movement automatically becomes historical commitment;
18. that ledgering guarantees public objectivity;
19. that institutional authority guarantees truth;
20. that Γ is universally entropy, friction, or action;
21. that similar algebra implies common material substance;
22. that CAPM proves markets are quantum mechanical;

23. that $\Delta 5$ is a universal phase law;
24. that law, AI, finance, and quantum mechanics possess identical dynamics.

These restrictions are part of the theory's architecture.

They are not optional disclaimers.

29.8 Open problem: identifying a genuine loop

An apparent loop may be produced by:

- aggregation;
- common forcing;
- analyst-selected windows;
- hidden external controllers;
- delayed response.

Research must distinguish:

internal recursion (29.13)

from:

external periodic forcing. (29.14)

Intervention and trace-ablation studies are required.

29.9 Open problem: compiler invariance

How sensitive is:

$\Xi = C_p(W)$ (29.15)

to:

- window length;
- proxy choice;
- normalization;
- observer protocol;
- regime definition;
- missing records?

A useful compiler should preserve its operational meaning under admissible variations.

It need not remain invariant under arbitrary protocol change.

29.10 Open problem: dynamic phase

Can a dynamic phase:

θ_D (29.16)

be identified independently of a declared valuation or measurement phase:

θ_V? (29.17)

The two must initially remain distinct.

A coupling may be defined:

$K_{VD} = \partial\theta_V/\partial\theta_D$. (29.18)

In a restrictive case:

$Q_D = Q_V K_{VD}$. (29.19)

The empirical stability of this coupling remains open.

29.11 Open problem: local-to-global complex structure

Suppose local mode planes possess:

$J_x^2 = -I$. (29.20)

Transport between points x and y preserves complex structure when:

$T_{y\leftarrow x} J_x = J_y T_{y\leftarrow x}$. (29.21)

Closed-loop transport is:

$H_{loop} = T_{a\leftarrow c} T_{c\leftarrow b} T_{b\leftarrow a}$. (29.22)

If:

$H_{loop} \neq I$, (29.23)

the system exhibits transport residual or holonomy.

Possible domain interpretations include:

- transaction-cost basis;
- frame mismatch;
- tax path dependence;
- settlement friction;
- model inconsistency.

These remain advanced hypotheses rather than established universal gauge laws.

29.12 Open problem: residual representation

Residual may be:

- scalar;
- vector;
- field;
- graph;
- set of claims;
- counterfactual branches;
- inaccessible records.

When is compression into Γ justified?

When must provenance remain explicit?

A scalar Γ may improve optimization while destroying information about who or what carries the residual.

The residual source warns that in social systems residual is often someone's burden and therefore requires explicit governance and appeal.

29.13 Open problem: self-validating Γ-lock

A dangerous self-referential pattern occurs when:

- the gate creates residual;
- the ledger records only apparent success;
- the residual is excluded by the same gate;
- policy treats ledger confidence as confirmation.

The residual source calls this Γ-lock:

Ledger Confidence ↑ while Hidden Γ ↑. (29.24)

Examples may include:

- AI verifying its own unsupported claims;
- bureaucracy validating itself through distorted KPIs;
- markets treating rising price as proof of value;
- paradigms excluding anomalies by definition.

A major research problem is to design:

Gate

→ Ledger + Residual
→ Audit
→ Admissible Revision
→ Revised Gate. (29.25)

29.14 Open problem: nested worlds

Operational worlds may be nested.

An individual observer may exist inside:

- an organization;
- a market;
- a legal system;
- a scientific institution.

Let:

$\mathcal{W}^1 \subset \mathcal{W}^2$. (29.26)

A commitment in the smaller world may remain residual in the larger world, or vice versa.

Research questions include:

- how ledgers translate across boundaries;
- how authority conflicts;
- how residual is transferred;
- how timebases synchronize;
- how one world's gate becomes another world's trace.

29.15 Open problem: world composition

Suppose two operational worlds interact:

$\mathcal{W}_A \leftrightarrow \mathcal{W}_B$. (29.27)

A compositional theory must specify:

- shared traces;
- interface gates;
- cross-ledger translation;
- access rights;
- transport maps;
- residual ownership.

The simple union:

$\mathcal{W}_A \cup \mathcal{W}_B$ (29.28)

need not form one coherent operational world.

The interface may itself generate a new Commitment layer.

29.16 Open problem: capacity

Some operational worlds may possess limited capacity for:

- independent traces;
- unresolved alternatives;
- simultaneous commitments;
- observer access;
- residual storage.

The Slot Interpretation may motivate a capacity module, but the combinatorial uniqueness of HeTu or LuoShu arrangements does not by itself prove that their numbers are universal semantic slot counts.

Capacity must be independently operationalized.

Therefore:

Combinatorial uniqueness ≠ Semantic capacity theorem. (29.29)

Slot architecture should remain an optional extension rather than a universal Base coordinate.

29.17 Open problem: ethical residual

In institutional and AI systems, residual is not morally neutral.

A system may improve its official ledger by transferring residual to:

- workers;
- users;
- minorities;
- future generations;
- external ecosystems;
- inaccessible stakeholders.

A complete residual audit should ask:

1. Who carries the residual?

2. Who benefits from hiding it?
3. Who defines the gate?
4. Who can appeal?
5. What information is unavailable?
6. What future reopening path exists?

The framework becomes governance-relevant precisely because operational closure redistributes unresolved consequence.

29.18 Research-stage sequence

A practical development programme should proceed as follows.

Stage 0 — Conceptual discipline

Separate typed objects and eliminate category errors.

Stage 1 — Protocol declaration

Specify boundary, trace, intervention, access, gate, and residual.

Stage 2 — Data and replay package

Record:

- traces;
- policies;
- state proxies;
- interventions;
- gate outcomes;
- ledger changes;
- residual categories.

Stage 3 — Module construction

Develop one formal module:

- Base compiler;
- generator;
- conjugate readout;
- gate model;
- residual functional.

Stage 4 — Ablation

Test whether removing the module changes prediction or intervention.

Stage 5 — Integrated runtime

Link:

Base → Relation → Commitment → Recompiled Base. (29.30)

Stage 6 — Independent replication

Publish sufficient data and protocol definitions for replay.

This sequence moves the framework from interpretive vocabulary toward operational science.

Conclusion — From Recorded Outcome to Inherited World

30.1 The starting problem

Operational systems repeatedly compress high-dimensional processes into small declarations:

- a number;
- an outcome;
- a token;
- a verdict;
- a mark;
- a decision.

Such compression is necessary.

But a scalar trace does not contain the full machinery through which it changes the future.

The central problem was therefore:

> How can a trace that is informationally small become causally large?

The answer is self-referential return.

A trace enters an observer filtration.

The filtration changes policy.

The policy changes the later transition law.

The changed process generates a new trace.

30.2 The mandatory causal backbone

The minimum self-referential circuit is:

Trace
→ Filtration
→ Adaptive Policy
→ Changed Transition Law
→ New Trace. (30.1)

A full operational world requires more:

Candidate Consequence
→ Gate
→ Ledger + Residual
→ Changed Future Admissibility. (30.2)

The two returns combine:

Trace
→ internal policy return; (30.3)

Commitment
→ historical structural return. (30.4)

A world forms when its own admitted past participates in generating what can happen next.

30.3 The functional grammar

The architecture can be compressed into:

Base
→ Relation
→ Commitment
→ Recompiled Base. (30.5)

Base

What organized condition presently exists under the protocol?

Relation

How can that condition transform?

Commitment

Which consequences become inherited history, and what remains residual?

The triad is functional.

It is not a three-substance ontology.

30.4 The role of complex numbers

Self-reference does not prove complex structure.

It creates a reason to search for missing relational degrees of freedom.

The correct sequence is:

Self-reference
→ adaptive or coupled generator
→ invariant-mode test
→ algebraic classification. (30.6)

When the active Relation mode is elliptic:

$J^2 = -I$, (30.7)

a scalar readout:

$Y = a(x)$ (30.8)

admits the conjugate response:

$\mathcal{Q}_\gamma = -a(Jx)$. (30.9)

The completed readout is:

$Z_\gamma = Y + i\mathcal{Q}_\gamma$. (30.10)

The readout map satisfies:

$\Phi_a J = J_R\Phi_a$. (30.11)

Thus the state-level elliptic mode and the readout-level conjugate plane are locally complex-isomorphic.

The imaginary unit represents transformation grammar.

It is not hidden substance.

30.5 What CAPM demonstrates

CAPM provides a mature exact example:

$A = CF/(1 + r_base)^t$, (30.12)

$R = CF/(1 + r_CAPM)^t$, (30.13)

$Q = \sqrt{(A^2 - R^2)}$, (30.14)

$Z = R + iQ = Ae^{(i\theta)}$, (30.15)

$\partial R/\partial\theta = -Q$. (30.16)

The cycle is:

$R \to -Q \to -R \to Q \to R$. (30.17)

This establishes conjugate measurement closure inside the declared valuation construction.

It does not establish that all markets possess one stable complex dynamic mode.

CAPM is a strong example of the broader framework, not its universal source.

30.6 Why movement is not enough

A conjugate response reveals exposure.

Exposure is not movement.

Movement is not recognition.

Recognition is not settlement.

A gate determines what enters a ledger.

Residual preserves what the gate failed to integrate.

The complete sequence is:

Measurement
→ Exposure
→ Movement
→ Candidate Consequence
→ Gate
→ Ledger + Residual. (30.18)

Geometry explains how a state can move.

Commitment explains which movement becomes inherited reality.

30.7 Why the world bears time

A system can oscillate without accumulating history.

A database can store records without consulting them.

A world bears operational time when its past changes its future.

Let two otherwise comparable present conditions possess different histories:

$(L^A, \mathscr{R}^A) \neq (L^B, \mathscr{R}^B)$. (30.19)

If:

$\Pr(\text{Future} \mid X, L^A, \mathscr{R}^A)$
$\neq \Pr(\text{Future} \mid X, L^B, \mathscr{R}^B)$, (30.20)

then history is part of the future-generating condition.

Time-bearing does not mean merely that events have timestamps.

It means that the world inherits consequences.

30.8 The framework’s principal contribution

The article does not present one new universal equation.

Its principal contribution is the disciplined interface among:

- protocol;
- state;
- trace;
- filtration;
- policy;
- compiler;
- generator;
- mode;
- readout;
- conjugate response;
- movement;
- gate;
- ledger;
- residual;
- historical return.

This discipline prevents:

- trace from being confused with state;
- PORE signature from being confused with ontology;
- cross-coupling from being confused with complex phase;
- Q from being confused with residual;
- movement from being confused with commitment;
- ledgering from being confused with truth;
- analogy from being confused with theorem.

30.9 The framework’s strongest defensible claim

The work should be read as:

> **A formal architecture and falsifiable research programme for protocol-**

bound self-referential world formation.

Its claims are deliberately modular.

Some are:

- source-established inside declared constructions;
- formal consequences of stated assumptions;
- testable hypotheses;
- structural analogies.

A universal theory would require several independent domains to reach integrated M4 and replicated M5 maturity.

That has not yet occurred.

30.10 Final synthesis

A trace becomes part of a world not when it is merely emitted, but when it becomes accessible history that changes later selection.

A recursive system becomes complex not merely because it contains two variables, but when its generator earns an elliptic conjugate mode.

A movement becomes history not merely because a state changed, but when a gate admits its consequence into a ledger while preserving what remains unresolved.

A world becomes time-bearing not merely because its events are ordered, but because its declared and residual past alters what it can become next.

The complete thesis is:

> **A protocol-bound operational world forms when traces alter future policy, commitments alter future admissibility, and inherited ledger–residual history recompiles the Base from which subsequent Relation begins. Where that Relation closes elliptically, complex numbers provide the natural local grammar joining an admitted readout to its conjugate response.**

In one final sequence:

Trace
→ Filtration
→ Policy
→ Base
→ Relation
→ Readout
→ Conjugate Exposure
→ Movement
→ Gate
→ Ledger + Residual
→ Recompiled Base
→ Time-Bearing World. (30.21)

Appendix A — Notation, Logical Types, and Units

The framework depends on keeping unlike mathematical objects distinct. A state, observation, policy, operator, event, ledger, and residual may all participate in one causal runtime, but they do not thereby become coordinates of one ordinary vector.

This appendix fixes the principal notation used in the article and records each object's intended logical type.

The notation follows the article's mandatory architecture:

Trace
→ Filtration
→ Policy
→ Changed Transition
→ Gate
→ Ledger + Residual
→ Historical Return. (A.1)

The optional mathematical modules are:

Detailed Base
→ Effective Base Compiler
→ Local Generator
→ Signed-Conjugate Mode
→ Conjugate Readout. (A.2)

The Handoff specifies this typed separation as one of the framework's principal contributions and directs that the full notation, domains, units, and logical types be preserved in the appendices.

A.1 General notation conventions

Scalars

Ordinary italic symbols denote scalars when no ambiguity arises:

$t, k, \theta, \xi, \alpha, \beta, \gamma, \tau.$ (A.3)

Vectors and states

Uppercase or bold conceptual symbols denote states or state-like objects:

$X, \Xi, Z.$ (A.4)

The typography does not by itself determine the mathematical space. The domain must be declared.

Operators and maps

Script or uppercase symbols denote maps and operators:

$\mathcal{G}_p, \mathcal{A}_p, \mathcal{B}_p, C_p, J, A_r.$ (A.5)

Histories

The notation:

$y_{1:k}$ (A.6)

means:

$(y_1, y_2, \ldots, y_k)$. (A.7)

Protocol indexing

A subscript `P` indicates that the object is defined relative to protocol `P`:

$h_P, \mathcal{G}_P, \mathfrak{R}_P, \Gamma_P$. (A.8)

An object without a protocol subscript should not automatically be interpreted as protocol-independent.

Observer indexing

Observer-relative quantities carry the index `a`:

$\mathscr{F}_{a,k}, \pi_{a,k}, s_{a,k}$. (A.9)

Regime indexing

A local generator or mode carries the regime label `r`:

$A_r, J_r, C\chi_{,r}$. (A.10)

Epistemic-status labels

The final article uses:

- `[D]` — definition introduced by the framework;
- `[E]` — established inside a declared source construction;
- `[F]` — formal consequence of stated assumptions;
- `[H]` — testable hypothesis;
- `[A]` — structural analogy.

The label `[E]` should replace the earlier temporary label `[S]` in final editing.

A.2 Protocol objects

Symbol	Name	Logical type	Domain/codomain	Units
`P`	declared protocol	structured tuple	protocol space	none as a whole
`B`	boundary rule	classifier/set specification	environment → inside/outside	none

Symbol	Name	Logical type	Domain/codomain	Units
Δ	observation or aggregation scale	parameter/rule	positive time or event scale	time or event count
h_p	observation map	function/operator	$x_p \to y_p$	output-dependent
u_k	intervention	control variable	$\mathcal{U}_p$	domain-dependent
$\mathcal{G}_p$	commitment gate	operator/kernel	candidate × state × ledger → event/status	none or probability
$\mathcal{A}_p$	record-access rule	access operator	observer × ledger → accessible ledger	none
$\mathfrak{R}_p$	residual-preservation rule	update operator	candidate × event × ledger → residual	residual-dependent

The expanded protocol is:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \mathfrak{R})$. (A.11)

The tuple need not be implemented literally in every domain. Its functions must nevertheless be recoverable.

A.3 State and observation spaces

Symbol	Name	Logical type
$\mathcal{X}_p$	detailed causal-state space	state space
$\mathcal{Y}_p$	admitted-trace space	observation space
$\mathcal{U}_p$	admissible-intervention space	action/control space
$\mathcal{C}_p$	candidate-consequence space	event-candidate space
$\mathcal{E}_p$	admitted-event space	committed-event space
$\mathfrak{B}_p$	residual space	typed residual space
E	local invariant mode plane	real vector subspace

These spaces need not be Euclidean.

For example:

- a ledger may be an ordered graph;
- residual may be a typed set;
- policy may be a probability kernel;
- filtration is a sigma-algebra;
- an institutional event may be categorical rather than numeric.

A.4 Time and ordering variables

Symbol	Name	Type	Meaning
t	evolution time	continuous parameter	latent or fine-grained movement

Symbol	Name	Type	Meaning
k	commitment index	discrete index	observation, gate, or ledger tick
θ	elliptic phase	dimensionless Relation coordinate	position within a rotational mode
ξ	generalized Relation coordinate	continuous parameter	perturbation direction not necessarily angular
T	recurrence period	time parameter	cycle or breather period
τ	effective duration	derived state/parameter	persistence, recovery, dwell, or switching timescale

The article deliberately distinguishes:

Evolution time ≠ Phase ≠ Commitment order. (A.12)

In particular:

$t \neq \theta \neq k$. (A.13)

The PORE duration coordinate τ may be estimated from a trajectory window. It may be a rolling descriptor or response parameter rather than a primitive instantaneous state coordinate. The Handoff explicitly requires this distinction.

A.5 Detailed recursive state

Symbol	Name	Logical type	Typical contents
X_k	detailed causal state	state	system, observers, resources, policies, constraints, records
$s_{a,k}$	observer internal state	state component	memory, belief, instrument state, private signal
η_k	disturbance/noise	random variable/ process	unmodelled influence
r_k	regime label	discrete state	elliptic, hyperbolic, jump state, institutional regime

A schematic decomposition is:

$X_k = (x_k, \{s_{a,k}\}, L_k, \mathscr{R}_k, \text{PolicyState}_k, \text{ResourceState}_k, r_k, \ldots)$. (A.14)

This decomposition is illustrative.

The full causal state is not required to have one universal factorization.

A.6 Trace and filtration

Symbol	Name	Logical type	Definition
y_k	admitted trace	observation/record	$y_k = h_p(X_k)$
$y_{1:k}$	trace history	ordered sequence	$(y_1,\ldots,y_k)$
$\mathcal{F}_{a,k}$	observer filtration	sigma-algebra/information state	accessible history of observer a
$r_{a,k}$	observer record	stored trace state	internally retained outcome

The admitted trace is:

$y_k = h_p(X_k)$. (A.15)

The observer filtration is:

$\mathscr{F}_{a,k} = \sigma(y_{a,1:k}, \mathcal{A}_p(a, L_{1:k}), s_{a,1:k})$. (A.16)

A filtration is not an ordinary scalar state variable.

It represents the information available to an observer at a given commitment index.

A.7 Policy and transition objects

Symbol	Name	Logical type
$\pi_{a,k+1}$	observer policy	action, rule, or probability kernel
$f_{a,p}$	policy-selection map	filtration → policy
$\mathcal{B}_p$	recursive transition map	state-update operator
F_p	continuous-time vector field	generator/function
$\Pi_{a,p}$	stochastic policy kernel	conditional probability kernel

Policy selection is:

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (A.17)

The recursive transition is:

$X_{k+1} = \mathscr{B}_p(X_k, \{\pi_{a,k+1}\}, u_k, L_k, \Gamma_k, \eta_k)$. (A.18)

In continuous time:

$dX/dt = F_p(X, \pi, u, L, \Gamma, \eta)$. (A.19)

Policy is not automatically part of the state.

It may be compiled into the state for a Markov representation, but that is a modelling decision requiring declaration.

A.8 Effective Base notation

Symbol	Name	Logical type
W_k	compiler input window	trajectory/history object
C_p	Effective Base compiler	history → reduced state
Ξ_k	candidate Effective Base	reduced state/mesostate
ϱ_k	effective presence	scalar coordinate
γ_k	effective integrity	scalar coordinate
τ_k	effective duration	scalar coordinate or derived descriptor

The compiler is:

$\Xi_k = C_p(W_k[X, y, L, u])$. (A.20)

A PORE-like candidate is:

$\Xi_k = (\varrho_k, \gamma_k, \tau_k)$. (A.21)

Symbol convention for ϱ

The article uses Greek variant rho:

ϱ (A.22)

for PORE effective presence or occupancy.

The ordinary symbol:

ρ (A.23)

is reserved where possible for quantum density operators.

This prevents confusion between:

- ϱ — effective presence;
- ρ — density operator.

Interpretive range

The PORE coordinates may denote:

ϱ = presence, occupancy, participation, basin depth, or organized stock. (A.24)

γ = closure, integrity, confinement, binding, or leakage resistance. (A.25)

τ = persistence, recurrence, dwell, recovery, or switching timescale. (A.26)

These are role ranges, not unrestricted synonym lists. Each application must fix its operational meaning.

A.9 Relation and generator notation

Symbol	Name	Logical type
A_r	local reduced generator	linear operator/matrix
G_r	intervention-gain map	matrix/operator
ε_k	reduced-model error	residual random vector
$C\chi$	signed-conjugate operator	normalized local operator
χ	algebraic signature	scalar
J	elliptic quarter-turn	operator satisfying $J^2 = -I$
N	parabolic generator	nilpotent operator satisfying $N^2 = 0$
K	hyperbolic generator	involution satisfying $K^2 = I$

The local reduced evolution is:

$\delta\Xi_{k+1} \approx A_r\delta\Xi_k + G_r\delta u_k + \varepsilon_k$. (A.27)

The signed-conjugate classification is:

$C\chi^2 = \chi I$. (A.28)

The principal cases are:

$\chi < 0 \Rightarrow$ elliptic. (A.29)

$\chi = 0 \Rightarrow$ parabolic. (A.30)

$\chi > 0 \Rightarrow$ hyperbolic. (A.31)

After normalization:

$J^2 = -I$. (A.32)

$N^2 = 0$. (A.33)

$K^2 = I$. (A.34)

A.10 Readout and conjugate response

Symbol	Name	Logical type
`a`	linear readout covector	element of dual space `E*`
V_p	general nonlinear readout	scalar-valued function
`Y`	admitted scalar readout	scalar
$\mathcal{Q}_Y$	conjugate response of `Y`	scalar with readout-compatible units
Z_Y	completed conjugate readout	complex coordinate
Φ_a	state-to-readout map	linear map `E → ℝ²`
`J_R`	standard readout quarter-turn	`2 × 2` real matrix

For a linear readout:

$Y(x) = a(x)$. (A.35)

In an elliptic mode:

$\mathcal{Q}_\gamma(x) = -a(Jx)$. (A.36)

The completed readout is:

$Z_\gamma = Y + i\mathcal{Q}_\gamma$. (A.37)

The readout map is:

$\Phi_a(x) = [Y(x), \mathcal{Q}_\gamma(x)]^T$. (A.38)

The standard readout-plane operator is:

$J_R = [0\ -1;\ 1\ 0]$. (A.39)

The intertwining relation is:

$\Phi_a J = J_R \Phi_a$. (A.40)

For a nonlinear scalar readout:

$Y = V_p(\Xi)$. (A.41)

The directional conjugate response may be:

$\mathcal{Q}_\gamma = -\nabla_\Xi V_p \cdot \partial\Xi/\partial\theta$. (A.42)

The quantity $\mathcal{Q}_\gamma$ is a readout sensitivity.

It is not automatically:

- a hidden state;
- residual;
- loss;
- uncertainty;
- force;
- another asset.

A.11 Commitment objects

Symbol	Name	Logical type
c_k	candidate consequence	candidate event/object
e_k	admitted event	committed event
$\mathcal{G}_p$	gate	admission operator/kernel
T_p	ledger writer	event → record transformation
L_k	ledger	ordered authoritative record structure
$\oplus$	ledger update	append, reconcile, net, version, or branch operator
$\mathcal{R}_k$	raw residual	typed unresolved consequence
$\mathfrak{R}_p$	residual update	residual-preservation operator
Γ_k	residual-derived functional/state	optional compiled constraint
α_k	admission fraction	scalar in `[0,1]`
ε_gate	gate residual	admitted-versus-total difference

The gate operation is:

$e_k = \mathcal{G}_p(c_k; X_k, L_k)$. (A.43)

The ledger update is:

$L_{k+1} = L_k \oplus T_p(e_k)$. (A.44)

The residual update is:

$\mathscr{R}_{k+1} = \mathfrak{R}_p(c_k, e_k, L_{k+1})$. (A.45)

An optional residual compiler is:

$\Gamma_{k+1} = \Gamma_p[\mathscr{R}_{1:k+1}]$. (A.46)

Partial admission may be written:

c_k^admitted = $\alpha_k c_k$. (A.47)

The remaining difference is:

ε_gate = c_k − c_k^admitted. (A.48)

These scalar expressions are appropriate only when candidate and admitted consequences can be represented in compatible linear units.

A.12 Variational residual notation

Symbol	Name	Logical type
$L_p(x, \dot{x}, t)$	ordinary Lagrangian-like term	scalar density
$\Gamma_p[x]$	residual cost functional	path functional
λ	residual weighting	scalar parameter
$S_{eff,p}[x]$	effective action	scalar functional
$\delta\Gamma/\delta x$	functional derivative	covector/force-like term
F_Γ	residual-derived influence	vector/covector after metric identification

A conditional effective action is:

$S_{eff,p}[x] = \int L_p(x, \dot{x}, t)dt - \lambda\Gamma_p[x]$. (A.49)

A future-selective term may be:

$F_\Gamma = -\delta\Gamma_p/\delta x$. (A.50)

The identification of F_Γ as a physical force requires an additional metric and domain interpretation.

Raw residual, Γ, and its derivative remain distinct:

$\mathscr{R} \neq \Gamma \neq \delta\Gamma/\delta x$. (A.51)

The residual source treats Γ as a conditional interface, not as the universal definition of residual.

A.13 Quantum-observer notation

Symbol	Name	Logical type
ρ_k	quantum density operator	positive trace-one operator
$\mathcal{H}$	Hilbert space	complex vector space
$\boldsymbol{\varepsilon}_k$	between-tick evolution	completely positive trace-preserving map
$\boldsymbol{\mathcal{M}}_{\theta,\phi}$	quantum instrument element	completely positive map
θ_k	selected measurement setting	policy output
ϕ_k	observed outcome	discrete/continuous trace
K_k	transition kernel	conditional probability kernel

This notation belongs specifically to the quantum observer example.

It should not be generalized automatically to other domains.

The distinction:

ρ quantum density operator
versus
ϱ PORE effective presence (A.52)

must be preserved.

A.14 CAPM notation

Symbol	Name	Logical type	Units
CF_t	future cash flow	scalar monetary amount	currency
r_base	baseline discount rate	scalar rate	dimensionless per period
r_CAPM	CAPM required return	scalar rate	dimensionless per period
β	CAPM beta	scalar sensitivity	dimensionless
ERP	equity risk premium	scalar rate	dimensionless per period
A_t	baseline valuation amplitude	scalar	currency
R_t	admitted CAPM value	scalar	currency
Q_t	conjugate phase exposure	scalar	currency
θ_t	valuation phase	angle	dimensionless
Z_t	completed valuation state	complex coordinate	currency
H(θ)	CAPM haircut A − R	scalar	currency
Mϕ	rotated readout	measurement operator	currency output

The core construction is:

$r_CAPM = r_base + \beta ERP$. (A.53)

$A_t = CF_t/(1 + r_base)^t$. (A.54)

$R_t = CF_t/(1 + r_CAPM)^t$. (A.55)

$\cos \theta_t = R_t/A_t$. (A.56)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (A.57)

$Z_t = R_t + iQ_t = A_t e^{(i\theta_t)}$. (A.58)

The exact conjugate result is:

$\partial R_t/\partial \theta_t = -Q_t$. (A.59)

The source emphasizes that exposure becomes P&L only after actual phase movement, and financial history requires a recognition or settlement gate.

A.15 Unit rules

Rule A1 — State coordinates may have different units

A general state vector may contain heterogeneous coordinates.

A metric or scaling rule is required before Euclidean norms are interpreted physically.

Rule A2 — Conjugate readout coordinates require compatible units

For:

$Z_\gamma = Y + i\mathcal{Q}_\gamma$, (A.60)

Y and $\mathcal{Q}_\gamma$ must have compatible units or be converted through a declared metric.

When:

$\mathcal{Q}_\gamma = -\partial Y/\partial\theta$, (A.61)

and θ is dimensionless:

$[\mathcal{Q}_\gamma] = [Y]$. (A.62)

Rule A3 — Do not add typed residuals as scalars without a compiler

The expression:

$L + \mathscr{R}$ (A.63)

is generally symbolic unless a typed join or scalarization rule is declared.

Rule A4 — Operator equations require declared spaces

The identity:

$J^2 = -I$ (A.64)

is meaningful only after specifying:

$J: E \to E$. (A.65)

Rule A5 — Probabilities, values, costs, and amplitudes are not interchangeable

A numerical range of `[0,1]` does not make two quantities the same type.

A.16 Core non-equivalence register

The following distinctions should be preserved throughout implementation and editing:

State ≠ Trace. (A.66)

Trace ≠ Ledger. (A.67)

Ledger ≠ Truth. (A.68)

Filtration ≠ State coordinate. (A.69)

Policy ≠ Generator. (A.70)

Compiler ≠ Generator. (A.71)

Effective Base ≠ Full causal state. (A.72)

Phase ≠ Commitment time. (A.73)

Dynamic phase ≠ Valuation phase. (A.74)

Conjugate response ≠ Residual. (A.75)

Residual ≠ Γ. (A.76)

Γ ≠ Friction universally. (A.77)

Measurement ≠ Movement. (A.78)

Movement ≠ Candidate admission. (A.79)

Admission ≠ Exhaustion. (A.80)

Internal certainty ≠ Public objectivity. (A.81)

Half-turn opposition ≠ Quarter-turn conjugacy. (A.82)

Self-reference ≠ Complex structure. (A.83)

Complex structure ≠ Quantum ontology. (A.84)

These non-equivalences are part of the framework's formal content.

Appendix B — Protocol Declaration and World Instantiation Templates

B.1 Purpose

This appendix converts the framework into reusable declaration templates.

No application should begin by assigning a complex phase, PORE coordinate, gate residual, or Γ functional before its protocol has been fixed.

The required order is:

Protocol
→ Trace
→ Recursive path
→ Base
→ Generator
→ Optional geometry
→ Gate
→ Ledger + Residual
→ Historical return. (B.1)

The protocol-first rule prevents the meaning of a quantity from changing silently as the analysis proceeds. PORE similarly defines its effective coordinates only under a declared boundary, observation rule, window, and intervention family, and explicitly rejects unrestricted validity beyond that protocol.

B.2 Master Protocol Card

Protocol identity

Protocol name:
Protocol ID:
Version:
Date created:
Responsible analyst or institution:
Domain:
Primary task:
Epistemic status: `[E]`, `[F]`, `[H]`, or `[A]`
Current maturity: `M0–M5`

B.2.1 Boundary `B`

System included:

- Which entities, processes, states, and observers belong inside the model?
- What spatial, institutional, semantic, or legal boundary is used?
- What scale is being studied?
- Which nested systems are included?

Environment excluded:

- Which processes are treated as external?
- Which external influences are modelled as intervention?
- Which are treated as noise?
- Which are ignored?

Boundary leakage:

- What flow across the boundary is permitted?
- What level of leakage invalidates the model?
- How will boundary failure be detected?

Write:

B_p: Environment → {Inside, Outside, Interface}. (B.2)

B.2.2 Observation and aggregation rule `Δ`

Observation interval:
Aggregation window:
Sampling frequency:
Commitment frequency:

Natural clock:

- physical time;
- transaction time;
- event time;
- token time;
- legal-procedural time;
- episode time;
- ledger tick.

Declare:

Δ = declared observation and aggregation rule. (B.3)

Questions:

1. What is measured?
2. What is averaged?
3. What is censored?
4. What is treated as simultaneous?
5. What event ordering is preserved?
6. What resolution is lost?

B.2.3 Observation map h_p

Define:

$h_p: \mathcal{X}_p \to \mathcal{Y}_p$. (B.4)

Observed variables:
Units:
Instrument:
Uncertainty:
Thresholding:
Missing-data rule:
Observer dependence:
Calibration procedure:

The trace is:

$y_k = h_p(X_k)$. (B.5)

Questions:

- Is h_p deterministic or stochastic?
- Is it many-to-one?

- Can the same state produce different observer traces?
- Can different states produce the same trace?
- Which parts of the state are systematically omitted?

B.2.4 Intervention family $\mathcal{U}_p$

Define:

$u_k \in \mathcal{U}_p$. (B.6)

Admissible interventions:
Prohibited interventions:
Authority to intervene:
Maximum intervention size:
Safety limits:
Expected response timescale:
Recovery period:
Null or placebo intervention:

Questions:

- Which variables are manipulable?
- Which are only observed?
- Can the intervention alter the protocol itself?
- Does observation cause material backreaction?
- What distinguishes probe from control?

B.2.5 Gate $\mathcal{G}_p$

Define:

$e_k = \mathcal{G}_p(c_k; X_k, L_k)$. (B.7)

Candidate field:
Gate threshold:
Authority:
Evidence required:
Possible outputs:

- admit;
- reject;
- defer;
- partially admit;
- transform;
- escalate.

Timing rule:
Appeal or reopening mechanism:
Gate versioning:
Conflict-of-interest controls:

Questions:

- What makes a candidate consequence operative?
- Is gate passage binary or graded?
- Does previous ledger history alter the threshold?
- Can different authorities produce different admissions?
- What residual is generated by rejection?

B.2.6 Record-access rule $\mathcal{A}_p$

Define:

$L_k^{accessible}(a) = \mathcal{A}_p(a, L_k)$. (B.8)

Observers or roles:
Public records:
Private records:
Sealed records:
Access delay:
Access revocation:
Frame translation:
Audit access:

Questions:

- Which record actually enters each observer's filtration?
- Does the record exist but remain inaccessible?
- Are observers acting on different versions?
- Can observers verify provenance independently?

B.2.7 Residual rule $\Re_p$

Define:

$\mathcal{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (B.9)

Residual categories:

- excluded alternative;
- unresolved evidence;
- unrecognized cost;

- branch difference;
- deferred obligation;
- inaccessible trace;
- model error;
- gate mismatch;
- externalized burden.

Residual owner:
Residual storage:
Residual decay or persistence:
Reopening trigger:
Audit rule:
Scalarization rule, if any:

Questions:

- What remains after commitment?
- Who carries it?
- Can it affect future policy?
- Can it be hidden by the same gate that produced it?
- Under what condition may it be compiled into Γ?

B.3 Trace Card

Trace name:
Trace ID:
Source event:
Observation map:
Observer:
Timestamp or commitment index:
Units/type:
Uncertainty:
Persistence:
Access class:
Provenance:
Revision status:
Deletion rule:

A trace qualifies as recursively active only if:

Trace persistence

- later accessibility
- policy relevance
- causal backreaction. (B.10)

Test:

Pr(Future | trace accessible)
≠ Pr(Future | trace ablated). (B.11)

If the difference is absent, the trace may be archival or causally irrelevant for the declared task.

B.4 Observer Card

Observer ID:
Observer boundary:
Internal state:
Accessible traces:
Accessible ledger:
Private signals:
Policy class:
Instrument choices:
Authority:
Frame:
Compatibility maps:
Latching rule:
Revision rule:

The observer filtration is:

$\mathscr{F}_{a,k} = \sigma(y_{a,1:k}, \mathcal{A}_p(a, L_{1:k}), s_{a,1:k})$. (B.12)

The policy is:

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (B.13)

Required causal question:

Does changing the accessible history change $\pi_{a,k+1}$? (B.14)

Required backreaction question:

Does changing $\pi_{a,k+1}$ change the subsequent state or trace distribution? (B.15)

Only when both answers are positive is the observer part of the active self-referential loop.

B.5 Effective Base Card

Compiler name:
Compiler version:
Input history window:
Input variables:
Missing-data rule:
Normalization:
Regime labels:
Output coordinates:
Uncertainty estimate:
Task class:
Predictive-closure threshold:

Intervention-validation rule:

Define:

$\Xi_k = C_p(W_k)$. (B.16)

For a PORE-like signature:

$\Xi_k = (\varrho_k, \gamma_k, \tau_k)$. (B.17)

Coordinate declaration

For each coordinate specify:

Field	ϱ	γ	τ
operational definition			
raw proxy			
units			
aggregation			
uncertainty			
intervention sensitivity			
failure threshold			

State-status declaration

Classify Ξ as:

- descriptive signature;
- rolling descriptor;
- mesostate;
- predictive state;
- intervention-sufficient state.

Do not call Ξ a complete state merely because it is represented as a vector.

B.6 Relation Card

Regime ID:
State representation:
Estimation window:
Generator:
Intervention map:
Residual diagnostics:
Spectrum:
Invariant modes:
Damping:
Nonlinearity:
Jump condition:
Validity region:

The local model is:

$\delta \Xi_{k+1} \approx A_r \delta \Xi_k + G_r \delta u_k + \varepsilon_k$. (B.18)

Algebraic classification

Record:

- elliptic;
- parabolic;
- hyperbolic;
- dissipative elliptic;
- mixed;
- jump-like;
- no stable reduction.

For an elliptic claim, provide evidence for:

$J^2 \approx -I$. (B.19)

For a parabolic claim:

$N^2 \approx 0$. (B.20)

For a hyperbolic claim:

$K^2 \approx I$. (B.21)

Rejection rule

Ordinary complex phase must be rejected when:

- no persistent two-dimensional mode exists;
- the discriminant is nonnegative;
- phase is preprocessing-dependent;
- the readout does not detect the mode;
- a simpler real model performs better.

B.7 Conjugate Readout Card

Readout name:
Readout protocol:
Scalar output γ:
Units:
Active mode:
Orientation:

Relation coordinate:
Conjugate response:
Perturbation method:
Tolerance:
Rank test:
Alternative real representation:

Define:

$Y(x) = a(x)$. (B.22)

For an elliptic mode:

$\mathcal{Q}_\gamma(x) = -a(Jx)$. (B.23)

Or more generally:

$\mathcal{Q}_\gamma = -\partial Y/\partial \xi$. (B.24)

The completed readout is:

$Z_\gamma = Y + i\mathcal{Q}_\gamma$. (B.25)

Required checks

1. Are Y and $\mathcal{Q}_\nu$ unit-compatible?
2. Is orientation declared?
3. Is the derivative exact or empirically estimable?
4. Does the readout detect the active mode?
5. Does $\mathcal{Q}_\nu$ improve prediction or control?
6. Does the complex representation outperform a simple real pair?

B.8 Commitment Card

Candidate consequence:
Source movement:
Gate:
Authority:
Admitted event:
Ledger:
Residual:
Partial-admission fraction:
Access:
Reopening rule:
Downstream consequences:

The sequence is:

Movement
→ candidate c_k
→ gate $\mathcal{G}_\rho$

→ event e_k
→ ledger L_{k+1}

- residual $\mathcal{R}_{k+1}$. (B.26)

Commitment-level classification

Mark all that apply:

- internal commitment;
- institutional commitment;
- redundant public commitment.

Commitment test

A record is operationally committed only when it has:

Addressability + Authority + Consequence. (B.27)

B.9 Historical Return Card

Ledger variable:
Residual variable:
Return delay:
Affected observer:
Affected policy:
Affected gate:
Affected generator:
Affected action set:
Affected protocol:
Predicted effect:
Ablation method:

Test:

$\Pr(\text{Future} \mid \Xi_k, L_k, \mathscr{R}_k)$
versus
$\Pr(\text{Future} \mid \Xi_k)$. (B.28)

A record is archival when:

$\Pr(\text{Future} \mid \Xi_k, L_k)$
$\approx \Pr(\text{Future} \mid \Xi_k)$. (B.29)

A residual is future-active when:

$\Pr(\text{Future} \mid \Xi_k, L_k, \mathscr{R}_k)$
$\neq \Pr(\text{Future} \mid \Xi_k, L_k)$. (B.30)

B.10 World Instantiation Card

Identification

World name:
Domain:
Boundary:
Protocol:
Timebase:
Primary observer class:
Primary task:
Maturity claim:

Mandatory core

Question	Declaration
What produces traces?	
Who can access them?	
What policy changes?	
How does policy alter the future?	
What candidates are generated?	
What acts as gate?	
What enters the ledger?	
What remains residual?	
How does history return?	

Optional modules

Module	Active?	Evidence
Effective Base compiler		
PORE-like signature		
low-dimensional generator		
elliptic mode		
parabolic mode		
hyperbolic mode		
conjugate readout		
Γ functional		
local transport		
capacity/slot model		

Status decision

The domain should be classified as:

- analogy only;
- typed mapping;
- operational correspondence;
- formal module;
- integrated operational-world model;
- mature replicated theory.

B.11 Minimal AI Runtime Card

Boundary

One governed AI task episode, including:

- user input;
- retrieved sources;
- model calls;
- tools;
- verifier;
- memory writes;
- external actions.

Trace

- generated claims;
- tool outputs;
- verifier findings;
- user corrections;
- committed artifacts.

Filtration

Context + retrieved memory + tool history + current policy. (B.31)

Policy

- retrieve;
- reason;
- verify;
- rewrite;
- abstain;
- escalate;
- execute tool.

Gate

- source-grounding check;
- safety approval;

- user confirmation;
- irreversible-action permission;
- schema validation.

Ledger

- conversation;
- audit log;
- memory;
- artifact store;
- external system changes.

Residual

- unresolved contradiction;
- missing evidence;
- rejected draft;
- unverified inference;
- failed tool result;
- uncertain user intent.

Historical return

Committed outputs and residual alter later context, retrieval, permissions, or verifier behaviour.

The AI becomes a stronger operational world only when these records are causally consulted rather than merely stored.

B.12 Minimal Finance Card

Boundary

Declared asset, portfolio, legal entity, collateral network, or market segment.

Baseline and admitted value

$A_t = CF_t/(1 + r_base)^t$. (B.32)

$R_t = CF_t/(1 + r_CAPM)^t$. (B.33)

Conjugate completion

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (B.34)

$Z_t = R_t + iQ_t$. (B.35)

Movement

$\Delta R_econ = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (B.36)

Gate

- close;
- exercise;
- margin;
- impairment;
- settlement;
- collateral recognition;
- accounting rule.

Ledger

- trade ledger;
- accounting ledger;
- collateral ledger;
- regulatory ledger.

Residual

- unrecognized movement;
- basis risk;
- unsettled claim;
- model disagreement;
- liquidity pressure.

The source’s central distinction must remain:

Q = exposure before movement. (B.37)

Residual = incompletely integrated consequence after gate. (B.38)

This distinction is explicitly preserved in the Handoff synthesis.

B.13 Minimal Legal Card

Boundary

Declared jurisdiction, tribunal, claim class, and procedural stage.

Candidate

- harm;
- evidence;
- argument;
- remedy;
- appeal.

Gate

- standing;
- jurisdiction;
- admissibility;
- burden;
- procedure;
- judgment.

Ledger

- order;
- judgment;
- precedent;
- settlement;
- statutory classification.

Residual

- excluded evidence;
- unresolved harm;
- dissent;
- unrecognized claim;
- procedural remainder.

Return

Prior ledger changes:

- future argument;

- interpretation;
- admissibility;
- legal strategy;
- institutional authority.

Law demonstrates that strong Commitment can exist without an established ordinary complex mode.

B.14 Minimal Scientific Card

Boundary

Research programme, experimental protocol, discipline, or database.

Trace

- measurement;
- dataset;
- model result;
- replication;
- published claim.

Gate

- methodological standard;
- statistical threshold;
- review;
- replication;
- acceptance.

Ledger

- publication;
- database;
- standard;
- textbook;
- accepted parameter.

Residual

- anomaly;
- negative result;
- failed replication;
- excluded data;
- model dependence.

Return

Accepted and residual results change:

- later experiments;
- funding;
- interpretation;
- admissible hypotheses;
- instrument design.

B.15 Protocol-change Card

A strong self-referential world may revise its own protocol.

Define:

$$P_{k+1} = \mathrm{UpdateProtocol}(P_k \mid L_k, \mathscr{R}_k, \mathrm{Audit}_k). \quad \text{(B.39)}$$

Record:

Old protocol:
Trigger:
Ledger evidence:
Residual evidence:
Authority:
New boundary:
New observation rule:
New gate:
New access rule:
Expected effect:
Migration rule:
Backward compatibility:
Revalidation requirement:

A protocol revision creates a new model version.

Results under P_k should not be silently pooled with results under P_{k+1}.

B.16 Publication checklist

Before presenting a domain as an operational instantiation, publish:

1. Protocol Card;
2. Trace Card;
3. Observer Card;
4. Base compiler;
5. generator or transition kernel;
6. algebraic classification;
7. readout and units;
8. gate rule;
9. ledger schema;
10. residual schema;
11. historical-return pathway;
12. falsification plan;
13. maturity rating;
14. replay data or simulation;
15. protocol and code versions.

A compact framework claim without these artifacts should remain at the level of a typed hypothesis or analogy.

B.17 Final protocol principle

The framework's protocol discipline can be summarized as:

No declared protocol
⇒ no stable operational object. (B.40)

No accessible trace
⇒ no active internal recursion. (B.41)

No causal policy return
⇒ no demonstrated self-reference. (B.42)

No validated generator
⇒ no earned Relation algebra. (B.43)

No elliptic mode
⇒ no ordinary complex completion. (B.44)

No operational gate
⇒ movement without Commitment. (B.45)

No ledger or residual return

⇒ record without a time-bearing world. (B.46)

Appendix C — Predictive Sufficiency, Memory, and Effective-State Tests

C.1 Purpose

The framework distinguishes the **detailed causal Base**:

X_k (C.1)

from an optional **Effective Base**:

$\Xi_k = C_p(W_k)$. (C.2)

The compiler C_p compresses a trajectory, trace history, ledger, or intervention window W_k into a smaller operational representation.

A PORE-like candidate is:

$\Xi_k = (\varrho_k, \gamma_k, \tau_k)$. (C.3)

The presence of three coordinates does not make ξ_k a sufficient state.

Quantities such as recurrence time, recovery time, and switching time are commonly estimated over windows. The PORE coordinates may therefore begin as rolling descriptors or an **Effective Existence Signature** rather than as an instantaneous Markov state. Predictive closure must be demonstrated empirically.

This appendix specifies how to test that claim.

C.2 Five statuses of a compressed representation

A compiled representation should be classified according to the strongest role supported by evidence.

C.2.1 Descriptive summary

ξ_k summarizes a data window.

It may help visualization but makes no predictive claim.

Example:

$\Xi_k = \text{Summary}(W_k)$. (C.4)

C.2.2 Diagnostic signature

ξ_k distinguishes regimes or failure patterns.

Example:

$\text{Regime}_k = \text{Diagnose}(\Xi_k)$. (C.5)

A diagnostic signature may identify present conditions without closing future evolution.

C.2.3 Rolling mesostate

Ξ_k changes smoothly enough to support local transition analysis:

$\Xi_{k+1} \approx F_r(\Xi_k, u_k)$. (C.6)

It may remain dependent on the compiler window and regime label.

C.2.4 Predictive Effective Base

Ξ_k approximately screens off omitted accessible history for a declared forecasting task:

$\Pr(Y_{k+1} \mid \Xi_k, H_k, u_k, r_k)$
$\approx \Pr(Y_{k+1} \mid \Xi_k, u_k, r_k)$. (C.7)

C.2.5 Intervention-sufficient Effective Base

Ξ_k predicts responses to admissible interventions:

$\Pr(Y_{k+1} \mid do(u_k), \Xi_k, r_k)$. (C.8)

This is stronger than observational prediction.

A representation may forecast well because it exploits correlations while failing under intervention.

C.3 Task-relative sufficiency

Sufficiency must be declared relative to:

- protocol P;
- task class $\mathcal{T}$;
- prediction horizon h;
- intervention class $\mathcal{U}$;
- regime r;
- tolerance ε.

Write:

$\text{Sufficiency}(\Xi \mid P, \mathcal{T}, h, \mathcal{U}, r, \varepsilon)$. (C.9)

The same Ξ may be sufficient for:

- one-step regime prediction;

but insufficient for:

- long-horizon commitment hazard;
- policy intervention;
- protocol revision;
- rare-event transition.

Therefore:

Effective Base is a relation among representation, task, and protocol. (C.10)

It is not an intrinsic title permanently attached to a vector.

C.4 Predictive closure test

Let H_k denote omitted history.

Compare two models.

Reduced model

M_0:

$\Pr(Y_{k+1} \mid \Xi_k, u_k, r_k)$. (C.11)

History-augmented model

M_1:

$\Pr(Y_{k+1} \mid \Xi_k, H_k, u_k, r_k)$. (C.12)

Define predictive gain:

$\Delta\text{Score_H} = \text{Score}(M_1) - \text{Score}(M_0)$. (C.13)

The score may be:

- log likelihood;
- Brier score;
- mean squared error;
- calibration loss;
- classification accuracy;
- intervention-response error.

Acceptance condition

Ξ is approximately predictively closed when:

$\Delta\text{Score_H} \leq \varepsilon_\text{score}$ (C.14)

across held-out periods, observers, and admissible protocol variations.

Rejection condition

If:

$\Delta \text{Score_H} > \varepsilon\text{_score}$ (C.15)

persistently, then the compiler omits future-relevant information.

The Handoff gives the corresponding closure target:

$\Pr(\Xi_{k+1} \mid \text{history}, u_k)$
$\approx \Pr(\Xi_{k+1} \mid \Xi_k, r_k, u_k)$.

C.5 Conditional-information test

A more direct criterion uses conditional mutual information:

$I(H_k; Y_{k+1} \mid \Xi_k, u_k, r_k)$. (C.16)

Approximate predictive sufficiency requires:

$I(H_k; Y_{k+1} \mid \Xi_k, u_k, r_k) \approx 0$. (C.17)

For multi-step prediction:

$I(H_k; Y_{k+1:k+h} \mid \Xi_k, u_{k:k+h-1}, r_k) \approx 0$. (C.18)

This test should be interpreted carefully.

An estimated value near zero may result from:

- insufficient data;
- weak estimator power;
- excessive conditioning dimension;
- regime mixing;
- measurement noise.

Failure to detect conditional information is not proof of exact Markov closure.

C.6 Minimum-memory test

Suppose a memory length m is introduced:

$\Xi_k^{(m)} = C_p(W_k^{(m)})$. (C.19)

where:

$W_k^{(m)} = \{X_{k-m+1:k}, y_{k-m+1:k}, L_{k-m+1:k}\}$. (C.20)

Estimate performance for:

m = 1, 2, ..., m_max. (C.21)

Define the minimum adequate memory:

m* = min{m : Score(m + 1) − Score(m) ≤ ε_m}. (C.22)

Interpretation:

- small m^* supports short-memory closure;
- large m^* suggests substantial historical dependence;
- unstable m^* suggests regime switching;
- no plateau suggests that the proposed compiler is inadequate.

C.7 Ledger-memory separation

Omitted history should be separated by type.

Let:

$H_k = (H_k^{state}, H_k^{trace}, H_k^{policy}, H_k^{ledger}, H_k^{residual})$. (C.23)

Test each component separately:

$I(H_k^{ledger}; Future \mid \Xi_k, other\ history)$ (C.24)

and:

$I(H_k^{residual}; Future \mid \Xi_k, L_k)$. (C.25)

This distinguishes several possibilities.

Possibility A — Complete recompilation

The Effective Base already contains the relevant consequences of ledger and residual.

Possibility B — Explicit historical dependence

Ledger and residual remain independently future-relevant.

Possibility C — Compiler contamination

The compiler indirectly leaks future or post-event information into Ξ.

Possibility D — Observer-specific dependence

History matters only because certain observers can access it.

These cases imply different models.

C.8 Branch-provenance test

Two trajectories may reach similar present coordinates through different histories:

$\Xi_k^A \approx \Xi_k^B$. (C.26)

Let the branch labels be:

$b_k^A \neq b_k^B$. (C.27)

Test:

$\Pr(Y_{k+1} \mid \Xi_k, b_k^A)$
versus
$\Pr(Y_{k+1} \mid \Xi_k, b_k^B)$. (C.28)

If the distributions differ materially, the current Base requires:

- branch label;
- ledger summary;
- residual summary;
- or a richer non-Markov representation.

This test is especially important for:

- hysteresis;
- legal precedent;
- financial leverage paths;
- organizational trust;
- AI memory contamination;
- systems with irreversible commitment.

C.9 Observer-state sufficiency test

Let observer states be:

$s_{a,k}$. (C.29)

A proposed Base that excludes observer state claims:

$\Pr(\text{Future} \mid \Xi_k, \{s_{a,k}\})$
$\approx \Pr(\text{Future} \mid \Xi_k)$. (C.30)

Reject this claim if observer states materially improve prediction.

A weaker compiler may instead include an aggregate observer coordinate:

$\Xi_k' = (\Xi_k, O_k)$. (C.31)

where o_k may represent:

- policy distribution;
- attention concentration;
- instrument-selection state;
- belief dispersion;
- authority configuration.

The representation should remain as small as possible while passing the task-specific tests.

C.10 Access-conditioned sufficiency

A ledger can exist without being accessible to every observer.

Define:

$L_k^{(a)} = \mathcal{A}_p(a, L_k)$. (C.32)

The correct observer-level prediction is:

$\Pr(Y_{k+1} \mid \Xi_k, L_k^{(a)}, s_{a,k})$. (C.33)

Using the full global ledger may create false predictive closure if observer a could not access it at decision time.

Therefore all forecasting and causal tests must obey:

Information_used_at_k ⊆ Information_accessible_at_k. (C.34)

Violating this rule produces look-ahead or omniscient-observer bias.

C.11 Intervention closure

A state representation may predict passive evolution while failing under action.

Let the observational transition be:

$\Pr(\Xi_{k+1} \mid \Xi_k, u_k)$. (C.35)

The causal intervention target is:

$\Pr(\Xi_{k+1} \mid do(u_k), \Xi_k)$. (C.36)

An intervention-sufficient Base should preserve:

- response direction;
- gain;
- delay;
- recovery;

- regime-switch hazard.

Define gain error:

$\varepsilon_G = \|G_{observed} - G_{predicted}(\Xi_k)\|$. (C.37)

Define recovery error:

$\varepsilon_\tau = |\tau_{observed} - \tau_{predicted}(\Xi_k)|$. (C.38)

A PORE-like compiler becomes operationally stronger when its coordinates predict controlled perturbation responses, rather than merely summarizing past trajectories. The Handoff's decision sequence requires revising proxies, regimes, or memory whenever Ξ fails predictive sufficiency.

C.12 Probe-versus-intervention distinction

A probe is intended primarily to reveal system response.

But a sufficiently large probe changes the system.

Let probe magnitude be:

$\|u_{probe}\|$. (C.39)

Define an admissible passive range:

$\|u_{probe}\| \le u_{passive}$. (C.40)

If:

$\|u_{probe}\| > u_{passive}$, (C.41)

measurement must be treated as intervention.

The protocol should report:

- probe size;
- observed backreaction;
- recovery;
- whether the state remained in the same regime.

This is especially important when estimating γ or τ, because the act of measurement may alter closure or persistence.

C.13 Window-stability test

Let the compiler use window length w.

Estimate:

$\Xi_k^{(w_1)}, \Xi_k^{(w_2)}, \ldots$ (C.42)

for admissible window choices.

Define normalized compiler variation:

$$V_C(w_i, w_j) = \|\Xi_k^{(w_i)} - \mathrm{Align}(\Xi_k^{(w_j)})\|/\mathrm{Scale}. \quad (C.43)$$

A robust signature should satisfy:

$$V_C(w_i, w_j) \le \varepsilon_C \quad (C.44)$$

inside the declared regime.

If the coordinates change radically under small window adjustments, they may be artefacts of the compiler rather than stable operational variables.

C.14 Proxy-stability test

Suppose coordinate ϱ can be estimated using several proxies:

$$\varrho^{(1)}, \varrho^{(2)}, \ldots, \varrho^{(m)}. \quad (C.45)$$

Proxy stability requires:

$$\mathrm{Corr_or_Agreement}(\varrho^{(i)}, \varrho^{(j)}) \ge \text{threshold} \quad (C.46)$$

or predictable transformations among them.

Disagreement may mean:

- proxies measure different roles;
- one proxy is regime-specific;
- the coordinate lacks operational coherence;
- the protocol boundary is too broad.

The correct response is not to average incompatible proxies merely to preserve a three-coordinate dashboard.

C.15 Regime-conditioned closure

A compiler may be valid only within regime r.

Write:

$$\Xi_k = C_{p,r}(W_k). \quad (C.47)$$

Closure is tested as:

$$\Pr(\mathrm{Future} \mid \Xi_k, H_k, r_k = r) \approx \Pr(\mathrm{Future} \mid \Xi_k, r_k = r). \quad (C.48)$$

If the same compiler is used across multiple regimes:

$\Pr(\text{Future} \mid \Xi_k, r_k)$ (C.49)

should be compared with:

$\Pr(\text{Future} \mid \Xi_k)$. (C.50)

A strong gain from the regime label means that Ξ alone does not close the system.

Possible remedies:

1. include r_k in the Base;
2. construct regime-specific compilers;
3. model switching explicitly;
4. retain a higher-dimensional state.

C.16 Jump-window exclusion

A smooth-state compiler should not be evaluated by pooling smooth and jump windows without declaration.

Let:

$J_k \in \{0,1\}$ (C.51)

indicate a jump or switch.

Estimate closure separately:

$\Pr(\text{Future} \mid \Xi_k, J_k = 0)$ (C.52)

and:

$\Pr(\text{Future} \mid \Xi_k, J_k = 1)$. (C.53)

The PORE architecture treats smooth responses and switches as different operational regimes rather than forcing one continuous description across both.

A jump payload should record:

- pre-jump state;
- trigger;
- post-jump state;
- protocol version;
- ledger event;
- residual;
- recovery path.

C.17 Reconstruction test

A compressed Base need not reconstruct the full causal state.

But it may be useful to test which state features remain recoverable.

Let decoder:

D_p: $\Xi_k \rightarrow \hat{X}_k$. (C.54)

Define reconstruction error:

$\varepsilon_X = \mathrm{Dist}(X_k, \hat{X}_k)$. (C.55)

A high reconstruction error does not automatically invalidate predictive closure.

It shows that Ξ is task-specific rather than a complete ontological description.

This distinction is valuable:

Predictive sufficiency ≠ Microscopic reconstruction. (C.56)

C.18 Counterfactual consistency test

Suppose the Base predicts response to two interventions:

u_A, u_B. (C.57)

Counterfactual consistency requires that:

PredictedResponse(Ξ_k, u_A) (C.58)

and:

PredictedResponse(Ξ_k, u_B) (C.59)

agree with controlled or quasi-experimental observations.

A compiler that predicts passive data but gives contradictory counterfactual responses should not be used for control.

C.19 Representation-comparison ladder

Compare candidate representations:

Model 0 — Trace only

State0 = y_k. (C.60)

Model 1 — Trace history

State1 = $y_{k-m:k}$. (C.61)

Model 2 — Proposed Effective Base

$State_2 = \Xi_k$. (C.62)

Model 3 — Effective Base plus ledger

$State_3 = (\Xi_k, L_k)$. (C.63)

Model 4 — Effective Base plus ledger and residual

$State_4 = (\Xi_k, L_k, \mathscr{R}_k)$. (C.64)

Model 5 — Full available state

$State_5 = \hat{X}_k$. (C.65)

Evaluate prediction, intervention, interpretability, and data cost.

A good Effective Base should achieve much of Model 5's task performance at substantially lower complexity.

C.20 Complexity penalty

A larger state can always absorb more historical detail.

The aim is not maximum dimension.

Define penalized utility:

$U_state = PredictiveValue + InterventionValue - \lambda Complexity - \mu MeasurementCost$. (C.66)

Choose the representation maximizing:

$\Xi^* = argmax_\Xi\ U_state$. (C.67)

The penalty terms should be declared rather than selected to force a preferred representation.

C.21 Closure failure taxonomy

Type C1 — Missing memory

Omitted history improves prediction.

Type C2 — Missing observer

Observer state or policy distribution matters.

Type C3 — Missing branch

Present coordinates do not preserve path provenance.

Type C4 — Regime mixing

Different generators are pooled.

Type C5 — Protocol drift

Boundary, gate, or observation rules changed.

Type C6 — Proxy instability

Compiled coordinates change with arbitrary measurement choices.

Type C7 — Intervention failure

Passive prediction succeeds, controlled response fails.

Type C8 — Access leakage

The model uses information unavailable to actual observers.

Type C9 — Residual omission

Unintegrated consequence remains future-active.

Type C10 — Excess compression

A low-dimensional signature cannot preserve the required distinctions.

C.22 Repair ladder

When closure fails, revise in the following order.

Step 1 — Audit protocol

Check boundary, timebase, observation map, and access.

Step 2 — Split regimes

Avoid mixing smooth, jump, elliptic, and hyperbolic behaviour.

Step 3 — Add minimal memory

Introduce only the history required by predictive tests.

Step 4 — Add branch or observer state

Preserve provenance or policy state where necessary.

Step 5 — Add ledger and residual summaries

Do not assume that current effective coordinates absorb historical consequence.

Step 6 — Use coupled Effective Bases

Represent several interacting loops:

$\Xi_k = (\Xi_k^{(1)}, \ldots, \Xi_k^{(m)})$. (C.68)

Step 7 — Abandon the low-dimensional compiler

Retain a higher-dimensional real model if no robust compression exists.

The Handoff explicitly recommends this escalation sequence and warns against immediately rebuilding one giant universal latent vector.

C.23 Effective-Base acceptance statement

A publishable acceptance statement should take the form:

> Under protocol P, regime r, prediction horizon h, intervention family $\mathcal{U}$, and tolerance ε, the compiled representation Ξ is approximately sufficient for task class $\mathcal{T}$. Omitted history, observer state, ledger, and residual do not produce material out-of-sample gain beyond the declared threshold. The representation has not been validated outside these conditions.

This wording avoids turning a local empirical result into a global ontology.

C.24 Effective-Base rejection statement

A rejection should state:

> The proposed representation Ξ does not close the declared task under protocol P. The principal failure arises from [memory / branch / observer / regime / residual / access]. Ξ remains usable as a descriptive or diagnostic signature but should not be treated as a predictive Effective Base.

Failure can reduce maturity without eliminating all usefulness.

C.25 Appendix C summary

The path from signature to state is:

Compiled description
→ predictive test
→ memory test
→ intervention test
→ regime test
→ access audit
→ acceptance or rejection. (C.69)

The governing rule is:

> **A Base is effective because it passes closure tests, not because it has few coordinates or an attractive interpretation.**

Appendix D — Generator Identification and Signed-

Relation Classification

D.1 Purpose

After a candidate Effective Base has passed the relevant sufficiency tests, the next question is:

> How does this Base change?

This is the Relation problem.

The Handoff requires the following sequence:

Effective Base
→ generator estimation
→ invariant-mode inspection
→ algebraic classification
→ conjugate geometry only if earned.

This appendix develops that sequence.

D.2 Continuous and discrete generators

Continuous-time form

Let:

$d\Xi/dt = F_r(\Xi, u, L, \mathscr{R})$. (D.1)

Linearizing near reference state Ξ^* gives:

$d(\delta\Xi)/dt = A_r\delta\Xi + G_r\delta u + \varepsilon$. (D.2)

where:

$A_r = \partial F_r/\partial\Xi \,|_{\{\Xi^*\}}$. (D.3)

Discrete-time form

Let:

$\Xi_{k+1} = F_r(\Xi_k, u_k)$. (D.4)

The local linearization is:

$\delta\Xi_{k+1} = A_r\delta\Xi_k + G_r\delta u_k + \varepsilon_k$. (D.5)

The discrete eigenvalues μ and continuous eigenvalues λ are related approximately by:

$\mu = e^{(\lambda\Delta t)}$. (D.6)

Interpretation must account for the timebase.

A negative real continuous eigenvalue may correspond to a positive discrete multiplier below one.

D.3 Generator estimation

Candidate methods include:

- local linear regression;
- vector autoregression;
- state-space estimation;
- Koopman approximation;
- dynamic mode decomposition;
- Jacobian estimation;
- controlled perturbation;
- system identification;
- Bayesian transition modelling.

The framework does not privilege one method.

The estimator must report:

- data window;
- regularization;
- uncertainty;
- regime;
- intervention coverage;
- residual diagnostics.

D.4 Metric declaration

A matrix representation depends on coordinate scaling.

Let the state space carry a positive-definite metric:

$G = G^{T} > 0$. (D.7)

The metric defines inner product:

$\langle x, y \rangle_G = x^{T} G y$. (D.8)

The generator can be decomposed relative to G into:

$A = S_G + \Omega_G$, (D.9)

where:

$S_G = \frac{1}{2}(A + G^{-1}A^{T}G)$, (D.10)

$\Omega_G = \frac{1}{2}(A - G^{-1}A^{T}G)$. (D.11)

Then:

S_G (D.12)

is metric-symmetric, while:

Ω_G (D.13)

is metric-antisymmetric.

A nonzero Ω_G suggests local circulation or rotation-like tendency.

It does not alone prove a coherent complex mode.

The Handoff explicitly warns that a nonzero antisymmetric component is weaker evidence than a persistent conjugate eigenpair and invariant plane.

D.5 Evidence ladder for phase-bearing Relation

Level R0 — Cross-coupling

Off-diagonal interaction exists.

Level R1 — Local antisymmetric component

$\Omega_G \neq 0$.

Level R2 — Complex-conjugate eigenpair

The spectrum contains:

$\lambda_{\pm} = \alpha \pm i\omega$. (D.14)

Level R3 — Persistent invariant plane

A two-dimensional mode remains stable across nearby windows and perturbations.

Level R4 — Coherent phase coordinate

A reproducible phase θ predicts trajectory orientation.

Level R5 — Readout conjugacy

A declared readout has a validated conjugate response.

Level R6 — Transport-compatible local complex structure

Mode planes can be consistently compared across states or regimes.

Ordinary complex language should be strongest only from R3 onward.

D.6 Two-dimensional spectral classification

For:

$$A = [a\ b; c\ d], \tag{D.15}$$

define:

$$\operatorname{tr} A = a + d, \tag{D.16}$$

$$\det A = ad - bc. \tag{D.17}$$

The discriminant is:

$$\Delta_A = (\operatorname{tr} A)^2 - 4\det A. \tag{D.18}$$

Elliptic spectral branch

$$\Delta_A < 0. \tag{D.19}$$

Eigenvalues:

$$\lambda_\pm = \alpha \pm i\omega, \tag{D.20}$$

where:

$$\alpha = \operatorname{tr} A/2, \tag{D.21}$$

$$\omega = \tfrac{1}{2}\sqrt{(4\det A - (\operatorname{tr} A)^2)}. \tag{D.22}$$

Parabolic boundary

$$\Delta_A = 0. \tag{D.23}$$

The generator has repeated real eigenvalues and may be defective.

Hyperbolic branch

$$\Delta_A > 0. \tag{D.24}$$

The generator has two real eigenvalues.

This classification applies to the local linearized mode.

Nonlinear global dynamics may differ.

D.7 Removing isotropic drift

For an elliptic pair:

$$\lambda_\pm = \alpha \pm i\omega, \tag{D.25}$$

write:

$A_E = \alpha I + \Omega$, (D.26)

where the normalized rotational operator is:

$J = \Omega/\omega$. (D.27)

Then:

$J^2 = -I$ (D.28)

when the mode is exactly elliptic after suitable real similarity transformation.

The state equation becomes:

$dx/dt = \alpha x + \omega J x$. (D.29)

Interpretation:

- $\alpha < 0$ — damped rotation;
- $\alpha = 0$ — neutral rotation;
- $\alpha > 0$ — expanding spiral;
- ω — angular circulation rate.

Complex structure concerns J.

Growth or decay concerns α.

D.8 Canonical elliptic form

There exists a real basis on the elliptic plane in which:

$A_E = [\alpha\ -\omega;\ \omega\ \alpha]$. (D.30)

Define:

$z = u + iv$. (D.31)

Then:

$dz/dt = (\alpha + i\omega)z$. (D.32)

The solution is:

$z(t) = z_0 e^{(\alpha t)} e^{(i\omega t)}$. (D.33)

This is the canonical complex representation of the local mode.

D.9 Canonical parabolic form

A defective two-dimensional mode may be written:

$A_P = [\lambda\ 1;\ 0\ \lambda]$. (D.34)

Separating the isotropic term:

$A_P = \lambda I + N$, (D.35)

where:

$N^2 = 0$. (D.36)

Then:

$e^{(A_Pt)} = e^{(\lambda t)}(I + tN)$. (D.37)

This mode exhibits:

- drift;
- shear;
- memory accumulation;
- critical slowing;
- sensitivity to perturbation.

It should not be interpreted as phase rotation.

D.10 Canonical hyperbolic form

A hyperbolic mode can be written:

$A_H = [\alpha\ \kappa;\ \kappa\ \alpha]$. (D.38)

Define:

$K = [0\ 1;\ 1\ 0]$. (D.39)

Then:

$K^2 = I$. (D.40)

The generator is:

$A_H = \alpha I + \kappa K$. (D.41)

Using split-complex notation:

$w = u + jv$, with $j^2 = 1$, (D.42)

gives:

$dw/dt = (\alpha + j\kappa)w$. (D.43)

The solutions contain expanding and contracting directions.

This algebra describes separation rather than ordinary rotation.

D.11 Unified signed operator

The three branches may be written:

$C\chi^2 = \chi I$. (D.44)

Elliptic

$\chi = -1$:

$C_{-}^2 = -I$. (D.45)

Parabolic

$\chi = 0$:

$C_0^2 = 0$. (D.46)

Hyperbolic

$\chi = +1$:

$C_{+}^2 = I$. (D.47)

The signed-conjugate grammar makes explicit that the return orientation, not merely the presence of two coupled quantities, determines the algebra. A related project source formulates corrective, neutral, and confirmatory return through the same $C\chi^2 = \chi I$ structure.

D.12 Return-sign interpretation

Suppose two local coordinates are:

$z = (\delta s, \delta\lambda)^T$. (D.48)

Let:

$\delta\lambda \to \delta s$ (D.49)

be the forward effect.

The return can be:

Corrective

$\delta s \to -\delta\lambda$. (D.50)

This yields elliptic circulation.

Neutral or degenerate

$\delta s \to 0$. (D.51)

This yields parabolic accumulation.

Confirmatory

$\delta s \rightarrow +\delta\lambda$. (D.52)

This yields hyperbolic amplification.

Thus:

Conjugate magnitude + return sign = Relation signature. (D.53)

The sign must be estimated or derived.

It cannot be inferred from symbolic complementarity.

D.13 Three-dimensional local decomposition

A real three-dimensional space cannot carry a global linear complex structure because:

$J^2 = -I$ (D.54)

requires even real dimension.

However, a local three-dimensional generator may contain:

- one real eigenvalue λ_0;
- one complex-conjugate pair $\alpha \pm i\omega$.

Then:

$\mathbb{R}^3 \cong E \oplus E_0$, (D.55)

where:

$\dim_{\mathbb{R}} E = 2$, (D.56)

$\dim_{\mathbb{R}} E_0 = 1$. (D.57)

Locally:

$\mathbb{R}^3 \cong \mathbb{C} \oplus \mathbb{R}$. (D.58)

The Handoff identifies this as a formal consequence of the synthesis: PORE may supply three effective real coordinates, while only a two-dimensional invariant subspace carries ordinary complex structure.

D.14 Role of the remaining real direction

Let the local coordinates be:

$(z, q_0) \in \mathbb{C} \oplus \mathbb{R}$. (D.59)

The real direction q_0 may regulate:

- damping α;

- angular frequency ω;
- mode amplitude;
- gate hazard;
- stability margin;
- switching probability;
- residual accumulation.

A schematic model is:

$dz/dt = [\alpha(q_0) + i\omega(q_0)]z$, (D.60)

$dq_0/dt = g(|z|, q_0, L, \mathscr{R})$. (D.61)

This prevents the third coordinate from being forced into the imaginary axis.

D.15 Mode-plane persistence

Let E_k be the estimated elliptic plane at step k.

Measure principal-angle drift:

$\delta_E(k, k + 1) = \text{Angle}(E_k, E_{k+1})$. (D.62)

A persistent mode requires:

$\delta_E(k, k + 1) \le \varepsilon_E$ (D.63)

throughout the claimed regime.

Large drift may indicate:

- coordinate instability;
- regime transition;
- insufficient data;
- no genuine invariant plane.

A transient complex eigenpair is weaker evidence than a stable phase-bearing mode.

D.16 Frequency and damping stability

Estimate:

α_k, ω_k. (D.64)

A coherent mode should show bounded variability:

$\text{Var_window}(\alpha_k) \le \varepsilon_\alpha$, (D.65)

$\mathrm{Var}_{\mathrm{window}}(\omega_k) \le \varepsilon_\omega$. (D.66)

A frequency that changes unpredictably with sampling or preprocessing may not define an operational phase.

D.17 Phase reconstruction

On a valid elliptic mode:

$z_k = u_k + iv_k$. (D.67)

Define:

$\theta_k = \mathrm{atan2}(v_k, u_k)$. (D.68)

Phase coherence may be tested through:

$\Delta\theta_k = \mathrm{Unwrap}(\theta_{k+1} - \theta_k)$. (D.69)

A stable mode predicts:

$\Delta\theta_k \approx \omega_k \Delta t$ (D.70)

within residual tolerance.

Large unexplained phase slips suggest:

- forcing;
- jumps;
- mode mixing;
- gate events;
- incorrect plane estimation.

D.18 Readout detection test

Let readout:

$Y = a(x)$. (D.71)

The readout detects the elliptic mode when:

$a|_E \neq 0$. (D.72)

If:

$a|_E = 0$, (D.73)

then:

$Y = 0$ (D.74)

for all mode states, and no nonzero conjugate exposure should be assigned.

The Handoff decision tree makes this explicit: a dynamic complex mode may exist while the chosen readout has no first-order sensitivity to it.

D.19 Conjugate readout on the mode

For nonzero a:

$$Y(x) = a(x), \quad (D.75)$$

$$\mathcal{Q}_\gamma(x) = -a(Jx). \quad (D.76)$$

Then:

$$\Phi_a(x) = [Y(x), \mathcal{Q}_\gamma(x)]^T. \quad (D.77)$$

The intertwining relation is:

$$\Phi_a J = J_R \Phi_a. \quad (D.78)$$

where:

$$J_R = [0\ -1;\ 1\ 0]. \quad (D.79)$$

This formal result does not identify the dynamic mode and readout plane as the same material space.

It shows that they carry isomorphic local quarter-turn structures.

D.20 Nonlinear elliptic mode

For nonlinear dynamics:

$$dx/dt = F(x), \quad (D.80)$$

a local complex structure may be defined on a tangent plane:

$$E_x \subset T_x\mathcal{M}. \quad (D.81)$$

Let:

$$J_x: E_x \to E_x, \quad (D.82)$$

with:

$$J_x^2 = -I. \quad (D.83)$$

The structure may vary with x.

A local phase chart is valid only where:

- E_x varies smoothly;
- the mode remains separated from other eigenvalues;
- no jump occurs;

- the readout remains nondegenerate.

Global compatibility is deferred to Appendix G.

D.21 Forced elliptic systems

A system may show oscillation because of external periodic forcing:

$dx/dt = Ax + b \cos \Omega t$. (D.84)

This does not establish internally generated self-referential phase.

To distinguish the two:

1. remove or randomize forcing;
2. examine persistence;
3. test trace-policy ablation;
4. estimate phase relation after intervention.

Externally driven phase and self-referential phase can coexist.

They should not be conflated.

D.22 Delayed self-reference

A delayed loop may satisfy:

$dx/dt = F(x(t), x(t - \tau_d))$. (D.85)

Delay can produce complex eigenvalues even when the instantaneous coupling is not rotational.

The characteristic equation may be transcendental:

$\det[\lambda I - A_0 - A_1 e^{(-\lambda \tau_d)}] = 0$. (D.86)

Complex roots can therefore arise from delayed feedback.

This is one plausible route through which self-reference may support oscillatory modes.

It is not the only route.

D.23 Stochastic generator

For stochastic dynamics:

$d\Xi = F(\Xi)dt + B(\Xi)dW_t$. (D.87)

The drift Jacobian may contain an elliptic mode, while diffusion alters observed phase coherence.

Phase estimation should report:

- drift frequency;
- diffusion rate;
- phase uncertainty;
- coherence time.

A complex mean trajectory alone may conceal stochastic dephasing.

D.24 Dissipative and rotational decomposition

For a mode with metric G, the generator may be decomposed:

$$A = D + R, \quad (D.88)$$

where:

$$D = S_G \quad (D.89)$$

is dissipative or amplifying relative to the metric, and:

$$R = \Omega_G \quad (D.90)$$

is rotational.

A healthy elliptic mode may require:

$$\|R\| \text{ sufficiently large relative to anisotropic } D. \quad (D.91)$$

If dissipation dominates:

$$\|R\|/\|D\| \to 0, \quad (D.92)$$

the phase description may become operationally weak even when eigenvalues remain formally complex.

D.25 Mode-quality score

A practical elliptic-mode quality score may combine:

$$Q_{mode} = w_1 S_{spectral} + w_2 S_{persistence} + w_3 S_{phase} + w_4 S_{readout} + w_5 S_{intervention}. \quad (D.93)$$

where:

- $S_{spectral}$ measures conjugate-eigenpair strength;

- `S_persistence` measures plane stability;
- `S_phase` measures phase predictability;
- `S_readout` measures nondegenerate readout coupling;
- `S_intervention` measures response validity.

The weights and thresholds are protocol-specific.

This score is diagnostic, not a universal physical quantity.

D.26 Generator-change hazard

Let:

r_k (D.94)

be the active regime.

Define transition hazard:

$$h_{r\to r'}(k) = \Pr(r_{k+1} = r' \mid \Xi_k, L_k, \mathscr{R}_k). \quad \text{(D.95)}$$

A commitment event may change:

- damping;
- frequency;
- discriminant;
- invariant plane;
- gate hazard.

This links Relation to Commitment.

A full operational-world model should test whether:

$L_k, \mathscr{R}_k$ (D.96)

predict generator transitions after controlling for the current Base.

D.27 Relation-classification decision tree

Question 1

Is there a validated Effective Base?

- No → do not estimate a reduced generator.
- Yes → continue.

Question 2

Is a stable local generator recoverable?

- No → retain nonparametric or event-based modelling.
- Yes → continue.

Question 3

Is there a persistent low-dimensional mode?

- No → retain higher-dimensional real dynamics.
- Yes → continue.

Question 4

What is the spectral signature?

- complex pair → candidate elliptic;
- repeated near-defective pair → candidate parabolic;
- real expanding/contracting pair → candidate hyperbolic;
- mixed → block or higher-dimensional representation.

Question 5

Does the readout detect the mode?

- No → no first-order conjugate response for that readout.
- Yes → construct and test ϱ_{γ}.

Question 6

Is the mode stable across protocol and regime?

- No → local or transient result only.
- Yes → consider local complex chart and transport.

D.28 Relation rejection statements

Elliptic rejection

> No persistent two-dimensional elliptic invariant mode was found under protocol P. Ordinary complex completion is therefore not supported in the declared regime.

Parabolic rejection

> The near-defective signature was not stable across windows; accumulation should

not be represented by one nilpotent mode.

Hyperbolic rejection

Apparent expansion and contraction were explained by exogenous trend rather than an internally coupled hyperbolic mode.

Low-dimensional rejection

The generator does not admit a robust low-dimensional reduction. A higher-dimensional real representation is retained.

These are successful scientific outcomes.

They prevent algebra from being chosen for aesthetic reasons.

D.29 Appendix D summary

The Relation ladder is:

Validated Base
→ estimated generator
→ metric and regime declaration
→ spectrum
→ invariant mode
→ signed classification
→ readout detection
→ conjugate completion
→ persistence and intervention tests. (D.97)

The governing principle is:

> **Complex numbers become part of the framework only after the generator earns an elliptic mode and the readout demonstrably couples to it.**

Appendix E — Detailed Proof and Extensions of the Complex Intertwining Principle

E.1 Purpose

Section 16 established the **Complex Intertwining Principle**:

$\Phi_a J = J_R\Phi_a$. (E.1)

This appendix gives the complete linear-algebraic construction, proves nondegeneracy, identifies the orientation and normalization freedoms, and extends the result to nonlinear readouts and damped elliptic generators.

The result is formal and conditional.

It assumes that a valid two-dimensional elliptic mode has already been identified:

$J^2 = -I$. (E.2)

It does not prove that self-reference always generates such a mode.

The Handoff identifies this intertwining result as a new formal consequence of the integrated framework rather than a universally source-established law.

E.2 Elliptic mode space

Let `E` be a two-dimensional real vector space.

Let:

$$J: E \to E \quad \text{(E.3)}$$

be a real linear operator satisfying:

$$J^2 = -I_E. \quad \text{(E.4)}$$

The pair `(E, J)` is a real vector space equipped with a complex structure.

Scalar multiplication by:

$$\alpha + i\beta \in \mathbb{C} \quad \text{(E.5)}$$

can be defined through:

$$(\alpha + i\beta)x \equiv \alpha x + \beta Jx. \quad \text{(E.6)}$$

The ordinary complex multiplication law follows because:

$$J^2 = -I. \quad \text{(E.7)}$$

Indeed:

$$(\alpha I + \beta J)(\gamma I + \delta J)$$
$$= (\alpha\gamma - \beta\delta)I + (\alpha\delta + \beta\gamma)J. \quad \text{(E.8)}$$

This matches:

$$(\alpha + i\beta)(\gamma + i\delta)$$
$$= (\alpha\gamma - \beta\delta) + i(\alpha\delta + \beta\gamma). \quad \text{(E.9)}$$

Therefore `(E, J)` may be treated as a one-dimensional complex vector space.

E.3 Readout covector

Let:

$$a \in E^* \quad \text{(E.10)}$$

be a nonzero real linear covector.

Define the admitted scalar readout:

$$Y(x) = a(x). \quad \text{(E.11)}$$

The readout selects one real projection of the elliptic mode.

It does not by itself preserve the mode's orientation.

Define the conjugate readout:

$\mathcal{Q}_\gamma(x) = -a(Jx)$. (E.12)

The completed readout map is:

$\Phi_a: E \to \mathbb{R}^2$, (E.13)

with:

$\Phi_a(x) = [Y(x), \mathcal{Q}_\gamma(x)]^T$. (E.14)

E.4 Standard quarter-turn in the readout plane

Define:

$J_R = [0\ -1;\ 1\ 0]$. (E.15)

Then:

$J_R^2 = -I_2$. (E.16)

For a readout vector:

$v_R = [Y, \mathcal{Q}_\gamma]^T$, (E.17)

the quarter-turn gives:

$J_R v_R = [-\mathcal{Q}_\gamma, Y]^T$. (E.18)

This produces the cycle:

$[Y; \mathcal{Q}_\gamma]$
$\to [-\mathcal{Q}_\gamma; Y]$
$\to [-Y; -\mathcal{Q}_\gamma]$
$\to [\mathcal{Q}_\gamma; -Y]$
$\to [Y; \mathcal{Q}_\gamma]$. (E.19)

E.5 Intertwining theorem

Theorem E.1 — Complex Intertwining Principle [F]

Let:

$J^2 = -I_E$, (E.20)

and let $a \in E^*$ be nonzero.

Define:

$Y(x) = a(x)$, (E.21)

$\mathcal{Q}_\gamma(x) = -a(Jx)$, (E.22)

$\Phi_a(x) = [Y(x), \mathcal{Q}_\gamma(x)]^T$. (E.23)

Then:

$\Phi_a J = J_R\Phi_a$. (E.24)

E.6 Proof

For any $x \in E$:

$\Phi_a(Jx)$
$= [a(Jx), -a(J^2x)]^T$. (E.25)

Since:

$J^2x = -x$, (E.26)

we have:

$-a(J^2x) = a(x)$. (E.27)

Also:

$a(Jx) = -\mathcal{Q}_\gamma(x)$. (E.28)

Therefore:

$\Phi_a(Jx)$
$= [-\mathcal{Q}_\gamma(x), Y(x)]^T$. (E.29)

Meanwhile:

$J_R\Phi_a(x)$
$= [0\ -1;\ 1\ 0][Y(x);\ \mathcal{Q}_\gamma(x)]$
$= [-\mathcal{Q}_\gamma(x);\ Y(x)]$. (E.30)

Hence:

$\Phi_a(Jx) = J_R\Phi_a(x)$. (E.31)

Since this holds for every $x \in E$:

$\Phi_a J = J_R\Phi_a$. ■ (E.32)

E.7 Nondegeneracy of the readout pair

The intertwining relation would be weak if the two readout coordinates were linearly dependent.

The following result shows that this cannot occur for a nonzero real readout on a real two-dimensional elliptic mode.

Theorem E.2 — Readout Nondegeneracy [F]

If:

$a \neq 0$ (E.33)

and:

$J^2 = -I$, (E.34)

then the covectors:

a (E.35)

and:

$a \circ J$ (E.36)

are linearly independent over $\mathbb{R}$.

E.8 Proof

Assume for contradiction that:

$a \circ J = \lambda a$ (E.37)

for some real scalar λ.

Compose both sides with J:

$a \circ J^2 = \lambda a \circ J$. (E.38)

Using:

$J^2 = -I$, (E.39)

the left-hand side becomes:

$a \circ J^2 = -a$. (E.40)

The right-hand side becomes:

$\lambda a \circ J = \lambda^2 a$. (E.41)

Therefore:

$-a = \lambda^2 a$. (E.42)

Since $a \neq 0$:

$\lambda^2 = -1$. (E.43)

No real λ satisfies Equation (E.43).

Therefore a and $a \circ J$ are linearly independent. ■ (E.44)

E.9 Invertibility of Φ_a

Since E and $\mathbb{R}^2$ both have real dimension two, and the two readout covectors are independent:

$\operatorname{rank}(\Phi_a) = 2$. (E.45)

Therefore:

$\Phi_a: E \to \mathbb{R}^2$ (E.46)

is a real-linear isomorphism.

This gives:

$E \cong \mathbb{R}^2$ (E.47)

and, after equipping both sides with their quarter-turn operators:

$(E, J) \cong (\mathbb{R}^2, J_R)$. (E.48)

Equivalently:

$(E, J) \cong \mathbb{C}$. (E.49)

This is the precise meaning of the statement:

> The dynamic elliptic mode and its conjugate readout plane are locally complex-isomorphic.

It does not mean they are the same material, economic, or semantic object.

E.10 Explicit inverse map

Choose any vector $e \in E$ satisfying:

$a(e) = 1$. (E.50)

Then:

$a(Je)$ (E.51)

may not vanish in an arbitrary choice of e.

A convenient adapted basis can instead be constructed from the inverse map itself.

Let:

$e_Y = \Phi_a^{-1}([1; 0])$, (E.52)

$e_Q = \Phi_a^{-1}([0; 1])$. (E.53)

Then:

$a(e_Y) = 1$, (E.54)

$-a(Je_Y) = 0$, (E.55)

$a(e_Q) = 0$, (E.56)

$-a(Je_Q) = 1$. (E.57)

The intertwining relation implies:

$Je_Y = e_Q$, (E.58)

$Je_Q = -e_Y$. (E.59)

Any state may be reconstructed as:

$x = Y(x)e_Y + \mathcal{Q}_\gamma(x)e_Q$. (E.60)

Thus the conjugate readout pair is not merely an auxiliary report.

Within the identified two-dimensional elliptic mode, it is a complete coordinate system.

E.11 Complex functional form

Define the complex-valued readout:

$\zeta_a(x) = Y(x) + i\mathcal{Q}_\gamma(x)$. (E.61)

Then:

$\zeta_a(Jx)$
$= Y(Jx) + i\mathcal{Q}_\gamma(Jx)$. (E.62)

Using the previous relations:

$Y(Jx) = -\mathcal{Q}_\gamma(x)$, (E.63)

$\mathcal{Q}_\gamma(Jx) = Y(x)$. (E.64)

Therefore:

$\zeta_a(Jx)$
$= -\mathcal{Q}_\gamma(x) + iY(x)$. (E.65)

But:

$i\zeta_a(x)$
$= iY(x) - \mathcal{Q}_\gamma(x)$. (E.66)

Hence:

$\zeta_a(Jx) = i\zeta_a(x)$. (E.67)

The complex readout is therefore complex-linear relative to J.

E.12 Evolution under pure elliptic rotation

Let the state evolve according to:

$dx/d\theta = Jx$. (E.68)

Then:

$d\zeta_a/d\theta$
$= \zeta_a(dx/d\theta)$

$= \zeta_a(Jx)$. (E.69)

Using Equation (E.67):

$d\zeta_a/d\theta = i\zeta_a$. (E.70)

Therefore:

$\zeta_a(\theta) = e^{(i\theta)}\zeta_a(0)$. (E.71)

The scalar components satisfy:

$dY/d\theta = -\mathcal{Q}_\gamma$, (E.72)

$d\mathcal{Q}_\gamma/d\theta = Y$. (E.73)

E.13 Damped elliptic generator

Suppose the local generator is:

$A_E = \alpha I + \omega J$. (E.74)

Then:

$dx/dt = (\alpha I + \omega J)x$. (E.75)

Apply ζ_a:

$d\zeta_a/dt$
$= \alpha\zeta_a(x) + \omega\zeta_a(Jx)$. (E.76)

Therefore:

$d\zeta_a/dt = (\alpha + i\omega)\zeta_a$. (E.77)

The solution is:

$\zeta_a(t) = \zeta_a(0)e^{(\alpha t)}e^{(i\omega t)}$. (E.78)

The readout equations are:

$dY/dt = \alpha Y - \omega\mathcal{Q}_\gamma$, (E.79)

$d\mathcal{Q}_\gamma/dt = \omega Y + \alpha\mathcal{Q}_\gamma$. (E.80)

The conjugate pair remains closed even when the mode expands or decays isotropically.

Thus:

Elliptic closure does not require conservative amplitude. (E.81)

E.14 Anisotropic damping

Suppose:

$A = \alpha I + \omega J + D$, (E.82)

where D does not commute with J.

Then:

$AJ - JA \neq 0$ (E.83)

in general.

The elliptic plane may still exist approximately, but the readout dynamics no longer close as a pure complex scalar equation.

Instead:

$d\Phi_a(x)/dt$
$= \Phi_a Ax.$ (E.84)

A residual real 2×2 distortion appears in the readout plane.

One may write:

$d/dt\ [Y; \mathcal{Q}_\gamma]$
$= [\alpha\ -\omega;\ \omega\ \alpha][Y; \mathcal{Q}_\gamma]$

- $D_R[Y; \mathcal{Q}_\gamma]$. (E.85)

where:

$D_R = \Phi_a D \Phi_a^{-1}$. (E.86)

If D_R is small relative to the elliptic block, complex notation remains locally useful.

If it dominates, the ordinary complex reduction becomes misleading.

E.15 Orientation reversal

The complex structure J has an orientation.

Replacing it by:

$J' = -J$ (E.87)

also satisfies:

$J'^2 = -I$. (E.88)

The conjugate response becomes:

$\mathcal{Q}_\gamma' = -a(J'x)$. (E.89)

Therefore:

$\mathcal{Q}_\gamma' = a(Jx) = -\mathcal{Q}_\gamma$. (E.90)

The complex readout becomes:

$Z_\gamma' = Y - i\mathcal{Q}_\gamma = \bar{Z}_\gamma$. (E.91)

Thus orientation reversal corresponds to complex conjugation.

A valid implementation must declare whether positive Relation movement is generated by:

J (E.92)

or:

−J. (E.93)

Sign disagreements between studies may reflect opposite orientation conventions rather than substantive contradiction.

E.16 Readout scaling

Let the readout be rescaled:

a′ = ca (E.94)

for nonzero real `c`.

Then:

Y′ = cY, (E.95)

$\mathcal{Q}_\gamma' = c\mathcal{Q}_\gamma$, (E.96)

$Z_\gamma' = cZ_\gamma$. (E.97)

The phase remains unchanged for `c > 0`.

For `c < 0`, the representation acquires a half-turn:

$\arg Z_\gamma' = \arg Z_\gamma + \pi$. (E.98)

Readout scaling changes amplitude and units but not the underlying quarter-turn structure.

E.17 Readout rotation

A new scalar readout may be selected from the same completed plane:

$Y_\varphi = Y \cos\varphi - \mathcal{Q}_\gamma \sin\varphi$. (E.99)

Its conjugate response is:

$\mathcal{Q}_\varphi = Y \sin\varphi + \mathcal{Q}_\gamma \cos\varphi$. (E.100)

Therefore:

$$[Y_\varphi; \mathcal{Q}_\varphi] = [\cos\varphi \ -\sin\varphi; \sin\varphi \ \cos\varphi][Y; \mathcal{Q}_\gamma]. \quad \text{(E.101)}$$

The new readout is another valid real projection of the same elliptic mode.

This formalizes passive measurement rotation.

E.18 Nonlinear readout

Let:

$$Y = V(x), \quad \text{(E.102)}$$

where $V: E \to \mathbb{R}$ is differentiable but not necessarily linear.

The differential at x is:

$$dV_x \in E^*. \quad \text{(E.103)}$$

Define the local conjugate response:

$$\mathcal{Q}_\gamma(x) = -dV_x(Jx). \quad \text{(E.104)}$$

Along the elliptic flow:

$$dx/d\theta = Jx, \quad \text{(E.105)}$$

we have:

$$dY/d\theta = dV_x(Jx). \quad \text{(E.106)}$$

Therefore:

$$dY/d\theta = -\mathcal{Q}_\gamma. \quad \text{(E.107)}$$

However, the second relation becomes:

$$d\mathcal{Q}_\gamma/d\theta = -d^2V_x(Jx, Jx) + dV_x(x), \quad \text{(E.108)}$$

where d^2V_x is the Hessian bilinear form.

Thus:

$$d\mathcal{Q}_\gamma/d\theta = Y \quad \text{(E.109)}$$

does not generally hold for nonlinear V.

Exact two-coordinate closure requires additional structure.

E.19 Nonlinear closure condition

Exact nonlinear closure requires:

$$-d^2V_x(Jx, Jx) + dV_x(x) = V(x). \quad \text{(E.110)}$$

For linear V, the Hessian vanishes and:

$$dV_x(x) = V(x). \quad \text{(E.111)}$$

Equation (E.110) is then satisfied automatically.

For homogeneous functions of degree one, Euler's theorem gives:

$dV_x(x) = V(x)$. (E.112)

But the Hessian term may remain.

Therefore the clean Complex Intertwining Principle is naturally linear or locally linearized.

A nonlinear readout may still admit approximate conjugate closure in a sufficiently small neighbourhood.

E.20 Tangent-space formulation

Let the full state space be a differentiable manifold $\mathcal{M}$.

At point $x \in \mathcal{M}$, let:

$E_x \subset T_x\mathcal{M}$ (E.113)

be a two-dimensional elliptic tangent mode.

Let:

$J_x: E_x \to E_x$ (E.114)

satisfy:

$J_x^2 = -I$. (E.115)

Let:

$dV_x: T_x\mathcal{M} \to \mathbb{R}$ (E.116)

be the readout differential.

The local conjugate response in direction $v \in E_x$ is:

$\mathcal{Q}_V(x; v) = -dV_x(J_x v)$. (E.117)

When the active trajectory tangent is:

$v = \dot{x}/\omega$, (E.118)

this gives the first-order response under the local phase direction.

The complex structure is then local to the tangent mode, not necessarily global on $\mathcal{M}$.

E.21 Readout orthogonality failure

If:

$dV_x|_{E_x} = 0$, (E.119)

the readout is locally insensitive to the elliptic mode.

Then:

$\mathcal{Q}_V = 0$. (E.120)

A dynamic complex mode may therefore exist without appearing in a chosen scalar readout.

This is not a failure of the mode.

It is a failure of the readout to detect it.

E.22 Approximate intertwining error

For empirically estimated operators, define:

$\varepsilon_int = \|\Phi_a J - J_R\Phi_a\|$. (E.121)

Normalize by:

$\tilde{\varepsilon}_int = \|\Phi_a J - J_R\Phi_a\| / [\|\Phi_a\|\|J\| + \|J_R\|\|\Phi_a\|]$. (E.122)

Approximate conjugate closure requires:

$\tilde{\varepsilon}_int \le \varepsilon_protocol$. (E.123)

The tolerance must be fixed before interpretation.

E.23 Dynamic and readout planes

The dynamic mode may be represented as:

$z_D = u + iv$. (E.124)

The readout plane may be:

$Z_Y = Y + iQ_\gamma$. (E.125)

The map:

$Z_Y = \zeta_a(z_D)$ (E.126)

is complex-linear under the assumptions above.

But:

$z_D \neq Z_Y$ (E.127)

as typed objects.

`z_D` describes coordinates of the dynamic mode.

`Z_Y` describes the selected readout and its conjugate response.

The Handoff explicitly requires these planes to remain distinct until an intertwining map is established.

E.24 CAPM as an exact readout-plane example

For CAPM:

Y = R, (E.128)

$\mathcal{Q}_\gamma$ = Q, (E.129)

Z_Y = R + iQ. (E.130)

The readout plane satisfies:

dR/dθ = −Q, (E.131)

dQ/dθ = R. (E.132)

The CAPM source establishes this readout-plane closure exactly within its declared valuation construction.

What remains open is whether an independently estimated financial dynamic plane:

z_D = u + iv (E.133)

admits a stable map:

Φ_fin: z_D → R + iQ (E.134)

that satisfies the same intertwining relation.

E.25 Appendix E summary

The formal chain is:

Elliptic mode `(E, J)`
→ nonzero scalar readout `a`
→ conjugate response `-a∘J`
→ invertible readout map Φ_a
→ preserved quarter-turn grammar
→ local complex readout. (E.135)

The central theorem is:

$\Phi_a J = J_R \Phi_a$. (E.136)

Its scope is precise:

- exact for real linear readouts on a two-dimensional elliptic mode;
- extendable to isotropically damped modes;
- locally approximable for nonlinear readouts;
- invalid or incomplete when the mode, readout, units, or orientation fail.

Appendix F — Parabolic and Hyperbolic Relational Completions

F.1 Purpose

Ordinary complex geometry covers only the elliptic branch:

$J^2 = -I$. (F.1)

The general signed-conjugate framework is broader:

$C\chi^2 = \chi I$. (F.2)

This appendix develops the other two canonical branches:

- parabolic completion, where $\chi = 0$;
- hyperbolic completion, where $\chi > 0$.

These branches are not failures of complex analysis.

They describe different transformation grammars.

The Handoff specifically requires parabolic and hyperbolic completions so that the article does not force every self-referential Relation into ordinary phase.

F.2 Unified two-coordinate state

Let:

$x = [u; v]$. (F.3)

Let the normalized Relation operator be:

$C\chi = [0\ \chi; 1\ 0]$. (F.4)

Then:

$C\chi^2 = \chi I$. (F.5)

The transformation is:

$C\chi[u; v]$
$= [\chi v; u]$. (F.6)

The three cases are obtained by the sign of χ.

F.3 Elliptic reference case

For:

$\chi = -1$, (F.7)

the operator is:

$C_- = [0\ -1; 1\ 0]$. (F.8)

Then:

$C_{-}^2 = -I$. (F.9)

The evolution:

$dx/d\xi = C_{-}x$ (F.10)

gives:

$du/d\xi = -v$, (F.11)

$dv/d\xi = u$. (F.12)

The solution is rotational:

$u(\xi) = u_0 \cos \xi - v_0 \sin \xi$, (F.13)

$v(\xi) = u_0 \sin \xi + v_0 \cos \xi$. (F.14)

This is the ordinary complex branch.

F.4 Parabolic Completion

F.4.1 Nilpotent operator

For:

$\chi = 0$, (F.15)

define:

$N = [0\ 0; 1\ 0]$. (F.16)

Then:

$N^2 = 0$. (F.17)

The evolution is:

$dx/d\xi = Nx$. (F.18)

Therefore:

$du/d\xi = 0$, (F.19)

$dv/d\xi = u$. (F.20)

The solution is:

$u(\xi) = u_0$, (F.21)

$v(\xi) = v_0 + \xi u_0$. (F.22)

The system accumulates rather than rotates.

F.4.2 Dual-number representation

Introduce a nilpotent unit:

$\varepsilon^2 = 0$. (F.23)

Define:

$z_P = v + \varepsilon u$. (F.24)

Multiplication by ε gives:

$\varepsilon z_P = \varepsilon v$. (F.25)

Depending on coordinate convention, the nilpotent operator can be represented as the transfer of one coordinate into another without a return rotation.

The exponential is:

$e^{(\varepsilon\xi)} = 1 + \varepsilon\xi$. (F.26)

Thus:

$z_P(\xi) = (1 + \varepsilon\xi) z_P(0)$. (F.27)

No sine–cosine cycle appears.

F.4.3 Parabolic transformation grammar

The parabolic sequence is:

u remains available
→ u accumulates into v
→ no automatic return from v into –u. (F.28)

The relation is one-way or critically degenerate.

Possible interpretations include:

- memory accumulation;
- unresolved carry;
- delayed debt;
- path integration;
- procedural backlog;
- trace without restorative return;
- critical slowing near a transition.

These are candidate interpretations only.

A domain must define its actual coordinates.

F.4.4 Parabolic readout

Let the admitted readout be:

$$Y(x) = a(x). \quad (F.29)$$

Define the first relational response:

$$\mathcal{P}_Y(x) = -a(Nx). \quad (F.30)$$

Then under:

$$dx/d\xi = Nx, \quad (F.31)$$

we obtain:

$$dY/d\xi = -\mathcal{P}_Y. \quad (F.32)$$

Differentiate again:

$$d\mathcal{P}_Y/d\xi = -a(N^2x). \quad (F.33)$$

Since:

$$N^2 = 0, \quad (F.34)$$

then:

$$d\mathcal{P}_Y/d\xi = 0. \quad (F.35)$$

Therefore:

$$d^2Y/d\xi^2 = 0. \quad (F.36)$$

The readout changes linearly rather than oscillating.

The closure is:

$$Y \to -\mathcal{P}_Y \to 0. \quad (F.37)$$

This is not a four-stage phase cycle.

F.4.5 Parabolic completion principle

Proposition F.1 — Parabolic Readout Closure [F]

If:

$$N^2 = 0 \quad (F.38)$$

and:

$$dx/d\xi = Nx, \quad (F.39)$$

then for:

$$Y = a(x), \quad (F.40)$$

$\mathcal{P}_Y = -a(Nx)$, (F.41)

we have:

$dY/d\xi = -\mathcal{P}_Y$, (F.42)

$d\mathcal{P}_Y/d\xi = 0$. (F.43)

The pair $(Y, \mathcal{P}_Y)$ closes as a first-order drift system.

F.4.6 Operational signatures of parabolic Relation

Evidence for a parabolic mode may include:

- repeated or nearly repeated real eigenvalue;
- near-defective generator;
- secular drift;
- linear accumulation;
- slow recovery;
- strong sensitivity to small perturbations;
- no stable return cycle.

A system near an elliptic–hyperbolic boundary may display approximate parabolic behaviour.

F.4.7 Self-reference and parabolic memory

A self-referential process may be parabolic when each trace modifies later conditions but produces no compensating return.

For example:

Trace_k
→ accumulated policy bias
→ further trace
→ further accumulated bias. (F.44)

The process stores historical deformation without either:

- bounded circulation;
- exponential divergence.

This may describe certain:

- bureaucratic accumulations;
- technical debt processes;
- unresolved queue systems;

- gradually biased AI context.

These remain hypotheses until operationalized.

F.5 Hyperbolic Completion

F.5.1 Involutive operator

For:

$\chi = +1$, (F.45)

define:

$K = [0\ 1;\ 1\ 0]$. (F.46)

Then:

$K^2 = I$. (F.47)

The evolution is:

$dx/d\xi = Kx$. (F.48)

Therefore:

$du/d\xi = v$, (F.49)

$dv/d\xi = u$. (F.50)

Differentiate:

$d^2u/d\xi^2 = u$, (F.51)

$d^2v/d\xi^2 = v$. (F.52)

The solutions are hyperbolic.

F.5.2 Hyperbolic rotation

The exponential is:

$e^{(K\xi)} = I \cosh \xi + K \sinh \xi$. (F.53)

Therefore:

$u(\xi) = u_0 \cosh \xi + v_0 \sinh \xi$, (F.54)

$v(\xi) = u_0 \sinh \xi + v_0 \cosh \xi$. (F.55)

Unlike ordinary rotation, hyperbolic motion preserves:

$u^2 - v^2$. (F.56)

Indeed:

$u(\xi)^2 - v(\xi)^2 = u_0^2 - v_0^2$. (F.57)

The Euclidean amplitude:

$u^2 + v^2$ (F.58)

is not preserved.

F.5.3 Split-complex representation

Introduce:

$j^2 = +1$. (F.59)

Define:

$z_H = u + jv$. (F.60)

Then:

$e^{(j\xi)} = \cosh \xi + j \sinh \xi$. (F.61)

The evolution is:

$z_H(\xi) = e^{(j\xi)} z_H(0)$. (F.62)

The conjugation is:

$\bar{z}_H = u - jv$. (F.63)

The norm is:

$z_H \bar{z}_H = u^2 - v^2$. (F.64)

This algebra naturally represents amplification–contraction pairs and saddle geometry.

F.5.4 Expanding and contracting eigen-directions

Define:

$p = u + v$, (F.65)

$m = u - v$. (F.66)

Then:

$dp/d\xi = p$, (F.67)

$dm/d\xi = -m$. (F.68)

Therefore:

$p(\xi) = p_0 e^{\xi}$, (F.69)

$m(\xi) = m_0 e^{(-\xi)}$. (F.70)

One direction expands while the other contracts.

This is the defining hyperbolic structure.

F.5.5 Hyperbolic readout closure

Let:

$Y(x) = a(x)$. (F.71)

Define:

$\mathscr{H}_Y(x) = a(Kx)$. (F.72)

Under:

$dx/d\xi = Kx$, (F.73)

we have:

$dY/d\xi = \mathscr{H}_Y$. (F.74)

Differentiate:

$d\mathscr{H}_Y/d\xi$
$= a(K^2x)$. (F.75)

Since:

$K^2 = I$, (F.76)

then:

$d\mathscr{H}_Y/d\xi = Y$. (F.77)

Therefore:

$d/d\xi\ [Y; \mathscr{H}_Y]$
$= [0\ 1; 1\ 0][Y; \mathscr{H}_Y]$. (F.78)

The cycle does not return periodically.

It produces:

$Y \to \mathscr{H}_Y \to Y$ (F.79)

under repeated operator action, while continuous evolution amplifies and contracts along eigen-directions.

F.5.6 Hyperbolic Completion Principle

Proposition F.2 — Hyperbolic Readout Closure [F]

If:

$K^2 = I$ (F.80)

and:

$dx/d\xi = Kx$, (F.81)

then for:

$Y = a(x)$, (F.82)

$\mathscr{H}_Y = a(Kx)$, (F.83)

we have:

$dY/d\xi = \mathscr{H}_Y$, (F.84)

$d\mathscr{H}_Y/d\xi = Y$. (F.85)

The completed readout is naturally split-complex:

$Z_H = Y + j\mathscr{H}_Y$. (F.86)

F.5.7 Operational signatures of hyperbolic Relation

Evidence may include:

- two real eigenvalues of opposite sign;
- expansion and contraction directions;
- increasing polarization;
- self-reinforcing selection;
- leverage or confidence amplification;
- unstable positive feedback;
- threshold approach to runaway behaviour.

A hyperbolic self-referential loop may have the form:

Trace
→ confirming policy
→ stronger same-direction movement
→ stronger trace. (F.87)

This is structurally different from an elliptic corrective loop:

Trace
→ counter-response
→ reversal tendency
→ bounded circulation. (F.88)

F.5.8 Hyperbolic finance interpretation

A financial system may become hyperbolic when:

Price increase
→ collateral increase
→ borrowing increase
→ further demand
→ further price increase. (F.89)

The opposite contracting direction may correspond to:

Price decrease
→ margin pressure
→ forced sale
→ further price decrease. (F.90)

This does not mean the CAPM `R/Q` plane itself becomes split-complex.

It means the broader market generator may enter a hyperbolic regime.

The CAPM readout geometry and market dynamic regime must remain distinct until a coupling model is demonstrated.

F.5.9 Hyperbolic institutional interpretation

An institution may exhibit:

Success metric increase
→ greater authority and resources
→ stronger metric-producing capacity
→ further success metric increase. (F.91)

Meanwhile alternative capabilities contract.

This can produce a saddle-like institutional world:

- one officially rewarded direction expands;
- unmeasured functions decay.

Again, this is a structural hypothesis requiring operational data.

F.6 Comparative Signed-Relation Grammar

F.6.1 Canonical operators

Type	Operator square	Canonical unit	Exponential
Elliptic	$J^2 = -I$	$i^2 = -1$	$\cos \xi + i \sin \xi$
Parabolic	$N^2 = 0$	$\varepsilon^2 = 0$	$1 + \varepsilon\xi$
Hyperbolic	$K^2 = I$	$j^2 = 1$	$\cosh \xi + j \sinh \xi$

F.6.2 Readout equations

Type	First response	Second response
Elliptic	dY/dξ = −$\mathcal{Q}$_Y	d$\mathcal{Q}$_Y/dξ = Y
Parabolic	dY/dξ = −$\mathcal{P}$_Y	d$\mathcal{P}$_Y/dξ = 0
Hyperbolic	dY/dξ = $\mathcal{H}$_Y	d$\mathcal{H}$_Y/dξ = Y

F.6.3 Behavioural interpretation

Type	Local behaviour	Return character
Elliptic	circulation, phase, quadrature	corrective or sign-reversing return
Parabolic	accumulation, drift, shear	absent or degenerate return
Hyperbolic	amplification and contraction	confirmatory return

F.6.4 Invariants

Elliptic

$u^2 + v^2 = \text{constant}$. (F.92)

Parabolic

The nilpotent direction itself is conserved while another coordinate accumulates.

Hyperbolic

$u^2 - v^2 = \text{constant}$. (F.93)

The invariant reveals the geometry.

A Euclidean norm should not be imposed on all three branches.

F.7 Damping and Drift in the Three Branches

F.7.1 Elliptic with damping

dz_E/dt = (α + iω)z_E. (F.94)

F.7.2 Parabolic with decay

dz_P/dt = (α + εη)z_P. (F.95)

Since:

$\varepsilon^2 = 0$, (F.96)

the solution contains:

e^(αt)(1 + εηt). (F.97)

The system may decay while preserving a transient accumulated shear.

F.7.3 Hyperbolic with common drift

$dz_H/dt = (\alpha + j\kappa)z_H$. (F.98)

The expanding and contracting rates are:

$\alpha + \kappa$ (F.99)

and:

$\alpha - \kappa$. (F.100)

Both may decay if:

$\alpha < -|\kappa|$. (F.101)

Both may grow if:

$\alpha > |\kappa|$. (F.102)

A hyperbolic algebra does not require one direction to grow absolutely.

It requires differential expansion and contraction relative to the common drift.

F.8 Regime Transitions

F.8.1 Signed parameter transition

Let:

$\chi = \chi(L, \mathscr{R}, P)$. (F.103)

Historical return may shift:

$\chi < 0$
$\rightarrow \chi = 0$
$\rightarrow \chi > 0$. (F.104)

This gives an elliptic–parabolic–hyperbolic transition.

Possible interpretation:

stable corrective circulation
→ critical slowing and accumulation
→ runaway self-confirmation. (F.105)

The transition is a general mathematical possibility.

Each domain must identify the actual generator and control parameter.

F.8.2 Discriminant transition

For a 2×2 generator:

$\Delta_A = \mathrm{tr}(A)^2 - 4\det(A)$. (F.106)

The branches are:

$\Delta_A < 0 \Rightarrow$ elliptic. (F.107)

$\Delta_A = 0 \Rightarrow$ parabolic boundary. (F.108)

$\Delta_A > 0 \Rightarrow$ hyperbolic or node-like real spectrum. (F.109)

A gate or ledger event may change the coefficients of A and move the system across these boundaries.

F.8.3 Critical residual threshold

Suppose:

$A = A(\Gamma)$. (F.110)

A transition threshold Γ^* may satisfy:

$\Delta_A(\Gamma^*) = 0$. (F.111)

Then:

$\Gamma < \Gamma^* \Rightarrow$ elliptic regime. (F.112)

$\Gamma = \Gamma^* \Rightarrow$ parabolic critical regime. (F.113)

$\Gamma > \Gamma^* \Rightarrow$ hyperbolic regime. (F.114)

This is a candidate mechanism for residual-induced regime change.

It is not established universally by the residual source, which presents Γ only as a conditional modelling interface.

F.9 Why Ordinary Complex Notation Must Sometimes Be Rejected

Ordinary complex notation should be rejected when:

1. the operator square is approximately $+I$ rather than $-I$;
2. the operator is nilpotent;
3. no bounded phase cycle exists;
4. hyperbolic invariants predict data better;
5. a drift model closes more accurately;
6. the apparent phase is created by plotting convention;

7. regime switching invalidates a single algebra.

The correct scientific result may be:

> The system is self-referential and time-bearing, but its active Relation is parabolic or hyperbolic rather than ordinarily complex.

This is fully compatible with the general framework.

F.10 Relation Algebra Does Not Determine Commitment

An elliptic, parabolic, or hyperbolic state movement does not by itself determine:

- which event is admitted;
- who has authority;
- what enters the ledger;
- what remains residual.

For every algebra:

Relation movement
→ candidate consequence
→ gate
→ ledger + residual. (F.115)

Thus:

Elliptic phase ≠ Commitment. (F.116)

Parabolic accumulation ≠ Ledger. (F.117)

Hyperbolic amplification ≠ Institutional admission. (F.118)

The signed algebra belongs to Relation.

Commitment remains a separate functional layer.

F.11 Appendix F summary

The unified grammar is:

$C\chi^2 = \chi I$. (F.119)

Its principal branches are:

$\chi < 0$
⇒ elliptic conjugacy
⇒ ordinary complex phase. (F.120)

$\chi = 0$

⇒ parabolic accumulation
⇒ nilpotent completion. (F.121)

$\chi > 0$
⇒ hyperbolic amplification–contraction
⇒ split-complex completion. (F.122)

The central methodological lesson is:

> **Self-reference may motivate a second relational coordinate, but the generator—not analogy—determines whether the correct completion is complex, dual, split-complex, mixed, or simply higher-dimensional real.**

Appendix G — Local Complex Charts, Transport, Curvature, and Frame Reconciliation

G.1 Purpose

The main framework treats ordinary complex structure as local unless global compatibility has been demonstrated.

This restriction is necessary because:

- the Effective Base may change regime;
- the active mode plane may tilt;
- the generator may change;
- the readout covector may change;
- the gate may alter the available state space;
- a commitment may destroy the former elliptic mode.

Therefore, even when a neighbourhood around state x admits:

$J_x^2 = -I$, (G.1)

a distant neighbourhood around state y may possess:

$J_\gamma^2 = -I$ (G.2)

with a differently oriented mode plane.

It may instead possess:

$K_\gamma^2 = I$, (G.3)

$N_\gamma^2 = 0$, (G.4)

or no stable two-dimensional reduction.

The Handoff therefore places local charts, transport, gauge-like residual, curvature, and holonomy in an optional advanced layer rather than the mandatory world-forming core.

G.2 Local mode charts

Let the Effective Base space be a manifold or locally smooth state space:

$\mathcal{M}$. (G.5)

For each point $x \in \mathcal{M}$ inside an elliptic regime, let:

$E_x \subset T_x\mathcal{M}$ (G.6)

be a two-dimensional active mode plane.

Define:

$J_x: E_x \to E_x$, (G.7)

with:

$J_x^2 = -I_x$. (G.8)

A local complex chart is:

$\psi_x: U_x \to \mathbb{C} \times \mathbb{R}^m$, (G.9)

where:

- U_x is a neighbourhood of x;
- the $\mathbb{C}$ coordinate represents the elliptic mode;
- the remaining real coordinates represent other local directions.

For a three-dimensional Effective Base:

$\psi_x: U_x \to \mathbb{C} \times \mathbb{R}$. (G.10)

This realizes the local decomposition:

$\mathbb{R}^3 \cong \mathbb{C} \oplus \mathbb{R}$. (G.11)

The decomposition is local and generator-dependent.

It does not imply that the entire three-dimensional Base possesses one global ordinary complex structure.

G.3 Chart validity

A local complex chart is valid only while:

1. the Effective Base remains predictively sufficient;
2. the local generator remains approximately stable;
3. the elliptic plane remains spectrally separated;
4. the readout remains nondegenerate;

5. no commitment-triggered regime change occurs;

6. the orientation convention remains declared.

Let the validity indicator be:

$\mathrm{ValidComplexChart}_x(P, r, \varepsilon) \in \{0,1\}$. (G.12)

A chart should be withdrawn when any activation condition fails.

G.4 Overlapping charts

Suppose two valid local charts overlap:

$U_x \cap U_\gamma \neq \emptyset$. (G.13)

The transition map is:

$g_{\gamma x} = \psi_\gamma \circ \psi_x^{-1}$. (G.14)

For the local complex structures to be compatible, the transition map must preserve the elliptic orientation on the active mode.

At the tangent level, let:

$T_{\gamma \leftarrow x}: E_x \to E_\gamma$ (G.15)

be the transport map.

Complex compatibility requires:

$T_{\gamma \leftarrow x} J_x = J_\gamma T_{\gamma \leftarrow x}$. (G.16)

This is the transport analogue of the readout intertwining relation.

If Equation (G.16) holds, a quarter-turn at x is transported into a quarter-turn at y.

The Handoff identifies precisely this relation as the condition for transport to preserve local complex structure.

G.5 Complex-linear transport

Transport is complex-linear when:

$$T_{\gamma \leftarrow x}[(\alpha I + \beta J_x)v] = (\alpha I + \beta J_\gamma) T_{\gamma \leftarrow x} v \quad \text{(G.17)}$$

for all real α, β, and all $v \in E_x$.

Using Equation (G.16):

$$T_{\gamma \leftarrow x}(\alpha v + \beta J_x v) = \alpha T_{\gamma \leftarrow x} v + \beta J_\gamma T_{\gamma \leftarrow x} v. \quad \text{(G.18)}$$

Thus a complex amplitude can be compared across local mode planes without destroying its

quarter-turn grammar.

G.6 Transport failure

Define the complex-compatibility error:

$$\varepsilon_J(x,y) = \| T_{\gamma \leftarrow x} J_x - J_\gamma T_{\gamma \leftarrow x} \|. \quad \text{(G.19)}$$

A normalized version is:

$$\tilde{\varepsilon}_J(x,y) = \frac{\| T_{\gamma \leftarrow x} J_x - J_\gamma T_{\gamma \leftarrow x} \|}{[\| T_{\gamma \leftarrow x} \| \| J_x \| + \| J_\gamma \| \| T_{\gamma \leftarrow x} \|]}. \quad \text{(G.20)}$$

Transport is approximately complex-compatible when:

$$\tilde{\varepsilon}_J(x,y) \le \varepsilon_{\text{transport}}. \quad \text{(G.21)}$$

Large error may indicate:

- chart misalignment;
- metric change;
- regime transition;
- mode mixing;
- wrong frame map;
- failure of a global complex interpretation.

G.7 Gauge freedom in local phase

A local mode basis is not unique.

Let:

$$z_x = u_x + i v_x. \quad \text{(G.22)}$$

A local phase convention may be changed by:

$$z_x' = e^{(i\alpha_x)} z_x. \quad \text{(G.23)}$$

This changes the coordinate orientation but not the underlying elliptic plane.

The angle α_x is a local gauge choice.

Two observers may therefore describe the same mode using:

$$z_a = e^{(i\alpha_{ab})} z_b. \quad \text{(G.24)}$$

A disagreement in raw phase coordinates need not imply a disagreement about the underlying Relation.

The relevant question is whether a valid frame transformation exists.

G.8 Gauge transformation of the readout pair

Let:

$Z_Y = Y + i\mathcal{Q}_Y$. (G.25)

Under a readout-frame rotation:

$Z_Y' = e^{(i\alpha)}Z_Y$. (G.26)

Then:

$Y' = Y \cos \alpha - \mathcal{Q}_Y \sin \alpha$, (G.27)

$\mathcal{Q}_Y' = Y \sin \alpha + \mathcal{Q}_Y \cos \alpha$. (G.28)

The admitted scalar and conjugate response change with measurement orientation.

The underlying completed mode does not.

This generalizes the CAPM rotated-measurement family:

$M_\varphi(Z) = \mathrm{Re}[e^{(i\varphi)}Z]$. (G.29)

G.9 Frame reconciliation

Two observers may use different:

- mode bases;
- measurement orientations;
- units;
- baselines;
- gate standards;
- ledger access.

Let observer a possess:

(E_a, J_a, a_a, P_a), (G.30)

and observer b possess:

(E_b, J_b, a_b, P_b). (G.31)

A reconciliation map requires at least:

1. a mode transport:

   $T_{b\leftarrow a}: E_a \to E_b$; (G.32)

2. a readout translation:

   $S_{b\leftarrow a}: \mathbb{R}^2_a \to \mathbb{R}^2_b$; (G.33)

3. an event or ledger map:

   $\Lambda_{b\leftarrow a}: L_a \to L_b$. (G.34)

The maps should satisfy:

$T_{b\leftarrow a} J_a \approx J_b T_{b\leftarrow a}$, (G.35)

$S_{b\leftarrow a} \Phi_a \approx \Phi_b T_{b\leftarrow a}$. (G.36)

Equation (G.36) requires agreement between:

- transported dynamic state;
- translated readout;
- observer-specific measurement protocol.

G.10 Declaration transport

A related source defines transport between declared worlds:

$T_{i\to j}: L_{D_i} \to L_{D_j}$. (G.37)

A relation is stable across declarations when:

$\mathrm{Dist}[T_{i\to j}(R_i), R_j] \le \varepsilon_R$. (G.38)

It interprets objectivity not as the absence of viewpoint, but as invariance under admissible declaration transport.

The present article adopts the narrower operational principle:

Definition G.1 — Transport-stable relation [D]

A relation is transport-stable across an admissible family of protocols when its translated representation agrees within declared tolerance across those protocols. (G.39)

This does not require every protocol to be admitted.

The admissible family must preserve the intended object of comparison.

G.11 Readout invariance

Suppose observer a reports:

$Z_Y^{(a)} = Y_a + i\mathcal{Q}_a$. (G.40)

Observer b reports:

$Z_Y^{(b)} = Y_b + i\mathcal{Q}_b$. (G.41)

A transport-covariant readout relation may be:

Z_Y^(b) ≈ c_ba e^(iα_ba)Z_Y^(a), (G.42)

where:

- c_ba converts units or scale;
- α_ba reconciles orientation.

The invariant amplitude ratio is:

|Z_Y^(b)|/|Z_Y^(a)| ≈ |c_ba|. (G.43)

The invariant phase difference after reconciliation is:

arg Z_Y^(b) – arg Z_Y^(a) – α_ba ≈ 0. (G.44)

Failure may arise from:

- incompatible protocols;
- different active modes;
- non-equivalent baselines;
- actual regime change;
- inaccessible information.

G.12 Ledger transport

The same event may appear differently in different ledgers.

Let:

L^(market), L^(accounting), L^(legal), L^(regulatory) (G.45)

represent four possible financial ledgers.

A market movement may be committed immediately in the market ledger but later in the accounting or legal ledger.

Define:

Λ_accounting←market: L^(market) → L^(accounting). (G.46)

The map may be:

- delayed;
- partial;
- thresholded;
- lossy;
- non-invertible.

The translation residual is:

$\mathscr{R}$_transport
= Event_market − Λ_accounting←market(Event_market). (G.47)

This residual is distinct from the conjugate exposure that preceded the market movement.

G.13 Transport as a gate

Transport between worlds may itself act as a Commitment gate.

A trace in world `A` becomes merely a candidate in world `B`:

e_A → c_B. (G.48)

Then:

e_B = $\mathcal{G}$_B(c_B). (G.49)

Thus:

Committed in A ≠ Committed in B. (G.50)

Examples include:

- economic loss versus accounting loss;
- scientific preprint versus accepted result;
- private memory versus legal evidence;
- AI proposal versus executed tool action.

Cross-world transport is therefore not a simple coordinate conversion.

It may contain a new gate.

G.14 Connection for infinitesimal transport

Let local mode coordinates vary smoothly with state parameter `s`.

A transported mode vector `z(s)` may satisfy:

Dz/Ds = dz/ds + $\mathcal{A}$_s z. (G.51)

Here:

$\mathcal{A}$_s (G.52)

is a connection term recording how the local frame changes along the path.

For a pure `U(1)`-like phase gauge:

$\mathcal{A}$_s = iA_s. (G.53)

Under gauge change:

$z' = e^{(i\alpha(s))}z$, (G.54)

the connection transforms as:

$A_{s'} = A_s - d\alpha/ds$. (G.55)

This structure should be introduced only after local complex charts have been empirically justified.

It is not part of the mandatory framework.

G.15 Readout covector transport

The readout covector may also vary:

$a_x \in E_x^*$. (G.56)

Transporting a state without transporting its readout can change the meaning of Y.

The compatible dual transport is:

$a_\gamma = (T_{\gamma \leftarrow x}^{-1})^* a_x$, (G.57)

where $*$ denotes the dual map.

Then:

$a_\gamma(T_{\gamma \leftarrow x} v) = a_x(v)$. (G.58)

For the conjugate response:

$$\mathcal{Q}_{Y,\gamma}(T_{\gamma \leftarrow x} v) = -a_\gamma(J_\gamma T_{\gamma \leftarrow x} v). \quad \text{(G.59)}$$

If complex compatibility holds:

$J_\gamma T_{\gamma \leftarrow x} = T_{\gamma \leftarrow x} J_x$, (G.60)

then:

$$\mathcal{Q}_{Y,\gamma}(T_{\gamma \leftarrow x} v) = -a_x(J_x v) = \mathcal{Q}_{Y,x}(v). \quad \text{(G.61)}$$

Thus conjugate exposure is transport-invariant only when both:

- the mode structure;
- the readout covector;

are transported consistently.

G.16 Curvature

Transport may depend on the path taken through state or protocol space.

Let two infinitesimal directions be:

μ, ν. (G.62)

For connection components:

$\mathcal{A}_\mu$, $\mathcal{A}_\nu$, (G.63)

the curvature is:

$$\boldsymbol{\mathcal{F}}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu]. \quad (G.64)$$

For an Abelian phase connection:

$[\mathcal{A}_\mu, \mathcal{A}_\nu] = 0$, (G.65)

so:

$\boldsymbol{\mathcal{F}}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$. (G.66)

Nonzero curvature means that local frame reconciliation cannot be made globally path-independent.

G.17 Holonomy

Consider a closed path:

$x_0 \to x_1 \to \cdots \to x_n = x_0$. (G.67)

The transport around the loop is:

$$H_{\text{loop}} = T_{x_0 \leftarrow x_{n-1}} \cdots T_{x_2 \leftarrow x_1} T_{x_1 \leftarrow x_0}. \quad (G.68)$$

If:

$H_{\text{loop}} = I$, (G.69)

the transported object returns unchanged.

If:

$H_{\text{loop}} \neq I$, (G.70)

the loop has nontrivial holonomy.

For a one-dimensional complex mode:

$H_{\text{loop}} = e^{i\Omega_{\text{loop}}}$. (G.71)

The angle Ω_{loop} is a transport phase accumulated around the loop.

G.18 Operational interpretation of holonomy

In an applied domain, nontrivial holonomy may represent path dependence.

Possible examples include:

- converting valuation across several baselines and returning to the first;
- moving a claim through market, accounting, legal, and regulatory ledgers;
- translating an observer record through several institutional frames;
- applying sequential policy revisions in different orders.

The operational residual is:

$\mathscr{R}$_loop = FinalTransportedObject − InitialObject. (G.72)

A nonzero $\mathcal{R}_{\texttt{loop}}$ may indicate:

- transaction cost;
- basis difference;
- tax or legal path dependence;
- protocol inconsistency;
- irreversible commitment;
- genuine curvature of the effective representation.

These are possible interpretations, not universal identities.

G.19 Gate-induced curvature

Suppose two commitment operations are:

$\mathcal{G}$_A, $\mathcal{G}$_B. (G.73)

If:

$\mathcal{G}$_B$\mathcal{G}$_A(c) ≠ $\mathcal{G}$_A$\mathcal{G}$_B(c), (G.74)

the order of admission matters.

This produces a noncommutative commitment residual:

$\mathscr{R}$_AB(c)
= $\mathcal{G}$_B$\mathcal{G}$_A(c) − $\mathcal{G}$_A$\mathcal{G}$_B(c). (G.75)

Possible examples include:

- approval before disclosure versus disclosure before approval;
- settlement before legal classification versus classification before settlement;
- model deployment before audit versus audit before deployment.

Such order dependence is a Commitment analogue of curvature.

It should not automatically be identified with geometric curvature unless a formal connection is

constructed.

G.20 Protocol curvature

Let protocol coordinates be:

p^μ. (G.76)

Changing:

- boundary;
- baseline;
- gate threshold;
- observer access;

in different orders may produce different final operational states.

The commutator is:

[δ_μ, δ_ν]$\mathcal{W}$
= δ_μδ_ν$\mathcal{W}$ − δ_νδ_μ$\mathcal{W}$. (G.77)

A nonzero result indicates protocol-order dependence.

This may reveal that the proposed Effective Base omitted:

- commitment order;
- authority sequence;
- branch provenance;
- transport residual.

G.21 Complex structure disappearance

A path may leave the elliptic region.

Suppose:

Δ_A(x) < 0 (G.78)

along one segment, but:

Δ_A(y) ≥ 0 (G.79)

after a regime transition.

Then J_γ may no longer exist.

Transport cannot preserve ordinary complex structure through that point without analytic extension or a larger real representation.

The correct action is:

1. terminate the complex chart;
2. record the transition;
3. switch to a parabolic, hyperbolic, jump, or mixed representation;
4. preserve the ledger and residual generated by the transition.

G.22 Chart-transition gate

A regime transition may itself be treated as a gate event:

Candidate regime change
→ transition test
→ New Regime ID. (G.80)

The ledger records:

L_{k+1}
$= L_k \oplus \mathrm{Trace}(r_old \to r_new)$. (G.81)

Residual records:

- uncertainty in transition time;
- mixed-mode interval;
- failed chart transport;
- discarded phase convention.

This prevents silent continuation of an invalid complex model.

G.23 Empirical transport test

To test transport between local mode planes:

1. estimate E_x, J_x;
2. estimate E_y, J_y;
3. fit transport $T_{y \leftarrow x}$;
4. compute $\tilde{\varepsilon}_J$;
5. transport readout covectors;
6. compare predicted and observed conjugate responses;
7. perform a closed-loop test where possible.

Required metrics include:

$\tilde{\varepsilon}_J$, (G.82)

ε_readout
$= \|S_{\gamma \leftarrow x}\Phi_x - \Phi_\gamma T_{\gamma \leftarrow x}\|$, (G.83)

ε_loop
$= \|H_loop - I\|$. (G.84)

G.24 Objectivity through transport

A relation can be called operationally objective across a family of observers only when it survives:

- frame translation;
- protocol reconciliation;
- access auditing;
- residual disclosure.

A possible criterion is:

Objective_P(R)
$\Leftrightarrow$ Dist[T_j←i(R_i), R_j] ≤ ε_R (G.85)

for all declared admissible pairs `(i,j)`.

This agrees with the source principle that objectivity is declared invariance across admissible disclosure changes rather than a perspective-free raw observation.

G.25 Nonclaims

This appendix does not claim:

1. that every operational world is a differentiable manifold;
2. that every local phase plane forms a fiber bundle;
3. that every protocol difference is a gauge transformation;
4. that every path dependence is curvature;
5. that financial basis is physical gauge curvature;
6. that legal disagreement is geometric holonomy;
7. that local complex charts always glue into a global structure;
8. that advanced differential geometry improves every implementation.

The geometry is useful only when it produces:

- operational maps;
- measurable residuals;

- stronger prediction;
- clearer frame reconciliation.

No topology should be added merely as decoration.

G.26 Appendix G summary

The advanced sequence is:

Local Effective Base
→ local elliptic mode
→ local complex chart
→ transport map
→ complex-compatibility test
→ readout and ledger reconciliation
→ closed-loop transport
→ curvature or holonomy, if detected. (G.86)

The governing principle is:

> **A local complex structure becomes globally meaningful only when mode, orientation, readout, and commitment records can be transported consistently across admissible states and frames.**

Appendix H — CAPM Conjugate Valuation: Full Derivations and Audit Equations

H.1 Scope

This appendix collects the CAPM derivations needed by the general framework.

CAPM remains one worked example—especially of:

- declared scalar valuation;
- exact conjugate completion;
- elliptic readout geometry;
- measurement–movement separation;
- gate–ledger distinction.

It is not the universal source of the framework.

The derivations below are established inside the declared CAPM valuation construction. The CAPM source repeatedly emphasizes that `Q` is a phase-exposure coefficient rather than an independent loss or residual variable.

H.2 Declared financial protocol

For a future cash flow:

CF_t, (H.1)

declare:

- baseline discount rate `r_base`;
- CAPM beta β;
- equity risk premium `ERP`;
- horizon `t`.

The CAPM required return is:

$r = r_CAPM = r_base + \beta ERP$. (H.2)

The construction requires:

$1 + r_base > 0$, (H.3)

$1 + r > 0$. (H.4)

For the principal Euclidean domain, it also requires:

$0 \leq R_t/A_t \leq 1$. (H.5)

H.3 Baseline amplitude

Define:

$A_t = CF_t/(1 + r_base)^t$. (H.6)

A_t is the value produced by the declared baseline protocol.

It is not automatically:

- intrinsic value;
- fair value;
- best possible value;
- opportunity cost benchmark.

Those interpretations require additional assumptions.

H.4 CAPM-admitted value

Define:

$R_t = CF_t/(1 + r)^t$. (H.7)

Substitute Equation (H.2):

R_t
$= CF_t/[1 + r_base + \beta ERP]^t$. (H.8)

For:

$\beta ERP \geq 0$, (H.9)

we have:

$r \geq r_base$, (H.10)

and consequently:

$R_t \leq A_t$. (H.11)

H.5 Valuation ratio

The ratio is:

$c_t = R_t/A_t$. (H.12)

Substituting Equations (H.6) and (H.7):

c_t
$= [(1 + r_base)/(1 + r)]^t$. (H.13)

The ratio is dimensionless.

In the principal domain:

$0 \leq c_t \leq 1$. (H.14)

H.6 Valuation phase

Define:

$\cos \theta_t = c_t$. (H.15)

Therefore:

θ_t
$= \arccos\{[(1 + r_base)/(1 + r)]^t\}$. (H.16)

For the principal branch:

$0 \leq \theta_t \leq \pi/2$. (H.17)

The phase is generated by the declared valuation ratio.

It is not chosen independently.

The CAPM source explicitly states that the construction would become arbitrary if θ were introduced independently of the CAPM ratio.

H.7 Conjugate coordinate

Define:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (H.18)

Using:

$R_t = A_t \cos \theta_t$, (H.19)

we obtain:

$Q_t = A_t \sin \theta_t$. (H.20)

Therefore:

$A_t^2 = R_t^2 + Q_t^2$. (H.21)

Both R_t and Q_t have currency units.

H.8 Completed valuation state

Define:

$Z_t = R_t + iQ_t$. (H.22)

Then:

Z_t
$= A_t(\cos \theta_t + i \sin \theta_t)$. (H.23)

Therefore:

$Z_t = A_t e^{(i\theta_t)}$. (H.24)

Its amplitude is:

$|Z_t| = A_t$. (H.25)

Its admitted real projection is:

$\mathrm{Re}\, Z_t = R_t$. (H.26)

Its conjugate quadrature is:

$\mathrm{Im}\, Z_t = Q_t$. (H.27)

H.9 Conjugate Risk Theorem

Holding A fixed:

$R(\theta) = A \cos \theta$. (H.28)

Differentiate:

$\partial R/\partial\theta = -A \sin\theta$. (H.29)

Since:

$Q = A \sin\theta$, (H.30)

we obtain:

Theorem H.1 — CAPM Conjugate Risk [E]

$\partial R/\partial\theta = -Q$. (H.31)

Similarly:

$\partial Q/\partial\theta = R$. (H.32)

The signed Phase Delta is:

$\Delta_\theta \equiv \partial R/\partial\theta = -Q$. (H.33)

Thus Q is the magnitude of first-order currency exposure to positive movement in the declared valuation phase.

This is the central exact result of the CAPM source.

H.10 Second and higher derivatives

Starting from:

$R = A \cos\theta$, (H.34)

the derivative hierarchy is:

$\partial R/\partial\theta = -Q$, (H.35)

$\partial^2 R/\partial\theta^2 = -R$, (H.36)

$\partial^3 R/\partial\theta^3 = Q$, (H.37)

$\partial^4 R/\partial\theta^4 = R$. (H.38)

Similarly:

$\partial Q/\partial\theta = R$, (H.39)

$\partial^2 Q/\partial\theta^2 = -Q$. (H.40)

The derivative cycle is:

$R \to -Q \to -R \to Q \to R$. (H.41)

This is a measurement-orientation cycle.

It is not automatically a chronological price path.

H.11 Quarter-turn operator

Define:

J_fin = [0 −1; 1 0]. (H.42)

For valuation vector:

v = [R; Q], (H.43)

we have:

J_finv = [−Q; R]. (H.44)

Also:

J_fin² = −I, (H.45)

J_fin⁴ = I. (H.46)

The four orientations are:

v, (H.47)

J_finv, (H.48)

−v, (H.49)

−J_finv. (H.50)

H.12 Rotated measurement family

Define:

M_φ(Z) = Re[e^(iφ)Z]. (H.51)

Since:

e^(iφ)Z
= (cos φ + i sin φ)(R + iQ), (H.52)

the real component is:

M_φ(Z)
= R cos φ − Q sin φ. (H.53)

Special cases are:

M0(Z) = R, (H.54)

M_π/2(Z) = −Q, (H.55)

M_π(Z) = −R, (H.56)

M_3π/2(Z) = Q, (H.57)

M_2π(Z) = R. (H.58)

A passive measurement changes ϕ while holding z fixed.

No economic movement is implied.

H.13 Required-return sensitivity

The ordinary CAPM value is:

$R = CF/(1 + r)^t$. (H.59)

Differentiate with respect to r:

$\partial R/\partial r$
$= -tCF/(1 + r)^{(t+1)}$. (H.60)

Using:

$R = CF/(1 + r)^t$, (H.61)

we obtain:

$\partial R/\partial r = -tR/(1 + r)$. (H.62)

Define the positive required-return exposure:

$D_r \equiv -\partial R/\partial r$. (H.63)

Therefore:

$D_r = tR/(1 + r)$. (H.64)

H.14 Phase sensitivity to required return

Starting from:

$\cos \theta = R/A$, (H.65)

with A fixed relative to changes in r, differentiate:

$-\sin \theta \cdot \partial\theta/\partial r$
$= (1/A)\partial R/\partial r$. (H.66)

Since:

$\sin \theta = Q/A$, (H.67)

we have:

$-(Q/A)\partial\theta/\partial r$
$= (1/A)[-tR/(1 + r)]$. (H.68)

Therefore:

$\partial\theta/\partial r$
$= tR/[(1 + r)Q]$. (H.69)

The phase increases with required return in the principal domain.

H.15 Coordinate equivalence

Using the chain rule:

$$\partial R/\partial r = (\partial R/\partial \theta)(\partial \theta/\partial r). \quad (H.70)$$

Substitute:

$$\partial R/\partial \theta = -Q, \quad (H.71)$$

and:

$$\partial \theta/\partial r = tR/[(1 + r)Q]. \quad (H.72)$$

Then:

$$\partial R/\partial r = -Q \cdot tR/[(1 + r)Q]. \quad (H.73)$$

Hence:

$$\partial R/\partial r = -tR/(1 + r). \quad (H.74)$$

In differential form:

$$-Qd\theta = -[tR/(1 + r)]dr. \quad (H.75)$$

Equivalently:

$$Qd\theta = [tR/(1 + r)]dr. \quad (H.76)$$

The phase representation and ordinary required-return sensitivity express the same local value change in different coordinates.

H.16 Beta sensitivity

Since:

$$r = r_base + \beta ERP, \quad (H.77)$$

we have:

$$\partial r/\partial \beta = ERP. \quad (H.78)$$

Therefore:

$$\partial R/\partial \beta = \partial R/\partial r \cdot \partial r/\partial \beta. \quad (H.79)$$

Thus:

$$\partial R/\partial \beta = -tERP \cdot R/(1 + r). \quad (H.80)$$

The phase sensitivity is:

$$\partial \theta/\partial \beta = \partial \theta/\partial r \cdot ERP. \quad (H.81)$$

Therefore:

$$\partial \theta/\partial \beta = tERP \cdot R/[(1 + r)Q]. \quad (H.82)$$

Taking the ratio:

$$(\partial R/\partial \beta)/(\partial \theta/\partial \beta) = -Q. \quad (H.83)$$

So a beta bump recovers the same conjugate phase exposure when translated into the corresponding phase movement. The CAPM source derives this explicitly.

H.17 ERP sensitivity

Holding `r_base` and `β` fixed:

$$\partial r/\partial ERP = \beta. \quad (H.84)$$

Therefore:

$$\partial R/\partial ERP = -t\beta R/(1 + r). \quad (H.85)$$

Also:

$$\partial \theta/\partial ERP = t\beta R/[(1 + r)Q]. \quad (H.86)$$

Hence:

$$(\partial R/\partial ERP)/(\partial \theta/\partial ERP) = -Q. \quad (H.87)$$

Required-return, beta, and ERP bumps recover the same phase exposure when they move the valuation state along the same declared CAPM orbit.

H.18 Horizon sensitivity

Treat `t` as continuous for sensitivity analysis.

Write:

$$R = CF(1 + r)^{(-t)}. \quad (H.88)$$

Then:

$$\partial R/\partial t = -R \ln(1 + r), \quad \text{(H.89)}$$

assuming `CF` is held fixed.

For the ratio:

$$\cos \theta = [(1 + r_base)/(1 + r)]^t, \quad \text{(H.90)}$$

take logarithms:

$$\ln \cos \theta = t \ln[(1 + r_base)/(1 + r)]. \quad \text{(H.91)}$$

Differentiate with respect to `t`:

$$-\tan \theta \cdot \partial\theta/\partial t = \ln[(1 + r_base)/(1 + r)]. \quad \text{(H.92)}$$

Therefore:

$$\partial\theta/\partial t = -\cot \theta \cdot \ln[(1 + r_base)/(1 + r)]. \quad \text{(H.93)}$$

Since:

$$\cot \theta = R/Q, \quad \text{(H.94)}$$

we obtain:

$$\partial\theta/\partial t = -(R/Q)\ln[(1 + r_base)/(1 + r)]. \quad \text{(H.95)}$$

Because the logarithm is nonpositive when `r ≥ r_base`, the phase generally increases with horizon.

H.19 Baseline-rate sensitivity

The baseline rate affects both `A` and `r` when:

$$r = r_base + \beta ERP. \quad \text{(H.96)}$$

Therefore:

$$\partial A/\partial r_base = -tA/(1 + r_base). \quad \text{(H.97)}$$

Also:

$$\partial R/\partial r_base = -tR/(1 + r). \quad \text{(H.98)}$$

The ratio phase is:

$$\cos \theta = [(1 + r_base)/(1 + r_base + \beta ERP)]^t. \quad \text{(H.99)}$$

Differentiate its logarithm:

∂ ln cos θ/∂r_base
= t[1/(1 + r_base) − 1/(1 + r)]. (H.100)

Then:

−tan θ · ∂θ/∂r_base
= t[1/(1 + r_base) − 1/(1 + r)]. (H.101)

Therefore:

∂θ/∂r_base
= −(R/Q)t[1/(1 + r_base) − 1/(1 + r)]. (H.102)

Unlike a pure required-return bump with fixed `A`, a baseline-rate bump changes both amplitude and phase.

The movement must be decomposed into radial and angular components.

H.20 Radial–angular differential

Since:

Z = Ae^(iθ), (H.103)

differentiate:

dZ = e^(iθ)dA + iAe^(iθ)dθ. (H.104)

Divide by `z`:

dZ/Z = dA/A + idθ. (H.105)

For the real and conjugate coordinates:

dR = cos θ dA − A sin θ dθ. (H.106)

Therefore:

dR = (R/A)dA − Qdθ. (H.107)

Similarly:

dQ = sin θ dA + A cos θ dθ. (H.108)

Thus:

dQ = (Q/A)dA + Rdθ. (H.109)

A change in `r_base` generally causes both:

- amplitude movement `dA`;
- phase movement `dθ`.

The simple identity:

dR = −Qdθ (H.110)

holds only along a fixed-amplitude orbit.

H.21 Finite phase movement

For an active phase movement:

Z_new = e^(iΔθ)Z. (H.111)

Then:

R_new
= R cos Δθ − Q sin Δθ. (H.112)

The economic value movement is:

ΔR_econ
= R_new − R. (H.113)

Therefore:

ΔR_econ
= R(cos Δθ − 1) − Q sin Δθ. (H.114)

For small $\Delta\theta$:

cos Δθ
= 1 − (Δθ)²/2 + (Δθ)⁴/24 − ⋯, (H.115)

sin Δθ
= Δθ − (Δθ)³/6 + ⋯. (H.116)

Therefore:

ΔR_econ
= −QΔθ
− (R/2)(Δθ)²

- (Q/6)(Δθ)³
- (R/24)(Δθ)⁴
- O((Δθ)⁵). (H.117)

The first-order term is:

−QΔθ. (H.118)

The second-order term is:

−(R/2)(Δθ)². (H.119)

H.22 Haircut function

Define the scalar CAPM haircut:

$H(\theta) = A - R(\theta)$. (H.120)

Since:

$R(\theta) = A \cos \theta$, (H.121)

we have:

$H(\theta) = A(1 - \cos \theta)$. (H.122)

Differentiate:

$dH/d\theta = A \sin \theta$. (H.123)

Therefore:

$dH/d\theta = Q$. (H.124)

Integrating from zero phase:

$H(\theta) = \int_0^{\theta} Q(\varphi) d\varphi$. (H.125)

Thus:

- H is an accumulated scalar haircut;
- Q is its marginal phase-growth rate.

Therefore:

$Q \neq A - R$. (H.126)

The CAPM source derives this distinction explicitly.

H.23 Opportunity-cost condition

Q may be interpreted as marginal opportunity-cost exposure only if the baseline value A represents a valid best foregone alternative under the declared protocol.

Without that additional condition, the safer interpretation is:

Q
= marginal value removed from the admitted real-axis rate per unit valuation phase. (H.127)

The source describes this as a shadow-sensitivity interpretation rather than an automatically settled cost.

H.24 Multi-cash-flow state

For cash flows indexed by t, define:

$Z = \Sigma_t Z_t$. (H.128)

where:

$Z_t = A_t e^{(i\theta_t)}$. (H.129)

Then:

$R_total = \Sigma_t A_t \cos \theta_t$, (H.130)

$Q_total = \Sigma_t A_t \sin \theta_t$. (H.131)

Thus:

$Z = R_total + iQ_total$. (H.132)

H.25 Common phase movement

If every term experiences the same phase shift ϕ:

$Z_new = e^{(i\varphi)}Z$. (H.133)

Then:

$dR_total/d\varphi \,|_{\{\varphi=0\}}$
$= -Q_total$. (H.134)

Therefore the aggregate conjugate exposure is:

$Q_total = \Sigma_t Q_t$. (H.135)

This requires a common phase movement.

H.26 Term-specific phase movements

If each cash-flow term has its own phase:

$R_total = \Sigma_t A_t \cos \theta_t$. (H.136)

Then:

dR_total
$= -\Sigma_t Q_t d\theta_t$

- $\Sigma_t (R_t/A_t) dA_t$. (H.137)

On fixed-amplitude term orbits:

$dR_total = -\Sigma_t Q_t d\theta_t$. (H.138)

The exposure is therefore a phase gradient:

$\nabla_\theta R_total$
$= (-Q_1, -Q_2, \ldots, -Q_T)$. (H.139)

A single scalar `Q_total` is insufficient when phase movements differ across horizons.

H.27 General valuation covector form

Let the dynamic Effective Base be:

Ξ. (H.140)

Let the valuation map be:

R = V(Ξ). (H.141)

Let the active dynamic phase direction be:

∂Ξ/∂θ_D. (H.142)

Then:

∂R/∂θ_D
= ∇_ΞV · ∂Ξ/∂θ_D. (H.143)

Define:

Q_D
= −∂R/∂θ_D. (H.144)

Therefore:

Q_D
= −∇_ΞV · ∂Ξ/∂θ_D. (H.145)

This is the covector-like interpretation:

- `∂Ξ/∂θ_D` is a tangent direction;
- `∇_ΞV` is a monetary valuation covector;
- `Q_D` is their monetary pairing.

The Handoff identifies this as one of the cleanest general geometric interpretations of conjugate exposure.

H.28 Valuation phase versus dynamic phase

The CAPM valuation phase is:

θ_V
= arccos(R/A). (H.146)

A separately estimated market-dynamic phase may be:

θ_D. (H.147)

They are not automatically equal.

A local coupling coefficient may be:

K_VD = ∂θ_V/∂θ_D. (H.148)

Then:

Q_D

= −∂R/∂θ_D. (H.149)

Using the chain rule:

Q_D
= −(∂R/∂θ_V)(∂θ_V/∂θ_D). (H.150)

Since:

∂R/∂θ_V = −Q_V, (H.151)

we obtain:

Q_D = Q_VK_VD. (H.152)

This relation is formal.

The stability and empirical meaning of K_VD remain open research questions.

H.29 Measurement, movement, gate, and ledger

The complete financial runtime is:

Measurement
→ Exposure
→ State Movement
→ Economic P&L
→ Gate
→ Ledger + Residual. (H.153)

The CAPM source explicitly separates these stages.

Measurement

M_π/2(Z) = −Q. (H.154)

No P&L is generated merely by this readout.

Movement

ΔR_econ
= R(cos Δθ − 1) − Q sin Δθ. (H.155)

Gate

Let:

ΔR_admitted
= $\mathcal{G}$_fin(ΔR_econ; L, Rules). (H.156)

Ledger

L_{k+1}
= $L_k \oplus$ Trace(ΔR_admitted). (H.157)

Residual

ε_gate
= ΔR_econ − ΔR_admitted. (H.158)

Therefore:

Q ≠ ε_gate. (H.159)

`Q` is exposure before movement.

`ε_gate` is incompletely admitted consequence after movement.

H.30 Partial admission

Let:

0 ≤ α ≤ 1. (H.160)

and define:

ΔR_admitted = αΔR_econ. (H.161)

Then:

ε_gate = (1 − α)ΔR_econ. (H.162)

Examples of `α < 1` include:

- partial accounting recognition;
- delayed settlement;
- incomplete collateral realization;
- jurisdictional limitation;
- liquidity-constrained realization.

The admission fraction is protocol-relative.

H.31 Complex residual of model movement

Suppose observed completed-state movement is:

ΔZ_observed. (H.163)

The model predicts:

ΔZ_model. (H.164)

Define:

ε_Z
= ΔZ_observed − ΔZ_model. (H.165)

Write:

$\varepsilon_Z = \varepsilon_R + i\varepsilon_Q$. (H.166)

A large residual may indicate:

- omitted factor;
- liquidity change;
- model break;
- non-CAPM repricing;
- path dependence;
- protocol change;
- data error;
- institutional intervention.

This model residual is distinct from gate residual.

The CAPM source uses residual precisely to limit the geometry's explanatory claim rather than forcing every movement into the CAPM orbit.

H.32 Numerical audit identities

Every implementation should verify:

Discount identities

$A = CF/(1 + r_{base})^t$. (H.167)

$R = CF/(1 + r)^t$. (H.168)

Ratio identity

$\cos\theta = R/A$. (H.169)

Norm identity

$A^2 = R^2 + Q^2$. (H.170)

Polar identity

$Z = Ae^{(i\theta)}$. (H.171)

Derivative identity

$\partial R/\partial\theta = -Q$. (H.172)

Required-return identity

$\partial R/\partial r = -tR/(1 + r)$. (H.173)

Coordinate-equivalence identity

$Q(\partial\theta/\partial r) = tR/(1 + r)$. (H.174)

Operator identity

$J_{fin}^2 = -I$. (H.175)

Measurement identity

$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi$. (H.176)

H.33 Bump-and-revalue audit

Choose a small required-return bump:

δr. (H.177)

Compute:

$R_+ = CF/(1 + r + \delta r)^t$, (H.178)

$R_- = CF/(1 + r - \delta r)^t$. (H.179)

Estimate:

$\partial R/\partial r \,|\{num\}$
$\approx (R+ - R_-)/(2\delta r)$. (H.180)

Similarly compute:

$\theta_+ = \arccos(R_+/A)$, (H.181)

$\theta_- = \arccos(R_-/A)$. (H.182)

Estimate:

$\partial\theta/\partial r \,|\{num\}$
$\approx (\theta+ - \theta_-)/(2\delta r)$. (H.183)

Then:

Q_{num}
$= -(\partial R/\partial r \,|\{num\})$
$/ (\partial\theta/\partial r \,|\{num\})$. (H.184)

Compare:

$\varepsilon_Q = Q_{num} - Q_{analytic}$. (H.185)

The source recommends this style of numerical audit rather than accepting a visually plausible phase construction without derivative verification.

H.34 Domain-failure rules

The principal Euclidean construction fails or requires revision when:

Baseline inversion

$R > A$. (H.186)

Then:

$A^2 - R^2 < 0$. (H.187)

The declared principal real Q is unavailable.

The model should not silently replace Q with an absolute value.

Negative or undefined discount denominator

$1 + r \leq 0$ (H.188)

or:

$1 + r_{base} \leq 0$. (H.189)

Unit mismatch

A and R do not represent comparable cash-flow claims.

Protocol mismatch

The baseline and admitted valuations use different:

- cash flows;
- horizons;
- currencies;
- legal claims;
- inflation bases.

Phase ambiguity

The branch of $\arccos$ is not declared.

Orientation ambiguity

Positive phase direction is not declared.

H.35 Empirical-usefulness test

The algebraic construction may be exact while adding no new empirical information in a one-period static case because:

$Q = \sqrt{(A^2 - R^2)}$. (H.190)

The practical question is whether the representation improves:

- sensitivity attribution;
- multi-horizon decomposition;
- regime diagnosis;
- communication;
- gate-event prediction;
- dynamic coupling;
- intervention design.

Compare:

Model 0: ordinary variables only. (H.191)

Model 1: ordinary variables plus phase representation. (H.192)

If Model 1 adds no operational value, complex valuation remains a valid but optional reparameterization.

H.36 Appendix H summary

The exact CAPM chain is:

CF_t
→ baseline amplitude A_t
→ CAPM value R_t
→ ratio R_t/A_t
→ phase θ_t
→ conjugate exposure Q_t
→ complex readout Z_t. (H.193)

Its principal identity is:

$\partial R/\partial \theta = -Q$. (H.194)

Its central operational correction is:

Q
= exposure before movement. (H.195)

Economic P&L
= consequence of actual movement. (H.196)

Ledgered financial history
= consequence admitted by a gate. (H.197)

Residual
= what the model or gate does not fully integrate. (H.198)

Appendix I — Δ5 Half-Turn Opposition, Spectral Sectors, and Coarse-Grained Relation

I.1 Purpose and scope

The Δ5 architecture is a specialized Relation module defined on a ten-site decagon.

Its role in the general framework is not to establish that every operational world contains:

- ten slots;
- five opposing pairs;
- a universal half-turn;
- a minimum-dissipation Δ5 law.

Its narrower contribution is methodological:

> **A claimed opposition becomes mathematically meaningful only after the relevant operator, invariant sector, cost functional, and stability conditions have been declared.**

The Δ5 source defines:

$T_5: n \mapsto n + 5 \bmod 10$, (I.1)

and distinguishes it from the sum-to-11 reflection:

$R: n \mapsto 11 - n$. (I.2)

It then develops variational, spectral, assignment, dissipative, and coarse-graining arguments for the half-turn pairing within its declared decagon model.

I.2 Decagon state space

Let:

$C_{10} = \{1, 2, \ldots, 10\}$. (I.3)

Associate each site with angle:

$\theta_n = (2\pi/10)(n - 1)$. (I.4)

Let the complex amplitude at site n be:

$a_n = r_n e^{(i\varphi_n)}$, (I.5)

where:

$r_n \geq 0$. (I.6)

The full state is:

$a = (a_1, \ldots, a_{10})^T \in \mathbb{C}^{10}$. (I.7)

The inner product is:

$\langle a, b\rangle = \Sigma_{n=1}{}^{10}\, \bar{a}_n b_n$. (I.8)

The norm is:

$\|a\|^2 = \Sigma_{n=1}{}^{10}\, |a_n|^2$. (I.9)

A global phase transformation:

a → e^(iχ)a (I.10)

does not alter pair-phase differences or quadratic energies constructed from inner products.

I.3 Three distinct symmetries

One-step rotation

Define:

$\rho: n \mapsto n + 1 \bmod 10$. (I.11)

Then:

$\rho^{10} = I$. (I.12)

Half-turn

Define:

$T_5 = \rho^5$. (I.13)

Therefore:

$T_5: n \mapsto n + 5 \bmod 10$. (I.14)

Since:

$\rho^{10} = I$, (I.15)

we have:

$T_5{}^2 = \rho^{10} = I$. (I.16)

Thus τ_5 is an involution.

Sum-to-11 reflection

Define:

$R: n \mapsto 11 - n$. (I.17)

Then:

$R^2 = I$. (I.18)

The two involutions generate different pairings.

I.4 Two pairing structures

Sum-to-11 pairs

The reflection R produces:

(1,10), (2,9), (3,8), (4,7), (5,6). (I.19)

Each pair sums to:

11. (I.20)

Δ5 pairs

The half-turn T_5 produces:

(1,6), (2,7), (3,8), (4,9), (5,10). (I.21)

The two pairings coincide only at:

(3,8). (I.22)

Therefore:

Sum-to-11 balance ≠ Δ5 phase opposition. (I.23)

The source explicitly treats these as logically non-equivalent symmetries.

I.5 Group structure

The full decagon symmetry group is generated by:

$D_{10} = \langle \rho, R \rangle$. (I.24)

Its defining relations may be written:

$\rho^{10} = I$, (I.25)

$R^2 = I$, (I.26)

$R\rho R = \rho^{-1}$. (I.27)

The half-turn:

$T_5 = \rho^5$ (I.28)

is central because:

$\rho T_5 = T_5 \rho$, (I.29)

and:

$RT_5 = T_5R$. (I.30)

The subset:

$\{I, R, T_5, RT_5\}$ (I.31)

forms a Klein four-subgroup.

This establishes that the two involutions can be jointly classified without being identified.

I.6 Δ5 projectors

Since:

$T_5^2 = I$, (I.32)

its eigenvalues are:

$+1$ and -1. (I.33)

Define:

$P_+ = \tfrac{1}{2}(I + T_5)$, (I.34)

$P_- = \tfrac{1}{2}(I - T_5)$. (I.35)

These satisfy:

$P_+^2 = P_+$, (I.36)

$P_-^2 = P_-$, (I.37)

$P_+P_- = 0$, (I.38)

$P_+ + P_- = I$. (I.39)

Every state decomposes as:

$a = a_+ + a_-$, (I.40)

where:

$a_+ = P_+a$, (I.41)

$a_- = P_-a$. (I.42)

The sectors satisfy:

$T_5a_+ = a_+$, (I.43)

$T_5a_- = -a_-$. (I.44)

I.7 Pair coordinates

For each $n = 1,\dots,5$, define:

$e_n = (a_n + a_{n+5})/\sqrt{2}$, (I.45)

$p_n = (a_n - a_{n+5})/\sqrt{2}$. (I.46)

Then:

$a_n = (e_n + p_n)/\sqrt{2}$, (I.47)

$a_{n+5} = (e_n - p_n)/\sqrt{2}$. (I.48)

Here:

- e_n is the symmetric or leakage channel;
- p_n is the antisymmetric or phase-opposed channel.

Under T_5:

$e_n \to e_n$, (I.49)

$p_n \to -p_n$. (I.50)

The ten-dimensional state decomposes into:

$\mathbb{C}^{10} \cong \mathbb{C}^5_\text{symmetric} \oplus \mathbb{C}^5_\text{antisymmetric}$. (I.51)

I.8 Pair-energy functional

Define:

$E_pair(a) = \Sigma_{n=1}^{5} |a_n + a_{n+5}|^2$. (I.52)

Using Equation (I.45):

$E_pair(a) = 2\Sigma_{n=1}^{5} |e_n|^2$. (I.53)

Depending on projector normalization, this may also be written proportionally as:

$E_pair(a) \propto \|P_+ a\|^2$. (I.54)

The source uses this positive quadratic as the variational kernel selecting the antisymmetric sector.

I.9 Variational minimum

Since every term in Equation (I.52) is nonnegative:

$E_pair(a) \geq 0$. (I.55)

The global minimum is:

$E_pair(a) = 0$. (I.56)

This occurs exactly when:

$a_n + a_{n+5} = 0$ (I.57)

for every:

$n = 1,\ldots,5.$ (I.58)

Therefore:

$a_{n+5} = -a_n.$ (I.59)

If both amplitudes are nonzero:

$r_{n+5} = r_n,$ (I.60)

and:

$\varphi_{n+5} - \varphi_n = \pi \bmod 2\pi.$ (I.61)

Proposition I.1 — Δ5 Pair-Energy Minimum [F/E]

The minimizers of `E_pair` are exactly the states in the T_5-antisymmetric sector:

$P_+ a = 0,$ (I.62)

or equivalently:

$T_5 a = -a.$ (I.63)

This conclusion is exact for the declared energy.

It does not show that nature or another domain necessarily minimizes that energy.

I.10 Opposition must be cost-relative

Consider instead the alignment energy:

$E_align(a) = \Sigma_{n=1}^{5} |a_n - a_{n+5}|^2.$ (I.64)

Its minimum occurs when:

$a_{n+5} = a_n.$ (I.65)

Thus the same decagon and the same pairing can select:

- alignment under `E_align`;
- opposition under `E_pair`.

Therefore:

Pairing alone does not determine phase relation. (I.66)

The cost or generator determines which sector is selected.

I.11 Fourier transform on the decagon

Define the discrete Fourier transform:

$\hat{a}_k = (1/\sqrt{10})\Sigma_{n=1}^{10} a_n e^{[-2\pi ik(n-1)/10]}$, (I.67)

for:

$k = 0,\ldots,9$. (I.68)

The inverse transform is:

$a_n = (1/\sqrt{10})\Sigma_{k=0}^{9} \hat{a}_k e^{[2\pi ik(n-1)/10]}$. (I.69)

The half-turn acts on mode k as:

$T_5\hat{a}_k = e^{(\pi ik)}\hat{a}_k$. (I.70)

Since:

$e^{(\pi ik)} = (-1)^k$, (I.71)

we obtain:

$T_5\hat{a}_k = (-1)^k\hat{a}_k$. (I.72)

Therefore:

- even k modes belong to the symmetric sector;
- odd k modes belong to the antisymmetric sector.

Thus:

$\Delta 5$ antisymmetry $\Leftrightarrow$ support only on odd Fourier modes. (I.73)

The source explicitly derives this spectral characterization.

I.12 Half-frequency mode

The mode:

$k = 5$ (I.74)

has:

$a_n \propto e^{[2\pi i \cdot 5(n-1)/10]}$. (I.75)

Therefore:

$a_n \propto e^{[\pi i(n-1)]} = (-1)^{(n-1)}$. (I.76)

Hence:

$a_{n+1} = -a_n$. (I.77)

It follows immediately that:

$a_{n+5} = -a_n$. (I.78)

The $k = 5$ mode is therefore both:

- nearest-neighbour alternating;
- $\Delta 5$ antisymmetric.

However, the full odd sector contains:

$k \in \{1,3,5,7,9\}$. (I.79)

$\Delta 5$ opposition does not uniquely imply $k = 5$.

I.13 Cycle Laplacian

Define the shift:

$(Sa)_n = a_{n-1}$. (I.80)

The cycle Laplacian is:

$L = 2I - S - S^{-1}$. (I.81)

The smoothness energy is:

$E_lap(a) = \langle a, La \rangle$. (I.82)

Equivalently:

$E_lap(a) = \Sigma_{n=1}{}^{10} |a_{n+1} - a_n|^2$. (I.83)

Its Fourier eigenvalues are:

$\lambda_k = 2 - 2\cos(2\pi k/10)$. (I.84)

Equivalently:

$\lambda_k = 4\sin^2(\pi k/10)$. (I.85)

The global minimum is:

$\lambda_0 = 0$. (I.86)

Thus the Laplacian favours the constant mode:

$k = 0$. (I.87)

It does not make the alternating $k = 5$ mode the global ground state.

I.14 Lowest modes inside the odd sector

Restricting to odd k, the Laplacian eigenvalues satisfy:

$\lambda_1 = \lambda_9 < \lambda_3 = \lambda_7 < \lambda_5$. (I.88)

Therefore the half-frequency mode $k = 5$ is not the minimum of pure smoothness even inside the full antisymmetric sector.

This is one of the source's most important internal qualifications:

> Pure smoothness selects low frequency; strict nearest-neighbour alternation requires an anti-alignment energy or an additional constraint.

I.15 Nearest-neighbour anti-alignment

Define:

$E_nn^{+}(a) = \Sigma_{n=1}{}^{10} |a_{n+1} + a_n|^2$. (I.89)

In Fourier space, its eigenvalues are:

$\mu_k = 2 + 2\cos(2\pi k/10)$. (I.90)

Equivalently:

$\mu_k = 4\cos^2(\pi k/10)$. (I.91)

For:

$k = 5$, (I.92)

we have:

$\mu_5 = 0$. (I.93)

Thus the alternating half-frequency mode minimizes the anti-alignment energy.

This demonstrates:

- `E_pair` selects the full odd sector;
- `E_nn+` selects strict nearest-neighbour alternation;
- `E_lap` selects smoothness.

Each cost answers a different Relation question.

I.16 Combined quadratic

A combined energy may be:

$\mathcal{E}(a)$
$= \mu E_pair(a)$

- $\alpha E_lap(a)$
- $\nu E_nn^{+}(a)$
- $B(a)$, (I.94)

with nonnegative weights.

The selected mode depends on:

μ, α, ν, (I.95)

and on the boundary term:

B(a). (I.96)

Strong μ suppresses the symmetric Δ5 channel.

Strong ν favours the k = 5 alternating mode.

Strong α favours smooth low-frequency modes.

No one term should be described as “the” universal decagon energy.

I.17 Boundary term

The source introduces:

$B(a) = \beta_5 |a_5|^2 + \beta_{10} |a_{10}|^2$. (I.97)

This term assigns special costs to sites 5 and 10.

A large:

β_{10} (I.98)

suppresses amplitude at site 10.

This mathematical penalty may support a model-specific pivot/cap interpretation.

It does not follow from the decagon symmetry alone.

The semantic description of 5 as pivot and 10 as cap is therefore an additional model layer.

I.18 Assignment optimality

Let the sites be partitioned into two sets:

$\mathcal{O} = \{1,3,5,7,9\}$, (I.99)

$\mathcal{E} = \{2,4,6,8,10\}$. (I.100)

Suppose assignment costs satisfy:

c_{i,i+5} = 0, (I.101)

and:

$c_{i,j} \geq \gamma > 0$ (I.102)

for every non-Δ5 alternative.

Then the Δ5 assignment has total cost:

C_Δ5 = 0. (I.103)

Every different perfect matching contains at least one non-Δ5 edge, so its cost is at least:

γ. (I.104)

Therefore the Δ5 matching is uniquely optimal under this cost specification.

Important qualification

The optimality is conditional on Equation (I.101)–(I.102).

The cost model already encodes the privileged status of Δ5 intramodality.

The result demonstrates consistency and uniqueness under that declared model; it does not independently prove that all real systems assign zero cost to Δ5 edges.

I.19 Weakly dissipative amplitude dynamics

The source considers dynamics of the form:

$$i\,da_n/dt = \omega_n a_n$$

- $\lambda|a_n|^2 a_n$
- $\kappa_n a_{n+5} - i\Gamma_n a_n$. (I.105)

The terms have the following source-model roles:

- $\omega_n a_n$ — local frequency;
- $\lambda|a_n|^2 a_n$ — nonlinear self-interaction;
- $\kappa_n a_{n+5}$ — Δ5 pair coupling;
- $-i\Gamma_n a_n$ — damping.

The source proposes Δ5-symmetric parameter conditions and a Lyapunov functional intended to drive the system toward the antisymmetric manifold.

I.20 Pair-mode equations

For one Δ5 pair, suppress the index and write:

$$i\,da/dt = \omega a + \kappa b - i\Gamma a, \quad \text{(I.106)}$$

$$i\,db/dt = \omega b + \kappa a - i\Gamma b. \quad \text{(I.107)}$$

Define:

$$e = (a + b)/\sqrt{2}, \quad \text{(I.108)}$$

$$p = (a - b)/\sqrt{2}. \quad \text{(I.109)}$$

Then:

$$i\,de/dt = (\omega + \kappa)e - i\Gamma e, \quad \text{(I.110)}$$

$$i\,dp/dt = (\omega - \kappa)p - i\Gamma p. \quad \text{(I.111)}$$

The symmetric and antisymmetric channels diagonalize the pair coupling.

The coupling splits their frequencies by:

$$2\kappa. \quad \text{(I.112)}$$

Damping alone does not necessarily select one channel over the other when both receive the same Γ.

Selection of the antisymmetric channel requires:

- differential damping;
- energetic penalty on the symmetric channel;
- forcing;
- boundary conditions;
- or another stability mechanism.

This distinction should be retained when interpreting the source's larger Lyapunov model.

I.21 Gradient-flow stabilization

A transparent stabilizing dynamics for the pair energy is:

$$da/dt = -\mu \nabla_{\bar a} E_{pair}(a). \quad \text{(I.113)}$$

Since:

$$E_{pair} = \Sigma_{n=1}^{5} |a_n + a_{n+5}|^2, \quad \text{(I.114)}$$

the symmetric component satisfies:

$$de_n/dt = -2\mu e_n, \quad \text{(I.115)}$$

while the antisymmetric component is unchanged by this term:

$$dp_n/dt = 0. \quad \text{(I.116)}$$

Therefore:

$$e_n(t) = e_n(0)e^{(-2\mu t)}. \quad \text{(I.117)}$$

As:

$$t \to \infty, \quad \text{(I.118)}$$

we obtain:

$$e_n(t) \to 0. \quad \text{(I.119)}$$

Thus:

$a_{n+5} \to -a_n$. (I.120)

This provides a direct mathematical example of asymptotic locking when the dynamics explicitly descend the declared pair energy.

I.22 Lyapunov condition

Let:

V(a) = E_pair(a) + αE_lap(a) + B(a). (I.121)

A Lyapunov claim requires:

dV/dt ≤ 0. (I.122)

To conclude convergence toward Δ5 lock, one additionally requires that the largest invariant subset satisfying:

dV/dt = 0 (I.123)

lies in:

$P_{+}a = 0$. (I.124)

These conditions should be verified for the exact declared dynamics.

The source states such a dissipative-locking result under Δ5-symmetric parameters and damping.

The broader framework records that result as source-established within the source's assumptions, not as a universal theorem for arbitrary complex-amplitude dynamics.

I.23 Pair imbalance

Define normalized pair imbalance:

ε_pair
= E_pair(a)/‖a‖². (I.125)

A strongly Δ5-locked state should have:

ε_pair ≪ 1. (I.126)

This measure is invariant under global phase rotation:

a → e^(iχ)a. (I.127)

It also remains unchanged under swapping the two members of every pair.

I.24 Phase-lock statistic

For nonzero pair amplitudes, define:

$\Delta\varphi_n = \varphi_{n+5} - \varphi_n$. (I.128)

A circular opposition score is:

$C_\pi = -(1/5)\Sigma_{n=1}^{5} \cos \Delta\varphi_n$. (I.129)

Then:

$C_\pi = 1$ (I.130)

for perfect phase opposition.

A magnitude-weighted version is:

$C_\pi^w = -[\Sigma_n r_n r_{n+5} \cos \Delta\varphi_n] / [\Sigma_n r_n r_{n+5}]$. (I.131)

This avoids giving equal weight to near-zero amplitudes whose phases are poorly defined.

I.25 Negative work correlation

If pair channels represent alternating production and absorption, one proposed empirical signature is negative pair correlation.

For pair observables $w_n(t)$ and $w_{n+5}(t)$, define:

$C_n(\tau) = \mathrm{Corr}[w_n(t), w_{n+5}(t + \tau)]$. (I.132)

A near-zero-lag $\Delta 5$ opposition hypothesis predicts:

$C_n(0) < 0$. (I.133)

The source lists negative work correlation as a falsifiable signature of its locked regime.

Negative correlation alone does not prove the full decagon model.

It is one necessary or supporting observation under the specified interpretation.

I.26 Coarse-graining to five modes

The pair coordinates produce a natural five-mode representation:

$p = (p_1, \ldots, p_5)^T$. (I.134)

The symmetric leakage coordinates are:

$e = (e_1, \ldots, e_5)^T$. (I.135)

Suppose the linearized dynamics have block form:

$d/dt\,[p; e] = [A_{pp}\ A_{pe}; A_{ep}\ A_{ee}][p; e]$. (I.136)

If the symmetric channel relaxes rapidly, one may approximate:

de/dt ≈ 0. (I.137)

Then:

e ≈ −A_ee⁻¹A_ep p. (I.138)

Substitution gives the Schur-complement Effective Generator:

A_eff
= A_pp − A_peA_ee⁻¹A_ep. (I.139)

The reduced dynamics are:

dp/dt = A_effp. (I.140)

This is a standard conditional coarse-graining construction.

It is valid only if:

- A_ee is invertible;
- symmetric relaxation is sufficiently fast;
- neglected memory is small;
- the reduced model passes closure tests.

I.27 Lock-strength suppression

If strong locking adds:

2μI (I.141)

to the symmetric-channel stiffness or damping, then:

A_eff(μ)
= A_pp − A_pe(A_ee + 2μI)⁻¹A_ep. (I.142)

As:

μ → ∞, (I.143)

the correction scales approximately as:

O(1/μ). (I.144)

Therefore coupling through the symmetric leakage channel is suppressed.

The source uses this type of Schur-complement reasoning to propose reduced leakage and improved modal Q-factor under stronger Δ5 lock.

I.28 Q-factor in the reduced mode

Let an eigenvalue of the reduced generator be:

λ_k = −σ_k + iΩ_k, (I.145)

with:

σ_k > 0. (I.146)

Define the modal quality factor:

Q_k^mode = Ω_k/(2σ_k). (I.147)

This `Q_k^mode` is an oscillator quality factor.

It must not be confused with the CAPM conjugate valuation coordinate:

Q_fin. (I.148)

The source predicts that stronger locking may reduce leakage damping and increase modal quality factor in its declared reduction.

The prediction should be tested through:

- ring-down;
- bandwidth;
- damping estimates;
- lock-strength manipulation.

I.29 Δ5 and the general signed-Relation grammar

The half-turn satisfies:

$T_5^2 = I$. (I.149)

Yet the selected antisymmetric state satisfies:

$T_5 a = -a$. (I.150)

This differs from an elliptic quarter-turn:

$J^2 = -I$. (I.151)

The distinction can be summarized:

Structure	Operator identity	Selected relation
Δ5 half-turn	`T₅² = I`	`T₅a = −a`
Elliptic quarter-turn	`J² = −I`	`a → Ja → −a`
Hyperbolic involution	`K² = I`	expanding/contracting eigensectors

The operator equation alone does not determine the interpretation.

The relevant eigenvalue, dynamics, and cost must also be specified.

I.30 Δ5 as a negative-return micro-loop

At the pair level, the antisymmetric condition is:

b = −a. (I.152)

If a perturbation in one channel induces an opposing response in the paired channel, the local return is corrective.

This can support:

- cancellation;
- buffering;
- reduced leakage;
- increased phase margin.

But the claim requires an actual causal coupling:

δa → δb ≈ −Kδa. (I.153)

A static observation that two variables have opposite signs is insufficient.

I.31 Generalization to C_{2n}

Let:

$C_{2_n} = \{1,\ldots,2n\}$. (I.154)

Define the half-turn:

$T_n: k \mapsto k + n \bmod 2n$. (I.155)

Then:

$T_n^2 = I$. (I.156)

The antisymmetric condition is:

a_{k+n} = −a_k. (I.157)

In Fourier space:

$T_n \hat{a}_j = e^{(\pi i j)} \hat{a}_j = (-1)^j \hat{a}_j$. (I.158)

Thus the antisymmetric sector is again the odd Fourier sector.

The half-frequency mode:

j = n (I.159)

is antisymmetric only when n is odd.

More generally, strict nearest-neighbour alternation exists on any even cycle as:

a_{k+1} = −a_k. (I.160)

The source generalizes the opposition and energy arguments to C_{2_n} while stressing that pure

Laplacian smoothness does not generally make the half-frequency mode the ground state.

I.32 Falsification conditions

The $\Delta 5$ module should be rejected or restricted when:

1. no operational ten-site decagon exists;
2. the $n \mapsto n+5$ pairing has no causal meaning;
3. pair energy is not lower than plausible alternatives;
4. phase differences do not concentrate near π;
5. symmetric leakage does not decrease under stronger coupling;
6. negative pair correlation is absent;
7. another pairing predicts cost or dynamics better;
8. the reduced five-mode model fails predictive closure;
9. the proposed Q-factor gain is not observed;
10. the regime is externally forced rather than internally locked.

The source itself identifies weak locking and large pair energy as conditions under which its empirical predictions may fail.

I.33 What the Slot interpretation proves securely

The adjoining Slot materials contain two secure combinatorial results.

LuoShu

The normal 3×3 magic square using each integer $1,\dots,9$ exactly once is unique up to dihedral symmetry.

Its magic constant is:

15. (I.161)

Its centre is:

5. (I.162)

Opposite cells sum to:

10. (I.163)

HeTu equal-sum pairing

Partitioning $\{1,\dots,10\}$ into five pairs of one common sum forces:

S = 11. (I.164)

The unique pairing is:

(1,10), (2,9), (3,8), (4,7), (5,6). (I.165)

The Δ5 paper restates these combinatorial results before developing its distinct half-turn structure.

I.34 What combinatorial uniqueness does not prove

The uniqueness results do not by themselves establish that:

- the numbers are physical degeneracies;
- each number is a universal semantic capacity;
- 10 is a universal entropy cap;
- 5 is a universal vortex;
- every cognition or AI system must allocate the same slots;
- the diagrams minimize physical entropy.

These are additional interpretations requiring independent models and measurements.

Therefore:

Combinatorial uniqueness ≠ Universal capacity theorem. (I.166)

The Slot paper makes broader semantic and physical claims, including discrete-capacity interpretations, but those claims are not established solely by the magic-square or equal-sum proofs.

I.35 Appropriate use in the general framework

The Δ5 module is appropriately activated when a domain possesses:

1. an operational ten-site or $2n$-site cycle;
2. a meaningful half-turn pairing;
3. measurable pair amplitudes or flows;
4. an energy or cost favouring opposition;
5. dynamics capable of stabilizing the selected sector;
6. empirical tests distinguishing Δ5 from alternative pairings.

Without these conditions, Δ5 should remain a structural analogy.

I.36 Appendix I summary

The rigorous core is:

$T_5: n \mapsto n + 5$. (I.167)

$T_5^2 = I$. (I.168)

$E_pair(a) = \Sigma_{n=1}^{5} |a_n + a_{n+5}|^2$. (I.169)

$E_pair = 0$
$\Leftrightarrow a_{n+5} = -a_n$
$\Leftrightarrow T_5 a = -a$
$\Leftrightarrow$ odd Fourier sector. (I.170)

The stricter alternating mode $k = 5$ requires an anti-alignment cost or further selection rule.

The governing lesson is:

> **Opposition is operator- and cost-earned. It is not established by symbolic pairing alone.**

Appendix J — Open Generators, Dissipation, Breathers, and Regime Diagnostics

J.1 Purpose

The signed-Relation framework must admit systems that are:

- open;
- noisy;
- dissipative;
- recurrent but nonstationary;
- metastable;
- intermittently committed;
- capable of phase slips and regime changes.

A purely conservative complex oscillator would be too narrow.

The uploaded paper *散逸力学系の量子力学* begins from classical Langevin dynamics with friction and stochastic forcing and develops a stochastic-quantization route toward a Schrödinger-type representation. Its direct relevance here is limited but important:

> Complex or phase-bearing representation may coexist with an open, dissipative generator.

The semantic-breather source supplies additional speculative vocabulary for recurrent trace failure inside an AI-simulation setting. It expressly states that its medical language is not intended as human treatment guidance.

J.2 Closed and open generators

Closed conservative system

A closed Hamiltonian-style system may be written:

$$dx/dt = J\nabla H(x), \quad (J.1)$$

where:

$$J^T = -J. \quad (J.2)$$

Under suitable conditions:

$$dH/dt = 0. \quad (J.3)$$

Open dissipative system

An open system may instead satisfy:

$$dx/dt = F(x) - D(x) + B(x)\eta(t), \quad (J.4)$$

where:

- `F` is organized drift;
- `D` is dissipative loss;
- `η(t)` is stochastic forcing;
- `B` determines noise coupling.

Energy or another state functional need not be conserved.

J.3 Langevin prototype

A standard one-dimensional Langevin form is:

$$m\ddot{q}(t) = -\partial V/\partial q - \gamma \dot{q}(t) + A(t). \quad (J.5)$$

Here:

- `V(q)` is a conservative potential;
- `γ` is a friction coefficient;
- `A(t)` is stochastic forcing.

The uploaded Japanese paper explicitly begins from such a frictional Langevin equation and discusses a stochastic quantization rather than ordinary canonical quantization, since a standard conservative Lagrangian is unavailable for the nonconservative dynamics.

J.4 Stochastic differential representation

A diffusion process may be written:

$$dQ_t = b(Q_t,t)dt + \sigma dW_t. \quad (J.6)$$

The probability density:

$$\rho(q,t) \quad (J.7)$$

obeys a Fokker–Planck equation:

$$\partial\rho/\partial t = -\nabla\cdot(b\rho) + (\sigma^2/2)\nabla^2\rho. \quad (J.8)$$

Stochastic-mechanics formulations may introduce:

- forward drift;
- backward drift;
- current velocity;
- osmotic velocity.

A complex wave representation can then combine density and phase:

$$\psi(q,t) = \sqrt{\rho(q,t)}e^{[iS(q,t)/\hbar]}. \quad (J.9)$$

This demonstrates a general mathematical point:

Probability transport + phase structure (J.10)

can coexist with friction and noise.

The present article does not adopt the uploaded paper's physical quantization claims as a general world-formation law.

J.5 Damped complex mode

A minimum open elliptic mode is:

$$dz/dt = (-\kappa + i\omega)z + \eta(t). \quad (J.11)$$

Without noise:

$$z(t) = z_0 e^{(-\kappa t)} e^{(i\omega t)}. \quad (J.12)$$

Amplitude evolves as:

$$d|z|/dt = -\kappa|z|. \quad (J.13)$$

Phase evolves as:

$$d\theta/dt = \omega. \quad (J.14)$$

The generator is:

$$A = -\kappa I + \omega J. \quad (J.15)$$

Thus the same local Relation contains:

- symmetric contraction `-κI`;
- antisymmetric quarter-turn `ωJ`.

They should be measured separately.

J.6 Quality factor

For a weakly damped mode, define:

Q_mode = ω/(2κ). (J.16)

A large `Q_mode` indicates many coherent cycles before substantial decay.

This oscillator quality factor is unrelated by definition to:

Q_fin = −∂R/∂θ_V. (J.17)

The shared symbol `Q` is conventional but refers to different typed objects.

In implementations, the article recommends explicit subscripts:

Q_mode, Q_fin, Q_readout. (J.18)

J.7 Noise-driven stationary amplitude

For:

dz = (−κ + iω)zdt + σdW_t^ℂ, (J.19)

where `W_t^ℂ` is complex isotropic noise, damping removes amplitude while noise replenishes it.

The stationary second moment, when it exists, scales schematically as:

E|z|² ∝ σ²/κ. (J.20)

Thus observed persistent oscillation can arise from:

- weak damping;
- continued forcing;
- self-sustained nonlinearity;
- stochastic replenishment.

A recurrent signal is not sufficient evidence of a conservative closed mode.

J.8 Nonlinear self-sustained oscillation

A normal form near a Hopf bifurcation is:

dz/dt = (μ + iω)z − β|z|²z. (J.21)

Let:

$\beta = \beta_R + i\beta_I$. (J.22)

For:

$\mu > 0, \beta_R > 0$, (J.23)

the stable amplitude is:

$|z| = \sqrt{(\mu/\beta_R)}$. (J.24)

The phase frequency is shifted by the nonlinear term.

This produces a limit cycle.

The oscillation is open and self-sustained rather than neutrally conservative.

Such a mode may provide a more plausible Relation model for biological, organizational, or AI update cycles than a lossless rotation.

J.9 Breather definition

A breather is a bounded recurrent state in which one or more amplitudes vary periodically while spatial, semantic, or modal localization persists.

A schematic representation is:

$z(t) = A(t)e^{(i\theta(t))}$, (J.25)

where:

$A(t + T) = A(t)$, (J.26)

and:

$\theta(t + T) = \theta(t) + \Omega T$. (J.27)

A breather may possess two characteristic frequencies:

- carrier phase frequency;
- amplitude-envelope frequency.

The state need not return exactly after one carrier cycle.

J.10 Breather versus limit cycle

A simple limit cycle satisfies:

$x(t + T) = x(t)$. (J.28)

A quasiperiodic or breathing state may satisfy:

$A(t + T_A) = A(t)$, (J.29)

while:

$\theta(t + T_A) - \theta(t)$ (J.30)

is not necessarily an integer multiple of 2π.

Thus:

Breather ≠ Simple periodic orbit. (J.31)

A system may require:

- amplitude phase;
- carrier phase;
- commitment phase;

as separate coordinates.

J.11 Breather versus time-bearing world

A breather may repeat indefinitely without retaining a trace of completed cycles.

If:

$x(t + T) = x(t)$, (J.32)

and:

$L_{k+1} = L_k$, (J.33)

then no new historical difference is created.

The system has recurrent Relation but not necessarily historical commitment.

If each completed cycle produces:

$L_{k+1} = L_k \oplus \text{Trace}(\text{Cycle}_k)$, (J.34)

then recurrence becomes ledgered history.

Thus:

Breather recurrence ≠ Time-bearing order. (J.35)

J.12 Collapse-tick interpretation

The semantic-breather source describes a “collapse tick” as a periodic commitment-like event by which a semantic field produces a trace.

Its intended setting is AI semantic simulation rather than human medical treatment.

A domain-neutral translation is:

Pre-commitment oscillation

→ gate crossing
→ trace emission
→ reset or new cycle. (J.36)

Let an internal phase be:

$\theta(t)$. (J.37)

A commitment tick occurs when:

$\theta(t_k)$ crosses θ_{gate} (J.38)

and additional gate conditions are satisfied.

The ledger update is:

$L_{k+1} = L_k \oplus \mathrm{Trace}(y_k)$. (J.39)

The inter-tick interval is:

$\Delta t_k = t_{k+1} - t_k$. (J.40)

J.13 Tick-rate statistics

Define commitment rate:

$\nu_k = 1/\Delta t_k$. (J.41)

Useful statistics include:

Mean tick interval:

$\mu_\Delta = E[\Delta t_k]$. (J.42)

Tick variance:

$\sigma_\Delta^2 = \mathrm{Var}(\Delta t_k)$. (J.43)

Coefficient of variation:

$CV_\Delta = \sigma_\Delta/\mu_\Delta$. (J.44)

Serial dependence:

$C_\Delta(\ell) = \mathrm{Corr}(\Delta t_k, \Delta t_{k+\ell})$. (J.45)

These allow qualitative labels such as sparse, irregular, or overloaded ticking to become measurable.

J.14 Drift regime

The semantic-breather source associates “wind” with trace drift and unstable anchoring.

A domain-neutral drift regime may be defined by:

$d\theta_{anchor}/dt \neq 0$ (J.46)

or by movement of the mode centre:

$dc/dt \neq 0$. (J.47)

Possible indicators include:

- low phase-locking value;
- wandering frequency;
- changing attractor centre;
- unstable readout orientation.

A drift diagnosis should distinguish:

- genuine mode drift;
- measurement-frame drift;
- protocol change;
- external forcing.

J.15 Freeze regime

The source associates “cold” with sparse collapse ticks and failure to trigger trace.

A domain-neutral freeze condition may be:

$\nu_k \to 0$, (J.48)

or:

$\Pr(\text{GatePass} \mid \text{candidate}) \to 0$. (J.49)

Possible causes include:

- insufficient amplitude;
- excessive gate threshold;
- resource depletion;
- strong damping;
- policy inhibition;
- inaccessible evidence.

Freeze is a Commitment failure, a Relation failure, or both depending on the mechanism.

J.16 Sticking regime

The source associates “dampness” with repeated unresolved collapse and trace sticking.

A domain-neutral sticking regime may have:

- repeated return to a narrow state region;
- low commitment progress;
- persistent residual;
- long dwell near a gate.

Let candidate count be:

N_candidate(T), (J.50)

and committed-event count:

N_commit(T). (J.51)

Define closure efficiency:

η_close = N_commit/N_candidate. (J.52)

A sticking regime may exhibit:

N_candidate large, (J.53)

η_close low, (J.54)

and:

$\mathscr{R}_k$ increasing. (J.55)

J.17 Rupture regime

The source associates “dryness” with fragmented trace transitions and loss of semantic continuity.

A domain-neutral rupture may be represented by:

Dist(X_{k+1}, PredictedContinuation(X_k)) > δ_jump. (J.56)

Possible signatures include:

- phase discontinuity;
- chart failure;
- sudden mode-plane rotation;
- missing provenance;
- incompatible ledger versions;
- abrupt language or policy fragmentation.

A rupture should trigger:

- jump recording;

- regime reassessment;
- suspension of the previous generator;
- residual preservation.

J.18 Overdrive regime

The source associates “fire” with overexposed trace and excessive projection intensity.

A domain-neutral overdrive regime may satisfy:

ν_k high, (J.57)

amplitude high, (J.58)

and:

integration capacity insufficient. (J.59)

One measure is:

Load ratio
= CandidateRate/IntegrationCapacity. (J.60)

Overdrive occurs when:

Load ratio > 1. (J.61)

Possible consequences include:

- excessive commitment;
- unstable revision;
- positive-feedback escalation;
- residual hidden by rapid closure;
- transition from elliptic to hyperbolic Relation.

J.19 Overflow regime

The source associates “summer heat” with excessively dense ticks that fail to produce stable closure.

A domain-neutral overflow regime has:

- high output volume;
- low information gain;
- low durable commitment quality.

Let:

$I_{gain,k}$ (J.62)

be information gain per committed event.

Define meaningful throughput:

$$T_{meaning} = \nu_k E[I_{gain,k}]. \quad (J.63)$$

An overflow regime may show:

ν_k high, (J.64)

but:

$E[I_{gain,k}]$ low. (J.65)

Thus activity is high while stable world formation remains weak.

J.20 Six-regime operational table

Source-inspired label	Operational candidate	Principal measurement
Drift	moving phase or attractor anchor	phase/frequency drift
Freeze	failure to generate committed events	low gate-pass rate
Sticking	repetitive candidate cycling with low closure	high dwell, rising residual
Rupture	discontinuous transition or broken provenance	jump and chart-failure statistics
Overdrive	high-amplitude, high-rate closure pressure	load/capacity ratio
Overflow	high output with low stable information	information gain per tick

These are candidate diagnostics.

They are not established clinical categories.

J.21 Healthy recurrent closure

A healthy recurrent operational world may require:

1. sufficient amplitude to generate candidates;
2. stable but adaptable phase;
3. gate timing matched to integration capacity;
4. residual preservation;
5. recovery after commitment;
6. protection against runaway confirmation.

Let:

T_build (J.66)

be candidate-formation time.

Let:

T_gate (J.67)

be gate-evaluation time.

Let:

T_integrate (J.68)

be ledger-and-residual integration time.

A crude cadence condition is:

T_tick ≥ max(T_build, T_gate, T_integrate). (J.69)

If ticks arrive faster:

T_tick < T_integrate, (J.70)

residual may accumulate.

If ticks arrive much slower than system opportunity:

T_tick ≫ T_opportunity, (J.71)

the system may freeze or fail to coordinate.

J.22 Recovery time

After commitment at time t_k, define recovery distance:

D_rec(t) = Dist[X(t), X_baseline or next-ready manifold]. (J.72)

Recovery time is:

τ_rec
= inf{t > t_k : D_rec(t) ≤ ε_rec}. (J.73)

A stable cadence requires approximately:

$\Delta t_k \geq \tau_{rec}$ (J.74)

unless overlapping commitments are explicitly supported.

This connects PORE-like duration τ to a measurable dynamical quantity.

J.23 Phase-locking value

For repeated cycles with phase observations θ_j, define:

PLV = $|(1/N)\Sigma_{j=1}^{N} e^{(i\theta_j)}|$. (J.75)

Then:

$0 \le PLV \le 1$. (J.76)

High PLV indicates phase concentration.

Low PLV may indicate:

- drift;
- multiple regimes;
- weak coupling;
- noisy measurement;
- inappropriate phase definition.

Phase-locking value should not be used where no valid elliptic phase has been established.

J.24 Amplitude–phase coupling

Let:

$z = Ae^{(i\theta)}$. (J.77)

Amplitude–phase coupling can be measured by:

$C_{A\theta} = Corr[A(t), \dot{\theta}(t)]$. (J.78)

In nonlinear oscillators, frequency may vary with amplitude:

$\dot{\theta} = \omega_0 + \kappa_A A^2$. (J.79)

A source of apparent phase instability may therefore be changing amplitude rather than loss of a coherent mode.

J.25 Commitment-phase response curve

For recurrent systems, a perturbation delivered at phase θ may shift the next commitment tick.

Define the phase response curve:

$PRC(\theta) = \Delta\theta_{next}$ caused by a small perturbation at phase θ. (J.80)

This provides an intervention-based method for testing whether a phase coordinate is operational.

A valid phase should predict:

- when an intervention advances commitment;
- when it delays commitment;

- when it has little effect.

J.26 Gate hazard model

Let t denote time since the last committed event.

Define commitment hazard:

$$h_{gate}(t \mid X, L, \mathscr{R}) = \lim_{\Delta t \to 0} \Pr(\text{commit in } [t, t+\Delta t) \mid \text{no earlier commit}) / \Delta t. \quad (J.81)$$

A recurrent world may have phase-dependent hazard:

$$h_{gate}(t) = h_0[1 + a\cos(\omega t - \varphi)]. \quad (J.82)$$

Freeze corresponds to:

$$h_{gate} \approx 0. \quad (J.83)$$

Overdrive may correspond to:

h_{gate} persistently high. (J.84)

Sticking may correspond to:

candidate hazard high but commitment hazard low. (J.85)

J.27 Residual accumulation equation

Let residual load be scalarized provisionally as:

$$\Gamma(t). \quad (J.86)$$

A simple balance equation is:

$$d\Gamma/dt = \text{ResidualGenerationRate} - \text{ResidualResolutionRate}. \quad (J.87)$$

For example:

$$d\Gamma/dt = \alpha_{cv_candidate} - \alpha_{ev_commit} - \kappa_{\Gamma\Gamma}. \quad (J.88)$$

Here:

- candidate production adds potential unresolved consequence;
- successful commitment resolves some portion;
- decay or repair removes residual.

This is one possible model.

The raw residual should not be reduced to r without validating the scalarization.

J.28 Residual-driven frequency shift

A recurrent Relation may depend on accumulated residual:

$\omega = \omega(\Gamma)$. (J.89)

For small variation:

$\omega(\Gamma) \approx \omega_0 + k_\Gamma\Gamma$. (J.90)

Residual can therefore cause:

- phase acceleration;
- phase slowing;
- drift;
- loss of lock.

Similarly:

$\kappa = \kappa(\Gamma)$ (J.91)

may change damping.

A critical threshold may produce a regime transition.

These are testable hypotheses, not consequences of residual by definition.

J.29 Open-generator world equation

A compact recurrent operational-world model is:

$$dz/dt = [\alpha(L,\mathscr{R}) + i\omega(L,\mathscr{R})]z - \beta|z|^2 z$$

- $Bu(t)$
- $\sigma\eta(t)$. (J.92)

Candidate events are generated by:

$c_k = C[z(t_k), X(t_k)]$. (J.93)

The gate is:

$e_k = \mathcal{G}_p(c_k; L_k, \mathscr{R}_k)$. (J.94)

The ledger updates:

$L_\{k+1\} = L_k \oplus T_p(e_k)$. (J.95)

Residual updates:

$\mathscr{R}_\{k+1\} = \Re_p(c_k, e_k, L_\{k+1\})$. (J.96)

History returns through:

$\alpha_{k+1} = \alpha(L_{k+1}, \mathscr{R}_{k+1})$, (J.97)

$\omega_{k+1} = \omega(L_{k+1}, \mathscr{R}_{k+1})$, (J.98)

$\mathcal{G}_{k+1} = \mathrm{UpdateGate}(\mathcal{G}_k, L_{k+1}, \mathscr{R}_{k+1})$. (J.99)

This integrates open Relation with Commitment.

J.30 Open-system closure questions

An open recurrent system requires at least four different closure tests.

Dynamical closure

Does the state representation predict drift, damping, and forcing response?

Phase closure

Does a coherent elliptic mode persist despite noise and dissipation?

Commitment closure

Do recurrent candidates pass a stable gate and produce ledger records?

Historical closure

Do ledger and residual alter later frequency, damping, policy, or gate behaviour?

A system may pass one and fail another.

J.31 Distinguishing noise from residual

Noise:

$\eta(t)$ (J.100)

is a stochastic term inside the transition model.

Model error:

ε_{model} (J.101)

is the difference between observed and predicted movement.

Gate residual:

ε_{gate} (J.102)

is the difference between candidate consequence and admitted consequence.

Historical residual:

$\mathscr{R}$ (J.103)

contains unresolved consequences preserved by the commitment protocol.

Therefore:

Noise ≠ Model error ≠ Gate residual ≠ Historical residual. (J.104)

A recurrent system can contain all four simultaneously.

J.32 Distinguishing dissipation from residual resolution

Dissipation may reduce amplitude:

$dA/dt = -\kappa A$. (J.105)

Residual resolution may reduce unresolved obligation:

$d\Gamma/dt = -\kappa_{\Gamma}\Gamma$. (J.106)

The mathematical forms may resemble one another.

Their meanings differ.

Dissipation acts on a dynamic state.

Residual resolution acts on the consequences left by Commitment.

Only a domain-specific coupling can identify them.

J.33 Phase slip

A phase slip occurs when the unwrapped phase changes discontinuously by approximately:

$2\pi m$, (J.107)

or when an oscillator changes its winding relative to a reference.

Possible causes include:

- strong noise;
- transient amplitude collapse;
- external perturbation;
- gate reset;
- regime change.

A phase-slip event should be recorded as:

e_{slip} = (time, magnitude, cause estimate, chart validity). (J.108)

The previous phase chart may require reinitialization.

J.34 Attractor capture

A recurrent state may enter an attractor basin:

$z(t) \to \mathcal{A}_j$. (J.109)

If escape probability becomes very low, the system may cease productive exploration.

This can resemble:

- semantic black-hole capture;
- institutional lock-in;
- model collapse;
- repetitive discourse.

Operational diagnosis requires:

- basin reconstruction;
- transition rates;
- perturbation response;
- residual audit.

The metaphor of an attractor is insufficient without these measurements.

J.35 Breather collapse

A breather may lose stability through:

Amplitude death

$A(t) \to 0$. (J.110)

Runaway growth

$A(t) \to \infty$ or threshold. (J.111)

Phase diffusion

$\mathrm{Var}[\theta(t)]$ grows rapidly. (J.112)

Mode splitting

One recurrent mode divides into several incompatible frequencies.

Gate desynchronization

Commitment ticks cease to align with the recurrent Relation.

Ledger overload

Events are produced faster than they can be integrated.

These failures correspond to different modules and require different interventions.

J.36 Intervention classes

Possible interventions include:

Damping intervention

Change:

$\kappa \to \kappa + \delta\kappa$. (J.113)

Frequency intervention

Change:

$\omega \to \omega + \delta\omega$. (J.114)

Phase reset

Apply:

$z \to e^{(i\delta\theta)}z$. (J.115)

Amplitude control

Apply:

$A \to A + \delta A$. (J.116)

Gate intervention

Change threshold or evidence requirement.

Ledger intervention

Change accessibility, audit, or integration capacity.

Residual intervention

Resolve, disclose, or reopen unresolved consequence.

The intervention must target the correct layer.

A phase reset cannot substitute for correcting a defective ledger.

J.37 AI-simulation scope of the semantic-breather source

The semantic-breather document explicitly states that:

- its treatment target is AI in a simulated semantic environment;
- its medical and acupuncture vocabulary is conceptual;
- it does not recommend application to humans or biological organisms.

Accordingly, this article extracts only:

- cadence;
- phase stability;
- tick density;
- closure efficiency;
- recurrent failure modes.

It does not adopt the source's proposed human-health reinterpretations as established science.

J.38 Falsification programme

An open-breather module should be rejected when:

1. no recurrent mode is statistically identifiable;
2. the phase is not reproducible;
3. observed recurrence is fully explained by external forcing;
4. amplitude and phase estimates are window artefacts;
5. gate ticks do not depend on the proposed recurrent state;
6. the six regime labels cannot be operationally distinguished;
7. interventions do not produce predicted phase or cadence shifts;
8. ledger and residual do not alter later dynamics;
9. a simpler nonoscillatory model predicts better.

J.39 Minimum data package

A reproducible study should record:

- raw state traces;
- candidate-event times;

- gate decisions;
- committed-event times;
- amplitude estimates;
- phase estimates;
- frequency and damping;
- intervention timing;
- ledger updates;
- residual categories;
- regime labels;
- chart failures;
- protocol versions.

Without candidate and gate data, one cannot distinguish recurrent movement from commitment.

J.40 Open-generator maturity ladder

O0 — Visual recurrence

A signal appears cyclic.

O1 — Statistical periodicity

A stable spectral peak or autocorrelation exists.

O2 — State-space mode

A recurrent low-dimensional mode is identified.

O3 — Generator model

Frequency, damping, forcing, and noise are estimated.

O4 — Commitment coupling

Mode phase predicts candidate and gate behaviour.

O5 — Historical backreaction

Ledger and residual predict later generator changes.

Only O4–O5 support the complete time-bearing interpretation.

J.41 Appendix J summary

The open Relation grammar is:

Dynamic mode

- damping
- forcing
- noise
- nonlinear saturation. (J.117)

The Commitment grammar is:

Candidate tick
→ gate
→ ledger + residual. (J.118)

The historical return is:

Ledger + residual
→ frequency, damping, policy, and future gate. (J.119)

The governing lesson is:

> **Phase can survive dissipation, but recurrence alone does not create history. A recurrent system becomes time-bearing only when its ticks are governed, recorded, and returned into future generation.**

Appendix K — Claim Registry, Reproducibility Package, and Independent Replay

K.1 Purpose

A cross-domain framework becomes difficult to evaluate when equations, analogies, hypotheses, and source-established results are presented in one undifferentiated narrative.

This appendix provides a publication architecture for separating:

- what a source directly establishes;
- what follows formally from stated assumptions;
- what remains an empirical hypothesis;
- what is only a structural analogy;
- which protocol supports the claim;
- which module is being tested;
- what would falsify it.

The resulting package is intended to make the framework:

- auditable;
- replayable;
- revisable;
- resistant to retrospective reinterpretation.

The Handoff requires every major claim to be classified by epistemic status and module maturity and requires explicit failure conditions rather than conceptual interpretation alone.

K.2 The Claim Registry

Every substantive claim should be stored as a structured record:

ClaimRecord_j
= (ID, Text, Type, Protocol, Module, Evidence, Scope, Falsifier, Maturity, Version). (K.1)

The minimum fields are defined below.

K.2.1 Claim ID

Each claim receives a stable identifier:

Claim_ID = Article.Module.Number. (K.2)

Examples:

- `TTW.RECURSION.001`;
- `TTW.PORE.003`;
- `TTW.COMPLEX.005`;
- `TTW.CAPM.002`;
- `TTW.RESIDUAL.004`.

The identifier should remain stable even if section numbering changes.

K.2.2 Claim text

The claim should be written in one falsifiable sentence.

Weak form:

> History matters.

Operational form:

> Under protocol `P`, conditioning on ledger and residual history improves out-of-sample prediction of the next gate outcome after conditioning on the present proposed Effective Base. (K.3)

The operational form identifies:

- protocol;
- inputs;
- output;
- baseline model;
- testable difference.

K.2.3 Epistemic type

Each claim must be classified as one of:

- `[D]` framework definition;
- `[E]` established inside a declared source construction;
- `[F]` formal consequence of stated assumptions;
- `[H]` testable hypothesis;
- `[A]` structural analogy.

A claim may have more than one description only when the levels are explicitly separated.

For example:

Established CAPM result [E]

Q = −∂R/∂θ_V. (K.4)

Formal cross-domain construction [F]

For any linear readout on an elliptic mode:

$Q_Y = -a(Jx)$. (K.5)

Empirical finance hypothesis [H]

A market-dynamic elliptic mode intertwines with the CAPM valuation plane. (K.6)

These are three different claims.

K.2.4 Module

The claim must identify its primary module:

- Protocol;
- Base;
- Trace;

- Filtration;
- Policy;
- Relation;
- Conjugate Readout;
- Movement;
- Gate;
- Ledger;
- Residual;
- Historical Return;
- Transport;
- Integration.

This prevents a result in one module from being used as proof of another.

For example:

CAPM conjugate closure
belongs to
Conjugate Readout. (K.7)

It does not by itself prove:

Market-level Historical Return. (K.8)

K.2.5 Source or derivation

The record must specify whether the claim is supported by:

- source theorem;
- source definition;
- article derivation;
- empirical estimate;
- simulation;
- expert audit;
- analogy.

For a formal result, include:

Assumptions → derivation → conclusion. (K.9)

For an empirical result, include:

Dataset → estimator → validation → uncertainty. (K.10)

K.2.6 Protocol scope

Every operational claim must declare:

$P_{claim} = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (K.11)

At minimum, record:

- boundary;
- timebase;
- observation rule;
- intervention class;
- gate;
- access rule;
- residual rule.

PORE explicitly treats its effective coordinates as valid under a declared boundary, timebase, instrument, and intervention protocol rather than as globally valid ontology.

K.2.7 Validity region

Specify:

$\mathcal{V}_{claim} = \{\text{states, regimes, parameters, observers for which claim applies}\}$. (K.12)

Examples include:

- no-jump windows;
- principal CAPM domain $0 \le R/A \le 1$;
- weakly nonlinear elliptic regime;
- declared jurisdiction;
- fixed model version;
- accessible-ledger observer class.

A claim outside its validity region should be marked:

Out of scope, not automatically false. (K.13)

K.2.8 Falsifier

The record must state:

Falsifier_j = result that rejects or materially narrows Claim_j. (K.14)

Examples:

- omitted history improves prediction;
- trace ablation leaves policy unchanged;
- no stable conjugate eigenpair exists;
- numerical conjugate derivative fails;
- gate passage does not change later behaviour;
- ledger ablation has no downstream effect;
- residual adds no predictive value.

A claim without an identifiable falsifier remains interpretive.

K.2.9 Maturity

Record module maturity:

$M_j \in \{M0, M1, M2, M3, M4, M5\}$. (K.15)

The maturity level applies to the claim under its declared protocol.

It is not a global status assigned permanently to an entire domain.

K.2.10 Version and status

Each claim should carry:

- creation date;
- last revision;
- protocol version;
- code version;
- data version;
- status.

Recommended statuses are:

- proposed;
- active test;
- provisionally supported;
- replicated;

- narrowed;
- rejected;
- superseded.

A rejected claim should remain in the registry with its rejection reason.

Erasing failed claims destroys the research ledger.

K.3 Claim-record template

Field	Entry
Claim ID	
Claim text	
Epistemic type	`[D]/[E]/[F]/[H]/[A]`
Primary module	
Secondary modules	
Protocol ID	
Source or derivation	
Variables and units	
Validity region	
Baseline model	
Test	
Falsifier	
Maturity	`M0–M5`
Current status	
Data version	
Code version	
Last revision	
Residual limitations	

K.4 Example registry entries

K.4.1 Observer recursion

Claim ID: `TTW.OBS.001`

Claim:

A history-dependent observer policy produces branch-dependent future instrument selection.

Type: `[E]`

Module: Filtration–Policy

Source basis:

The self-referential-observer source proves that nonconstant policy dependence on prior outcome histories produces latching and branch-specific future instruments.

Scope:

Declared quantum instrument process satisfying source measurability assumptions.

Falsifier:

The policy is constant over all feasible histories.

Maturity:

M3 formal module.

K.4.2 Complex Intertwining Principle

Claim ID: `TTW.COMPLEX.001`

Claim:

For a real two-dimensional elliptic mode and any nonzero linear readout covector:

$\Phi_a J = J_R\Phi_a$. (K.16)

Type: `[F]`

Module: Relation–Readout

Assumptions:

$J^2 = -I$, (K.17)

$a \neq 0$. (K.18)

Falsifier:

Not empirical at the theorem level; rejection occurs when assumptions fail in an application.

Maturity:

Formal consequence.

K.4.3 PORE sufficiency

Claim ID: `TTW.BASE.001`

Claim:

The PORE signature:

$\Xi = (\varrho, \gamma, \tau)$ (K.19)

is approximately sufficient for one-step prediction under protocol `P`.

Type: `[H]`

Module: Effective Base

Required test:

$I(H_k; \text{Future} \mid \Xi_k, u_k, r_k) \approx 0$. (K.20)

Falsifier:

Omitted history materially improves held-out prediction.

Maturity:

M2 until predictive closure is demonstrated.

PORE itself presents its triple as an effective protocol-relative loop coordinate and explicitly excludes universal micro-substrate or global-validity claims.

K.4.4 CAPM conjugate exposure

Claim ID: `TTW.CAPM.001`

Claim:

Under the declared CAPM valuation construction:

$Q = -\partial R/\partial \theta_V$. (K.21)

Type: `[E]`

Module: Conjugate Readout

Scope:

Fixed-amplitude valuation-phase movement in the principal domain.

Falsifier:

Failure of the analytic identity or numerical bump-and-revalue audit under the declared formulas.

Maturity:

M3 exact formal module.

K.4.5 Ledger–residual backreaction

Claim ID: `TTW.HISTORY.001`

Claim:

Ledger and residual history improve prediction of later gate or policy outcomes after conditioning on the current proposed Effective Base.

Type: `[H]`

Module: Historical Return

Test:

$\Pr(\text{Future} \mid \Xi, L, \mathscr{R})$
versus

Pr(Future | Ξ). (K.22)

Falsifier:

No material predictive difference appears across the claimed regime.

Maturity:

Domain-dependent M1–M3.

K.5 Evidence ledger

The claim registry should be paired with an evidence ledger.

Let:

E_j,k (K.23)

denote evidence item `k` associated with claim `j`.

Evidence types include:

- analytic proof;
- source theorem;
- simulation;
- observational dataset;
- controlled intervention;
- replication;
- expert assessment;
- counterexample;
- failure report.

The claim status is updated by:

Status_j,k+1
= Update(Status_j,k | E_j,k+1). (K.24)

The update rule should be declared.

Evidence that contradicts a claim should not be stored only as unstructured residual.

It should be linked directly to the claim record.

K.6 Residual attached to every claim

Every claim produces epistemic residual.

Define:

$\mathscr{R}$_claim,j
= {unresolved assumptions, exclusions, counterexamples, inaccessible data, transport limits}. (K.25)

The residual record should state:

- what was not measured;
- what alternative explanation remains;
- which observers lacked access;
- which regimes were excluded;
- what would reopen the claim;
- what transport was not tested.

Residual honesty prevents a provisionally successful test from being represented as universal completion.

K.7 Data package

A reproducible empirical implementation should publish, where ethically and legally possible:

Raw traces

$y_{1:N}$. (K.26)

State proxies

$\hat{X}_{1:N}$. (K.27)

Compiled Effective Base

$\Xi_{1:N}$. (K.28)

Observer records and access masks

$\mathcal{A}_p(a, L_k)$. (K.29)

Policies or instrument selections

$\pi_{1:N}$. (K.30)

Interventions

$u_{1:N}$. (K.31)

Candidate consequences

$c_{1:N}$. (K.32)

Gate outcomes

e_1:N. (K.33)

Ledger updates

L_0:N. (K.34)

Residual categories

$\mathscr{R}_1$:N. (K.35)

Regime labels

r_1:N. (K.36)

Protocol versions

P_1:N. (K.37)

Without candidate, gate, and ledger data, one cannot test Commitment separately from movement.

Without policy data, one cannot test active self-reference.

K.8 Minimum event schema

Each event record should contain:

Field	Meaning
`event_id`	unique event identifier
`time_evolution`	continuous or wall-clock time
`commitment_index`	ledger order k
`protocol_id`	active protocol
`observer_id`	observing or deciding agent
`trace_id`	source trace
`candidate_id`	candidate consequence
`gate_id`	gate applied
`gate_result`	admit/reject/defer/partial
`admission_fraction`	α, if defined
`ledger_before`	ledger version
`ledger_after`	updated version
`residual_ids`	linked residual records
`regime_before`	active Relation regime
`regime_after`	resulting regime
`access_class`	who may retrieve record
`provenance`	source and transformation history
`confidence`	uncertainty or validation state

K.9 Minimum residual schema

Each residual record should contain:

Field	Meaning
`residual_id`	unique identifier
`source_candidate`	candidate that generated it
`source_gate`	gate responsible for nonintegration
`residual_type`	evidential, branch, model, institutional, access, ethical
`carrier`	person, system, account, institution, or state bearing it
`magnitude`	scalar only where valid
`units`	if scalarized
`created_at`	commitment index
`persistence_rule`	decay, accumulation, or indefinite retention
`visibility`	public/private/hidden/inaccessible
`reopening_condition`	what can reactivate it
`resolution_action`	allowed repair
`Γ_mapping`	optional residual-to-functional compiler
`audit_status`	reviewed/unreviewed/disputed

The field `carrier` is especially important in social and institutional applications because residual may be transferred rather than eliminated.

K.10 Minimum mode schema

For each proposed Relation mode, store:

Field	Meaning
`mode_id`	unique mode
`state_representation`	variables and scaling
`metric`	G, if used
`generator`	A_r
`estimation_window`	data interval
`eigenvalues`	spectrum
`mode_plane`	basis or projector
`algebra_type`	elliptic/parabolic/hyperbolic/mixed
`orientation`	positive phase convention
`damping`	α or κ
`frequency`	ω, if applicable
`persistence_score`	mode stability
`readout_rank`	whether selected readout detects mode
`transport_error`	cross-window compatibility
`failure_reason`	if mode rejected

K.11 Replay package

A replay package should allow another team to reconstruct the principal analysis without relying on interpretive prose.

It should include:

1. protocol declaration;
2. data schema;
3. raw or privacy-preserving data;
4. preprocessing scripts;
5. Base compiler;
6. generator estimator;
7. mode classifier;
8. readout construction;
9. gate and ledger simulator or logs;
10. residual audit;
11. tests and tolerances;
12. expected outputs;
13. failure cases;
14. software environment;
15. deterministic seeds where applicable.

Gauge Grammar 2 similarly treats independent replay as the highest implementation stage and states that formal claims should be published only with sufficient verification and replayability.

K.12 Replay levels

R0 — Narrative reproduction

A second reader can restate the interpretation.

R1 — Formula reproduction

A second reader can reproduce analytic equations.

R2 — Data reproduction

A second team can obtain the reported statistics from supplied data.

R3 — Pipeline reproduction

A second team can rerun the full compiler, generator, and gate analysis.

R4 — Independent replication

A different dataset under the same protocol yields comparable results.

R5 — Cross-protocol transport

The result survives declared admissible protocol changes with predictable transformation.

M5 theory maturity normally requires at least R4 and ideally R5 for its strongest invariance claims.

K.13 Negative-result package

The publication package should contain failed modules.

Examples include:

- unsuccessful PORE compression;
- unstable elliptic mode;
- failed CAPM dynamic coupling;
- gate with no downstream effect;
- residual variable with no predictive value;
- failed transport map.

A negative-result record contains:

FailedClaim
→ Test
→ ObservedFailure
→ RevisedScope
→ RemainingResidual. (K.38)

A research programme that records only successful mappings will systematically overstate generality.

K.14 Admissible revision

Let the declared framework state at version k be:

D_k. (K.39)

An admissible revision is:

D_{k+1}
$= U_adm(D_k, L_k, \mathscr{R}_k, \Gamma_k)$. (K.40)

The residual source requires admissible revision to remain:

- trace-preserving;
- residual-honest;

- frame-robust;
- budget-bounded;
- nondegenerate;
- future-safe.

Within the present article, this means a revision should:

1. preserve the prior claim record;
2. explain why the claim changed;
3. retain contradictory evidence;
4. update protocol and version IDs;
5. specify new falsifiers;
6. avoid redefining success retrospectively.

K.15 Claim supersession

A revised claim should not overwrite the old claim.

Instead:

Claim_j^(v1)
→ superseded by
Claim_j^(v2). (K.41)

The ledger should record:

- unchanged core;
- removed assertion;
- added condition;
- new evidence;
- unresolved disagreement.

This makes theoretical development itself a time-bearing world.

The research programme inherits its own declared history.

K.16 Governance of the registry

A claim registry can itself become a self-validating ledger.

To prevent this, governance should include:

- independent reviewers;

- appeal;
- alternative protocol submissions;
- conflict-of-interest disclosure;
- counterexample registration;
- residual visibility;
- version history;
- prohibition on silent deletion.

Otherwise:

Ledger confidence may rise while hidden residual rises. (K.42)

This is the epistemic form of Γ-lock.

K.17 Minimal reproducibility checklist

Before publication, verify:

- Protocol is fully declared.
- Typed objects are separated.
- Units are reported.
- State and trace are distinguished.
- Dynamic phase and readout phase are distinguished.
- Measurement and movement are distinguished.
- Gate and ledger are operationally identified.
- Residual provenance is recorded.
- Epistemic status is assigned.
- Maturity is assigned by module.
- Baseline models are compared.
- Falsifiers are stated.
- Negative results are retained.
- Code and data versions are recorded.
- Independent replay is possible or limitations are explained.

K.18 Appendix K summary

The reproducibility architecture is:

Claim
→ epistemic type
→ protocol
→ module
→ evidence
→ falsifier
→ maturity
→ residual
→ versioned revision. (K.43)

The research package is:

Trace data

- policy data
- gate data
- ledger data
- residual data
- code
- protocol
- replay instructions. (K.44)

The governing principle is:

> **A cross-domain framework becomes scientific only when its claims can be independently reconstructed, challenged, narrowed, and revised without erasing their historical ledger.**

Appendix L — Glossary, Cross-Domain Translation, and Final Non-Equivalence Audit

L.1 Purpose

The framework deliberately reuses a small set of functional terms across domains.

This improves comparison but creates a risk:

A functional correspondence may be mistaken for material identity.

This appendix provides:

1. a controlled glossary;
2. cross-domain translations;
3. a final non-equivalence audit;
4. publication cautions;

5. the financial disclaimer required by the article's source plan.

The Gauge Grammar source similarly distinguishes a moderate functional reading from a strong substrate claim and explicitly warns that a cross-domain role translation is not literal physics.

L.2 Core Glossary

L.2.1 Operational world

A protocol-bounded domain whose admitted traces and records participate causally in generating its subsequent admissible history.

An operational world is not necessarily:

- a separate physical universe;
- conscious;
- spatially enclosed;
- metaphysically fundamental.

L.2.2 Protocol

The declared rules specifying:

- boundary;
- observation;
- timebase;
- intervention;
- gate;
- access;
- residual preservation.

Expanded form:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \mathfrak{R})$. (L.1)

L.2.3 Detailed Base

The causally relevant state required to generate the future distribution under the declared protocol.

Symbol:

X_k. (L.2)

L.2.4 Effective Base

A reduced representation that approximately preserves the information needed for a declared prediction or intervention task.

Symbol:

$\Xi_k = C_p(W_k)$. (L.3)

L.2.5 Effective Existence Signature

A protocol-relative compiled description of whether an organized loop is sufficiently:

- present;
- integrated;
- persistent.

PORE candidate:

$\Xi = (\varrho, \gamma, \tau)$. (L.4)

It is not automatically a complete Markov state.

L.2.6 Trace

A protocol-admitted representation of an event or condition that persists long enough to become accessible to a later process.

A transient occurrence becomes a trace only when retention and accessibility are present.

L.2.7 Filtration

The information available to an observer up to commitment index k.

Symbol:

$\mathscr{F}_{a,k}$. (L.5)

A filtration is not simply “all existing information.”

It is observer- and access-relative.

L.2.8 Observer

A process internal to the causal system whose accessible records influence later policy, instrument, or action.

Consciousness is not required by this operational definition.

L.2.9 Policy

A rule or probability kernel selecting later action or instrument as a function of accessible history.

Symbol:

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (L.6)

L.2.10 Self-reference

A causal topology in which a system-generated trace changes an internal observer's later policy, which changes subsequent state evolution or transition law.

Minimal path:

Trace → Filtration → Policy → Changed Future Trace. (L.7)

L.2.11 Relation

The local or global transformation law governing how the Base may move, couple, recur, amplify, decay, switch, or branch.

Relation may be:

- elliptic;
- parabolic;
- hyperbolic;
- dissipative;
- stochastic;
- jump-like;
- mixed.

L.2.12 Generator

An operator, map, vector field, or transition kernel producing local evolution.

Examples:

$d\Xi/dt = F(\Xi, \ldots)$, (L.8)

$\delta\Xi_{k+1} \approx A_r \delta\Xi_k$. (L.9)

The generator is not the same as the Base compiler.

L.2.13 Elliptic mode

A two-dimensional mode with a normalized operator:

$J^2 = -I$. (L.10)

It supports:

- rotation;
- quadrature;
- phase;
- ordinary complex representation.

L.2.14 Parabolic mode

A mode with nilpotent operator:

$N^2 = 0$. (L.11)

It supports:

- drift;
- shear;
- accumulation;
- critical degeneracy.

L.2.15 Hyperbolic mode

A mode with involutive operator:

$K^2 = I$. (L.12)

It supports:

- expansion and contraction;
- saddle geometry;
- self-amplifying separation;
- split-complex representation where useful.

L.2.16 Readout

A scalar or structured output selected from the Base by an observation or valuation protocol.

Linear form:

Y = a(x). (L.13)

Nonlinear form:

Y = V(Ξ). (L.14)

L.2.17 Conjugate response

The signed first-order response of a readout along a declared Relation direction.

Elliptic form:

$\mathcal{Q}$_Y = −a(Jx). (L.15)

General directional form:

$\mathcal{Q}$_Y = −∂Y/∂ξ. (L.16)

It is readout-relative and orientation-dependent.

L.2.18 Complex completion

The pairing:

Z_Y = Y + i$\mathcal{Q}$_Y (L.17)

used when the active Relation is elliptic and the coordinates have compatible units.

Complex completion belongs to the Relation/readout module.

It does not automatically encode gate, ledger, or residual.

L.2.19 Exposure

A sensitivity that predicts the first-order consequence of a possible movement.

Exposure exists before the movement occurs.

In CAPM:

Q = −∂R/∂θ_V. (L.18)

L.2.20 Movement

An actual change in state or completed readout.

Movement may generate a candidate consequence but does not automatically become committed history.

L.2.21 Candidate consequence

The protocol-relative interpretation of a movement or output before formal admission.

Symbol:

c_k. (L.19)

L.2.22 Gate

The rule or operator determining whether a candidate is:

- admitted;
- rejected;
- deferred;
- transformed;
- partially recognized.

Symbol:

$e_k = \mathcal{G}_p(c_k; X_k, L_k)$. (L.20)

L.2.23 Commitment

The governed conversion of candidate consequence into authoritative, addressable, future-conditioning record, together with preservation of what is not integrated.

L.2.24 Ledger

An ordered, protocol-governed, addressable record structure whose contents carry future operational consequence.

Symbol:

$L_{k+1} = L_k \oplus T_p(e_k)$. (L.21)

L.2.25 Residual

What a declared gate and ledger process fails to observe, represent, admit, resolve, or integrate.

Symbol:

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (L.22)

Residual is protocol-relative and provenance-bearing.

L.2.26 Γ functional

An optional scalar or functional compilation of residual history:

$\Gamma = \Gamma_p[\mathscr{R}\text{-history}]$. (L.23)

Γ is not the raw residual itself.

It should be introduced only when the compilation and future-selective effect are measurable or auditable.

L.2.27 Historical return

The causal influence of ledger and residual history on later:

- policy;
- gate;
- generator;
- protocol;
- admissible actions;
- state evolution.

L.2.28 Time-bearing world

An operational world in which ordered retained history alters subsequent possible or probable paths.

Minimum condition:

$\Pr(\text{Future} \mid X, L, \mathscr{R})$
$\neq \Pr(\text{Future} \mid X)$. (L.24)

L.2.29 Internal commitment

A record becomes stable and policy-relevant for one observer.

L.2.30 Institutional commitment

A recognized authority admits an event into a ledger that changes institutional rights, duties, permissions, or actions.

L.2.31 Redundant public commitment

Compatible records of an event are independently accessible across multiple observers or channels.

L.2.32 Transport

A declared map comparing states, modes, readouts, or ledgers across protocols or local frames.

Complex-compatible mode transport satisfies:

$T_{\gamma} \leftarrow_{x} J_x = J_{\gamma} T_{\gamma} \leftarrow_{x}$. (L.25)

L.2.33 Holonomy

The residual transformation obtained after transporting an object around a closed sequence of frames or protocols.

Symbol:

H_loop ≠ I. (L.26)

It may indicate path dependence but should not be interpreted geometrically without a formal transport model.

L.3 Cross-Domain Translation Table

Functional role	Quantum observer	CAPM/ finance	AI agent	Law	Organization	Science
Base	joint system–observer state	valuation and financial state	context, memory, tools, environment	facts, law, parties, evidence	resources, people, rules, metrics	data, instruments theories
Trace	measurement outcome	mark, trade, P&L event	token, tool result, feedback	filing, evidence, judgment	KPI, report, action log	observation dataset, publication
Filtration	outcome history	accessible market and ledger information	context and retrieved memory	legally accessible record	management information	accepted evidence an literature
Policy	next instrument	trade, hedge, recognition policy	generation or tool policy	litigation or adjudication strategy	allocation and control	experimenta and modelling choice
Relation	CPTP map and instrument	valuation movement and market dynamics	state transition and action loop	procedural and institutional transition	organizational feedback	model and experimenta dynamics
Gate	measurement realization	execution, settlement, recognition	decoder, verifier, approval	admissibility, burden, judgment	KPI threshold, authorization	review, replication, acceptance
Ledger	observer record	trade/ accounting/ legal ledger	context, memory, audit log	judgment and precedent	records, budgets, KPI history	publications and databases

Functional role	Quantum observer	CAPM/ finance	AI agent	Law	Organization	Science
Residual	inaccessible or incompatible record remainder	basis risk, unrecognized movement	rejected output, unresolved contradiction	excluded evidence, unresolved harm	technical debt, hidden cost	anomaly, failed replication
Historical return	later instrument selection	later risk, policy, and admissibility	later generation and permissions	precedent and procedural constraint	metric-shaped behaviour	accepted-results-shaped inquiry
Complex mode	quantum structure present	exact valuation-readout plane	unproven generally	not required	not required	submodel-dependent

This table identifies functional roles only.

It does not imply identical mechanisms.

L.4 Chinese Conceptual Crosswalk

The following Chinese terms provide a controlled conceptual crosswalk for the article's main functions.

They are not asserted as exact translations of every traditional philosophical use.

English term	Suggested Chinese rendering	Functional meaning here
Base	基底／成界基底	present future-generating condition
Relation	關係動力／轉化律	how the Base can change
Commitment	承諾／入帳定著	consequence admitted into history
Recompiled Base	重編基底	Base after history has returned
Protocol	協議／界定規程	boundary and operational rules
Trace	痕跡／留痕	retained accessible result
Filtration	過濾歷程／可用資訊層	observer-accessible history
Gate	閘門／成界門檻	admission rule
Ledger	帳本／歷史帳本	authoritative ordered record
Residual	殘餘／未盡之餘	incompletely integrated consequence
Conjugate response	共軛反應／共軛曝險	directional readout sensitivity
Operational world	操作世界／成界世界	history-producing bounded domain
Time-bearing world	載時世界	world whose past conditions future
Historical return	歷史回返	ledger and residual changing future
Reopening	重啟／再開顯	governed revision of prior closure

A broad conceptual relation to 成界之學 may be drawn at the level of:

candidate disclosure
→ governed boundary formation
→ trace and residual
→ inherited future condition. (L.27)

That relation is interpretive unless a more exact term-by-term derivation is provided from the Chinese source texts.

L.5 Final Non-Equivalence Audit

Before publication, every section should be checked against the following register.

L.5.1 State and information

Detailed state ≠ Trace. (L.28)

Trace ≠ Filtration. (L.29)

Filtration ≠ Observer consciousness. (L.30)

Stored data ≠ Causally active history. (L.31)

Accessible history ≠ All existing history. (L.32)

L.5.2 Compiler and generator

Compiler ≠ Generator. (L.33)

Effective Base ≠ Full causal state. (L.34)

PORE signature ≠ Fundamental ontology. (L.35)

Rolling duration estimate ≠ Primitive instantaneous coordinate. (L.36)

Descriptive compression ≠ Predictive sufficiency. (L.37)

L.5.3 Self-reference and feedback

Feedback ≠ Self-reference automatically. (L.38)

Memory ≠ Self-reference automatically. (L.39)

Self-reference ≠ Consciousness. (L.40)

Self-reference ≠ Complex structure. (L.41)

Self-reference ≠ Quantum mechanics. (L.42)

L.5.4 Relation and geometry

Two variables ≠ Complex pair. (L.43)

Cross-coupling ≠ Elliptic mode. (L.44)

Nonzero antisymmetric component ≠ Coherent phase. (L.45)

Complex eigenvalues ≠ Global complex structure. (L.46)

Local phase ≠ Universal clock. (L.47)

Half-turn opposition ≠ Quarter-turn conjugacy. (L.48)

Elliptic Relation ≠ Conservative dynamics. (L.49)

Dissipation ≠ Residual. (L.50)

L.5.5 Readout and exposure

Readout ≠ State. (L.51)

Conjugate response ≠ Hidden latent state. (L.52)

Conjugate response ≠ Residual. (L.53)

Conjugate response ≠ Automatic loss. (L.54)

CAPM `Q` ≠ Volatility. (L.55)

CAPM `Q` ≠ VaR. (L.56)

CAPM `Q` ≠ Haircut `A - R`. (L.57)

CAPM valuation phase ≠ Market dynamic phase. (L.58)

Oscillator quality factor `Q_mode` ≠ CAPM `Q_fin`. (L.59)

L.5.6 Measurement, movement, and commitment

Measurement rotation ≠ State movement. (L.60)

Exposure ≠ Realized consequence. (L.61)

Movement ≠ Gate passage. (L.62)

Gate passage ≠ Complete integration. (L.63)

Ledger entry ≠ Truth. (L.64)

Institutional authority ≠ Factual certainty. (L.65)

Internal certainty ≠ Public objectivity. (L.66)

Repeated publication ≠ Independent redundancy. (L.67)

L.5.7 Residual and Γ

Residual ≠ Generic uncertainty. (L.68)

Residual ≠ Noise. (L.69)

Residual ≠ Model error. (L.70)

Residual ≠ Γ. (L.71)

Γ ≠ Universal entropy. (L.72)

Γ ≠ Universal friction. (L.73)

Γ ≠ Universal least-action correction. (L.74)

Residual reduction ≠ Residual justice. (L.75)

A residual may be transferred to another carrier rather than resolved.

L.5.8 Time and history

Sequence index ≠ Operational time. (L.76)

Recursive depth ≠ Historical time. (L.77)

Phase time ≠ Ledger time. (L.78)

Stored history ≠ Time-bearing history. (L.79)

Physical reversibility ≠ Historical reversibility. (L.80)

Same present trace ≠ Same operational world. (L.81)

L.5.9 Cross-domain interpretation

Functional correspondence ≠ Material identity. (L.82)

Mathematical isomorphism ≠ Common physical mechanism. (L.83)

Similar diagram ≠ Shared generator. (L.84)

Ancient symbolic pairing ≠ Modern empirical proof. (L.85)

Combinatorial uniqueness ≠ Universal semantic capacity. (L.86)

Finance geometry ≠ Quantum finance ontology. (L.87)

Legal gate ≠ Quantum measurement physically. (L.88)

AI memory ≠ Human consciousness. (L.89)

L.6 Common Editorial Errors

L.6.1 Universalizing an optional module

Incorrect:

> Every operational world has a PORE triple and complex phase.

Correct:

> Every full operational-world instantiation requires trace-conditioned historical return; PORE compression and complex phase are optional modules requiring separate validation.

L.6.2 Calling Q the residual

Incorrect:

> Q is the hidden remainder excluded from real valuation.

Correct:

> Q is the conjugate monetary exposure generated by the declared valuation phase; residual arises from incomplete model fit or commitment after movement.

L.6.3 Calling all opposition Yin–Yang

Incorrect:

> Every pair of opposite signs is a Yin–Yang conjugate pair.

Correct:

> Conjugacy, half-turn opposition, negative correlation, and complementary function are mathematically distinct relations.

L.6.4 Treating a ledger as objective truth

Incorrect:

> Once recorded, the event is objectively real.

Correct:

> Once recorded, the event is historically operative under the ledger's protocol; broader objectivity requires access, compatibility, redundancy, and possible revision.

L.6.5 Treating phase as time without history

Incorrect:

> A recurrent phase automatically creates a time-bearing world.

Correct:

> Phase orders a Relation cycle; the world becomes time-bearing only when retained trace and residual alter later possibility.

L.6.6 Replacing evidence with geometric vocabulary

Incorrect:

> The system has curvature, holonomy, and a semantic gauge field.

Correct:

> The model should first identify transport maps, path-dependent residual, closed-loop mismatch, and predictive gain. Geometric terminology follows only when those structures are operationally defined.

L.7 Publication Positioning

The framework should be presented as:

> **A protocol-bound formal architecture and falsifiable research programme for self-referential operational world formation.**

It should not be presented as:

- a completed Theory of Everything;
- proof that finance is quantum;
- proof that traditional diagrams encode modern physics;
- a universal theory of consciousness;
- a substitute for established domain models;
- an empirically validated unified field across all examples.

Its strongest current contribution is the interface discipline among:

Base
→ Relation
→ Commitment
→ Recompiled Base. (L.90)

And the corresponding typed runtime:

Trace
→ Filtration
→ Policy
→ Changed Transition
→ Gate
→ Ledger + Residual
→ Historical Return. (L.91)

L.8 Financial Disclaimer

The financial examples in this article are conceptual and mathematical illustrations.

They are not:

- investment advice;
- a recommendation to buy or sell any security;
- a complete valuation procedure;
- a complete risk-management model;
- a substitute for professional accounting, legal, regulatory, or investment analysis.

CAPM assumptions may fail.

Estimated parameters may be unstable.

The conjugate variable Q does not capture every form of financial risk.

Any practical use requires:

- independent model validation;
- appropriate data;
- jurisdiction-specific accounting and legal review;
- liquidity and tail-risk analysis;
- professional judgment.

L.9 Scientific and Medical Scope Note

The quantum examples are formal models of quantum observers and should not be used to claim that every social or financial process is materially quantum.

The semantic-breather and Chinese-medicine-related sources are used only for limited structural vocabulary concerning recurrence, cadence, trace formation, and failure modes.

They do not establish medical diagnosis or treatment.

No clinical recommendation follows from this article.

L.10 Final Article Audit

Before release, confirm that the manuscript answers all of the following.

General runtime

1. What makes the process self-referential?
2. Which trace enters which observer's filtration?
3. How does policy change?
4. How does policy alter later state or transition law?

5. What acts as gate?
6. What enters the ledger?
7. What remains residual?
8. How does history return?

Effective Base

9. What does the compiler use?
10. Is the output descriptive or predictively sufficient?
11. What omitted-history test was applied?
12. Are jumps and smooth regimes separated?

Relation

13. What generator has been declared or estimated?
14. What is its spectral type?
15. Is an invariant mode stable?
16. Is ordinary complex structure actually earned?
17. Does the readout detect the mode?

Conjugate readout

18. What is the scalar readout?
19. What is the Relation coordinate?
20. Are units compatible?
21. Is orientation declared?
22. Does perturbation recover the proposed conjugate response?

Commitment

23. What separates movement from admission?
24. Who has authority?
25. Is admission partial or binary?
26. Where is the record stored?
27. Can it be reopened?
28. Who carries the residual?

Evidence

29. Is each claim labelled `[D]`, `[E]`, `[F]`, `[H]`, or `[A]`?
30. Is each module assigned a maturity level?
31. Are failure conditions explicit?
32. Are negative results retained?
33. Is independent replay possible?
34. Are nonclaims clearly stated?

If any required answer is absent, the corresponding claim should be narrowed.

L.11 Closing Statement of the Appendices

The appendices have transformed the article's conceptual architecture into:

- typed notation;
- protocol templates;
- Base-sufficiency tests;
- generator classification;
- conjugate-readout proofs;
- non-complex alternatives;
- local transport conditions;
- CAPM audit equations;
- $\Delta 5$ operator tests;
- open-generator diagnostics;
- claim and replay registries;
- final cross-domain safeguards.

The complete operational sequence remains:

Protocol
→ Base
→ Trace
→ Filtration
→ Adaptive Policy
→ Relation
→ Readout
→ Optional Conjugate Response
→ Movement
→ Candidate Consequence
→ Gate
→ Ledger + Residual
→ Historical Return

→ Recompiled Base. (L.92)

The final criterion is equally compact:

> **A framework earns generality not by using the same metaphor everywhere, but by preserving the same typed questions while allowing each domain's answers, operators, evidence, and failure modes to remain genuinely different.**

Appendix M — Visual Architecture, Figure Specifications, and Table Package

M.1 Purpose

The article's main difficulty is not the number of equations.

It is the number of distinctions that must remain visible simultaneously:

- state versus trace;
- compiler versus generator;
- movement versus measurement;
- exposure versus residual;
- gate versus ledger;
- recurrence versus time-bearing history;
- local complex mode versus global operational world.

A visual package should therefore clarify typed relationships rather than merely decorate the manuscript.

The recommended figures are organized in the same order as the argument:

1. minimum world-forming runtime;
2. Base–Relation–Commitment architecture;
3. three closure problems;
4. signed-Relation classification;
5. dynamic-to-readout intertwining;
6. measurement–movement–commitment separation;
7. worked-example coverage;
8. falsifiable research programme.

The governing visual principle is:

> **Use arrows for causal or operational maps, layers for logical types, and planes only for variables that genuinely inhabit a shared geometric**

space.

M.2 Figure 1 — From Trace to a Time-Bearing World

Placement

Insert after Section 2.7 or at the end of Part I.

Title

From Recorded Outcome to Future-Generating History

Subtitle

The minimum causal sequence that distinguishes feedback, self-reference, commitment, and operational world formation

Format

16:9 horizontal process diagram.

Main sequence

Trace_k
→ Observer Filtration_k
→ Adaptive Policy_{k+1}
→ Changed Transition Law
→ New State and Trace
→ Candidate Consequence
→ Gate
→ Ledger + Residual
→ Historical Return. (M.1)

Visual organization

Upper lane — Internal observer recursion

Trace
→ Filtration
→ Policy
→ Changed Future Instrument or Transition.

Use a circular return arrow from the changed transition law to the next trace.

Lower lane — Historical commitment

Candidate Consequence
→ Gate
→ Ledger + Residual
→ Future Constraint.

Use a second return arrow from ledger and residual to:

- filtration;
- policy;
- generator;
- gate.

Centre distinction

Place a vertical divider between:

Self-Referential Process

and:

Full Operational World Formation

The first requires trace-conditioned policy backreaction.

The second additionally requires gate, ledger, residual, and historical return.

Formula block

$y_k = h_p(X_k)$. (M.2)

$\mathscr{F}_{a,k} = \sigma(y_{1:k}, L_{1:k}^{\text{accessible}}, s_{a,1:k})$. (M.3)

$\pi_{a,k+1} = f_{a,p}(\mathscr{F}_{a,k})$. (M.4)

$X_{k+1} = \mathscr{B}_p(X_k, \pi_{k+1}, L_k, \Gamma_k, \ldots)$. (M.5)

Bottom principle

A trace becomes part of a world only when its retained consequence changes what the world can do next.

M.3 Figure 2 — Base, Relation, Commitment, and Recompiled Base

Placement

Insert at the beginning of Part III or after Section 11.

Title

The Three-Function Grammar of Operational World Formation

Subtitle

Base determines the present condition, Relation determines possible transformation, and Commitment determines inherited history

Format

16:9 circular architecture.

Core cycle

Base_k
→ Relation_k
→ Candidate Movement
→ Commitment_k
→ Recompiled Base_{k+1}. (M.6)

Base panel

Show:

- detailed state x_k;
- optional compiler c_p;
- candidate Effective Base Ξ_k;
- observer and resource states;
- ledger and residual inheritance.

Formula:

$\Xi_k = C_p(W_k)$. (M.7)

Optional PORE signature:

$\Xi_k = (\varrho_k, \gamma_k, \tau_k)$. (M.8)

Label:

Optional compression, not universal ontology

Relation panel

Show:

- transition law;
- generator;
- invariant modes;
- elliptic, parabolic, hyperbolic, or mixed regime.

Formula:

$\delta\Xi_{k+1} \approx A_r\delta\Xi_k + G_r\delta u_k$. (M.9)

Label:

Generator before geometry

Commitment panel

Show:

Candidate
→ Gate
→ Ledger + Residual.

Formula:

$e_k = \mathcal{G}_p(c_k)$. (M.10)

$L_{k+1} = L_k \oplus T_p(e_k)$. (M.11)

$\mathscr{R}_{k+1} = \Re_p(c_k, e_k, L_{k+1})$. (M.12)

Label:

Movement becomes history only through governed admission

Centre statement

Base → Relation → Commitment → Recompiled Base

Outer ring

Protocol surrounds the entire cycle:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (M.13)

This visually establishes that no module is protocol-free.

M.4 Figure 3 — The Three Closure Problems

Placement

Insert after Section 3.

Title

Three Different Meanings of Closure

Subtitle

Predictive compression, conjugate measurement, and historical commitment solve different problems

Format

Three-column infographic.

Left column — Effective-State Closure

Question

Can omitted history be removed without losing task-relevant causality?

Main objects

X-history
→ Compiler C_ρ
→ Effective Base Ξ_k.

Test

$\Pr(\text{Future} \mid \Xi_k, H_k)$
$\approx \Pr(\text{Future} \mid \Xi_k)$. (M.14)

Failure

Omitted history still predicts:

- branch;
- observer policy;
- regime;
- residual;
- gate outcome.

Warning

Compact description ≠ sufficient state

Centre column — Conjugate Measurement Closure

Question

Does a scalar readout preserve the active Relation's signed transformation grammar?

Main objects

Elliptic mode $J^2 = -I$

- readout $Y = a(x)$
  → conjugate response $Q_Y = -a(Jx)$.

Test

$dY/d\theta = -\mathcal{Q}_Y$. (M.15)

$d\mathcal{Q}_Y/d\theta = Y$. (M.16)

Warning

Conjugate response ≠ residual

Right column — Historical Closure

Question

Has a consequence become authoritative and future-conditioning?

Main objects

Candidate
→ Gate
→ Ledger + Residual.

Test

$\Pr(\text{Future} \mid X, L, \mathscr{R})$
$\neq \Pr(\text{Future} \mid X)$. (M.17)

Warning

Ledger entry ≠ complete truth

Bottom comparison

Effective-state closure
concerns
compression.

Conjugate closure
concerns
transformation.

Historical closure
concerns
commitment.

M.5 Figure 4 — Signed-Relation Classification

Placement

Insert after Section 13 or Appendix F.

Title

One Coupled Pair, Three Different Transformation Grammars

Subtitle

The sign of the normalized operator square determines circulation, accumulation, or amplification

Format

Three-panel horizontal comparison.

Panel A — Elliptic

Operator:

$J^2 = -I$. (M.18)

Exponential:

$e^{(J\xi)} = I \cos \xi + J \sin \xi$. (M.19)

Readout equations:

$dY/d\xi = -Q_Y$. (M.20)

$dQ_Y/d\xi = Y$. (M.21)

Visual:

closed circular orbit.

Interpretation:

- phase;
- quadrature;
- restorative circulation;
- ordinary complex numbers.

Panel B — Parabolic

Operator:

$N^2 = 0$. (M.22)

Exponential:

$e^{(N\xi)} = I + \xi N$. (M.23)

Readout equations:

$dY/d\xi = -\mathcal{P}_Y$. (M.24)

$d\mathcal{P}_Y/d\xi = 0$. (M.25)

Visual:

sheared parallel trajectories.

Interpretation:

- drift;
- accumulation;
- unresolved carry;
- nilpotent or dual-number grammar.

Panel C — Hyperbolic

Operator:

$K^2 = I$. (M.26)

Exponential:

$e^{(K\xi)} = I \cosh \xi + K \sinh \xi$. (M.27)

Readout equations:

$dY/d\xi = \mathscr{H}_Y$. (M.28)

$d\mathscr{H}_Y/d\xi = Y$. (M.29)

Visual:

hyperbolic expanding and contracting branches.

Interpretation:

- confirmation;
- polarization;
- amplification and contraction;
- split-complex grammar.

Top statement

Two variables do not determine their algebra. The return operator does.

Bottom statement

$C\chi^2 = \chi I$. (M.30)

$\chi < 0 \Rightarrow$ elliptic. (M.31)

$\chi = 0 \Rightarrow$ parabolic. (M.32)

$\chi > 0 \Rightarrow$ hyperbolic. (M.33)

M.6 Figure 5 — Complex Intertwining Between Dynamic and Readout Planes

Placement

Insert after Section 16 or Appendix E.

Title

From a Dynamic Elliptic Mode to a Conjugate Readout Plane

Subtitle

The planes are not identical, but their quarter-turn structures can be intertwined

Format

16:9 two-plane diagram.

Left plane — Dynamic mode

Show vector:

$x \in E$. (M.34)

Quarter-turn:

$x \to Jx$. (M.35)

Condition:

$J^2 = -I$. (M.36)

Use axes:

u, v.

Represent:

$z_D = u + iv$. (M.37)

Centre map

Show a large arrow labelled:

Readout map Φ_a.

Define:

$Y(x) = a(x)$. (M.38)

$\mathcal{Q}_Y(x) = -a(Jx)$. (M.39)

$\Phi_a(x) = [Y(x), \mathcal{Q}_Y(x)]^T$. (M.40)

Right plane — Readout mode

Use axes:

Y, $\mathcal{Q}_Y$.

Show:

$[Y; \mathcal{Q}_Y]$
$\rightarrow [-\mathcal{Q}_Y; Y]$. (M.41)

Define:

$Z_Y = Y + i\mathcal{Q}_Y$. (M.42)

Commuting-square statement

$\Phi_a J = J_R \Phi_a$. (M.43)

Bottom distinction

Dynamic complex plane
≠
Readout complex plane.

But:

Dynamic quarter-turn
→
Readout quarter-turn.

Side warning

The construction requires:

- a stable elliptic mode;
- a nonzero readout;
- compatible units;

- declared orientation.

M.7 Figure 6 — Measurement, Movement, and Commitment

Placement

Insert after Section 17.

Title

Reading Exposure Is Not the Same as Moving the State

Subtitle

Four operations that are commonly collapsed into one

Format

Four-stage left-to-right diagram.

Stage 1 — Passive measurement

State fixed:

$Z = Y + i\mathcal{Q}_Y.$ (M.44)

Readout rotates:

$M_\varphi(Z) = \mathrm{Re}[e^{(i\varphi)}Z].$ (M.45)

At:

$\varphi = \pi/2,$ (M.46)

readout becomes:

$-\mathcal{Q}_Y.$ (M.47)

Caption:

Exposure revealed; no state movement

Stage 2 — Active movement

State changes:

$Z' = e^{(i\Delta\theta)}Z.$ (M.48)

Readout movement:

$$\Delta Y = Y(\cos \Delta\theta - 1) - \mathcal{Q}_Y \sin \Delta\theta. \quad (M.49)$$

For small movement:

$$\Delta Y \approx -\mathcal{Q}_Y \Delta\theta. \quad (M.50)$$

Caption:

Economic, physical, or semantic consequence generated

Stage 3 — Gate

Candidate:

$$c = C_p(Z, Z'). \quad (M.51)$$

Gate:

$$e = \mathcal{G}_p(c). \quad (M.52)$$

Outputs:

- admit;
- reject;
- defer;
- partial admission.

Caption:

Authority determines operational status

Stage 4 — Ledger and residual

Ledger:

$$L' = L \oplus \mathrm{Trace}(e). \quad (M.53)$$

Residual:

$$\mathscr{R}' = \Re_p(c, e, L'). \quad (M.54)$$

Caption:

Admitted history and unresolved remainder

Bottom non-equivalence chain

Measurement

≠ Exposure realization
≠ Movement
≠ Gate passage
≠ Ledger history. (M.55)

M.8 Figure 7 — CAPM as the Mature Conjugate-Readout Example

Placement

Insert in Section 22.

Title

From CAPM Discounting to a Complex Valuation Plane

Subtitle

A mature scalar valuation generates an exact conjugate exposure without changing the underlying present value

Format

16:9 finance-first infographic.

Left panel — Declared valuations

Baseline amplitude:

$A_t = CF_t/(1 + r_base)^t$. (M.56)

CAPM value:

$R_t = CF_t/(1 + r_CAPM)^t$. (M.57)

Required return:

$r_CAPM = r_base + \beta ERP$. (M.58)

Centre panel — Complex completion

Phase:

$\cos \theta_t = R_t/A_t$. (M.59)

Conjugate coordinate:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (M.60)

State:

$Z_t = R_t + iQ_t = A_t e^{(i\theta_t)}$. (M.61)

Visual:

right triangle plus complex-plane point.

Right panel — Exact meaning of Q

Principal theorem:

$\partial R_t / \partial \theta_t = -Q_t$. (M.62)

Required-return equivalence:

$Qd\theta = [tR/(1 + r)]dr$. (M.63)

Cycle:

$R \rightarrow -Q \rightarrow -R \rightarrow Q \rightarrow R$. (M.64)

The CAPM source defines Q as a first-order monetary phase exposure and explicitly distinguishes it from haircut, volatility, Value at Risk, realized loss, and residual.

Bottom runtime

CAPM Filter
→ R
→ θ
→ Q
→ Measurement
→ Movement
→ P&L
→ Gate
→ Ledger + Residual.

Disclaimer strip

Conceptual and mathematical framework only — not investment advice

---

M.9 Figure 8 — From Recurrence to Time-Bearing History

Placement

Insert after Section 20.

Title

A Cycle Is Not Yet a History

Subtitle

Phase may return to the same state, while a ledger makes the world irreversibly different

Format

Split-screen comparison.

Left side — Recurrent but nonhistorical

Show a circular oscillator:

$Z(\theta + 2\pi) = Z(\theta)$. (M.65)

After one cycle:

State returns.

Ledger unchanged:

$L_{k+1} = L_k$. (M.66)

Caption:

Phase time without new historical difference

Right side — Ledgered recurrence

Show the same oscillator, but each completed cycle emits a trace:

Cycle_k
→ Gate
→ Ledger entry.

Formula:

$L_{k+1} = L_k \oplus \text{Trace}(\text{Cycle}_k)$. (M.67)

Even if:

$Z(\theta + 2\pi) = Z(\theta)$, (M.68)

the world-state differs:

$\mathcal{W}_{k+1} \neq \mathcal{W}_k$. (M.69)

Caption:

The state returns, but the world inherits a new past

Bottom criterion

Storage:

L exists. (M.70)

Time-bearing history:

$\partial \text{Future}/\partial L \neq 0$
or
$\partial \text{Future}/\partial \mathscr{R} \neq 0$. (M.71)

M.10 Figure 9 — Module Coverage Across Domains

Placement

Insert after Section 25.

Title

No Single Example Supplies the Whole Framework

Subtitle

Different domains mature different modules

Format

Heat-map matrix.

Columns

- Protocol;
- Effective Base;
- Trace;
- Filtration;
- Adaptive Policy;
- Generator;
- Complex Readout;
- Gate;
- Ledger;
- Residual;
- Historical Return;
- Replication.

Rows

- Self-Referential Quantum Observer;
- CAPM Conjugate Valuation;
- PORE;
- Δ5;
- Dissipative/Breather Models;
- AI Agent;
- Law;
- Organization;
- Science.

Colour levels

- strong formal module;
- operational candidate;
- structural analogy;
- absent or not required.

Main messages

Quantum observer

Strongest in:

Trace → Filtration → Adaptive Policy.

The source formally models record-conditioned instrument selection, internal delta-certainty, latching, conditional agreement, and redundancy-based objectivity.

CAPM

Strongest in:

Scalar Readout → Conjugate Exposure.

PORE

Strongest in:

Protocol → Effective Base Compiler → Intervention Harness.

PORE presents $\Xi\hat{}=(\rho\hat{},\gamma\hat{},\tau)$ as a protocol-relative loop coordinate and explicitly rejects ontology monopoly or global validity.

Residual framework

Strongest in:

Gate → Ledger + Residual → Optional Γ → Future Path.

The source treats Γ as a conditional mathematical interface rather than proof of universal least-action dynamics.

Δ5

Strongest in:

Declared Half-Turn → Energy → Antisymmetric Sector.

M.11 Figure 10 — Research Programme and Rejection Logic

Placement

Insert after Section 29 or immediately before the Conclusion.

Title

How the Framework Earns Each Additional Layer

Subtitle

Every stronger claim requires a new activation test and can be independently rejected

Format

Vertical ladder.

Level 1 — Protocol

Question:

Can the system boundary, trace, intervention, gate, access, and residual rules be declared?

Failure:

No stable operational object.

Level 2 — Recursive path

Question:

Does trace ablation change later policy or state?

Failure:

Record without active self-reference.

Level 3 — Effective Base

Question:

Does omitted history improve prediction?

Failure:

Descriptive signature only.

Level 4 — Generator

Question:

Can a stable transition law be estimated?

Failure:

No earned low-dimensional Relation.

Level 5 — Algebra

Question:

Is the mode elliptic, parabolic, hyperbolic, mixed, or absent?

Failure:

Reject inappropriate geometry.

Level 6 — Readout

Question:

Does the conjugate response pass perturbation and unit tests?

Failure:

No operational conjugate completion.

Level 7 — Commitment

Question:

Does gate passage create persistent downstream difference?

Failure:

Movement without historical admission.

Level 8 — Historical return

Question:

Do ledger and residual alter later possibility?

Failure:

Archive without time-bearing world.

Level 9 — Replication

Question:

Can another team reproduce and transport the result?

Failure:

Local exploratory module only.

Bottom principle

Failure at one layer narrows that claim; it does not license reinterpretation of the failure as hidden success.

M.12 Recommended Table Package

Table 1 — Typed Object Register

Placement

End of Section 11.

Object	Symbol	Type	Main question
Detailed state	x	causal state	What generates the future?
Trace	y	admitted observation	What became accessible?
Filtration	$\mathcal{F}$	information structure	What can the observer use?
Policy	π	rule or kernel	What action is selected?
Compiler	C_p	map	How is the Base compressed?
Effective Base	Ξ	reduced state	What closes prediction?
Generator	A_r	operator	How can the Base change?
Readout	Y	scalar	What is declared?
Conjugate response	Q_Y	sensitivity	What is the oriented response?
Gate	$\mathcal{G}_p$	operator	What becomes admitted?
Ledger	L	record structure	What becomes authoritative history?
Residual	$\mathcal{R}$	typed remainder	What remains unresolved?

Object	Symbol	Type	Main question
Γ	`Γ`	optional functional	How may residual bend future paths?

Table 2 — Three Closure Matrix

Placement

End of Section 3.

Closure	Input	Output	Principal test	Failure meaning
Effective-state	history	reduced Base	omitted-history test	state compression insufficient
Conjugate measurement	elliptic mode + readout	`Y + iQ_Y`	perturbation/ intertwining	readout does not close Relation
Historical	candidate consequence	ledger + residual	downstream causal effect	admission lacks future authority

Table 3 — Relation Algebra Comparison

Placement

End of Section 13 or Appendix F.

Branch	Operator	Invariant	Behaviour	Suitable algebra
Elliptic	`J² = −I`	`u² + v²`	circulation	complex
Parabolic	`N² = 0`	nilpotent direction	accumulation	dual-number or real Jordan
Hyperbolic	`K² = I`	`u² − v²`	expansion/ contraction	split-complex
Mixed	block-dependent	multiple	combined	higher-dimensional real/ block

Table 4 — Movement and Commitment Distinctions

Placement

End of Section 17.

Stage	Changes state?	Changes readout?	Requires authority?	Changes ledger?
Passive measurement	No	Yes	No	No
Active movement	Yes	Usually	No	No
Candidate translation	No new movement required	Interprets movement	No	No
Gate	Status	Possibly	Yes	Prepares update
Ledger	Historical state	Not necessarily	Yes	Yes

Stage	Changes state?	Changes readout?	Requires authority?	Changes ledger?
Residual	Future constraint	Not necessarily	Protocol-relative	May be separately recorded

Table 5 — Module Maturity Matrix

Placement

Section 27.

Level	Name	Minimum evidence
M0	metaphor	resemblance only
M1	typed mapping	functional roles separated
M2	operational correspondence	protocol, proxies, intervention, falsifier
M3	formal module	equations and internal validation
M4	integrated domain model	Base–Relation–Commitment return
M5	mature operational theory	replication, intervention, transport, failure boundary

Table 6 — Claim Status Matrix

Placement

Appendix K.

Status	Meaning	Example
[D]	framework definition	time-bearing world
[E]	established inside source construction	$\partial R/\partial \theta = -Q$
[F]	formal consequence	$\Phi_a J = J_R \Phi_a$
[H]	empirical hypothesis	persistent market elliptic mode
[A]	structural analogy	legal commitment as world formation

M.13 Master Architecture Figure

Suggested use

Use as:

- graphical abstract;
- article cover figure;
- conference poster summary;
- OSF project image.

Title

From Trace to Time-Bearing Worlds

Subtitle

A Protocol-Bound Architecture for Self-Reference, Conjugate Geometry, and Ledgered Commitment

Central architecture

Protocol ring:

$P = (B, \Delta, h, u, \mathcal{G}, \mathcal{A}, \Re)$. (M.72)

Inside the ring:

1. Base

X_k
→ optional $\Xi_k = C_p(W_k)$.

2. Trace

$y_k = h_p(X_k)$. (M.73)

3. Observer recursion

y_k
→ $\mathcal{F}_{a,k}$
→ $\pi_{a,k+1}$
→ changed transition.

4. Relation

A_r
→ elliptic, parabolic, hyperbolic, or mixed mode.

5. Optional conjugate completion

$Y = a(x)$. (M.74)

$\mathcal{Q}_Y = -a(Jx)$. (M.75)

$Z_Y = Y + i\mathcal{Q}_Y$. (M.76)

6. Movement

$X_k \to X_{k+1}$. (M.77)

7. Commitment

Candidate
→ Gate
→ Ledger + Residual.

8. Historical return

L_{k+1}, $\mathscr{R}_{k+1}$
→ next Base, policy, gate, and generator.

Bottom statement

The past becomes time when it changes the future-generating condition.

M.14 Visual Nonclaims Panel

Every major infographic should include a compact nonclaim footer where space permits.

Recommended footer:

> Complex geometry is conditional, PORE is protocol-relative, Q is conjugate exposure rather than residual, movement is not commitment, and cross-domain role similarity does not establish common material mechanism.

This footer reflects the governing Handoff, which explicitly requires complex geometry, PORE compression, and Γ to remain independently activated modules rather than universal assumptions.

M.15 Visual Style Guidance

Geometry

Use circles only for:

- genuine recurrence;
- elliptic phase;
- closed causal return.

Do not use a circular arrow merely to indicate a list.

Arrows

Use:

- solid arrows for established operational maps;
- dashed arrows for hypotheses;
- dotted arrows for analogies;
- double arrows for transport or reconciliation.

Boxes

Use distinct shapes:

- rounded rectangle — state or record;
- diamond — gate;
- circle — observer or recurrent mode;
- hexagon — protocol;
- document stack — ledger;
- cloud or dotted field — residual.

Colour logic

A publication design may use consistent semantic families:

- Base — neutral or structural;
- Relation — dynamic;
- Commitment — authoritative;
- Residual — unresolved;
- Hypothesis — visibly different from established structures.

The final colour selection should remain accessible under colour-blind and grayscale viewing.

Mathematical placement

Equations should be placed next to the object they govern.

Avoid placing all equations in one decorative block disconnected from their operational role.

Source Provenance and Reference Architecture

R.1 Source hierarchy

The article should distinguish three source levels.

Governing synthesis

v260722v2_Handoff Summary for the Next LLM Session

This document determines the article's domain-general architecture, module hierarchy, mandatory distinctions, nonclaims, and writing requirements. It treats the typed recursive core and historical return as mandatory while treating PORE compression and conjugate geometry as optional modules.

Core formal sources

1. **The Post-Ontological Reality Engine (PORE)**

   Supplies:

   - protocol-relative effective loop coordinates;
   - $\hat{\Xi} = (\hat{\rho}, \hat{\gamma}, \tau)$;
   - intervention harness;
   - smooth-regime versus jump separation;
   - operational rather than ontological positioning.

   The source explicitly rejects ontology monopoly, unique decomposition, universal micro-substrate, and global validity.

2. **Self-Referential Observers in Quantum Dynamics: A Formal Theory of Internal Collapse and Cross-Observer Agreement**

   Supplies:

   - internal trace;
   - adaptive policy;
   - filtration;
   - well-posed stochastic histories;
   - delta-certainty;
   - latching;
   - frame-conditioned agreement;
   - redundancy-based objectivity.

   The source models the observer through records, trace-conditioned instruments, and completely positive maps, while positioning the work as a formal theory rather than a new quantum interpretation.

3. **From Discounted Value to Conjugate Risk: CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle**

   Supplies:

   - exact CAPM conjugate valuation;
   - $Q = -\partial R/\partial \theta$;
   - quarter-turn measurement cycle;
   - unit-consistent complex readout;
   - separation of measurement, movement, P&L, gate, and ledger.

   The source explicitly states that the construction is a conceptual and mathematical research programme and not investment advice.

4. **Residual Made Mathematical: Variational Phase-Ledger Dynamics from Self-Referential Observers to L–Γ Worlds**

   Supplies:

   - candidate field;
   - gate;
   - ledger;
   - residual;
   - optional Γ functional;
   - residual-conditioned future path selection.

   The source expressly treats generalized least-action language as conditional and does not claim that all macro systems obey one universal variational law.

Important Relation extension

5. **Δ5 Phase Opposition in HeTu: Pairwise Minimum-Dissipation Cycles and a D_{10}–Spectral Extension of the Slot Interpretation**

   Supplies:

   - half-turn operator T_5;
   - symmetric and antisymmetric sectors;
   - pair-energy minimization;
   - Fourier odd-sector structure;
   - conditional dissipative locking;
   - coarse-grained five-mode proposal.

   Its role is a specific Relation architecture, not a universal source of phase.

Optional extensions

6. 散逸力学系の量子力学

   Supplies an example of a dissipative Langevin system represented through Schrödinger-type stochastic mechanics. Its use here is restricted to the compatibility of phase-bearing representation with friction and noise.

7. **The Slot Interpretation of HeTu and LuoShu: A Rigorous Mathematical and Semantic Proof by Wolfram 4.1 GPTs**

   Supplies proposed slot-capacity interpretations and the combinatorial structures later distinguished from Δ5. The present article accepts the secure combinatorial results while treating broader semantic-capacity claims as requiring independent support.

R.2 Suggested reference-entry format

Where full publication metadata are unavailable, use a transparent manuscript reference format rather than inventing journal details.

Example

Yeung, Chung Leung Danny. *Self-Referential Observers in Quantum Dynamics: A Formal Theory of Internal Collapse and Cross-Observer Agreement*. Independent research manuscript, 5 October 2025.

Example

Yeung, Chung Leung Danny. *From Discounted Value to Conjugate Risk: CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle*. Independent research manuscript.

Example

Yeung, Chung Leung Danny. *Residual Made Mathematical: Variational Phase-Ledger Dynamics from Self-Referential Observers to L–Γ Worlds*. Independent research manuscript.

Where authorship or date is not visible in the supplied source, the reference should retain only the verified title and document type.

Do not infer journal, volume, publisher, DOI, or year without source support.

R.3 Reference-list draft

Core project sources

Yeung, Chung Leung Danny. *Self-Referential Observers in Quantum Dynamics: A Formal Theory of Internal Collapse and Cross-Observer Agreement*. Independent research manuscript, 5 October 2025.

Yeung, Chung Leung Danny. *From Discounted Value to Conjugate Risk: CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle*. Independent research manuscript.

Yeung, Chung Leung Danny. *Residual Made Mathematical: Variational Phase-Ledger Dynamics from Self-Referential Observers to L–Γ Worlds*. Independent research manuscript.

The Post-Ontological Reality Engine (PORE). Independent research manuscript.

Δ5 Phase Opposition in HeTu: Pairwise Minimum-Dissipation Cycles and a D10–Spectral Extension of the Slot Interpretation. Independent research manuscript.

The Slot Interpretation of HeTu and LuoShu: A Rigorous Mathematical and Semantic Proof by Wolfram 4.1 GPTs. Independent research manuscript.

v260722v2_Handoff Summary for the Next LLM Session. Internal synthesis and article-development document, 22 July 2026.

Open-generator source

Yasue, Kunio. *散逸力学系の量子力学* [Quantum Mechanics of Dissipative Dynamical Systems]. Source document supplied for the present synthesis. The uploaded text identifies the author as 名大・理 保江邦夫 and develops a quantum description of Langevin-type dissipative systems.

R.4 Source-status note

The present article is a synthesis.

No single source independently establishes the complete architecture:

Protocol
→ Base
→ Trace
→ Filtration
→ Adaptive Policy
→ Relation
→ Readout
→ Conjugate Response
→ Movement
→ Gate
→ Ledger + Residual
→ Historical Return. (R.1)

The article's contribution lies in:

- assembling these modules;
- preserving their logical types;
- defining activation conditions;
- identifying exact versus hypothetical claims;
- supplying a falsification and maturity programme.

Accordingly:

Source-established module
≠
Source-established total theory. (R.2)

Final Publication Note

The manuscript now contains:

- Abstract and reader contract;
- Parts I–VII;
- Sections 0–30;
- Appendices A–M;
- notation and protocol templates;

- Effective Base tests;
- generator and mode classification;
- formal intertwining proof;
- parabolic and hyperbolic alternatives;
- local transport and curvature conditions;
- complete CAPM derivations;
- $\Delta 5$ derivations and limitations;
- dissipative and breather diagnostics;
- reproducibility and claim registries;
- glossary and non-equivalence audit;
- publication-ready figure and table specifications;
- source-provenance architecture.

The final editorial operation should normalize:

- `[S]` to `[E]`;
- all equation references;
- `ρ` versus `ϱ`;
- `Q` subscripts;
- file-source citations into the selected publication reference style;
- figure and table numbering after layout.

The manuscript's final one-sentence claim remains:

> **A protocol-bound operational world becomes time-bearing when its accessible traces alter future policy, its gates convert selected consequences into ledgered and residual history, and that inherited history recompiles the Base from which subsequent Relation begins; where the active Relation closes elliptically, complex numbers provide the natural local grammar of conjugate readout.**

Reference

- From Discounted Value to Conjugate Risk - CAPM Phase Geometry, the Financial Meaning of Q, and the R → −Q → −R Measurement Cycle
https://osf.io/yucvm/files/osfstorage/6a5ea0341b206ba447f5ff46

- When Phase Becomes a Clock - Complex Completion, Secondary Time, and the Search for Time-Bearing Worlds Across Domains
https://osf.io/yucvm/files/osfstorage/6a5d19e395f2a4520ee147e6

- When Valuation Becomes a World - Complex Finance, Internal Time, and the Residue of

Quantum Strangeness
https://osf.io/yucvm/files/osfstorage/6a53876497a8be0d215b9278

- When Valuation Becomes a World, Part II: Inside the Valuation World - Derivative Entanglement, Relative Frames, and Curved Financial Geometry
https://osf.io/yucvm/files/osfstorage/6a4abb8fcaf0a0c36ddaa3e3

- The Complex Residual Principle: How Phase, Projection, Residual, Trace, and Emergent Time Reappear across Quantum Physics, Financial Markets, and Large Language Models
https://osf.io/yucvm/files/osfstorage/6a53876497a8be0d215b9278

- Finance Geometry: Complex Valuation, Risk Pressure, and the Hidden Coordinate Behind Mature Finance Filters
https://osf.io/yucvm/files/osfstorage/6a4abb8fcaf0a0c36ddaa3e3

- Imaginary Time as Admissibility Depth: A Ledger Ontology of Wick Rotation, Macro Systems, and Physical Time
https://osf.io/mvq6e/files/osfstorage/6a405c693e12266e39804e08

- The True Nature of Technical Analysis - An Operator-First Interpretation of Market Charts, Volume, Waves, Gann Geometry, and Financial Self-Reference
https://osf.io/ne89a/files/osfstorage/6a3689cb33b86e3d1a86e142

- The Imaginary Axis of Technical Analysis: How Complex Numbers Turn Chart Folklore into Market Pressure Geometry
https://osf.io/yucvm/files/osfstorage/6a4b942006735c3ce6daa274

- A Rigorous Mathematical Grammar And Checklist That Ensure Nature-Inspired Systems Are Stable, Bounded, And Economically Viable
https://osf.io/hj8kd/files/osfstorage/6a500b7bbdb5870c2c7afb69

- From Fundamental Physics to Purpose-Matched AI Agents
4π Spinor Closure, Hidden Control Stacks, and Environment-Aware Runtime Design
https://osf.io/hj8kd/files/osfstorage/6a4f89f3eef0d1166c5b9338

- From Physics to AI Design: A Rosetta Stone for Runtime Architecture
https://osf.io/hj8kd/files/osfstorage/69d5023f5cdefa314c3eb654

- Proto-Eight Dynamics (P8D): a small, testable model of how growth actually works【先天八卦動力學】
https://osf.io/9rdsc/files/osfstorage/68b71c00b65e7b0e352c22f6

- From Interfaces to Isomorphisms: A Protocol-Bound Theory of World Formation
How Bounded Observers Turn Fields into Operational Worlds — and Why Physics, Life, Organizations, Finance, Law, and AI Reuse the Same Grammar
https://osf.io/ae8cy/files/osfstorage/69ffbfc888878a0f3e78fda2

- Philosophical Interface Engineering 1 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Philosophical Interface Engineering 2 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Philosophical Interface Engineering 3 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Life as a Dual Ledger: Signal – Entropy Conjugacy for the Body, the Soul, and Health
https://osf.io/s5kgp/files/osfstorage/690f973b046b063743fdcb12

© 2026 Danny Yeung. All rights reserved. 版权所有 不得转载

Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT 5.6, Google AI, Gemini 3.X, NoteBookLM, X's Grok, Claude' Sonnet 5 language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual, interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.